

Chapter I

Power System Generation, Transmission and Distribution

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Basics of power system
- Structure of power systems
- Basic power generation concepts
- Thermal station
- Coal and ash handling plant
- Types of boilers
- Types of super heaters
- Types of condensers
- Types of hydro power stations
- Mass curve

BASICS OF POWER SYSTEM

Complex Power

The total power in a single-phase AC system at any instant is the product of the instantaneous voltage and instantaneous current. For a single-phase AC circuit, the voltage and current phasors are conventionally written as

$$u_{an} = V_{\max} \cos \omega t$$

$$i_{an} = I_{\max} \cos(\omega t - \theta)$$

The instantaneous power is

$$S = u_{an} i_{an} = V_{\max} I_{\max} \cos \omega t \cos(\omega t - \theta)$$

The angle ' θ ' is positive for current lagging the voltage and negative for leading current.

$$'S' = \frac{V_{\max} I_{\max}}{2} \cos \theta (1 + \cos 2\omega t) + \frac{V_{\max} I_{\max}}{2} \sin \theta \sin^2 \omega t$$

The above equation shows that the term containing ' $\cos \theta$ ' is always positive and has an average value of $P = \frac{V_{\max} I_{\max}}{2} \cos \theta$

RMS value of current and voltage will give the power expression $P = |V| |I| \cos \theta$,

This average power ' P ' is also called the real or active power. Unit for this power is watt.

The term containing $\sin \theta$ component of the instantaneous power ' S ' is called the 'instantaneous reactive power'. The maximum value of the pulsating power, designated Q is called reactive power or reactive volt ampere. The units for Q is VARS (Volt Ampere reactive), the reactive power is

$$Q = \frac{V_{\max} I_{\max}}{2} \sin \theta$$

$$Q = |V| |I| \sin \theta$$

The square root of the sum of the squares of P and Q is equal to the product of $|V|$ and $|I|$.

$$\sqrt{P^2 + Q^2} = \sqrt{(|V| |I| \cos \theta)^2 + (|V| |I| \sin \theta)^2} = |V| |I|$$

In a simple series circuit, when ' Z ' is equal to $R + jX$

$$P = |I|^2 |Z| \cos \theta = |I|^2 R$$

$$Q = |I|^2 |Z| \sin \theta = |I|^2 X$$

$$\therefore |Z| \cos \theta = R$$

$$\therefore |Z| \sin \theta = X$$

The cosine of the phase angle ' θ ' between the voltage and the current is called the power factor. An inductive circuit is said to have a lagging power factor and a capacitive circuit is said to have a leading power factor. Power factor indicated whether the current

is lagging or leading the applied voltage, the power factor is therefore

$$\cos \theta = \cos \left\{ \tan^{-1} \left(\frac{Q}{P} \right) \right\} \text{ or}$$

$$\cos \theta = \frac{P}{\sqrt{P^2 + Q^2}}$$

Phasor representation of the complex power is given as

$$|S| = \sqrt{P^2 + Q^2}$$

$$\theta = \tan^{-1} \left(\frac{Q}{P} \right)$$

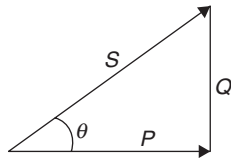


Figure 1 Power triangle of inductive load

Solved Examples

Example 1: A 15 HP, 440 V, three-phase motor is operating at full load, 85% efficiency and 90% power factor lagging, the real and reactive power drawn by the motor is

- (A) 13,169 W, 14,662 VAR
 (B) 13,169 W, 6365 VAR
 (C) 6378 W, 13169 VAR
 (D) 14,662 W, 13,169 VAR

Solution: (B)

$$\text{Current } (I) = \frac{15 \text{ HP} \times 746}{\sqrt{3} \times 440 \times 0.85 \times 0.9} = 19.2 \text{ A}$$

Real power drawn by the motor

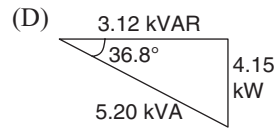
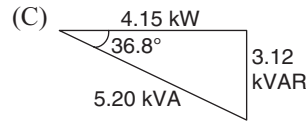
$$= \sqrt{3} V_L I_L \cos \theta = \sqrt{3} \times 440 \times 19.2 \times 0.9 = 13,169.12 \text{ W}$$

Reactive power drawn by the motor $= \sqrt{3} V_L I_L \sin \theta$

$$P = \sqrt{3} \times 440 \times 19.2 \times 0.435 = 6365 \text{ VAR}$$

Example 2: A 25 kVA, 200 V, 50 Hz, 3- ϕ alternator supplies a line current of 15 A per phase at 0.8 lagging power factor and at rated voltage under balance steady-state condition. For this operating condition, the power triangle is

- (A) (B)



Solution: (C)

Real power

$$\sqrt{3} V_L I_L \cos \theta = \sqrt{3} \times 200 \times 15 \times \cos \theta = 4.15 \text{ kW}$$

Reactive power

$$\sqrt{3} V_L I_L \sin \theta = \sqrt{3} \times 200 \times 15 \times \sin \theta = 3.12 \text{ kVAR}$$

$$\text{Complex power} = \sqrt{P^2 + Q^2} = 5.20 \text{ kVA}$$

Example 3: A single-phase load is supplied by a single-phase voltage source. If the current flowing from the load to the source is $10 \angle -160^\circ \text{ A}$ and if the voltage at the load terminals is $100 \angle 60^\circ \text{ V}$ then the

- (A) Load absorbs both real and reactive power
 (B) Load delivers both real and reactive power
 (C) Load absorbs real power and delivers reactive power
 (D) Load delivers real power and absorbs reactive power

Solution: (A)

$$\text{Voltage across load } (V) = 100 \angle 60^\circ \text{ V}$$

$$\begin{aligned} \text{Current entering load } (I) &= -10 \angle -160^\circ \text{ V} \\ &= 10 \angle 20^\circ \end{aligned}$$

Complex power $= VI$

$$= 100 \angle 60^\circ \times 10 \angle 20^\circ = 173.6 + j984.8$$

P and Q are positive, so load absorbs both P and Q .

STRUCTURE OF POWER SYSTEMS

The flow of the electric power from the generating station to the consumer is called an electric power system or electric supply systems. It consists of the following important components.

1. Generating station
2. Transmission network
3. Distribution network

The figure shows the line diagram of a typical transmission, distribution scheme and the complete flow of an electric power from generation to the consumer premises.

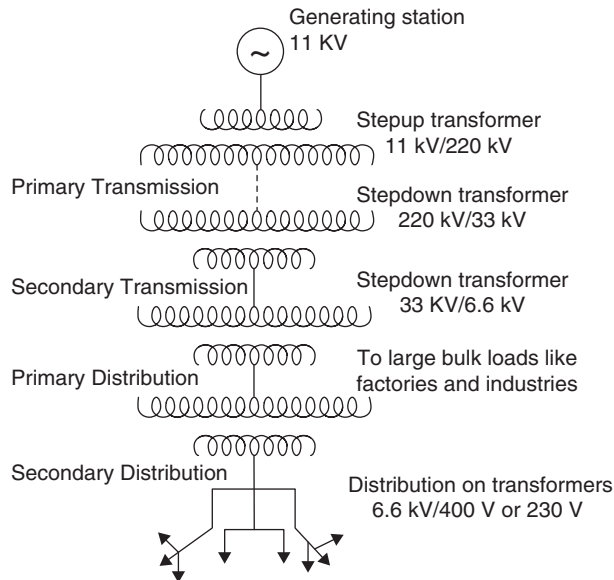


Figure 2 Line diagram of typical transmission and distribution on system

At the generating stations, an electric power is generated with the help of three-phase alternators running in parallel. In the scheme shown in figure, the voltage level is 11 kV but the voltage level can be 6.6 kV, 22 kV or 33 kV depending upon the capacity of the generating station. After generating station, actual transmission and distribution starts, the overall scheme can be divided into four sections which are,

Primary Transmission

It is basically with the help of overhead transmission lines. For the economic aspects, the voltage level is increased to 132 kV, 22 kV or more, with the help of step-up transformers. Hence, this is also called high voltage transmission.

Secondary Transmission

The primary transmission line continuous transmission towers till the receiving stations. At the receiving station, the voltage level is reduced to 22 kV or 33 kV using the step-down transformers. This is secondary transmission; the conductors used for the secondary transmission are called feeders.

Primary Distribution

At substations the voltage level is reduced to 6.6 kV, 3.3 kV or 11 kV with the help of step-down transformers. It uses three-phase three wire underground systems. And the power is further transmitted to the local distribution. For the large consumers like factories and industries, the power is directly transmitted to such loads from a substation.

Secondary Distribution

At the local centres, there are step down distribution transformers. The voltage levels of 6.6 kV, 11 kV is further reduced to 400 V using distribution transformers and

service mains to the consumers. This is secondary distribution also called low voltage distribution.

BASIC POWER GENERATION CONCEPTS

Electricity is not often generated at a power station by electromechanical generators, primarily driven by heat engines fuelled by chemical combustion or nuclear fission but also may be by other means such as kinetic energy of flowing water and wind. The types of electric generation concepts are as follows:

1. Thermal power plants
2. Hydroelectric power plants
3. Nuclear power plants
4. Diesel power plants
5. Wind plants
6. Solar power generation, etc.

Thermal Station

Definition

A generating station which converts heat energy of coal combustion into electrical energy is known as steam station (thermal station).

Advantages

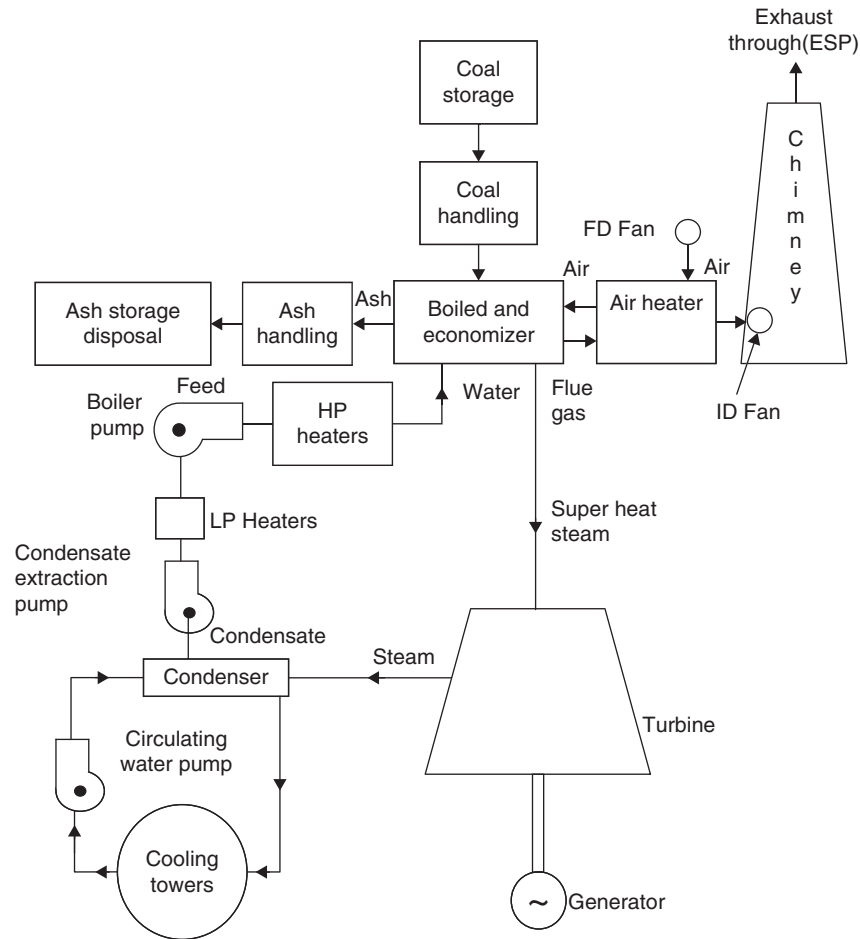
1. Less initial cost.
2. It requires less space as compared to hydroelectric power station.
3. Cost of generation is less than that of diesel power station.

Disadvantages

1. It pollutes the atmosphere due to production of large amount of smoke and fumes.
2. Its running cost is more compared to hydro electric plant.

Points to decide the selection of site for thermal power station

1. Large quantity of cooling water should be available for the condensers, etc.
2. The fuel to be used in the power station should be cheap and available nearby or ample facilities for its transportation to the site should be available.
3. The soil should be such that, it should provide good and firm foundation to the building and the plants.
4. The site should not be surrounded by residential buildings.
5. The future expansion of power station should be possible.
6. Ample accommodation for the operational and maintenance staff should be available nearby at reasonable rates.
7. Facility for the disposal of ash, etc. should be available.
8. The cost of land should be reasonable.



Schematic Arrangement of Steam Power Station

1. Coal and ash handling arrangement
2. Steam-generating plant
3. Steam turbine
4. Alternator
5. Feed water
6. Cooling arrangement

Coal and Ash Handling Plant

The coal is transported to the power station by road or rail and is stored in a coal storage plant. Storage of coal is primarily a matter of protection against coal strikes, failure of transportation system and general coal shortages.

From coal storage plant, coal is delivered to the coal handling plant where it is pulverized. The pulverized coal is fed to the boiler by conveyor belt. The coal is burnt in the boiler and the ash produced after complete combustion of coal is moved to the ash handling plant and then delivered to the ash storage plant for disposal.

Steam-generating Plant

1. **Boiler:** The heat of combustion of coal in the boiler is utilized to convert water into steam at high temperature and pressure.

2. **Superheater:** The steam produced in the boiler is wet and is passed through a superheater where it is superheated (i.e. steam temperature increased above that of boiling point of water).
 - Advantage is that efficiency is increased
3. **Economizer:** An economizer is a feed water heater, i.e. increases temperature of feed water from extracting heat of flue gases.
4. **Air Preheater:** It increases temperature of the air supplied for coal burning by delivering heat from flue gases.

Advantages

- Thermal efficiency increases.
- Steam capacity per square metre of boiler surface is increased.

Steam Turbine

The dry and superheated steam from the superheater is fed to the steam turbine and it converts into mechanical energy. After giving heat energy to the turbine, the steam is exhausted to the condenser, which condenses exhausted steam.

Alternator

The alternator converts mechanical energy into electrical energy. The electrical energy is delivered to busbars.

Feed Water

The condensate from the condenser is used as feed water to the boiler. The feed water on its way to the boiler is heated by water heaters and economizer. This helps in raising the overall efficiency of the plant.

Cooling Arrangement

Water is circulated through the condenser, which takes up heat of the exhausted steam. In case availability of water throughout the year is not assured, cooling tower is used.

Types of Boilers

1. Water tube boiler
 2. Fire tube boiler
- In water tube boiler, water flows through the tubes and the hot gases of combustion flow over these tubes. Advantages are less space requirement, high pressure and less liable to explosion, etc.
 - In fire tube boiler, the hot products of combustion pass through the tubes surrounded by water. Disadvantage is low pressure.

Types of Superheaters

1. Radiant superheater
 2. Convection superheater
- In Radiant superheater, it is placed in the furnace between water walls and receives heat from the burning fuel through radiation process. Disadvantage is that it may get over heated and temperature of superheater falls with increase in steam output.
 - In convection superheater, it is placed in the boiler tube bank and receives heat from flue gases entirely through the convection process.

Advantage is temperature of superheater increases with increase in steam output.

Types of Air-pre Heaters

1. Recuperative type
 2. Regenerative type
- In recuperative type, air-heater consists of a group of steel tubes. The flue gases are passed through the tubes while air flows externally to the tubes.
 - In regenerative type, air-heater consists of slowly moving drum. The flue gases flow continuously on one side of the drum and air on the other side.

Types of Condensers

1. Jet Condenser
 2. Surface condenser
- In a jet condenser, cooling water and exhausted steam are mixed together. Therefore, the temperature of

cooling water and condensate is the same when leaving the condenser.

Advantages

Low initial cost, less floor area required, less cooling water required and low maintenance charges.

Disadvantages

Condensate is wasted and high power is required for pumping water.

- In a surface condenser, there is no direct contact between cooling water and exhausted steam. It consists of a bank of horizontal tubes enclosed in a cast iron shell. The cooling water flows through the tubes and exhausted steam flows over the surface of the tubes.

Advantages

Condensate can be used as feed water, less pumping power required and creation of better vacuum at turbine exhaust.

Disadvantages

High initial cost, large floor area requirement and high maintenance charges.

Types of Steam Turbines

1. Impulse turbine
 2. Reaction turbine
- In an impulse turbine, the steam expands completely in the stationary nozzles (or flexi blades), the pressure over the moving blades remaining constant. In doing so, the steam attains a high velocity and impinges against the moving blades.
 - In reaction turbine, the steam is partially expanded in the stationary nozzles; the remaining expansion takes place during its flow over the moving blades.

Efficiency of Steam Power Station

Thermal Efficiency

The ratio of heat equivalent to the mechanical energy transmitted to the turbine shaft, to the heat of combustion of coal is known as thermal efficiency.

$$\eta_{\text{Thermal}} = \frac{\text{Heat equivalent to the mechanical energy transmitted to the turbine shaft}}{\text{Heat of coal combustion}}$$

- The thermal efficiency of a modern steam power station is about 30%

Overall Efficiency

The ratio of heat equivalent of the electrical output to the heat of combustion of coal is known as overall efficiency.

$$\eta_{\text{overall}} = \frac{\text{Heat equivalent of the electrical output}}{\text{Heat of combustion of coal}}$$

- The overall efficiency of steam power station is about 29%
- Overall efficiency = Thermal efficiency $\times$ Electrical efficiency

Hydroelectric Power Station

Definition

A generating station which utilizes the potential energy of water at a high level for the generation of electrical energy is known as a hydroelectric power station.

Advantages

1. It is quiet and clean.
2. It requires less maintenance.
3. No fuel requirement.
4. It can be put into service instantly.
5. It has longer life.
6. In addition to the generation of electrical energy, they also help in irrigation and controlling floods.

Disadvantages

1. It involves high capital cost.
2. Availability of huge amount of water.
3. Skilled and experienced hands are required to build the plant.
4. Increased cost of transmission lines because of remote location of plants.
5. Long dry season may affect the delivery of power.

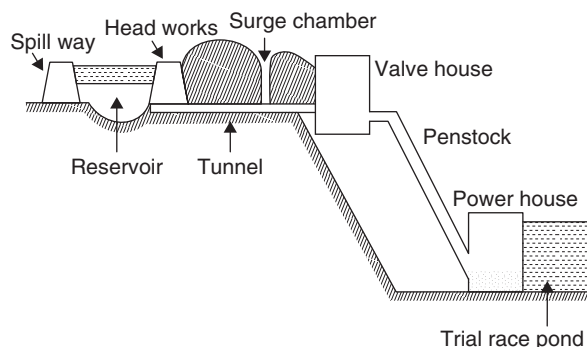


Figure 3 A typical layout of a Hydro plant

Selection of Site for Hydroelectric Station

1. Sufficient quantity of water at reasonable level should be available.
2. The site should allow for strong foundation with minimum cost.
3. No possibility of future sources of leakage of water.
4. Reservoir should have large catchment area.
5. Local supplies of sand, gravel, etc., should be available.
6. The land for the construction of the plant should be available at reasonable price.

7. The site selected for hydroelectric plant should be accessible by rail and road so that necessary equipment and machinery could be easily transported.

Components of Hydroelectric Plant

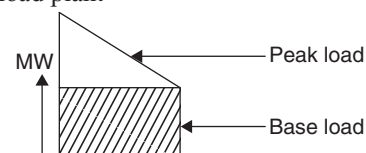
1. **Dam:** A dam is a barrier which stores water and creates water head.
2. **Spillways:** It discharges the surplus water to downstream side of the dam.
3. **Surge tank:** A surge tank is located near the beginning of the conduit. It overcomes the abnormal pressure in the conduit when load on the turbine falls. It acts as a reservoir during increase of load on the turbine.
4. **Penstocks:** Penstocks are open or closed conduits which carry water to the turbines.
 - Low-pressure penstocks
E.g.: Cement canals
 - High pressure penstocks
E.g.: Steel pipes
5. **Trash rack:** It is used to avoid the entry of debris into the pen stock.
6. **Forebay:** It is a small pondage where extra water is stored in it and regulates the load variations.
7. **Reservoir:** The area to store water.
8. **Turbines:** It converts energy of falling water into mechanical energy.
9. **Alternator:** It converts mechanical energy into electrical energy.

Types of Turbines

1. Impulse turbine
It has high power output and low efficiency.
E.g.: Pelton wheel
2. Reaction turbines
It has low power output and high efficiency turbines.
E.g.: Kaplan, Francis and Propeller turbines.

Types of Hydro Power Stations

- Based on the water availability
 1. Run off river plants
 - (a) Without pondage
 - (b) With pondage
- Based on the head available
 1. Low head (<30 m)
 2. Medium head (30–50 m)
 3. High head (>50 m)
- Based on the nature of load
 - (1) Base load plant
 - (2) Peak load plant



Type of Alternator used for Hydroelectric Power Station

Air cooled salient pole construction, vertical configuration, large diameter and small axial length alternator is used.

Types of turbines used for hydroelectric power station based on head

1. **High head:** Pelton wheel turbines
2. **Medium head:** Francis, propeller turbines
3. **Low head:** Kaplan turbines.

Power

$$P = \frac{0.736 QWh\eta}{75} \text{ kW}$$

where

Q = Discharge in m^3/s

W = Water density in $1000 \text{ kg}/\text{m}^3$

h = Height of water in metre

η = Efficiency

Specific Speed

With available head of 1 m to develop 1 HP power output, the speed at which the turbine will rotate is called specific speed.

$$N_s = \frac{N\sqrt{P}}{H^{\frac{5}{4}}}$$

where

P = Power in HP

N = Actual speed in rpm

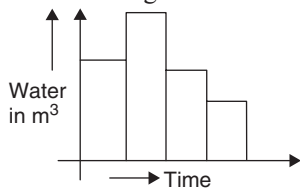
H = Head of water

$$N_s = \frac{1.145N\sqrt{P}}{H^{\frac{5}{4}}}$$

if P is in kW.

Hydrograph

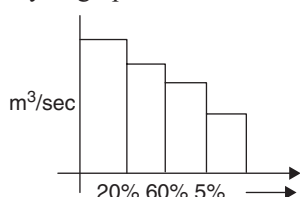
It is a graph between discharge of water in m^3 to time.



- It gives the variation of water discharge from time to time.
- The area under this graph gives the total quantity of water available at a particular period.

Flow Duration Curve

It is drawn from hydrograph.



It is drawn in descending order of water discharge.

Mass Curve

A mass curve is a graph drawn between cumulative volume of water versus time.

- A mass curve gives the storage requirement.

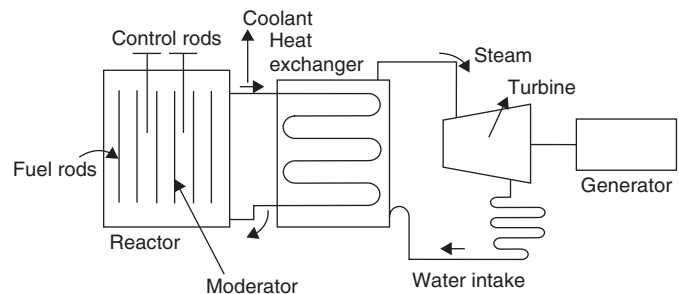
Nuclear Power Station

Definition

A generating station in which nuclear energy is converted into electrical energy is known as nuclear power station.

Advantages

1. Fuel requirement is quite small.
2. It requires less space.
3. Running charges are low.
4. It can be located near the load centres because it does not require large quantities of water and it need not be near coal mines.
5. Such plants can ensure continued supply because of large deposits of fuel available.
6. Reliability operation.



Disadvantages

1. Fuel is expensive and difficult to recover.
2. Capital cost is very high.
3. Radioactive pollution problems.
4. It requires greater technical staff.
5. Not suited for varying loads.

Selection of site for nuclear power station

1. Plant should be located where ample quantity of water is available.
2. Plant should have adequate arrangement for disposal of radioactive waste.
3. It should be quite distant from populated areas as there is a danger of presence of radioactivity in the atmosphere near the plant.
4. Transportation facilities should be available.

Schematic Arrangement of Nuclear Power Station

1. Nuclear reactor
2. Heat exchanger
3. Steam turbine
4. Alternator

Components of Nuclear Reactor

1. Reactor core: It consists of fuel rods.
2. Moderator
 - The purpose of moderator is to slow down the fast neutrons.
 - It should have low molecular weight.
 - Materials that can be used as moderator are Beryllium, Graphite, H_2O and Heavy water (D_2O).
3. Control rods: The purpose of control rods is to absorb the fast neutrons and they control the value of K (multiplication factor).
 - Materials that can be used for control rods are cadmium, boron and Hafnium.
4. Reflector: The purpose of this reflector is to bounce back escaping neutrons into reactor core.
Materials: Reactor grade graphite
5. Shielding: The purpose of shielding is to avoid α , β , γ radiations outside the reactor core.
Materials: Stainless steel, etc.
6. Coolant: The purpose of coolant is to absorb the heat from the reactor core and exchange with water in heat exchanger. Materials are
Liquid coolants $\rightarrow H_2O, D_2O$
Liquid metals $\rightarrow Na$ and K
Gases $\rightarrow CO_2, H_2$

Heat Exchanger

The coolant gives up heat to the heat exchanger which is utilized in rising the steam. After giving up heat, the coolant is again fed to the reactor.

Steam Turbine

The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to the condenser. The condenser condenses the steam which is fed to the heat exchanger through feed water pump.

Alternator

Alternator is a device which converts mechanical energy into electrical energy.

Multiplication factor (K)

$$K = \frac{\text{No. of neutrons released in fission}}{\text{No. of neutrons released in preceding fission}}$$

- The value of K can be controlled by control rods.
- $K > 1$, for increasing generation
 $K = 1$, for constant generation
 $K < 1$, for decreasing generation
 $K = 0$, for shut-down generation

Fission Material

It is the material used as fuel in nuclear power station.

1. Natural Uranium

$$\begin{cases} 99.3\% U^{238} \\ 0.7\% U^{235} \end{cases}$$

2. Enriched Uranium
3. Thorium
4. Plutonium

- ‘Enriched Uranium’ is used in Indian nuclear power stations
- Fission materials can be divided into two types.
 1. Fissile materials
 2. Fertile materials
- Fissile materials are those with which chain reactions start directly.
Ex: U_{235} , U_{233} and Pu_{239}
- Fertile materials are those with which chain reactions start indirectly.
Ex: U^{238} , Th^{232}

Critical Mass

It is the mass to be added in the chain reaction to maintain the ‘ K ’ value at ‘1’ only.

Comparison of Various Power Plants

S. No.	Item	Steam Power station	Hydroelectric power plant	Diesel power plant	Nuclear power plant
1.	Site	Located at water and coal available places	Located at where reservoirs can be obtained by constructing dam Example: hill areas	Located at any place	Located away from thickly populated areas
2.	Initial cost	Lower than hydro and nuclear plants	Very high	Less compared to other plants	Highest
3.	Running cost	Higher than hydro and nuclear plants	Very less	Highest	Except hydro, it has minimum running cost
4.	Efficiency	Least about 25%	About 85%	About 35%	More efficient than steam power station
5.	Maintenance cost	Quite high	Quite low	Less	Very high
6.	Transmission and distribution	Quite low	Quite high	Least	Quite low
7.	Starting	Requires lot of time	Started instantly	Started quickly	Started easily

ECONOMIC ASPECTS

Commonly used terms in system operation are discussed in the following sections.

Connected Load

It is the sum of the continuous ratings of all load-consuming apparatus connected to the system.

Maximum Demand

It is the highest load demand on the power system during a given period (day, month, and year).

Firm Power

It is the power intended to be always available.

Cold Reserve

It is the reserve generating capacity available for service but is not in operation.

Hot Reserve

It is the reserve generating capacity in operation but not in service.

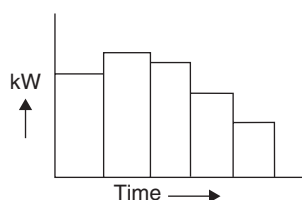
Spinning Reserve

It is the generating capacity which is connected to the bus and is ready to take load.

Factors Affecting the Cost of Generation

Load Curve

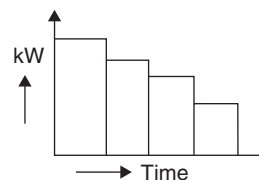
The curve showing the variation of load on power station with reference to time is known as a load curve.



- The variation of quantity w.r.t time, it is known as 'chronological curve'
- Load curve can have -ve or +ve slopes.
- The area under the curve gives the total number of units generated.
- Average demand = $\frac{\text{Total no. of units generated in a period}}{\text{No. of hours in that period}}$

Load Duration Curve

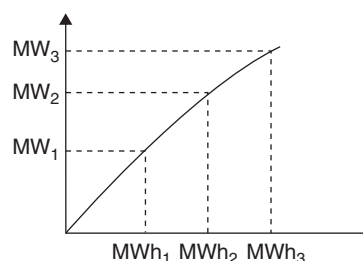
It is a graph drawn from load curve by rearranging load elements in descending order w.r.t their duration of occurrence.



- The curve will have only negative slope.
- The area under this curve gives the no. of units generated.

Integrated Load Duration Curve

It is a graph drawn between MW versus MWh.



- It gives the cumulative value of energy when load is changing.
- By using integrated load curve, we can estimate the number of units generated when the load is changing from MW_1 to MW_2 .

Load Factor (LF)

It is the ratio of average demand to maximum demand

$$\text{Load factor} = \frac{\text{Average demand}}{\text{Maximum demand}}$$

- Load factor is less than 1.
- Higher is the load factor, lesser will be the cost per unit.

Demand Factor (DF)

It is the ratio of maximum demand to connected load

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Connected load}}$$

- It is usually less than 1.
- Low demand factor results in less capital cost.

Diversity Factor (Div.F)

$$\text{Diversity factor} = \frac{\text{Sum of individual maximum demands}}{\text{Maximum demand on power station}}$$

- It is usually more than 1 only.
- Div.F ↑, Max. Demand ↓, Installed capacity ↓ cost ↓

Utilization Factor (UF)

$$\text{U.F} = \frac{\text{Maximum Demand}}{\text{Installed capacity}}$$

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- It is always ≤ 1 .
- If UF is high, i.e. maximum demand is high, then the power station is having proper utilization.
- Reserve capacity = Installed capacity – Maximum demand.
- If reserve capacity is zero, utilization factor is one.

Capacity Factor (CF)

$$\text{Capacity factor} = \frac{\text{Average demand}}{\text{Installed capacity}}$$

$$= \frac{\text{Actual energy produced}}{\text{Max energy that could have been produced}}$$

- $CF = LF \times UF$

Plant use Factor

$$\text{Plant use factor} = \frac{\text{Station output in kWh}}{\text{Plant capacity} \times \text{Hours of use}}$$

- Plant use factor is more than or equal to plant capacity factor.

Example 4: A generating station has a connected load of 550 MW. The maximum demand is 380 MW. The number of units generated per annum is 20×10^8 . The demand factor and load factor are

- (A) 0.69 and 0.60 (B) 0.60 and 0.69
(C) 1.45 and 0.60 (D) 0.69 and 1.66

Solution: (A)

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Connected load}}$$

$$= \frac{380}{550} = 0.69$$

$$\text{Average demand} = \frac{\text{No. of units generated per annum}}{8760} \text{ kW}$$

$$= \frac{20 \times 10^8}{8760} = 228.31 \text{ MW}$$

$$\text{Load factor} = \frac{\text{Average demand}}{\text{Maximum demand}}$$

$$= \frac{228.31}{380} = 0.60$$

Example 5: A generating station has a maximum demand of 1000 MW. The annual load factor is 70% and plant capacity factor is 60%. The reserve capacity is

- (A) 700 MW (B) 1.166 MW
(C) 167 MW (D) 1000 MW

Solution: (C)

$$\text{Average demand} = 0.7 \times 1000 = 700 \text{ MW}$$

$$\text{Plant capacity} = \frac{\text{Average demand}}{\text{Capacity factor}}$$

$$= \frac{700}{0.5} = 1166.6 \text{ MW}$$

$$\text{Reserve capacity} = \text{Plant capacity} - \text{Maximum demand}$$

$$= 1166.6 - 1000 = 166.6 \text{ MW} \approx 167 \text{ MW}$$

Example 6: Which of the following stations has minimum running cost?

- (A) Hydroelectric (B) Nuclear
(C) Diesel (D) Coal

Solution: (A)

No fuel cost

Example 7: For a hydroelectric power plant with an available head of 60 m, which of the following turbines is most likely to be used?

- (A) Pelton wheel (B) Francis
(C) Kaplan (D) None of the above

Solution: (A)

Pelton wheels are typically used for high heads.

POWER SYSTEM REPRESENTATION

Single Line Diagram

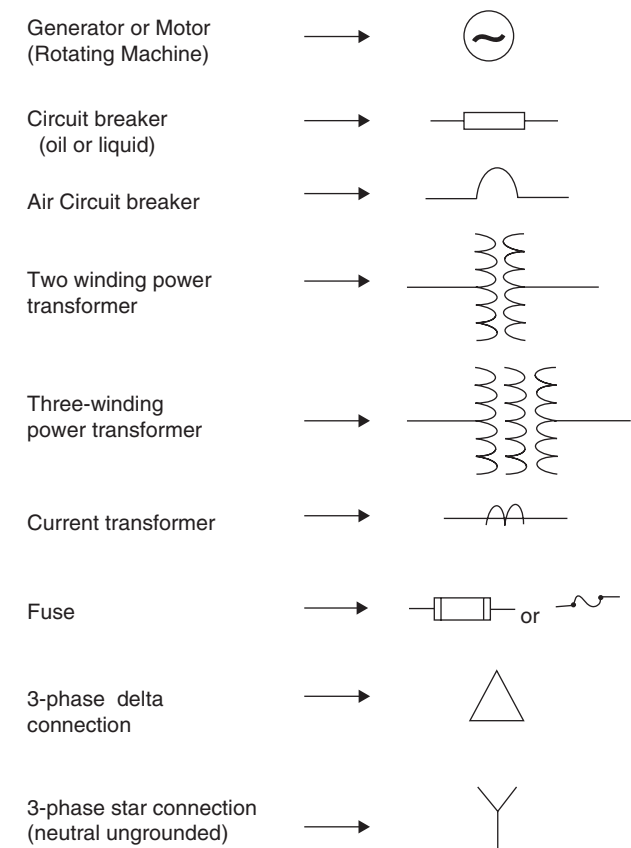


Figure 4 (Continued)

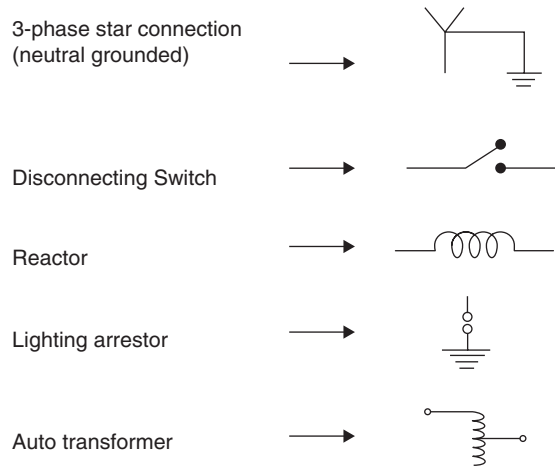


Figure 4 Symbols for representing single line diagram

Almost all modern power systems are three-phase systems with the phases of equal magnitude and equal phase difference (i.e. 120°). These three-phase balanced systems are always solved as a single-phase circuit composed of one of the three lines and neutral return and this is sufficient to give a complete analysis. Such a simplified diagram of an electric system is called a 'one-line diagram' or 'single-line diagram'. Combined with a standard set of symbols for electric components, such one-line diagrams provide a compact way to represent information. The purpose of the one-line diagram is to supply in concise form, the significant information about the system, the importance of different features of a system varies with the problem under consideration, and the amount of information included on the diagram depends on the purpose for which the diagram is intended. Figure 3 below shows the symbols for representing the components of a three-phase power system.

Above Figure 4 shows the single line diagram of an electrical power system. Two generators grounded through reactors are connected to a bus and through a step-up transformer to a transmission line. Another generator, grounded through reactor, is connected to a bus and through a transformer to the opposite end of the transmission line. A load is connected at each bus.

Impedance and Reactance Diagrams

The impedance diagram on single-phase basis for use under balanced operating conditions can be easily drawn from the one-line diagram. For the system of Figure 4, the impedance diagram is shown in Figure 5.

No currents flow in the ground under balanced conditions and the neutral of the generators are at the potential of the neutral of the system, so the impedance diagram does not include the current limiting impedances shown in the one line diagram between the neutral of the generator and ground.

Since the shunt current of a transformer is usually insignificant compared with the full load current, the shunt admittance is usually omitted in the equivalent circuit of the transformer.

The inductive reactance of a system is much larger than its resistance. So, the resistance is neglected in fault calculations.

Synchronous motor loads are always included in making fault calculations, since their generated emfs contribute to the short-circuit current. Induction motors are represented by a generated emf in series with an inductive reactance if the diagram is to be used to determine the current immediately after the occurrence of a fault. Induction motors are ignored in computing the current a few cycles after the fault occurrence because the current contributed by an induction

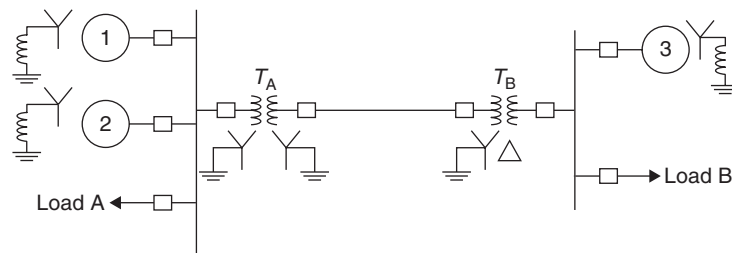


Figure 5 Single-line diagram of an electrical power system

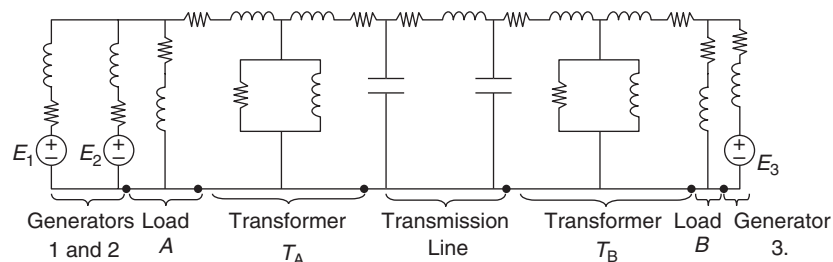


Figure 6 The per-phase impedance diagram

motor dies out very quickly after the induction motor is short circuited. Per-phase reactance diagram after neglecting all static loads, resistances, shunt admittances of each transformer and the capacitance of the transmission line is shown in Figure 6. The per-phase impedance and reactance diagrams are sometimes called the per-phase positive sequence diagram.

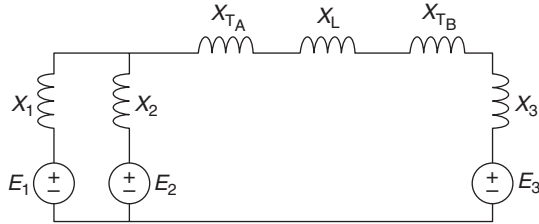


Figure 7 Simplified reactance diagram

PER UNIT SYSTEM

The quantities in a power system (i.e., voltage, current, volt amperes and impedance) are often expressed as a per cent or per unit of a base or reference value specified for each. The per-unit value of any quantity is defined as the ratio of the actual quantity to its base quantity.

$$\text{Per Unit Value} = \frac{\text{The actual value in any units}}{\text{The base or reference value in the same units}}$$

The ratio in per cent is 100 times the value in per unit. Both the per cent and per-unit methods of calculation are similar and more informative than the use of actual quantities. The per-unit method has advantages over per cent calculation method because the product of two quantities expressed in per unit is expressed in per unit itself, but the product of two quantities expressed in per cent must be divided by 100 to obtain the result in per cent.

There are several reasons for using a per-unit system.

1. Similar apparatus (generators, transformers, lines) will have similar per-unit impedances and losses expressed on their own ratings, regardless of their absolute size.
2. Use of the constant $\sqrt{3}$ is reduced in three-phase calculations.
3. Per-unit quantities are the same on either side of a transformer, independent of voltage level.
4. By normalizing quantities to a common base, the calculations are simplified.

A per unit system provides units for power, voltage, current and impedance. Only two of these are independent, usually power and voltage. Generally base values of power and voltage are chosen. Once the base power and the base voltage are chosen, the base current and the impedances are determined by the natural laws of electrical circuits.

The relationship between quantities in a per-unit system depends on whether the system is single phase or three phase.

Single Phase

Assuming that the independent base values are power and voltage,

$$\text{Base volt amperes} = S_{\text{base}} = 1 \text{ p.u.}$$

$$\text{Base voltage} = V_{\text{base}} = 1 \text{ p.u.}$$

$$\text{Base active power} = P_{\text{base}} = S_{\text{base}} \cdot \cos \phi$$

$$\text{Base reactive power} = Q_{\text{base}} = S_{\text{base}} \cdot \sin \phi$$

$$\text{Base current} = I_{\text{base}} = \frac{(VA)_{\text{base}}}{V_{\text{base}}} = \frac{S_{\text{base}}}{V_{\text{base}}} = 1 \text{ p.u.}$$

$$\text{Base impedance} = Z_{\text{base}} = \frac{V_{\text{base}}}{I_{\text{base}}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}$$

$$\text{Base admittance} = Y_{\text{base}} = \frac{1}{Z_{\text{base}}}$$

If the actual impedance is Z (ohms), its per-unit value is given by

$$\begin{aligned} Z_{(\text{p.u.})} &= \frac{Z}{Z_{\text{base}}} = \frac{Z(\text{ohms}) \times (VA)_{\text{base}}}{V_{\text{base}}^2} \\ &= \frac{Z(\text{ohms}) \times S_{\text{base}}}{V_{\text{base}}^2} \end{aligned}$$

Three Phase

Power and voltage are specified in the same way as single-phase system. But the difference is that the power is specified as total power (not per phase), and voltage is line-to-line voltage.

In three-phase systems the relations

$$P_{\text{base}} = S_{\text{base}} \cdot \cos \phi \text{ and } Q_{\text{base}} = S_{\text{base}} \sin \phi \text{ holds good.}$$

The apparent power S equals

$$S_{\text{base}} = \sqrt{3} V_{\text{base}} I_{\text{base}}$$

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3} V_{\text{base}}} = 1 \text{ p.u.}$$

$$Z_{\text{base}} = \frac{\frac{V_{\text{base}}}{\text{phase}}}{I_{\text{base}}} = \frac{V_{\text{base}}^2}{S_{\text{base}}} = 1 \text{ p.u.}$$

$$Y_{\text{base}} = \frac{1}{Z_{\text{base}}} = \frac{S_{\text{base}}}{V_{\text{base}}^2} = 1 \text{ p.u.}$$

If the actual impedance is Z (ohms), its per-unit value is given by

$$Z_{\text{p.u.}} = \frac{Z}{Z_{\text{base}}} = \frac{Z(\text{ohms}) \times S_{\text{base}}}{V_{\text{base}}^2}$$

If the impedance has to be represented in a new base value denoted as $Z_{\text{base new}}$ (Referred to $S_{\text{base new}}$ and $V_{\text{base new}}$)

$$Z_{\text{p.u. new}} = Z_{\text{p.u. old}} \times \left(\frac{V_{\text{base old}}}{V_{\text{base new}}} \right)^2 \times \frac{S_{\text{base new}}}{S_{\text{base old}}}$$

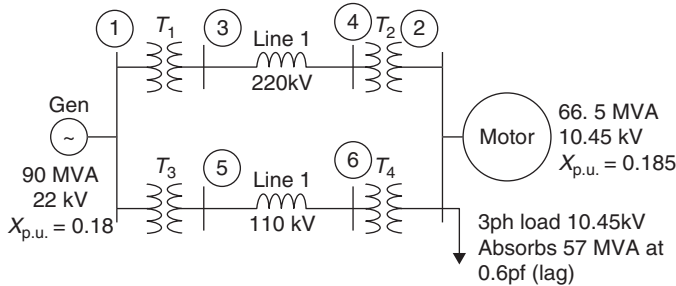
Note:

$$Z_{p.u.} \propto S_{base}$$

$$Z_{p.u.} \propto \frac{1}{V_{base}^2}$$

Example:

Single line diagram of a power system is shown in the figure. The system contains one generator, four transformers and two transmission lines. The system ratings and reactances are indicated in the figure. Draw the equivalent impedance diagram for the given system and per unit equivalent diagram with $S_{base} = 100$ MVA and $V_{base} = 22$ kV.



T_1 : 50 MVA, 22/220 kV. $X_{p.u.} = 0.1$,

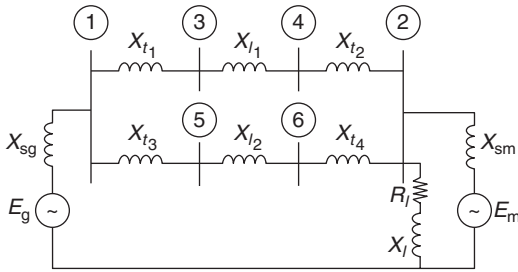
T_2 : 40 MVA, 220/11 kV. $X_{p.u.} = 0.06$,

T_3 : 40 MVA, 22/110 kV. $X_{p.u.} = 0.064$,

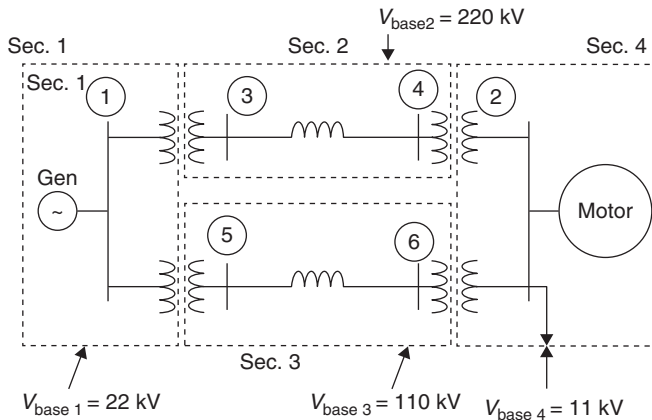
T_4 : 40 MVA, 110/11 kV. $X_{p.u.} = 0.08$,

Line 1 : 48.4 Ω , Line 2 : 65.43 Ω

Reactance Diagram



Per Unit Representation



Base impedance with given base values

$$Z_{base} = \frac{V_{base}^2}{S_{base}}$$

For Section I $Z_{base_1} = \frac{22^2 \times 10^6}{100 \times 10^6} = 4.84 \Omega$

For Section II $Z_{base_2} = \frac{220^2 \times 10^6}{100 \times 10^6} = 484 \Omega$

For Section III $Z_{base_3} = \frac{110^2 \times 10^6}{100 \times 10^6} = 121 \Omega$

For Section IV $Z_{base_4} = \frac{11^2 \times 10^6}{100 \times 10^6} = 1.21 \Omega$

Per unit impedance calculations $Z_{p.u.} = \frac{Z_{actual}}{Z_{base}}$

$$Z_{p.u., new} = Z_{p.u., old} \times \frac{S_{base new}}{S_{base old}} \times \left(\frac{V_{base old}}{V_{base new}} \right)^2$$

For generator 1, new per unit reactance

$$X_{sg} = X_{sg, old} \times \frac{100}{90} \times \left(\frac{22}{22} \right)^2 = 0.2 \text{ p.u.}$$

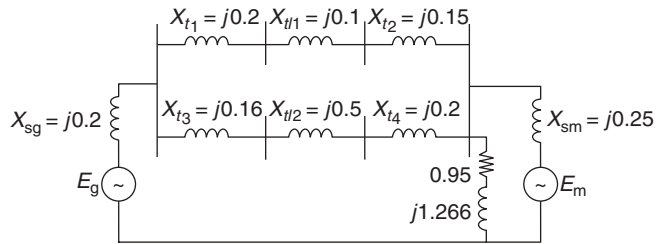


Figure 8 Per-unit impedance diagram

For Transmission line 1, $X_{l_{p.u.}} = \frac{48.4}{484} = 0.1 \text{ p.u.}$

For Transmission line 2, $X_{l_{p.u.}} = \frac{65.43}{121} = 0.5 \text{ p.u.}$

For Transformer 1, $X_{t_1} = 0.1 \times \frac{100}{50} \times \left(\frac{22}{22} \right)^2 = 0.2 \text{ p.u.}$

For Transformer 2,

$$X_{t_2} = 0.06 \left(\frac{100}{40} \right) \left(\frac{22}{22} \right)^2 = 0.15 \text{ p.u.}$$

For transformer 3,

$$X_{t_3} = 0.064 \left(\frac{100}{40} \right) \left(\frac{220}{220} \right)^2 = 0.16 \text{ p.u.}$$

For transformer 4,

$$X_{t_4} = 0.08 \left(\frac{100}{40} \right) \left(\frac{220}{220} \right)^2 = 0.2 \text{ p.u.}$$

For motor, $X_{sm} = 0.185 \left(\frac{100}{66.5} \right) \left(\frac{10.45}{11} \right)^2 = 0.25 \text{ p.u.}$

For three-phase load:

Power factor: $\cos^{-1}(0.6) = 53.13^\circ$

Load $S_{3\phi} = 57 \angle 53.13$

$$Z_{act} = \frac{V_{rated}^2}{S^*} = \frac{10.45^2}{57 \angle -53.13} = 1.1495 + j1.5326 \Omega$$

$$\text{Per unit impedance of 3-}\phi \text{ load} = \frac{1.1495 + j1.532}{1.21} = 0.95 + j1.2667 \text{ p.u.}$$

Example 8: The base impedance and base voltage of a 345 kV system are chosen to be 3000 A and 300 kV, respectively, the base impedance of the system is

- (A) 115 Ω (B) 100 Ω
(C) 10 Ω (D) 0.01 Ω

Solution: (B)

$$\text{Base impedance} = \frac{300 \times 10^3}{3000} = \frac{V_b}{I_b} = 100 \Omega$$

Example 9: Let a 10 kVA, 400/200-V transformer be approximately represented by a 4 Ω reactance referred to the low-voltage side, considering the rated values as base quantities, what is the transformer reactance as a per unit quantity.

- (A) 1 p.u. (B) 0.5 p.u.
(C) 0.25 p.u. (D) 1.5 p.u.

Solution: (A)

$$\text{Base impedance } (Z_B) = \frac{V^2}{S_B} = \frac{200^2}{10,000} = 4 \Omega$$

The per unit reactance referred to the low-voltage side is

$$\text{Per unit reactance} = \frac{4}{4} = 1 \text{ p.u.}$$

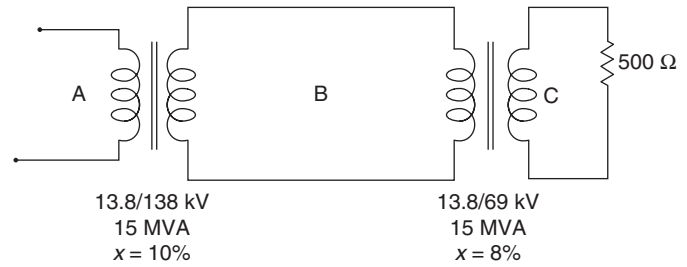
Example 10: The per-unit impedance of an alternator corresponding to base values 13.2 kV and 25 MVA is 0.18 p.u. The per unit value of the impedance for base values are 13.8 kV and 60 MVA in p.u. will be

- (A) 0.395 p.u.
(B) 0.472 p.u.
(C) 0.329 p.u.
(D) 0.082 p.u.

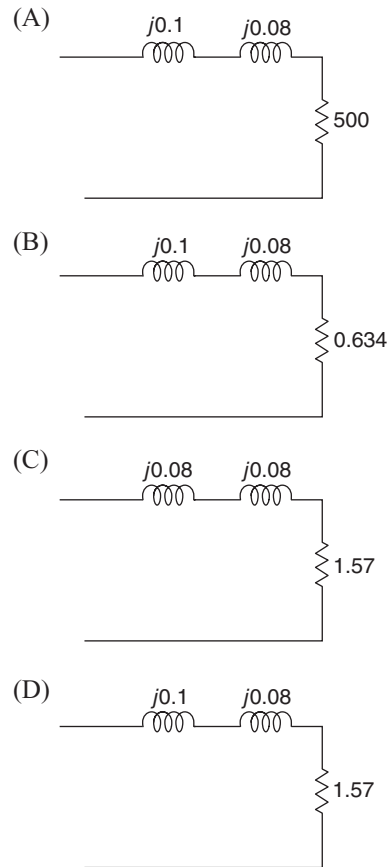
Solution: (A)

$$Z_{p.u.new} = Z_{p.u.old} \left(\frac{V_{base.new}}{V_{base.old}} \right)^2 \times \left(\frac{VA_{base.old}}{VA_{base.new}} \right) = 0.18 \times \left(\frac{60}{25} \right) \left(\frac{13.2}{13.8} \right)^2 = 0.395 \text{ p.u.}$$

Example 11: A single-phase system is shown in figure below:



With the base in 'A' circuit chosen as 13.8 kV and 15 MVA, the impedance diagram is



Solution: (D)

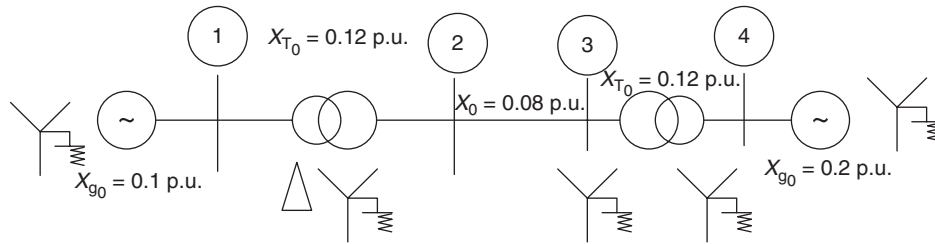
Chosen base values 13.8 kV and 15 MVA

Transformer 1 $X_{p.u.} = 0.1 \text{ p.u.}$ [old and new base values are same]

Transformer 2 $X_{p.u.} = 0.08 \text{ p.u.}$ [old and new values are same]

$$\text{Load } Z_{p.u.} = Z_{actual} \times \frac{S_{base}}{V_{base}^2} = 500 \times \frac{15 \times 10^6}{(69 \times 10^3)^2} = 1.57 \text{ p.u.}$$

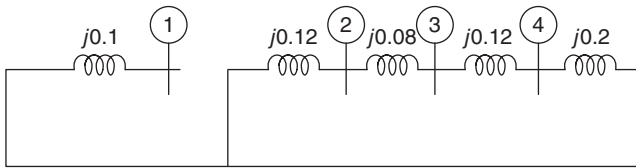
Example 12: The zero sequence reactance in p.u. are indicated in the network shown in the given below. The zero sequence driving point reactance of node-3 will be



- (A) $j0.2$ p.u. (B) $j0.32$ p.u.
(C) $j0.123$ p.u. (D) $j0.52$ p.u.

Solution: (C)

Reactance diagram is given by



$$\text{Driving point reactance at node 3} = j(0.12 + 0.08) \parallel j(0.12 + 0.2) = j \frac{0.2 \times 0.32}{0.52} = 0.123 \text{ p.u.}$$

MECHANICAL DESIGN OF TRANSMISSION LINES

Introduction

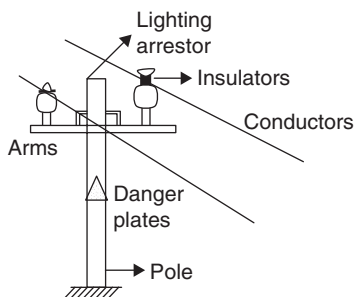
Electric power can be transmitted or distributed by means of overhead lines and underground cables, the underground cables are used rarely because of two main reasons. They are (i) high initial cost (ii) high Insulation cost. Hence, the power is transmitted by overhead lines by proper mechanical factors like strength weather conditions and other external interferences taken into consideration because of the continuity of operations in the line.

Components required for overhead lines

The successful operation of an overhead line depends to a great extent upon the mechanical design of the line.

The components of the OH lines are

1. Conductors
2. Poles
3. Insulators



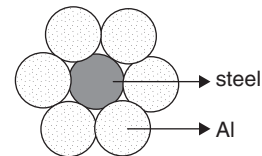
Conductor Materials

The conductor is one of the important items as most of the capital outlay is invested for it. The proper choice of material and size of the conductor is of considerable importance. The conductors should have the following properties

1. High electrical conductivity
2. High tensile strength
3. Low cost
4. Low specific gravity

All above properties are not found in single material. So, combination of different materials will give above properties.

The common materials used for overhead lines are copper, aluminium steel-cored, aluminium galvanized steel and cadmium copper. Conductors are preferably stranded in order to maintain flexibility. In stranded conductors one central wire around this some no of conductors are used based on the voltage designed according to the 6, 12, 18, 24.....wires.



If there are 'n' layers, then, $3n(n+1)+1$ is total no of individual layers. In the manufacture of stranded conductors, the layers of wires are twisted in opposite directions so that layers are bounded together.

Copper

Copper is an ideal material for OH lines due to its high conductivity and tensile strength.

- High electrical conductivity
- Greater tensile strength
- High current density
- High cost
- Non-availability

Specific gravity of copper is 8.9 gm/cc. Due to disadvantage of copper, they are rarely used.

Aluminium

- Aluminium is cheap and light in weight compared to copper.

3.862 | Power Systems

- Smaller conductivity and low tensile strength.
- The conductivity of aluminium is 60% that of copper.
- The diameter of Al conductor is 1.26 times the diameter of copper for equal resistances.
- The increased cross sections of Al give greater surface to wind pressure, so supporting towers should be in greater in strength.
- Specific gravity of Al is 2.71 gm/cc.

Galvanized Steel

- Galvanized steel conductors can be used for extremely long spans.
- They are suitable in rural areas where cheapness is the main consideration.
- It has poor conductivity due to high resistance of steel.

Cadmium Copper

Conductor material in this case is copper alloyed with cadmium. Addition of 2% cadmium to copper increases 50% tensile strength and 15% reduction of conductivity. Therefore, these conductors can be useful for exceptionally long spans.

ACSR (Aluminium Conductor Steel Reinforced)

- The aluminium conductor is reinforced with a core of galvanized steel wires.
- The diameter of both steel and Aluminium is same.
- The cross section of two metals is generally in the ratio of 1:6.
- They are represented by x/y .
- Steel-cored aluminium spans can be used for towers of smaller heights.
- Steel-cored aluminium conductor will produce smaller Sag, and hence longer spans can be used with smaller tower heights.

Example 13: An ACSR conductor consists of 30 Al and 7 steel strands

Solution: $\frac{30}{7}$ Ans.

Example 14: ACSR conductors represented by 7/54

Solution: 7 steel and 54 Al

Diameter of Stranded Conductor

$$D = (2n - 1)d$$

$\therefore d$ = diameter of each strand

n = no of layers

Poles

The supporting structure for overhead lines are poles and towers.

The poles or towers should have following properties

1. High mechanical strength
2. Cheap in cost

3. Longer life
4. Light in weight

Different types of poles

1. Wooden poles
2. Steel poles and steel towers
3. RCC (Reinforced concrete) poles

Insulators

The overhead line conductors are supported on the poles in such a way that the current should not flow to earth by poles, so insulators are properly fixed to conductor to provide necessary insulation between line conductors and support.

The Insulators should have following properties

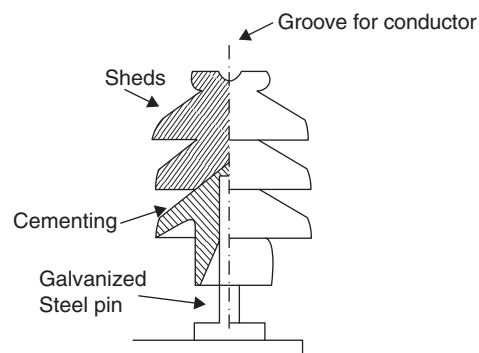
1. High mechanical strength
2. High relative permittivity
3. High electrical

Different types of Insulators

1. Pin type
2. Suspension type
3. Strain type
4. Shackle type

Pin-type Insulator

- The pin type is secured (fixed) to cross arm on the pole.
- Pin-type insulators are used for transmission and distribution of electrical power at voltage up to 33 kV.



Causes of Insulator Failure

- Insulators are required to withstand both mechanical and electrical stress.
- The electrical breakdown of the insulator can occur either by flash – over or puncture.

Flashover: an arc occurs between the live conductor and insulator pin.

Puncture: the discharge occurs from the conductor to pin through the body of the insulator.

$$\text{Safety factor of insulator} = \frac{\text{Puncture strength}}{\text{Flash – Overvoltage}}$$

Mechanical–Electrical

1. Flashover: Arc occurs between line conductor and insulator or pin (i.e. Earth) and the discharge jumps over air gap, following the shortest path. After flashover,

insulator continues to work properly, unless destroyed by the heat produced during flashover.

2. Puncture: Discharge focus from pin to the body of insulator. This destroys the insulator and it must be replaced.

Suspension-type Insulator

- Suspension-type insulator consists of a number of porcelain discs connected in series by metal links in the form of a string. The conductor is suspended at the bottom end of this string while the other end of the string is secured to the cross arm of the tower.
- Each disc is designed for low voltage 11 kV. For example, 6 discs $\rightarrow$ 66 kV.
- These insulators are cheaper than pin-type insulators for voltages beyond 33 kV.
- We can replace the damaged disc from the whole string and we can add the desired number of discs.
- These insulators are used with steel towers.

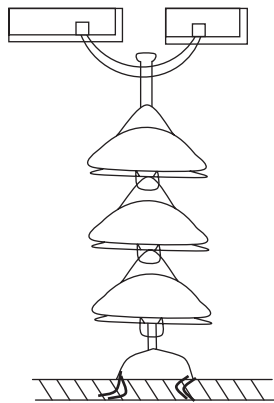
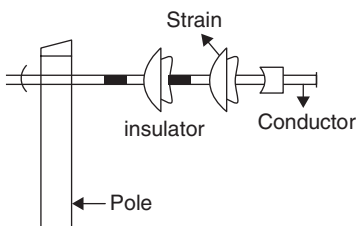


Figure 9 Suspension-type insulators

Strain Insulators

When there is a dead end of the line or there is corner or sharp curve, the line is subjected to greater tension. Strain insulator is used for low voltages line (<11 kV).

- By assembling suspension insulators, we can use as the strain insulator.



Potential Distribution Over a String of Insulators

A string of suspension insulators consist of a number of porcelain discs connected in series through metallic links.

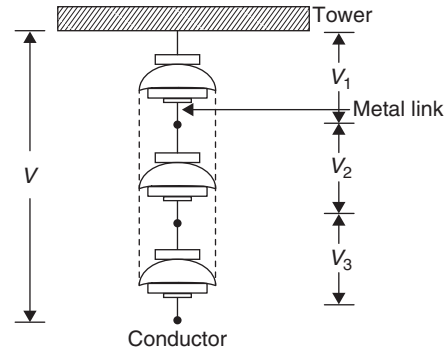


Figure 10 A typical suspension-type Insulator

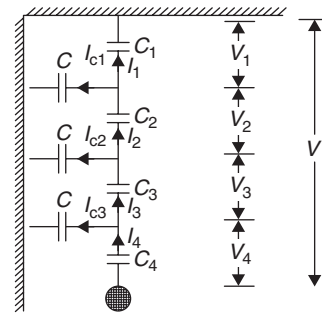


Figure 11 Potential distribution across a string of Insulators

Porcelain portion of each disc between two metallic links forms a capacitance. This capacitance is known as self-capacitance. If there were mutual capacitance alone, the potential distribution across each unit would have been the same. But in actual case, capacitance also exists between metal fitting and tower or earth. This capacitance is known as shunt capacitance. Due to the shunt capacitance, charging current is not the same through all the discs of the string. Therefore, voltage across each disc will be different. Presence of self-capacitance and shunt capacitance results in the

1. Non-uniform potential distribution of voltage across the individual discs of suspension-type insulators.
2. The disc nearest to the conductor has maximum voltage across it.
3. The disc nearest to the conductor is under maximum electrical stress and is likely to be punctured.
4. Since the insulator capacitances are ineffective for DC, the voltage across each disc would be same for DC voltage across the string.

The unequal potential distribution is generally expressed in terms of string efficiency:

String Efficiency

The ratio of voltage across the whole string to the product of discs and the voltage across the disc nearest to the conductor is known as 'string efficiency'.

String efficiency

$$= \frac{\text{Voltage across the string}}{n \times \text{Voltage across disc nearest to conductor}}$$

n = no. of discs in the string.

$$\text{Let } K = \frac{\text{Capacitance per insulator}}{\text{Capacitance to ground}} = \frac{KC}{C}$$

where $C_1 = C_2 = C_3 = C_4 = KC$

Let V be the operating voltage (Line to ground) and V_1, V_2, V_3 and V_4 the voltage drops across the units as shown in the figure.

$$V = V_1 + V_2 + V_3 + V_4.$$

From the figure, we can write

$$I_2 = I_1 + I_{c1}$$

$$\omega kc V_2 = \omega kc V_1 + \omega c V_1 \quad (\because \omega = \text{supply frequency})$$

$$V_2 = V_1 \left(\frac{1+K}{K} \right) = V_1 \left(1 + \frac{1}{K} \right).$$

$$V_3 = V_1 \left(1 + \frac{3}{K} + \frac{1}{K^2} \right)$$

$$V_4 = V_1 \left(1 + \frac{6}{K} + \frac{5}{K^2} + \frac{1}{K^3} \right).$$

V_1 can be expressed in terms of V and from this V_2, V_3 , etc. can be obtained normally.

If $k = 5$, then $V_2 = 1.2 V_1, V_3 = 1.64 V_1, V_4 = 2.40 V_1$

$$\therefore V_1 < V_2 < V_3 < V_4.$$

As ' k ' increases, the division of voltage becomes uniform. The 100% string efficiency can be achieved by increasing ' K ' value till the potential distribution across discs become uniform.

Methods of Improving String Efficiency

1. Selection of K
2. Grading of units (discs)
3. Static shielding

Selection of K

Performance of string (string efficiency) depends on the value of ' K '. As K is increased the division of voltage becomes more equalized. To increase the ' K ' value, longer cross arm for the tower to be used.

Grading of Units

Insulators of different dimensions are chosen so that each disc has different capacitance. The insulators are capacitance graded such a way that the top unit has the minimum capacitance and the capacitance increases progressively as the bottom unit is reached. This method has a disadvantage that a large number of different sized insulators are required.

Static Shielding

The string efficiency in this method is improved by providing grading in the form of a large metal ring surrounding

the bottom unit and connected to the metal work at the bottom of this unit and therefore to the line. The guard ring introduces capacitance between the metal fittings and the line conductor. The guard ring is connected in such a way that shunt capacitance currents i_{c1}, i_{c2}, i_{c3} , etc. are equal to metal fitting line Capacitance currents i_{m1}, i_{m2}, i_{m3} , etc. The result is that same charging current ' I ' flows through each unit of string. Consequently, there will be uniform potential distribution across the units.

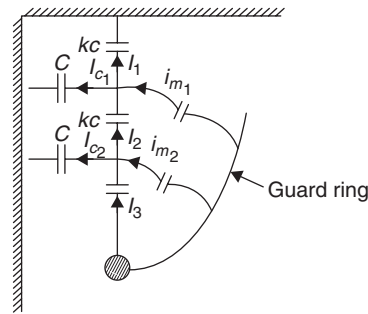


Figure 12 Static shielding

Common Data for Examples

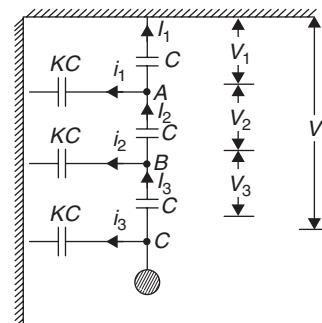
A 3- ϕ overhead transmission line is being supported by three disc insulators. The potential drop across top unit and middle unit are 10 kV and 12 kV, respectively.

Example 15: The ratio of capacitance between pin and earth to self-capacitance of each unit is

- (A) 0.2 (B) 5
(C) 0.833 (D) 1.2

Solution: (A)

The equivalent circuit of string insulators is shown below



Conductor location

Let ' K ' be the ratio of capacitance between pin and earth to self-capacitance.

' C ' is self-capacitance of each unit.

Capacitance between pin to earth = KC

Apply KCL at node A

$$I_2 = I_1 + i_1$$

$$V_2 \omega c = V_1 \omega c + V_1 k \omega c$$

$$V_2 = V_1(1 + k)$$

$$k = \frac{V_2 - V_1}{V_1} = \frac{12 - 10}{10} = 0.2$$

Example 16: The string efficiency of the system is

- (A) 10.6% (B) 12%
(C) 78% (D) 92%

Solution: (C)

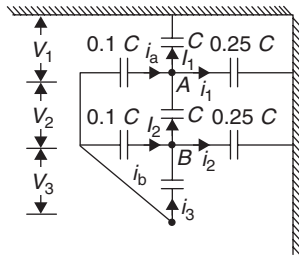
From the given data,

$$\begin{aligned} V_1 &= 10 \text{ kV}, V_2 = 12 \text{ kV} \\ V_3 &= V_2 + (V_1 + V_2) k \\ &= 12 + (10 + 12) \times 0.2 = 16.4 \text{ kV} \end{aligned}$$

$$\begin{aligned} \text{Voltage between line and earth} &= V_1 + V_2 + V_3 \\ &= 38.4 \text{ kV} \end{aligned}$$

$$\begin{aligned} \text{String efficiency} &= \frac{\text{Voltage across string}}{\text{No. of insulators} \times V_3} \times 100 \\ &= \frac{38.4}{3 \times 16.4} \times 100 = 78\% \end{aligned}$$

Example 17: For the following figure, the voltage across middle insulator as a percentage of line voltage to earth



- (A) 29.85% (B) 40.65%
(C) 70.15% (D) 59.35%

Solution: (A)

Applying Kirchhoff's law at node 'A'

$$\begin{aligned} I_2 + i_a &= I_1 + i_1 \\ V_2 \omega C + 0.1 \omega C (V_2 + V_3) &= V_1 \omega C + 0.25 V_1 \omega C \\ V_2 + 0.1 V_2 + 0.1 V_3 &= 1.25 V_1 \\ 1.1 V_2 + 0.1 V_3 &= 1.25 V_1 \end{aligned}$$

$$\begin{aligned} \text{At node 'B'} \quad I_3 + i_b &= I_2 + i_2 \\ 0.25 V_2 + 1.25 V_2 &= 1.1 V_3 \end{aligned}$$

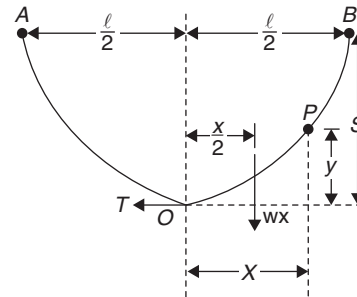
From the above two equations, we have

$$\begin{aligned} V_1 &= 0.988 V_2 \\ V_3 &= 1.362 V_2 \\ V &= V_1 + V_2 + V_3 \\ V &= 0.988 V_2 + V_2 + 1.362 V_2 = 3.35 V_2 \\ V_2 &= \frac{1}{3.35} V = 29.85\% V \\ V_2 &= 29.85\% \text{ of } V \end{aligned}$$

Sag Calculation in OH Lines

While erecting an overhead line, it is very important that conductors are under safe tension. In order to permit safe tension in the conductors, they are not fully stretched but are allowed to have a dip or sag.

The difference in level between points of supports and the lowest point on the conductor is called sag.



Consider a conductor between two equilateral supports A and B with 'O' as the lowest point as shown above.

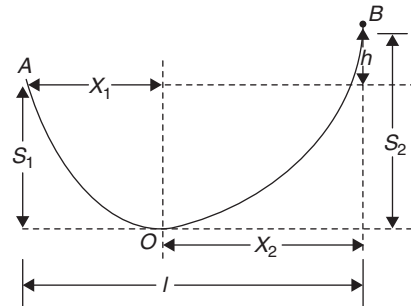
$$\text{Sag } S = \frac{Wl^2}{8T}$$

where l = Length of span

W = Weight per unit length of conductor

T = Tension in the conductor

Let A and B be at different levels from the ground.



$$S_1 = \frac{Wx_1^2}{2T} \quad \text{where } x_1 = \frac{l}{2} - \frac{Th}{Wl}$$

$$S_2 = \frac{Wx_2^2}{2T} \quad \text{where } x_2 = \frac{l}{2} + \frac{Th}{Wl}$$

where h = Difference in levels between two supports.

$$S_2 - S_1 = \frac{Wl}{2T} [x_2 - x_1]$$

and

$$l = x_2 + x_1$$

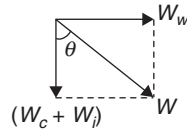
Effect of Wind and Ice

The effect of ice covering (w_i) is to increase the weight of the conductor (w_c) and thus increases vertical sag. To guard against this, the tension on the conductors at the time of erection is suitably decreased.

If the wind pressure is also considered, the wind pressure acting on the projected area of the conductor per metre converted to local should be added to the weight. If W_w is wind pressure.

Resultant weight (per conductor)

$$W = \sqrt{(W_c + W_i)^2 + W_w^2}$$



Example 18: If the ultimate strength is 5750 kg, sag is 2 m and factor of safety is 2 and the overhead line has a span of 250 m. The weight of the conductor is

- (A) 0.736 kg (B) 184 kg
(C) 92 kg (D) 23 kg

Solution: (B)

Allowable maximum tension

$$(T) = \frac{\text{Ultimate strength}}{\text{Safety factor}} = \frac{5750}{2} = 2875 \text{ kg}$$

Total sag(s) = 2 m

$$S = \frac{WL^2}{8T}$$

$$W = \frac{8ST}{L^2} = \frac{2 \times 8 \times 2875}{(250)^2}$$

$$W = 0.737 \text{ kg/m}$$

$$\text{Half-span length} = l + \frac{w^2 l^2}{6T^2}$$

$$= \frac{250}{2} + \frac{(0.737)^2 \left(\frac{250}{2}\right)^2}{6 \times (2875)^2} = 125 \text{ m}$$

Total length = 250 m

$$\text{Weight of conductor} = (0.737 \times 250) \text{ kg} = 184 \text{ kg}$$

Example 19: An overhead line conductor is subjected to a horizontal wind load of 1.78 kg/m and vertical ice loading of 1.08 kg/m. If the maximum possible sag is 6 m. Safety factor is 2. Weight of conductor is 0.844 kg/m². The tension is 7950 kg. The permissible span between two supports is

- (A) 135 m (B) 275 m
(C) 13 m (D) 27 m

Solution: (B)

$$\begin{aligned} \text{Total weight } w &= \sqrt{(w_c + w_i)^2 + w_w^2} \\ &= \sqrt{(0.844 + 1.08)^2 + (1.78)^2} \\ &= 2.621 \text{ kg/m} \end{aligned}$$

$$\text{Maximum possible sag } D = \frac{wl^2}{2T}$$

$$6 = \frac{2.621 \times l^2}{2 \times (7950/2)}$$

$$l = \sqrt{\frac{12 \times 3975}{2.621}} = 135 \text{ m}$$

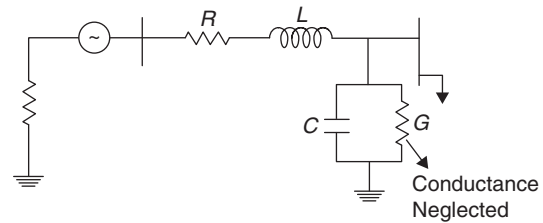
The span length = $2 \times 135 = 270 \text{ m}$.

ELECTRICAL DESIGN OF OVERHEAD LINES

An AC transmission has resistance inductance and capacitance uniformly distributed along its length. These four are called parameters. These parameters will determine the efficiency of transmission system.

Parameters of Transmission line:

1. Resistance (R)
2. Inductance (L)
3. Capacitance (C)
4. Conductance (G)



The transmission line consists of a series combination of resistance, inductance and a parallel combination of capacitance, conductance. Conductance is normally neglected due to leakage over line insulation is almost always in overhead transmission line.

The series line parameters, i.e., inductance resistance are uniformly distributed along the line and they together form series impedance. Inductance is the most dominant line parameter for a power system. It is the inductive reactance which limits the transmission capacity of line.

Resistance

It opposes the flow of current in a line conductor. It is the cause for power loss in the Transmission line

$$R = \frac{\rho l}{a}$$

R = Resistance

ρ = Resistivity

l = Length of conductor

a = Area of conductor

The variation of resistance of metallic conductor with temperature is practically linear.

$$R_2 = R_1 [1 + \alpha_1(t_2 - t_1)]$$

$$\alpha_1 = \frac{\alpha_0}{1 + \alpha_0 t_1}$$

α_0 = temperature coefficient

Skin Effect

An alternating current flowing through the conductor does not distribute uniformly. Rather it has the tendency to concentrate near the surface of the conductor. This effect is known as skin effect.

The skin effect depends on

1. Nature of material
2. Diameter of wire
3. Frequency
4. shape of wire

When frequency < 50 Hz and conductor diameters < 1 cm, skin effect can be negligible.

Flux Linkages

Inductance of a circuit is defined as the flux linkages per unit current.

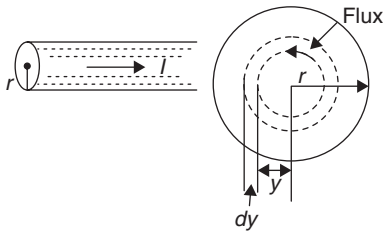
There are two important cases of flux linkages.

Flux linkages due to a single current carrying conductor

The magnetic lines of force will exist inside the conductor and outside of the conductor. Contribution of both fluxes will form the inductance

For example

A long straight cylindrical conductor of radius ' r ' metre and carrying a current of ' I ' Amperes



Internal Flux

Figure shows the cross-sectional view of a long cylindrical conductor carrying current I

The mmf around a concentric closed circular path of radius (internal) of the conductor is

[$\because H_y = \text{magnetic field intensity}$] (AT/m)

$\oint H_y ds = I_y$ (i) (Amperes law), where

$I_y = \text{current enclosed (A)}$

By symmetry, H_y is constant and is in direction of ds all along the circular path. Therefore from equation (1)

$$2\pi y H_y = I_y \quad (2)$$

Assuming uniform current density

$$I_y = \left(\frac{\pi y^2}{\pi r^2} \right) I = \left(\frac{y^2}{r^2} \right) I \quad (3)$$

From Equations (2) and (3), we obtain

$$H_y = \frac{yI}{2\pi r^2} \text{ AT/m} \quad (4)$$

The flux density (B_y) at ' y ' metre from the centre of conductor is

$$B_y = \mu H_y = \frac{\mu y I}{2\pi r^2} \text{ Wb/m}^2 \quad (5)$$

where μ is the permeability of the conductor.

Consider now an infinitesimal tabular element of thickness dy and length 1 metre. The flux in the tabular element

($d\phi = B_y dy$) weber links the fractional turn $\frac{\ell y}{\ell} = \frac{y_2}{r_2}$ resulting in flux linkages of

$$d\lambda = \left(\frac{y^2}{r^2} \right) d\phi = \left(\frac{y^2}{r^2} \right) \frac{\mu y I}{2\pi r^2} dy \text{ Weber-turns} \quad (6)$$

Integrating, we get the total internal flux linkages as

$$\lambda_{\text{int}} = \int_0^r \frac{\mu I}{2\pi r^4} y^3 dy = \frac{\mu I}{8\pi} \text{ Wb-T/m}$$

For relative permeability $\mu_r = 1$ (non-magnetic conductor)

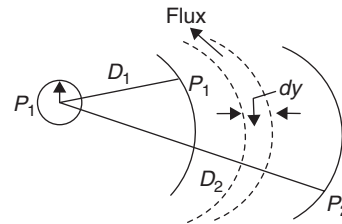
$$\mu = 4\pi \times 10^{-7} \text{ Wb-T/m}$$

and

$$L_{\text{int}} = \frac{1}{2} \times 10^{-7} \text{ H/m}$$

Flux Linkage Due to Flux Between Two Points External to Conductor

Figure shows two points P_1 and P_2 , at distance D_1 and D_2 from a conductor which carries current of I amperes. The magnetic field external to the conductor is concentric circles around the conductor and therefore all the flux between P_1 and P_2 lies within the concentric cylindrical surfaces passing through P_1 and P_2 .



Magnetic field intensity at distance y from the conductor is

$$H_y = \frac{1}{2\pi y} \text{ AT/m}$$

The flux $d\phi$ contained in the tabular element of thickness dy is

$$d\phi = \frac{\mu I}{2\pi y} dy \text{ Wb}$$

The flux $d\phi$ links all the current in the conductor once and only once.

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Flux linkages,

$$d\lambda = 1 \times d\phi = \frac{\mu I}{2\pi y} dy$$

Total flux linkages of the conductor due to flux between points

$$\lambda_{12} = \int_{D_1}^{D_2} \frac{\mu I}{2\pi y} dy = \frac{\mu}{2\pi} \ln \frac{D_2}{D_1} \text{ Wb-turns/m}$$

where 'ln' stands for natural logarithm.

Since $\mu_r = 1$ $\mu = 4\pi \times 10^{-7}$

$$\lambda_{12} = 2 \times 10^{-7} \ln \frac{D_2}{D_1} \text{ Wb-turns/m}$$

The inductance of the conductor by the flux included between points P_1 and P_2 is then

$$L_{12} = 2 \times 10^{-7} \ln \frac{D_2}{D_1} \text{ H/m}$$

Or

$$L_{12} = 0.461 \log \frac{D_2}{D_1} \text{ mH/km}$$

Flux linkages due to External flux:

Let the external point be at distance D from the centre of the conductor, due to external flux, these linkages can be obtained by substituting $D_1 = r$ and $D_2 = D$, i.e.

$$\lambda_{\text{ext}} = 2 \times 10^{-7} \ln \frac{D}{r}$$

Total flux linkages of the conductor due to internal and external flux are

$$\begin{aligned} \lambda &= \lambda_{\text{int}} + \lambda_{\text{ext}} \\ &= \frac{I}{2} \times 10^{-7} + 2 \times 10^{-7} \ln \frac{D}{r} \\ &= 2 \times 10^{-7} \left(\frac{1}{4} + \ln \frac{D}{r} \right) = 2 \times 10^{-7} \ln \frac{D}{re^{-1/4}} \\ &= 2 \times 10^{-7} \ln \frac{D}{r'} \end{aligned}$$

where $r' = re^{-1/4} = 0.77888 r$

$$\lambda = 2 \times 10^{-7} \ln \frac{D}{r'} \text{ Wb-T/m}$$

Inductance of the conductor due to flux up to an external point

$$L = 2 \times 10^{-7} \ln \left(\frac{D}{r'} \right) \text{ H/m}$$

Inductance (L)

- It opposes sudden changes in currents. The voltage induced can be obtained by

$$e = \frac{d\phi}{dt} \text{ V}$$

ψ = flux linkages

$$e = \frac{d\phi}{di} \cdot \frac{di}{dt} V = L \frac{di}{dt}$$

where $L = \frac{d\phi}{di}$ is defined as the inductance of circuit in henry

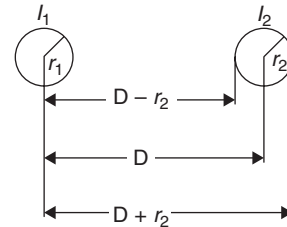
$$L = \frac{\phi}{i} \text{ Henry}$$

$$\psi = Li \text{ (Wb-T)}$$

If the current is alternative

$$\lambda = LI$$

Inductance of a single-phase two wire line



The flux linkage caused by current in Conductor 1 is given by

$$\lambda_1 = 2 \times 10^{-7} \ln \frac{D}{r'}$$

Inductance of the conductor due to current in conductor 1 only is

$$L_1 = 2 \times 10^{-7} \ln \frac{D}{r'_1}$$

Due to current in conductor 2 only is

$$L_2 = 2 \times 10^{-7} \ln \frac{D}{r'_2}$$

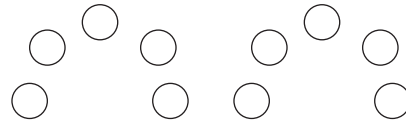
$$L = L_1 + L_2 = 4 \times 10^{-7} \ln \frac{D}{\sqrt{r'_1 r'_2}} \text{ H/m}$$

If $r'_1 = r'_2 = r'$; then

$$L = 4 \times 10^{-7} \ln D/r' \text{ H/m}$$

$$L = 0.921 \ln D/r' \text{ mH/m.}$$

Inductance of Composite Conductor Live



$$\lambda_i = 2 \times 10^{-7} \frac{I}{n} \left(\ln \frac{1}{D_{i1}} + \ln \frac{1}{D_{i2}} + \dots + \ln \frac{1}{D_{in}} \right)$$

$$= 2 \times 10^{-7} \frac{I}{m_1} \left(\ln \frac{1}{D_{i1}} + \ln \frac{1}{D_{i2}} + \dots + \ln \frac{1}{D_{in}} \right)$$

$$= 2 \times 10^{-7} I \ln \frac{(D_{i1} \dots D_{i2} \dots D_{in})^{1/m'}}{(D_{i1} D_{i2} D_{in})^{1/n}} \text{ Wb-T/m}$$

The inductance of filament is then

$$L_i = \frac{\lambda_i}{I/n} = 2n \times 10^{-7} L_n \frac{(D_{i1} \dots D_{ij} \dots D_{im})}{(D_{i1} D_{i2} \dots D_{ii} \dots D_{in})} \text{H/m}$$

$$L_{\text{avg}} = \frac{L_1 + L_2 + L_3 + \dots L_n}{n}$$

∴ Conductor A is composed of n filaments electrically in parallel

$$L_A = \frac{L_{\text{avg}}}{n} = \frac{L_1 + L_2 + \dots L_n}{n^2}$$

$$L_A = 2 \times 10^{-7}$$

$$L_n \frac{\left[(D_{11'} \dots D_{1j'} \dots D_{1m'}) \dots (D_{il'} \dots D_{lj'} \dots D_{im'}) \dots (D_{nl'} \dots D_{nj'} \dots D_{nm'}) \right]^{1/n^2}}{\left[(D_{11'} \dots D_{1j'} \dots D_{1m'}) \dots (D_{il'} \dots D_{lj'} \dots D_{im'}) \dots (D_{nl'} \dots D_{nj'} \dots D_{nm'}) \right]^{1/n^2}}$$

$$L_A = 2 \times 10^{-7} \ln \left(\frac{\text{Products of all possible mutual inductances}}{\text{Geometric mean radius}} \right)$$

$$L_A = 2 \times 10^{-7} L_n \frac{\text{DM}}{\text{DSA}} \text{H/m} = 0.461 L_{\text{ong}} \frac{\text{DM}}{\text{DSA}} \text{mH/km}$$

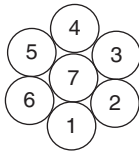
Total inductance of line

$$L = L_A + L_B$$

Problems on Self-GMD

Example 20: A conductor is composed of seven (7) identical copper strands each having a radius r , as shown in Figure 1. Find the self-GMD of the conductor.

$$\begin{aligned} D_{14} &= 4r \\ D_{12} &= 2r \\ D_{26} &= 2\sqrt{3}r \end{aligned}$$



Solution: The self-GMD of seven strand conductor is the 49th root of the 49 distances. Thus,

$$D_S = ((r')^7 (D_{12}^2 D_{26}^2 D_{14} D_{17})^6 (2r)^6)^{1/49}$$

Substituting the value of various distances,

$$D_S = \left((0.7788r)^7 (2^2 r^2 \times 3 \times 2^2 r^2 \times 2^2 r^2 \times 2^2 r \times 2r \times 2r)^6 (2r)^6 \right)^{1/49}$$

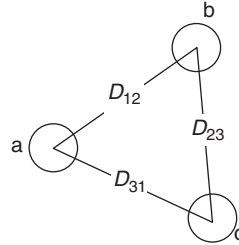
$$D_S = \frac{2r(3(0.7788))^{1/7}}{6^{1/49}}$$

$$D_S = 2.177r$$

Inductance of Three-phase Lines

The basic equations developed can however be easily adapted to the calculation of the inductance of three-phase lines.

Figure shows conductor of a three-phase live Unsymmetrical spacing:



Practically transposition is difficult but mathematical it can be assumed.

$$L/\text{phase} = 2 \times 10^{-7} \ln \left(\frac{\text{GMD}}{\text{GMR}} \right) \text{H/M}$$

$$L/\text{phase} = 2 \times 10^{-7} \ln \left(\frac{\text{DM}}{\text{DSA}} \right) \text{H/M}$$

$$\text{GMD or DM} = 3\sqrt{XYZ} \text{ if } D_{12} = X, D_{23} = Y, D_{31} = Z$$

$$\text{GMR or DSA} = r'$$

Symmetrical lines

$$X = Y = Z = d \text{ (equilateral spacing)}$$

$$L/\text{phase} = 0.2 \text{Ln} \left(\frac{\text{GMD}}{\text{GMR}} \right) \text{mH/km}$$

$$\text{GMD} = d$$

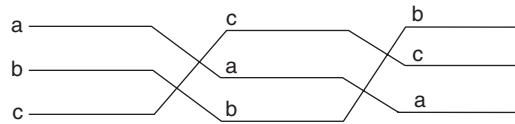
$$\text{GMR} = r'$$

Inductive reactance

$$X = 2\pi FL$$

Unsymmetrical Spacing

When three-phase line conductors are not equidistant from each other, the conductor spacing is said to be unsymmetrical. So, flux linkages are different. Interchanging the positions of the conductor at regular intervals along the line such an exchange of positions is known as transposition.



$$I_a + I_b + I_c = 0$$

$$I_a = I(1 + j0)$$

$$I_b = I(-0.5 - j0.866)$$

$$I_c = I(-0.5 + j0.866)$$

$$\lambda_{a1} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r'_a} + I_b \ln \frac{1}{D_{12}} + I_c \ln \frac{1}{D_{31}} \right) \text{Wb-T/m}$$

$$\lambda_{a2} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r'_a} + I_b \ln \frac{1}{D_{23}} + I_c \ln \frac{1}{D_{12}} \right) \text{Wb-T/m}$$

$$\lambda_{a3} = 2 \times 10^{-7} b \left(I_a \ln \frac{1}{r'_a} + I_b \ln \frac{1}{D_{31}} + I_c \ln \frac{1}{D_{23}} \right) \text{Wb-T/m}$$

Average flux linkages of conductor 'a' are

$$\lambda_a = \frac{\lambda_{a1} + \lambda_{a2} + \lambda_{a3}}{3}$$

$$= 2 \times 10^{-7} \left(I_a \ln \frac{1}{r'_a} + I_b \ln \frac{1}{(D_{12} D_{23} D_{31})^{1/3}} + I_c \ln \frac{1}{(D_{12} D_{23} D_{31})^{1/3}} \right)$$

$$I_b + I_c = -I_a$$

$$\lambda_0 = 2 \times 10^{-7} I_a \ln \frac{(D_{12} D_{23} D_{31})^{1/3}}{r'_a}$$

$$D_{eq} = (D_{12} D_{23} D_{31})^{1/3} = \text{equivalent equilateral spacing}$$

$$L_a = 2 \times 10^{-7} \ln \frac{D_{eq}}{r'_a} = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_s} \text{H/m}$$

$$D_m = D_{eq}$$

$$r'_a = r'_b = r'_c$$

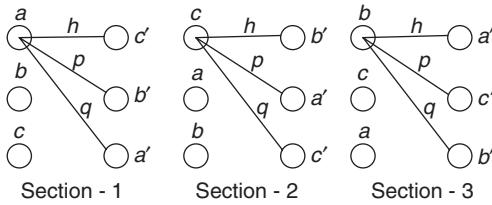
$$L_a = L_b = L_c$$

where D_s = self-GMD, D_m = Mutual GMD

Double-circuit Three-phase Line

The double-circuit three-phase line is built so as to increase transmission reliability, at enhanced cost.

The three sections of the transposition cycle of two parallel circuit three-phase line with vertical spacing are shown below.



$$D_{eq} = (D_{ab} D_{bc} D_{ca})^{1/3}$$

D_{ab} = mutual GMD between phases a and b in Section 1 of transposition cycle

$$= (D_p D_p)^{1/4} = (D_p)^{1/2}$$

D_{bc} = mutual GMD between phases b and c in Section 1 of transposition cycle

$$= (D_p)^{1/2}$$

D_{ca} = mutual GMD between phases a and c in

Section 1 of transposition cycle

$$= (2Dh)^{1/2}$$

$$D_{eq} = 2^{1/6} D^{1/2} p^{1/3} h^{1/6}$$

$$D_{sa} = (r'_q r'_q)^{1/4} = (r'_q)^{1/2}$$

$$D_{sb} = (r'_h r'_h)^{1/4} = (r'_h)^{1/2}$$

$$D_{sc} = (r'_q r'_q)^{1/4} = (r'_q)^{1/2}$$

Equivalent self-GMD $D_s = (D_{sa} D_{sb} D_{sc})^{1/3}$

$$= (r')^{1/2} q^{1/3} h^{1/6}$$

$$L = 2 \times 10^{-7} \ln \left[2^{1/6} \left(\frac{D}{r'} \right)^{1/2} \left(\frac{p}{q} \right)^{1/3} \right] \text{H/phase/m.}$$

Mutual inductance between the two circuits

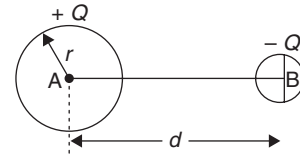
$$M = 2 \times 10^{-7} \ln \left(\frac{p}{q} \right)^{2/3}$$

$M \rightarrow 0$. $\therefore$ mutual impedance is zero.

$$L = 1 \times 10^{-7} \ln \frac{3\sqrt{2}D}{r'}$$

Capacitance

The capacitance and conductance together forms the shunt admittance of a transmission line. When an alternating voltage is applied to the line, the line capacitance draws a leading sinusoidal current called the charging current which is drawn even when the line is open circuited at the far end.



$$V_A = \int_r^\infty \frac{Q}{2\pi x \epsilon_0} dx + \int_d^\infty \frac{-Q}{2\pi x \epsilon_0} dx$$

$$= \frac{Q}{2\pi \epsilon_0} \left(\log_e \frac{\infty}{r} - \log_e \frac{\infty}{d} \right) V$$

$$= \frac{Q}{2\pi \epsilon_0} \log_e \frac{d}{r} V$$

Potential difference between conductor B and central 'infinite'

$$V_B = \int_r^\infty \frac{-Q}{2\pi x \epsilon_0} dx + \int_d^\infty \frac{Q}{2\pi x \epsilon_0} dx$$

$$= \frac{-Q}{2\pi \epsilon_0} \left[\log_e \frac{d}{r} \right] V$$

$$= \frac{-Q}{2\pi\epsilon_0} \log_e \frac{d}{r} V$$

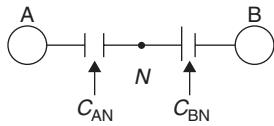
$$V_{AB} = 2V_A = \frac{2Q}{2\pi\epsilon_0} \log_e \frac{d}{r} V$$

$$C_{AB} = Q/V_{AB} = \frac{Q}{\frac{2Q}{2\pi\epsilon_0} \log_e \frac{d}{r}} \text{ F/m}$$

$$C_{AB} = \frac{\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

Capacitance to Neutral

The aim is to know the capacitance between one of the conductors and a neutral point between them. Since potential of the midpoint between the conductors is zero, the potential difference between each conductor and ground or neutral is half the potential differences between the conductors.



Capacitance of neutral

$$C_N = C_{AN} = C_{BN} = 2C_{AB}$$

$$C_N = \frac{2\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F/m.}$$

Capacitance of a Three-phase

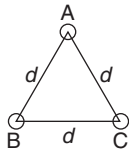
The capacitance of each conductor is considered instead of capacitance from conductor to conductor

Two cases:

1. Symmetrical spacing
2. Unsymmetrical spacing

Symmetrical Spacing

The three conductors A, B and C are having charges of Q_A , Q_B and Q_C per metre (equidistant of d metre) from each other, the capacitance from line conductor to central in the symmetrical spaced line.



$$V_A = \int_r^\infty \frac{Q_A}{2\pi x \epsilon_0} dx + \int_d^\infty \frac{Q_B}{2\pi x \epsilon_0} + \int_d^\infty \frac{Q_C}{2\pi x \epsilon_0} dx$$

$$= \frac{1}{2\pi\epsilon_0} \left(Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d} + Q_C \log_e \frac{1}{d} \right)$$

$$= \frac{1}{2\pi\epsilon_0} \left[Q_A \log_e \frac{1}{r} + (Q_B + Q_C) \log_e \frac{1}{d} \right]$$

Assume balanced supply, then

$$Q_A + Q_B + Q_C = 0$$

$$Q_B + Q_C = -Q_A$$

$$V_A = \frac{1}{2\pi\epsilon_0} \left(Q_A \log_e \frac{1}{r} - Q_A \log_e \frac{1}{d} \right) = \frac{Q_A}{2\pi\epsilon_0} \log_e \frac{d}{r} V$$

Capacitance of conductor A with respect to neutral,

$$C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi\epsilon_0} \log_e \frac{d}{r}} \text{ F/M} = \frac{2\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

Unsymmetrical Spacing

Three-phase transposed line having unsymmetrical spacing for balanced condition $Q_A + Q_B + Q_C = 0$

Capacitance from conductor to neutral is

$$C_A = \frac{Q_A}{V_A} = \frac{2\pi\epsilon_0}{\log_e \frac{3\sqrt{d_1 d_2 d_3}}{r}} \text{ F/m}$$

Example 21: A single-phase transmission line has two parallel conductor 4 m apart, radius of each conductor being 2 cm. calculate the capacitance of line per km .given that $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

Solution: Conductor radius = 2 cm

Spacing of conductors = 4 m = 400 cm

Capacitance of the line

$$= \frac{\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

$$= \frac{\pi \times 8.85 \times 10^{-12}}{\log_e \frac{400}{2}} \text{ F/m}$$

$$= 5.247 \times 10^{-12} \text{ F/m}$$

$$= 5.247 \times 10^{-3} \mu\text{F/km}$$

Example 22: A three-phase overhead transmission line has its conductors arranged at the corners of an equilateral triangle of 8 m side. Calculate the capacitance of each line conductor per km. diameter of each conductor is 2.5 cm.

Solution: Conductor radius $r = 2.5/2 = 1.25 \text{ cm}$

Spacing of conductors $d = 8 \text{ m} = 800 \text{ cm}$

Capacitance of each line conductor

$$= \frac{2\pi\epsilon_0}{\ln \frac{d}{r}} \text{ F/m} = \frac{2 \times \pi \times 8.854 \times 10^{-12}}{\ln \left(\frac{800}{1.25} \right)} \text{ F/m}$$

$$= 8.609703 \times 10^{-12} \text{ F/m}$$

$$= 8.6097 \times 10^{-3} \mu\text{F/km}$$

Example 23: Calculate the capacitance of a 400 km long three-phase 50 Hz overhead transmission line consisting of 3 conductors of each diameter 2 cm and spaced 4 m at the corner of an equilateral triangle.

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Solution: Equilateral spacing $d = 4 \text{ m} = 400 \text{ cm}$

$$R = 2/2 = 1 \text{ cm}$$

Capacitance of each conductor to neutral

$$\frac{2\pi\epsilon_0}{\log_e \frac{d}{r}} = \frac{2\pi\epsilon_0}{\log_e 400/1} \text{ F/m} = \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e 400/1} \text{ F/m}$$

$$= 9.28 \times 10^{-9} \text{ F/km} = 9.28 \times 10^{-3} \mu\text{F/km}$$

Example 24: A long two-wire line composed of solid ground conductors is 0.5 cm and the distance between their centres is 2 m. If this distance is double, then the inductance per unit length.

- (A) Halves
- (B) Increases but does not double
- (C) Doubles
- (D) Decreases but does not halve

Solution: (B)

Inductance of conductors per phase $= 0.2 \ln \left(\frac{d}{r'} \right) \text{ mH/km}$

$$L_1/\text{phase} = 0.2 \ln \left(\frac{2}{0.778 \times 0.5 \times 10^{-2}} \right) \text{ mH/km}$$

After distance is doubled

$$L_2/\text{phase} = 0.2 \ln \left(\frac{4}{0.778 \times 0.5 \times 10^{-2}} \right) \text{ mH/km}$$

$L_2/\text{phase} > L_1/\text{phase}$ but $L_2/\text{phase} \neq 2L_1/\text{phase}$
Inductance per unit length has increased but not doubled

Example 25: The conductor of 15 km long, single-phase, two-wire line is separated by a distance of 1.5 m. The diameter of each conductor is 0.75 cm. If the conductors are of copper, the inductance of the circuit is

- (A) 2.22 mH
- (B) 22.8 mH
- (C) 1.10 mH
- (D) 11.0 mH

Solution: (B)

Inductance of 1- ϕ circuit $= 2 L$ per phase

$$= 2 \times 0.2 \ln \frac{d}{r'}$$

Distance (d) = 1.5 m,

Radius (r') = $0.778 \times 0.75 \times 10^{-2} \text{ m}$

$$L = 0.4 \ln \left(\frac{1.5}{0.778 \times 0.75 \times 10^{-2}} \right) \text{ mH/km}$$

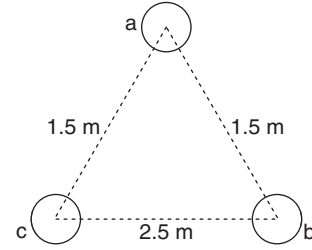
$$= 2.2L \text{ mH/km}$$

Length of the line is 10 km

Total inductance $= L \times 10 \text{ km}$

$$L = 22.2 \text{ mH}$$

Example 26: A 100 km long three-phase transmission lines are shown in the figure. The diameter of each conductor is 0.5 cm. The value of the inductance per phase is



- (A) 1.36 mH
- (B) 136 mH
- (C) 6.817 mH
- (D) 681 mH

Solution: (B)

Inductance per phase $= 2 \times 10^{-7} \ln \frac{\text{GMD}}{\text{GMR}}$

$$\text{GMD} = \sqrt[3]{D_{ab} D_{bc} D_{ca}} = \sqrt[3]{1.5 \times 1.5 \times 2.5} = 1.778 \text{ m}$$

$$\text{GMR} = 0.778 r = 0.778 \times 0.25 \text{ cm}$$

$$= 0.1945 \times 10^{-2} \text{ m}$$

Inductance per phase $= 0.2 \ln \frac{\text{GMD}}{\text{GMR}} \text{ mH/km}$

$$= 0.2 \ln \left(\frac{1.778}{0.1945 \times 10^{-2}} \right)$$

$$= 1.36 \text{ mH/km}$$

Total inductance per phase $= 1.36 \times 100 \text{ km} = 136 \text{ mH}$

Example 27: A 125 km long transmission line is loaded at 110 kV. If the loss of line is 20 MW and the load is 175 MVA the resistance of the line is

- (A) 7.9 Ω/phase
- (B) 4.56 Ω/phase
- (C) 13.68 Ω/phase
- (D) 45.6 Ω/phase

Solution: (A)

Given line loss expression is given by

$$P_{\text{loss}} = I_L^2 R$$

$$I_L = \frac{P_L}{V} = \frac{175 \times 10^6}{110 \times 10^3} = 1.6 \text{ kA}$$

$$P_{\text{loss}} = 20 \times 10^6 = I_L^2 R = (1.6 \times 10^3)^2 R$$

$$R = 7.9 \Omega/\text{phase}$$

Common Data Questions

Example 28: A 100 km long, three-phase, 110 kV 50 Hz overhead transmission line has three conductors each diameter 1.5 cm. The conductors are spaced 2 m at the corners of equilateral triangle. The capacitance of the line is

- (A) $1 \times 10^{-3} \mu\text{F}$
- (B) 2.29 F
- (C) 1.5 F
- (D) 1.5 μF

Solution: (A)

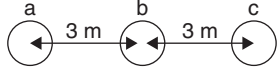
Capacitance of the line (C) $= \frac{2\pi \epsilon_0}{\ln \frac{d}{r}}$

$$= \frac{2\pi \times 8.854 \times 10^{-12}}{\ln \left(\frac{200}{0.75} \right)} = 10 \times 10^{-12} \text{ F/m}$$

Capacitance of 100 km line = $10 \times 100 \times 10^{-12}$
 $C = 1 \times 10^{-9}$

Example 29: The capacitance to neutral is
 (A) $8.4 \mu\text{F/km}$ (B) $0.0119 \mu\text{F/km}$
 (C) 8.4 F/km (D) 0.0084 F/km

Solution: (B)



$$C_n = \frac{0.242}{\ln \frac{D_{eq}}{r}}$$

$$D_{eq} = (3 \times 3 \times 6)^{1/3} = 3.78 \text{ m}$$

$$C_n = \frac{0.0242}{\ln \frac{3.78}{0.5}} = 0.0119 \mu\text{F/km}$$

Example 30: The charging current per kilometre line is
 (A) 0.379 A/km (B) 0.167 A/m
 (C) 0.379 A/m (D) 0.23 A/km

Solution: (D)

$$X_n = \frac{1}{\omega C_n} = \frac{1}{2 \times \pi \times 50 \times 0.0119 \times 10^{-6}}$$

$$= 267.48 \times 10^3 \Omega/\text{km}$$

$$\text{Charging current} = \frac{V_n}{X_n} = \frac{\left(\frac{100}{\sqrt{3}}\right) \times 1000}{267.4 \times 10^3}$$

$$= 0.23 \text{ A/km}$$

PERFORMANCE OF LINES

Performance of transmission lines is to find the efficiency and Regulation.

The efficiency of the transmission line is %

$$\text{Efficiency} = \frac{\text{Power delivered at the receiving end}}{\text{Power sent from sending end}} \times 100$$

$$\frac{\text{Power delivered at the receiving end}}{\text{Power sent from sending end}} \times 100$$

Receiving end in a transmission line is where the load is connected and the supply is from the sending end.

Regulation

The regulation of the line is defined as the change in the receiving end voltage, expressed as the percentage of full load voltage from no - load to full load, keeping the sending end voltage and frequency constant.

$$\% \text{ Regulation} = \frac{V_{R1} - V_{R2}}{V_{R2}} \times 100$$

where V_{R1} is the receiving end voltage under No-load condition.

V_{R2} is the receiving end voltage under full load condition. Note that V_{R1} and V_{R2} are magnitudes only Regulation is calculated on

1. Speed
2. Voltage

Performance of rotating equipment is evaluated based on efficiency and speed regulation.

Performance of static equipment is evaluated based on Efficiency and voltage regulation.

Since transmission line comes under static equipment, performance is evaluated when the system reaches steady state rather than in transient state.

- The loss in a transmission line is due to real power but not due to reactive power. Hence, the efficiency is calculated in terms of real power.

Classification of Overhead Transmission Lines

- In steady-state condition, for a line whose lengths is less than 160 km, the voltage or current does not vary much and for the line lengths of about 160 km, the parameters can be assumed as lumped but not distributed. These lines are called as electrically short transmission lines. These electrically short transmission lines are again divided as short and medium transmission lines.

Short Transmission Lines

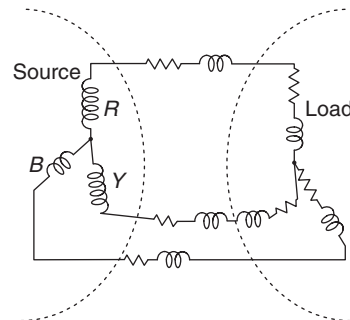
The lines of lengths up to 80 km are known as short transmission lines in which the effect of shunt capacitance is neglected.

Medium Transmission Lines

The lines of lengths above 80 km and below 160 km are known as medium transmission lines in which the shunt capacitance is assumed to be lumped at the centre (or) half of the capacitance is lumped at each end of the line

Long Transmission Lines

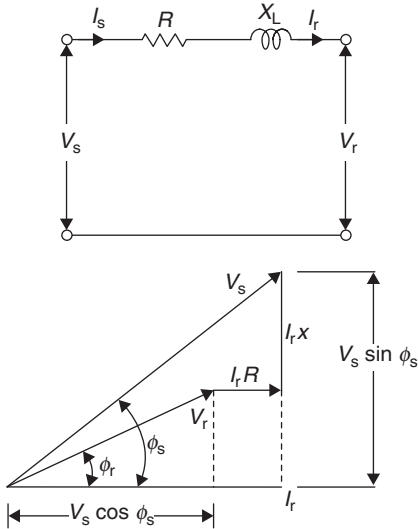
For line lengths more than 160 km, all the parameters are distributed and in some cases where the line length is less than 250 km, the analysis is done using nominal- π and T.



- The above figure shows a balanced 3- ϕ source which is connected to a balanced star load. As the sum of all the current in a balanced polyphase network is zero, hence the current through the wire connected between the star point of the load and neutral of the system is zero. Hence, the star point of the load and the neutral point of the source are at the source potential. Hence, for all purposes, a 3- ϕ balanced system can therefore be analysed on a single-phase basis.

Short Transmission Line

- The parameters considered here are the lumped parameters. Lumped parameters are those which can be placed at one point (or) two points to the maximum, and these parameters have physical and electrical separation. These parameters are used to study the steady-state behaviour of the long transmission lines.
- The inductance will be high and capacitance will be less in short transmission lines. Hence the capacitance is neglected. Therefore the parameters present are only resistance and inductance.
- The equivalent circuit and the vector diagram for a short transmission line are shown in the below figure.



The vector diagram is drawn taking I_r , the receiving end current, as the reference

From the vector diagram,

$$V_s \cos \phi_s = V_r \cos \phi_r + I_r R \quad (7)$$

$$V_s \sin \phi_s = V_r \sin \phi_r + I_r X \quad (8)$$

Squaring and adding equations (7) and (8),

$$V_s^2 = V_r^2 + 2 I_r V_r \cos \phi_r + 2 I_r X V_r \sin \phi_r + I_r^2 (R^2 + X^2)$$

$$\Rightarrow V_s = V_r \sqrt{1 + \frac{2 I_r R \cos \phi_r}{V_r} + \frac{2 I_r X \sin \phi_r}{V_r} + \frac{I_r^2}{V_r^2} (R^2 + X^2)}$$

The last term in the square root can be neglected.

$$V_s = V_r \left\{ 1 + \left(\frac{2 I_r R}{V_r} \right) \cos \phi_r + \frac{2 I_r X}{V_r} \sin \phi_r \right\}^{1/2}$$

Using Binomial expansion, and limiting only to two terms,

$$V_s \approx V_r + I_r R \cos \phi_r + I_r X \sin \phi_r$$

Here, V_s is the sending end voltage corresponding to a particular load current and power factor condition. From the equivalent circuit of short line, we can see that the receiving end voltage under no load V_r' is same as the sending end voltage under full-load condition, i.e.

$$V_r' = V_s$$

$$\therefore \% \text{ Regulation} = \frac{V_s - V_r}{V_r} \times 100$$

$$= \left(\frac{I_r R}{V_r} \cos \phi_r + \frac{I_r X}{V_r} \sin \phi_r \right) \times 100$$

$$\text{Per unit Regulation} = \frac{I_r R}{V_r} \cos \phi_r + \frac{I_r X}{V_r} \sin \phi_r$$

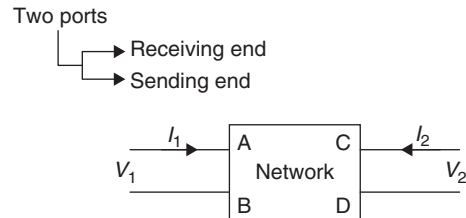
$$= u_r \cos \phi_r + u_x \sin \phi_r$$

where u_r and u_x are the per unit values of the resistance and reactance of the line, respectively.

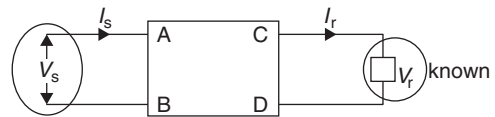
- The performance of transmission line can be analysed at a faster rate with the help of less no. of equations by representing the transmission line in a 2-port network model rather R-L-C model.

Two-port Network Model

Port: pair of terminals (phase and Neutral) is known as ports.



As we want V_1, I_1 from V_2, I_2 , we take the transmission line as A B C D networks.



The unknown sending end electrical quantities are expressed in terms of known receiving end electrical quantities along with certain parameters.

Those parameters are called A B C D parameters.

$$V_s = A V_r + B I_r \text{ kV/phase}$$

$$I_s = C V_r + D I_r \text{ kA/phase}$$

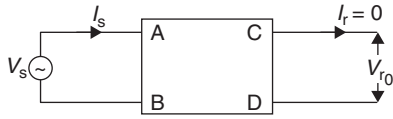
$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_r \\ I_r \end{bmatrix}$$

$$\begin{aligned} \text{If } A &= D \rightarrow \text{Symmetrical network} \\ \text{If } A D - B C &= 1 \\ \text{(or)} \quad -A D + B C &= 1 \end{aligned} \left. \vphantom{\begin{aligned} \text{If } A &= D \\ \text{If } A D - B C &= 1 \\ \text{(or)} \quad -A D + B C &= 1 \end{aligned}} \right\} \rightarrow \text{Reciprocal network}$$

- Electrical network may be symmetrical (or) unsymmetrical but always reciprocal network

Units for A, B, C, D

Open Circuit at the Receiving End



$$V_s = A V_{r0}$$

$$I_s = C V_{r0}$$

$$A = \frac{V_s}{V_{r0}} \Big|_{I_r=0} \text{ (unitless)}$$

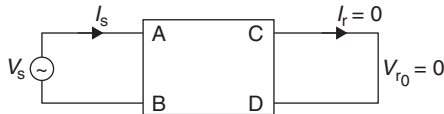
$$C = \frac{I_s}{V_{r0}} \Big|_{I_r=0}$$

'C' open circuit admittance in Mho's

$$\% \text{ Regulation} = \frac{|V_{r0}| - |V_r|}{|V_r|} \times 100$$

$$\% \varepsilon = \frac{\left| \frac{V_s}{A} \right| - |V_r|}{|V_r|} \times 100$$

Short Circuit at Receiving End



When the receiving end is short circuited $V_r = 0$. Then ABCD parameters are given by

$$V_s = B I_r$$

$$I_s = D I_r$$

$$\Rightarrow \frac{V_s}{I_s} = \frac{B}{D}$$

$$B = \frac{V_s}{I_r} \Big|_{V_r=0}$$

(Short-circuit impedance in ohms)

$$D = \frac{I_s}{I_r} \Big|_{V_r=0}$$

- For short transmission lines, regulation is equal to voltage drop but this is not possible in medium and long transmission lines, as shunt capacitance is present.
- In case of short transmission line, if load is open, then $|V_r| = |V_s|$ and $I_s = 0$. But this is not true in case of medium and long transmission lines as shunt capacitance is present and hence $I_s \neq 0$ but $I_r = 0$ and $V_s = V_r$.

Condition for Maximum Regulation

The expression is given by

$$\% \varepsilon = \% V_r \cos \phi_r + \% V_x \sin \phi_r$$

The condition for maximum regulation can be obtained from

$$\frac{d\% \varepsilon}{d\phi_r} = 0$$

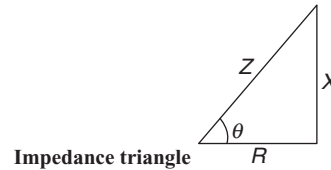
$$\Rightarrow -\% V_r \sin \phi_r + \% V_x \cos \phi_r = 0$$

$$\Rightarrow \tan \phi_r = \frac{\% V_x}{\% V_r} = \frac{\frac{I_r X}{V_r} \times 100}{\frac{I_r R}{V_r} \times 100} = \frac{X}{R}$$

$$\Rightarrow \phi_r = \tan^{-1} \left(\frac{X}{R} \right)$$

X – Line reactance Impedance triangle

R – Line resistance



Impedance triangle

$$\tan \theta = \frac{X}{R}$$

When the phase angle of load is equal to the impedance angle, the regulation is maximum.

$$\Rightarrow \tan \phi_r = \frac{X}{R} = \tan \theta$$

$$\Rightarrow \phi_r = \theta$$

Condition for Zero Regulation

$$\% \varepsilon = \% V_r \cos \phi_r - \% V_x \sin \phi_r = 0$$

$$\Rightarrow \tan \phi_r = \frac{\% V_r}{\% V_x} \frac{\frac{I_r R}{V_r} \times 100}{\frac{I_r X}{V_r} \times 100} = \frac{R}{X}$$

$$\Rightarrow \tan \phi_r = \frac{R}{X} = \cot \theta = \tan \left(\frac{\pi}{2} - \theta \right)$$

$$\Rightarrow \phi_r = \frac{\pi}{2} - \theta$$

$$\Rightarrow \phi_r + \theta = \frac{\pi}{2}$$

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- Max regulation occurs for lagging load,

$$\% \varepsilon = \% V_r \cos \phi_r + \% V_x \sin \phi_r$$

- Min regulation occurs for leading load,

$$\% \varepsilon = \% V_r \cos \phi_r - \% V_x \sin \phi_r$$

(negative value)

- We get % regulation as '0' only in the leading loads provided

$$(i) \frac{X}{R} = 1.0$$

(ii) 0.707 lead p.f.

Example 31: A single-phase overhead transmission line delivers 1100 kW at 33 kV at 0.8 p.f. lagging. The total resistance and inductive reactance of the line are 15 Ω and 20 Ω , respectively. Find the (i) sending end voltage (ii) sending end power factor and (iii) transmission efficiency.

Solution: Load power factor, $\cos \phi_r = 0.8$ lag

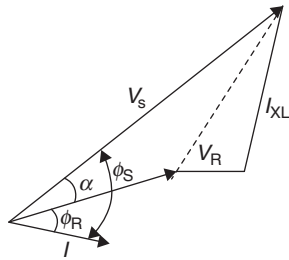
Total line impedance $\rightarrow Z = R + jX_L$
 $= 15 + j20$

Receiving end voltage, $V_R = 33$ kV

$$\begin{aligned} \therefore \text{Line current, } I &= \frac{\text{kW} \times 10^3}{V_R \times \cos \phi_R} \\ &= \frac{1100 \times 10^3}{33,000 \times 0.8} = 41.66 \end{aligned}$$

$$\cos \phi_R = 0.8$$

$$\Rightarrow \sin \phi_R = 0.6$$



$$\vec{V}_R = V_R + j0 = 33,000 \text{ V}$$

$$\begin{aligned} I &= I(\cos \phi_R - j \sin \phi_R) \\ &= 41.67 (0.8 - j 0.6) \\ &= 33.33 - j25 = 41.67 \angle -36.86^\circ \end{aligned}$$

$$\begin{aligned} (i) \text{ Sending end voltage, } \vec{V}_S &= \vec{V}_R + \vec{I} * \vec{Z} \\ &= 33,000 + (33.33 - j25.0) (15 + j20) \\ &= 33,999.95 + 291.6 j \end{aligned}$$

$$\begin{aligned} \therefore \text{ Magnitude of } V_S &= \sqrt{(33,999.95)^2 + (291.6)^2} \\ &= 34,001.20 \text{ V} \end{aligned}$$

(ii) Angle between $\vec{V}_S$ and $\vec{V}_R$

$$\alpha = \tan^{-1} \left(\frac{291.6}{33,999.95} \right) = 0.49^\circ$$

$\therefore$ Sending power factor angle,

$$\phi_s = \phi_r + \alpha = 36^\circ 86' + 0^\circ 49' = 37^\circ 35'$$

$\therefore$ Sending end power factor, $\cos \phi_s = \cos(37^\circ 35') = 0.79$ lagging

(iii) Line losses = $I^2 R = (41.67)^2 \times 15 = 26.045$ kW

Output = 1100 kW

$$\begin{aligned} \text{Power sent} &= 1100 + 26.045 \text{ kW} \\ &= 1126.045 \text{ kW} \end{aligned}$$

$\therefore$ Transmission efficiency

$$\begin{aligned} &= \frac{\text{Power delivered}}{\text{Power sent}} \times 100 \\ &= \frac{1100}{1126.045} \times 100 = 97.68\% \end{aligned}$$

Example 32: What is the maximum length in km for a single-phase transmission line having copper conductor of 0.775 cm² cross-section over which 200 kW at unity power factor and at 3300 V are to be delivered? The efficiency of transmission is 90%. Take specific resistance as 1.725 $\mu\Omega$ cm.

Solution: Receiving end power = 200 kW

Transmission efficiency = 0.9

$$\begin{aligned} \therefore \text{ Sending end power} &= \frac{2,00,000}{0.9} \\ &= 2,22,222 \text{ W} \end{aligned}$$

$$\begin{aligned} \therefore \text{ Line losses} &= 2,22,222 - 2,00,000 \\ &= 22,222 \text{ W} \end{aligned}$$

$$\text{Line current } I = \frac{200 \times 10^3}{3300 \times 1} = 60.6 \text{ A}$$

Let $R \Omega$ be the resistance of the 2 conductors, hence line losses = $2 I^2 R$

$$22,222 = 2 I^2 R$$

$$\Rightarrow R = \frac{22,222}{2 \times (60.6)^2} = 3.025 \Omega$$

$$\begin{aligned} \therefore R &= \frac{\rho I}{A} \Rightarrow l = \frac{R \times A}{\rho} \\ &= \frac{3.025 \times 0.775}{1.725 \times 10^{-6}} = 1.36 \times 10^6 \text{ cm} \\ &= 13.6 \text{ km} \end{aligned}$$

Medium Transmission Lines

We know that the transmission lines with lengths between 80 km and 160 km are categorized as medium length lines, in which the parameters are assumed to be lumped.

The shunt capacitance is assumed to be either at the middle of the line or half of the total capacitance is concentrated at each end of the line. These two configurations are classified as—nominal-T and nominal- π , respectively. The circuits are shown below.

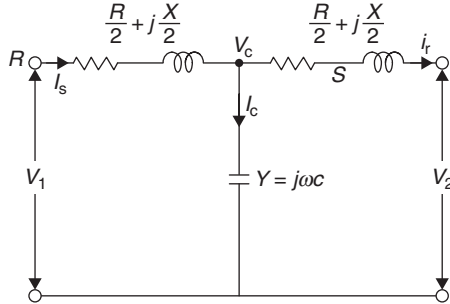


Figure 13 Nominal-T network

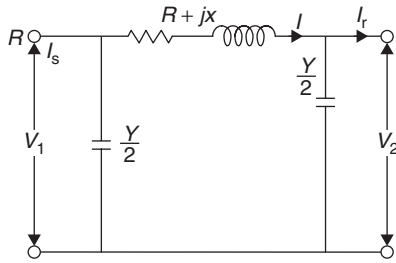


Figure 14 Nominal- π network

Nominal-T Network

To analyse the medium length lines by this network, it would be convenient to take receiving end current as the reference vector rather than taking the receiving end voltage (V_r) as the reference. The vector diagram for lagging power factor load is given below.

From the network diagram, in order to calculate the regulation, we have to first calculate the sending end voltage V_s .

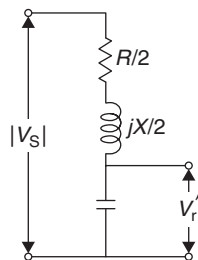
$$V_c = (|V_r| \cos \phi_r + j |V_r| \sin \phi_r) + I_r \left(\frac{R}{2} + j \frac{X}{2} \right)$$

$$I_c = j\omega C V_c$$

$$I_s = I_c + I_r = I_r + j\omega C V_c$$

$$V_s = V_c + I_s \left(\frac{R}{2} + j \frac{X}{2} \right)$$

$$= (|V_r| \cos \phi_r + j |V_r| \sin \phi_r) + I_r \left(\frac{R}{2} + j \frac{X}{2} \right) + I_s \left(\frac{R}{2} + j \frac{X}{2} \right)$$



To calculate the regulation, we have to find V_r' , the receiving end no load voltage keeping the sending end voltage V_s fixed in magnitude. The above nominal-T network reduces to the network beside.

$$V_r' = \frac{|V_s| \left(\frac{-j}{\omega C} \right)}{\frac{R}{2} + j \frac{X}{2} - \frac{j}{\omega C}}$$

∴ The regulations for the nominal-T network is obtained as

$$\% \text{ regulations} = \frac{V_r' - V_r}{V_r} \times 100$$

The efficiency is obtained as

$$\% \eta = \frac{\text{Power at receiving end}}{\text{Power at receiving + Loss}} \times 100$$

In order to determine A, B, C, D constants for nominal-T,

$$V_c = V_r + I_r \left(\frac{Z}{2} \right)$$

$$I_c = V_c Y$$

$$I_s = I_r + I_c = I_r + V_c Y \Rightarrow I_r + \left(V_r + I_r \frac{Z}{2} \right) Y$$

$$V_s = V_c + I_s \frac{Z}{2} = V_r + I_r \frac{Z}{2} + \left\{ I_r + \left(V_r + I_r \frac{Z}{2} \right) Y \right\} \frac{Z}{2}$$

$$= V_r \left(1 + \frac{YZ}{2} \right) + I_r \left(\frac{Z}{2} + \frac{Z}{2} + \frac{YZ^2}{4} \right)$$

$$= V_r \left(1 + \frac{YZ}{2} \right) + I_r \left(Z + \frac{YZ^2}{4} \right)$$

$$I_s = I_r \left(1 + \frac{YZ}{2} \right) + V_r Y = Y V_r + \left(1 + \frac{YZ}{2} \right) I_r$$

By comparing the above equations with the standard ones,

$$V_s = A V_r + B I_r$$

$$I_s = C V_r + D I_r$$

$$\Rightarrow A = 1 + \frac{YZ}{2}, B = Z \left(1 + \frac{YZ}{4} \right), C = Y, D = \left(1 + \frac{YZ}{2} \right)$$

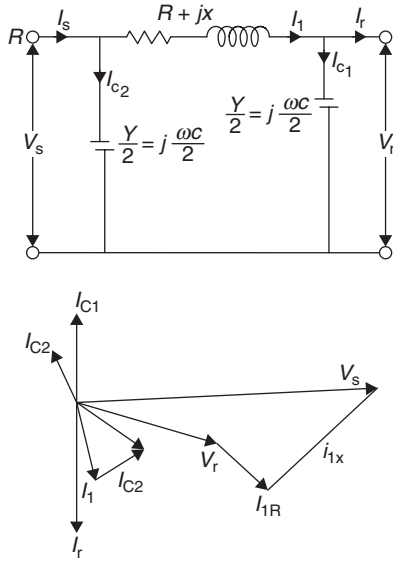
We can see that $A = D$,

$$\text{and } AD - BC = \left(1 + \frac{YZ}{2} \right)^2 - YZ \left(1 + \frac{YZ}{4} \right)$$

$$= 1 + \frac{Y^2 Z^2}{4} - YZ - YZ - \frac{Y^2 Z^2}{4} = 1$$

Nominal- π Representation

Nominal- π representation of medium length transmission lines and their phasor diagram are shown below.



From the above representation,

$$I_{c1} = jV_r \frac{\omega C}{2}, I_1 = |I_r| (\cos \phi_r - j \sin \phi_r) + jV_r \frac{\omega C}{2}$$

$$V_s = V_r + I_1 Z$$

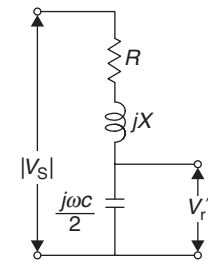
$$= V_r + \left\{ |I_r| (\cos \phi_r - j \sin \phi_r) + jV_r \frac{\omega C}{2} \right\} (R + jX)$$

and $I_s = I_1 + I_{c2} = I_1 + j \frac{V_s \omega C}{2}$

$$I_s = |I_r| (\cos \phi_r - j \sin \phi_r) + \frac{jV_r \omega C}{2} + \frac{jV_s \omega C}{2}$$

$$V_s = V_r + \left\{ |I_r| (\cos \phi_r - j \sin \phi_r) + \frac{jV_r \omega C}{2} \right\} (R + jX)$$

To calculate the regulation, we have to find no load receiving end voltage by keeping the sending end voltage constant. The nominal- π circuit for no-load reduces as follows.



$$V_r' = \frac{|V_s| \left(\frac{-2j}{\omega C} \right)}{R + jX - \frac{2j}{\omega C}}$$

$$\therefore \% \text{ Regulation} = \frac{V_r' - V_r}{V_r} \times 100$$

and $\% \eta = \frac{P}{P + 3I_1^2 R} \times 100$

To find the A, B, C, D constants for nominal- π network,

$$I_c = V_r \frac{Y}{2}$$

$$I_1 = I_r + I_c = I_r + \frac{V_r Y}{2}$$

$$V_s = V_r + I_1 Z = V_r + \left(I_r + \frac{V_r Y}{2} \right) Z = \left(1 + \frac{YZ}{2} \right) V_r + ZI_r$$

$$I_s = I_1 + I_{c2} = I_1 + \frac{V_s Y}{2} = I_r + \frac{V_r Y}{2} + \left\{ V_r \left(1 + \frac{YZ}{2} \right) + ZI_r \right\} \frac{Y}{2}$$

$$= V_r \left(Y + \frac{Y^2 Z}{4} \right) + \left(1 + \frac{YZ}{2} \right) I_r$$

Comparing the coefficients of equations with the standard ones,

$$A = 1 + \frac{YZ}{2},$$

$$B = Z$$

$$C = Y \left(1 + \frac{YZ}{4} \right)$$

$$D = \left(1 + \frac{YZ}{2} \right)$$

It is seen that $A = D$,

and $AD - BC = \left(1 + \frac{YZ}{2} \right)^2 - ZY \left(1 + \frac{YZ}{4} \right) = 1$

Example 33: A three-phase 50 Hz overhead transmission line 100 km long has the following constants:

Resistance/km/phase = 0.1 Ω

Inductive reactance/km/phase = 0.2 Ω

Capacitive susceptance/km/ phase = 0.04×10^{-4} mho

Determine (i) sending end current (ii) sending end voltage (iii) sending end power factor and (iv) transmission efficiency when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0.8 lagging, use nominal-T method.

Solution: Total resistance/phase, $R = 0.1 \times 100 = 10 \Omega$

Total resistance/phase, $X_L = 0.2 \times 100 = 20 \Omega$

Capacitive susceptance,

$$Y = 0.04 \times 10^{-4} \times 100 = 4 \times 10^{-4} \Omega$$

Receiving end voltage/phase

$$V_R = \frac{66,000}{\sqrt{3}} = 38,105 \text{ V}$$

Load current, $I_R = \frac{10,000 \times 10^3}{\sqrt{3} \times 66 \times 0.8} = 109 \text{ A}$

$$\cos \phi_r = 0.8, \Rightarrow \sin \phi_r = 0.6$$

Impedance per phase, $\bar{Z} = R + jX_L = 10 + j20$

Taking receiving end voltage as the reference vector, we have,

Receiving voltage, $\bar{V}_R = V_R + j0 = 38,105 \text{ V}$

Load current, $\bar{I}_R = I_R (\cos \phi_r - j \sin \phi_r)$
 $= 109 (0.8 - j0.6) = 87.2 - j65.4$

Voltage across C, $\bar{V}_1 = \bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2}$
 $= 38,105 + (87.2 - j65.4) (5 + j10)$
 $= 38,105 + 436 + j872 - j327 + 654$
 $= 39,195 - j545$

Charging current, $\bar{I}_c = jY\bar{V}_1$
 $= j4 \times 10^{-4} (39,195 - j545)$
 $= 0.218 + j15.6$

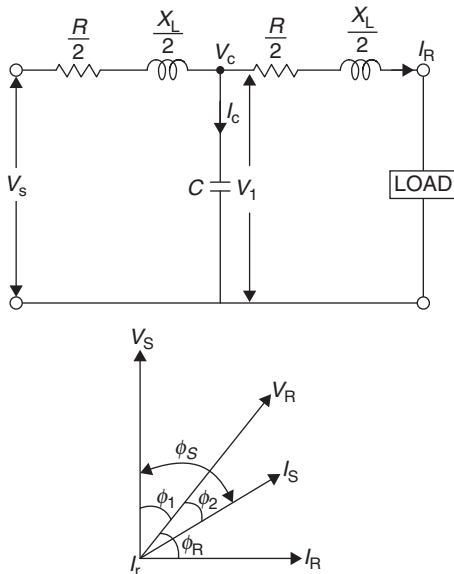
Sending end current

$\bar{I}_s = \bar{I}_R + \bar{I}_c$
 $= (87.2 - j65.4) + (0.218 + j15.6)$
 $= 87.4 - j49.8 = 100 \angle -29^\circ 40' = 100 \text{ A}$

Sending end voltage,

$\bar{V}_s = \bar{V}_1 + \bar{I}_s \frac{\bar{Z}}{2}$
 $= (39,195 - j545) + (87.4 - j49.8) (5 + j10)$
 $= 40130 + j80$

$\therefore$ Line-to-line sending end voltage $= 40130 \times \sqrt{3}$
 $= 69,507 \text{ V} = 69.507 \text{ kV}$



$\phi_s = \theta_1 + \theta_2 = 6' + 29^\circ 40' = 29^\circ 46'$

$\therefore$ Sending end power factor, $\cos \phi_s$
 $= \cos 29^\circ 46 = 0.868$

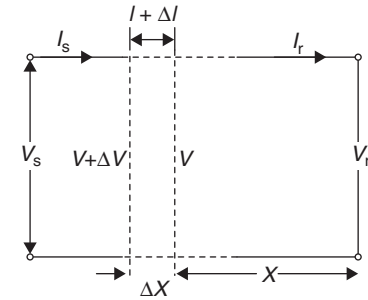
$\therefore$ Sending end power $= 3 V_s I_s \cos \phi_s$
 $= 3 \times 40,130 \times 100 \times 0.868 = 10,449.85 \text{ kW}$

Power delivered $= 10,000 \text{ kW}$

$\therefore$ Transmission efficiency $= \frac{10000}{10,449.855} \times 100 = 95.7\%$

Long Transmission Lines

If the lines are more than 160 km in length, for the analysis to be accurate, the parameters must be taken as distributed rather than lumped for the voltage and currents.



Let Z = Series impedance per unit length

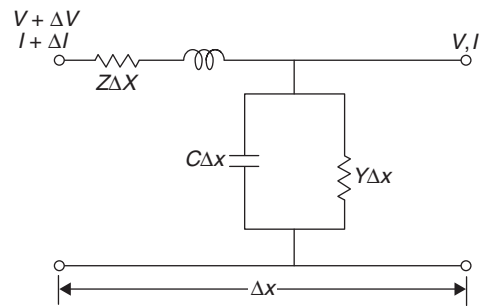
y = Shunt admittance per unit length

l = Length of the line

$Z = zl$ = Total series impedance

$Y = yl$ = Total shunt admittance

For analysis, the elemental length dx is redrawn



We take the receiving end as the reference for measuring distances. Take an elemental length dx of the line at a distance ' x ' from the receiving end. Let the voltage and current at a distance ' x ' are V and I and at a distance $x + dx$, $V + \Delta V$ and $I + \Delta I$, respectively.

$$\Delta V = I_z \Delta x \Rightarrow \Delta I = V_y \Delta x$$

$$\Rightarrow \frac{\Delta V}{\Delta x} I_z \Rightarrow \frac{\Delta I}{\Delta x} = V_y$$

When $\Delta x \rightarrow 0$, the equations become,

$$\frac{dV}{dx} = Iz \quad (9)$$

$$\frac{dI}{dx} = Vy \quad (10)$$

Differentiating Equation (9), we get

$$\Rightarrow \frac{d^2V}{dx^2} = Z \frac{dI}{dx} = Z \cdot Vy \cdot V \Rightarrow \frac{d^2V}{dx^2} = Z y V = 0$$

The solution of the above equation is

$$V = A \exp(\sqrt{yz} \cdot x) + B \exp(-\sqrt{yz} \cdot x)$$

Let $Z_c = \sqrt{\frac{z}{y}}$ and $\gamma = \sqrt{yz} = \alpha + j\beta$

where Z_c is known as characteristic impedance and ' γ ' is the propagation constant.

The above equations can be rewritten as

$$V = A e^{\gamma x} + B e^{-\gamma x}$$

$$I = \frac{1}{Z_c} (A e^{\gamma x} - B e^{-\gamma x})$$

The constants can be determined by the boundary conditions, at $x = 0$,

$$V = V_r \text{ and } I = I_r \Rightarrow V_r = A + B$$

$$I_r = \frac{1}{Z_c} (A - B) \Rightarrow A = \frac{V_r + I_r Z_c}{2} \text{ and } B = \frac{V_r - I_r Z_c}{2}$$

Substituting 'A' and 'B' in above equations

$$V = \frac{V_r + I_r Z_c}{2} e^{\gamma x} + \frac{V_r - I_r Z_c}{2} e^{-\gamma x}$$

and
$$I = \frac{1}{Z_c} \left[\frac{V_r + I_r Z_c}{2} e^{\gamma x} - \frac{V_r - I_r Z_c}{2} e^{-\gamma x} \right]$$

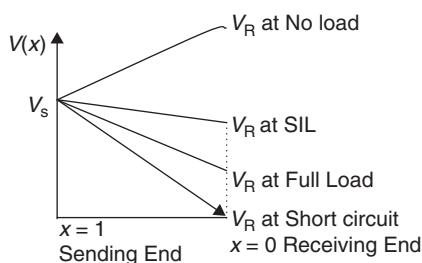
We can observe that V and I are the functions of the distance x , receiving end voltage V_r and current I_r and the parameters of the line, which means they will vary as we move from receiving end towards the sending end.

$$Z_c \text{ and } \gamma \text{ are complex} \Rightarrow Z_c = \sqrt{\frac{z}{y}} = \sqrt{\frac{r + j\omega L}{g + j\omega c}}$$

$$\text{For a loss-less line, } r = 0, g = 0, \Rightarrow Z_c = \sqrt{\frac{L}{c}}$$

The Surge impedance (Z_c) is purely resistive and is known as surge impedance of the line.

Voltage Profiles



During high frequencies (or) surges, normally the losses are neglected and hence, the characteristic impedance becomes surge impedance. 'Surge impedance loading' of a line is the power transmitted when the line is terminated through a resistance equal to surge impedance. A line terminated in its characteristic impedance is called a flat line or an infinite line.

The lower value of surge impedance in case of cables is due to relatively large capacitance and low inductance of the cables.

The propagation constant $\gamma = \alpha + j\beta$, the real part is known as attenuation constant (α) and the quadrature component β is known as the phase constant and is measured in radians per unit length.

In the above equation, the first part, $\frac{V_r + I_r Z_c}{2} e^{\alpha x} e^{j\beta x}$ is called the incident voltage wave as its value increases as ' x ' is increased, this means that the magnitude of voltage wave decreases in magnitude as it travels towards the receiving end. Similarly, the second part of the equation $\frac{V_r - I_r Z_c}{2} e^{-\alpha x} e^{-j\beta x}$ is called reflected voltage. At any point on the line, the voltage wave is the sum of the incidence and reflected waves. The above result is true even for the current equation.

The above equations for voltage and current can be re-arranged as follows,

$$V = V_r \frac{e^{\gamma x} + e^{-\gamma x}}{2} + I_r Z_c \frac{e^{\gamma x} - e^{-\gamma x}}{2}$$

$$= V_r \cosh \gamma x + I_r Z_c \sinh \gamma x$$

and
$$I = \frac{1}{Z_c} \left[V_r \frac{e^{\gamma x} + e^{-\gamma x}}{2} + I_r Z_c \frac{e^{\gamma x} - e^{-\gamma x}}{2} \right]$$

$$= \frac{1}{Z_c} [V_r \sinh \gamma x + I_r Z_c \cosh \gamma x]$$

$$= \frac{V_r}{Z_c} \sinh \gamma x + I_r \cosh \gamma x$$

The above equations can be re-written for $x = l$, where $V = V_s$ and $I = I_s$

$$V_s = V_r \cosh \gamma l + I_r Z_c \sinh \gamma l$$

$$I_s = \frac{V_r \sinh \gamma l}{Z_c} + I_r \cosh \gamma l$$

Comparing the above equations with standard equations,

$$A = \cosh \gamma l$$

$$B = Z_c \sinh \gamma l$$

$$C = \frac{\sinh \gamma l}{Z_c}, D = \cosh \gamma l$$

We can see that $A = D$ and

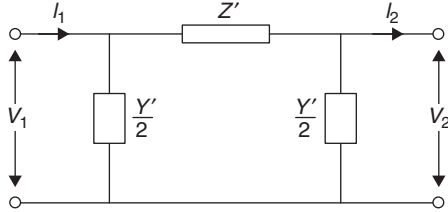
$$AD - BC = \cosh^2 \gamma l - \sinh^2 \gamma l = 1$$

Equivalent Representation of a Long Line

For Line lengths more than 160 km, the parameters are assumed to be distributed. The equivalent-T and Equivalent- π networks are represented. For long lines, nominal-T and nominal- π networks may not be used as these networks consider the lines with lumped parameters.

Equivalent- π Representation

The sending end and receiving end voltages and currents are found from this equivalent network.



V_1 – Sending end voltage
 V_2 – Receiving end voltage
 I_1 – Sending end current
 I_2 – Receiving end current.

$$V_1 = \left(1 + \frac{Y'Z'}{2}\right)V_r + Z'I_r \quad (11)$$

$$I_1 = Y'\left(1 + \frac{Y'Z'}{4}\right)V_r + \left(1 + \frac{Y'Z'}{2}\right)I_r \quad (12)$$

Comparing the above two equations with the standard ones, i.e.,

$$V_1 = V_2 \cosh \gamma l + \frac{T}{2} Z_c \sinh \gamma l$$

$$I_1 = V_2 \frac{\sinh \gamma l}{Z_c} + \frac{T}{2} \cosh \gamma l$$

$$\Rightarrow 1 + \frac{Y'Z'}{2} = \cosh \gamma l, Z' = Z_c \sinh \gamma l,$$

$$Y'\left(1 + \frac{Y'Z'}{4}\right) = \frac{\sinh \gamma l}{Z_c} \text{ and } 1 + \frac{Y'Z'}{2} = \cosh \gamma l$$

$$\text{Consider } Z' = Z_c \sinh \gamma l$$

$$= \sqrt{\frac{z}{y}} \times \sinh \gamma l \times \frac{l}{l\sqrt{yz}} \times \sqrt{yz}$$

$$= Zl \frac{\sinh \gamma l}{\gamma l} = Z \frac{\sinh \gamma l}{\gamma l}.$$

From the above expression, we can observe that the equivalent series impedance, the lumped impedance must be multiplied with $\frac{\sinh \gamma l}{\gamma l}$.

In order to get the shunt parameters of the equivalent- π network, substitute Z' from the above equation, i.e.,

$$1 + \frac{Y'Z'}{2} = \cosh \gamma l \Rightarrow 1 + \frac{Y'Z'}{2} \times Z_c \sinh \gamma l = \cosh \gamma l$$

(as $Z = z l$ and $\gamma = \sqrt{yz}$)

$$\Rightarrow \frac{Y'}{2} Z_c \sinh \gamma l = \cosh \gamma l - 1$$

$$= \cosh^2 \frac{\gamma l}{2} + \sinh^2 \frac{\gamma l}{2} - \cosh^2 \frac{\gamma l}{2} + \sinh^2 \frac{\gamma l}{2}$$

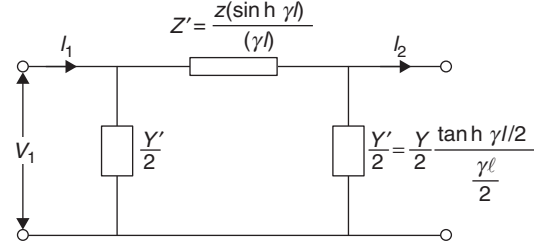
$$\Rightarrow Y' Z_c \sinh \frac{\gamma l}{2} \cosh \frac{\gamma l}{2} = 2 \sinh^2 \frac{\gamma l}{2}$$

$$\Rightarrow \frac{Y'}{2} = \frac{1}{Z_c} \tanh \frac{\gamma l}{2} = \sqrt{\frac{y}{z}} \times \frac{\sqrt{yz} l}{2} \times \frac{\tanh \gamma l / 2}{\sqrt{yz} l / 2}$$

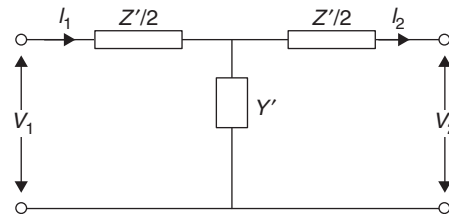
$$= \frac{Y \tanh \gamma l / 2}{\gamma l / 2}.$$

where ' Y ' is the total shunt admittance of the nominal- π network. Hence, we can conclude that the total shunt admittance of the nominal- π network should be multiplied with $\frac{\tanh \gamma l / 2}{\gamma l / 2}$

Hence, the equivalent circuit can be redrawn as



Equivalent T-Representation of Long Line



V_1 and V_2 are the sending end and receiving end voltages, respectively, whereas I_1 and I_2 are the sending end and receiving end currents, respectively.

From the network of nominal-T,

We have,

$$V_s = V_r \left(1 + \frac{YZ}{2}\right) + I_r Z \left(1 + \frac{YZ}{4}\right)$$

$$I_s = Y V_r + \left(1 + \frac{YZ}{2}\right) I_r$$

As the above network is similar to nominal-T network, we can use the above equations to analyse the equivalent-T network.

$$V_1 = \left(1 + \frac{Y'Z'}{2}\right)V_2 + Z' \left(1 + \frac{Y'Z'}{4}\right)I_2$$

$$I_1 = \left(1 + \frac{Y'Z'}{2}\right)I_2 + Y'V_2$$

Comparing the above equations with the equations of long transmission lines, i.e.

$$V_1 = V_s \cosh \gamma l + I_2 Z_c \sinh \gamma l$$

$$I_1 = V_2 \frac{\sinh \gamma l}{Z_c} + I_s \cosh \gamma l$$

$$\left(1 + \frac{YZ}{2}\right) = \cosh \gamma l, \quad Z' \left(1 + \frac{Y'Z'}{4}\right) = Z_c \sinh \gamma l$$

$$1 + \frac{Y'Z'}{2} = \cosh \gamma l, \quad Y' = \frac{\sinh \gamma l}{Z_c}.$$

To find the shunt branch parameters of the equivalent-T circuit,

$$Y' = \frac{1}{Z_c} \sinh \gamma l = \sqrt{\frac{y}{z}} \times \sinh \gamma l \times \frac{\sqrt{yz} \times l}{\sqrt{yz} \times l}$$

$$= Y \frac{\sinh \gamma l}{\gamma l} \quad (\text{where } \gamma = \sqrt{yz}).$$

Hence, to find the equivalent shunt admittance, the lumped shunt admittance should be multiplied with $\frac{\sinh \gamma l}{\gamma l}$.

To find the series impedance, consider the equation.

$$\left(1 + \frac{Y'Z'}{2}\right) = \cosh \gamma l$$

Substitute $Y' = \frac{\sinh \gamma l}{Z_c}$ in the above equation,

$$\Rightarrow 1 + \frac{Z'}{2} \times \frac{\sinh \gamma l}{Z_c} = \cosh \gamma l$$

$$\Rightarrow \frac{Z'}{2} \times \frac{2 \sinh \frac{\gamma l}{2} \cosh \frac{\gamma l}{2}}{Z_c} = \cosh \gamma l - 1 = 2 \sinh^2 \frac{\gamma l}{2}$$

$$\Rightarrow \frac{Z'}{2} = Z_c \tanh \frac{\gamma l}{2} = \sqrt{\frac{z}{y}} \tanh \frac{\gamma l}{2}$$

Multiply and divide by $\frac{\sqrt{yz} \cdot l}{2}$

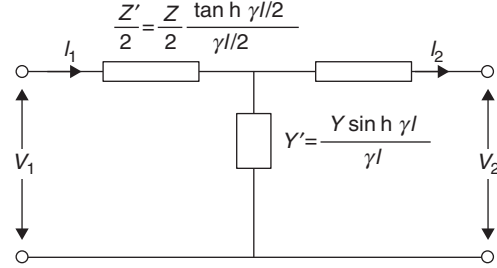
$$\Rightarrow \frac{Z'}{2} = \sqrt{\frac{z}{y}} \times \frac{\sqrt{yz} \cdot l}{2} \times \frac{\tanh(\gamma l/2)}{\frac{\sqrt{yz} \cdot l}{2}}$$

$$= \frac{Z \tanh \frac{\gamma l}{2}}{\frac{\gamma l}{2}} \quad (\text{where } z = x \cdot l \text{ and } \gamma = \sqrt{zy})$$

Therefore to get the series equivalent impedance, the lumped series impedance of the nominal-T network should

be multiplied with $\frac{\tanh \frac{\gamma l}{2}}{\frac{\gamma l}{2}}$.

The equivalent T-representation is as follows.



Calculation of Receiving End Voltages and Currents

We know that

$$V_s = AV_r + B I_r \quad (13)$$

$$I_s = CV_r + D I_r \quad (14)$$

where V_s, I_s are sending end voltage and current, respectively, while V_r, I_r are receiving end voltage and current, respectively.

Multiply (13) with C and (14) with A

$$\Rightarrow CV_s = CA V_r + CB I_r \quad (15)$$

$$AI_s = AC V_r + AD I_r \quad (16)$$

Subtract (15) from (16),

$$AI_s - CV_s = AC V_r + AD I_r - CA V_r - CB I_r$$

$$\Rightarrow AI_s - CV_s = I_r(AD - BC)$$

We know that $AD - BC = 1$ and $A = D$, receiving end current is expressed as

$$\Rightarrow I_r = -CV_s + DI_s \quad (17)$$

Multiply (13) with 'D' and (14) with 'B'

$$DV_s = AD V_r + DB I_r \quad (18)$$

$$BI_s = BC V_r + BD I_r \quad (19)$$

Subtract (18) from (19),

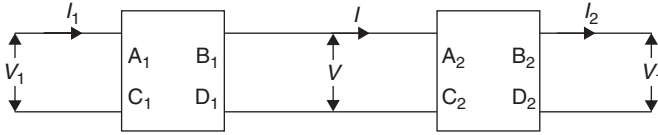
$$DV_s - BI_s = (AD - BC) V_r$$

We know that $AD - BC = 1$, receiving end voltage is expressed as

$$\Rightarrow V_r = DV_s - BI_s \quad (20)$$

Constants of Two Networks in Series

Two networks are said to be connected in series when the output of one network is connected to the input of the other network. Let the constants of the network be A_1, B_1, C_1, D_1 and the constants of the networks be A_2, B_2, C_2, D_2 , provided there are two networks connected in series as shown in the below figure.



From the figure,

$$V = D_1 V_1 - B_1 I_1$$

$$I = -C_1 V_1 + A_1 I_1$$

And, $V = A_2 V_2 + B_2 I_2$

$$I = C_2 V_2 + D_2 I_2$$

From the above equations, we can see that

$$D_1 V_1 - B_1 I_1 = A_2 V_2 + B_2 I_2 \quad (21)$$

$$-C_1 V_1 + A_1 I_1 = C_2 V_2 + D_2 I_2 \quad (22)$$

Multiply Equation (21) with A_1 and (22) with B_1 and adding the equations,

$$\begin{aligned} & A_1 D_1 V_1 - A_1 B_1 I_1 = A_1 A_2 V_2 + A_1 B_2 I_2 \\ + & -B_1 C_1 V_1 + A_1 B_1 I_1 = B_1 C_2 V_2 + B_1 D_2 I_2 \end{aligned}$$

$$V_1 (A_1 D_1 - B_1 C_1) = (A_1 A_2 + B_1 C_2) V_2 + (A_1 B_2 + B_1 D_2) I_2 \quad (23)$$

Multiply (21) with C_1 and (22) by D_1 and the equations,

$$\begin{aligned} & C_1 D_1 V_1 - B_1 C_1 I_1 = A_2 C_1 V_2 + B_2 C_1 I_2 \\ + & -C_1 D_1 V_1 + A_1 D_1 I_1 = C_2 D_1 V_2 + D_1 D_2 I_2 \end{aligned}$$

$$I_1 (A_1 D_1 - B_1 C_1) = V_2 (A_2 C_1 + C_2 D_1) + I_2 (B_2 C_1 + D_1 D_2) \quad (24)$$

We know that $A_1 D_1 - B_1 C_1 = 1$

Equations (23) and (24) represent the combination of two networks and representing as a single network with a single constants.

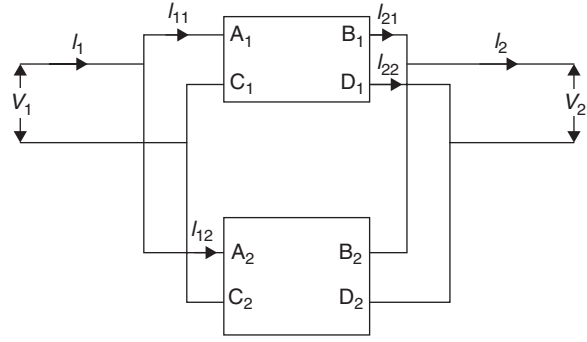
$$\begin{aligned} \therefore A &= A_1 A_2 + B_1 C_2 \\ B &= A_1 B_2 + B_1 D_2 \\ C &= A_2 C_1 + C_2 D_1 \\ D &= B_2 C_1 + D_1 D_2 \end{aligned}$$

Therefore the above relations represented in matrix form as

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

Constants for Two Networks in Parallel

The two networks connected in parallel are as shown below and the constants for the combination of two networks are calculated as follows.



From the above network, we can write that

$$V_1 = A_1 V_2 + B_1 I_{21} \quad (25)$$

$$V_1 = A_2 V_2 + B_2 I_{22} \quad (26)$$

And the combination of the two networks as a single network can be written in the form of equations as

$$V_1 = A V_2 + B I_2$$

where

$$I_2 = I_{21} + I_{22}$$

Multiply Equation (25) with B_2 and (26) with B_1 and adding these results in

$$\begin{aligned} & V_1 B_2 = A_1 B_2 V_2 + B_2 B_1 I_{21} \\ + & V_1 B_1 = A_2 B_1 V_2 + B_1 B_2 I_{22} \\ \hline & V_1 (B_1 + B_2) = (A_1 B_2 + A_2 B_1) V_2 + B_1 B_2 (I_{21} + I_{22}) \end{aligned}$$

$$\Rightarrow V_1 = \frac{A_1 B_2 + A_2 B_1}{B_1 + B_2} V_2 + \frac{B_1 B_2}{B_1 + B_2} I_2 \quad (27)$$

From Equation (27), we can infer that

$$A = \frac{A_1 B_2 + A_2 B_1}{B_1 + B_2} \text{ and } B = \frac{B_1 B_2}{B_1 + B_2}$$

As the transmission line in a symmetric network,

$$A = D = \frac{A_1 B_2 + A_2 B_1}{B_1 + B_2} = \frac{D_1 B_2 + D_2 B_1}{B_1 + B_2}$$

And $AD - BC = 1$ and substituting A, B, D in this equation, we get

$$\Rightarrow \left(\frac{A_1 B_2 + A_2 B_1}{B_1 + B_2} \right) \left(\frac{A_1 B_2 + A_2 B_1}{B_1 + B_2} \right) - \frac{B_1 B_2}{B_1 + B_2} \times C$$

$$\Rightarrow C = C_1 + C_2 + \frac{(A_1 - A_2)(D_2 - D_1)}{B_1 + B_2}$$

Example 34: A 220 kV, 20 km long, 3- ϕ transmission line has the following A, B, C, D constants.

$$A = D = 0.96 \angle 3^\circ$$

$$B = 55 \angle 65^\circ \Omega/\text{phase}$$

$$C = 0.5 \times 10^{-4} \angle 90^\circ \text{ S}/\text{phase}$$

Its charging current per phase is

(A) 6.3 A

(B) 11 A

(C) 127 A

(D) 220 A

Solution: (A)

Admittance per phase (Y) = C

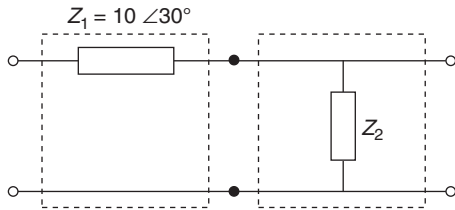
$$= 0.5 \times 10^{-4} \angle 90^\circ$$

Charging current per phase = $V_{pn} \times Y$

$$= \frac{220}{\sqrt{3}} \times 10^3 \times 0.5 \times 10^{-4}$$

$$= \frac{11}{\sqrt{3}} \text{ A} = 6.35 \text{ A}$$

Example 35: Two networks are connected in cascade as shown in figure with the usual notation the equivalents A , B , C and D constants are obtained. Given that $C = 0.025 \angle 40^\circ$, the value of Z_2 is



(A) $40 \angle 40^\circ$

(B) $20 \angle 40^\circ$

(C) $10 \angle 20^\circ$

(D) $20 \angle 20^\circ$

Solution: (A)

For the given two part network

$$\text{The open-circuit admittance} = \frac{1}{Z_2}$$

$$\therefore C = \frac{1}{Z_2}$$

$$Z_2 = \frac{1}{C} = 40 \angle -40^\circ \Omega$$

Example 36: The total reactance and total susceptance of a lossless overhead EHV line operating at 50 Hz are given by 0.04 p.u. and 1.1 p.u., respectively. If the velocity of wave propagation is 30×10^5 km/s, then the approximate length of line is

(A) 200 km

(B) 2000 km

(C) 20 km

(D) 2 km

Solution: (B)

$$\text{Velocity of wave propagation } (v) = \frac{1}{\sqrt{(L/\text{km}) / (C/\text{km})}}$$

Total reactance of line (X_L) = 0.04 p.u.

$$L/\text{km} = \frac{X_L}{2\pi fl} = \frac{0.04}{2\pi \times 50 \times l}$$

Total susceptance of line = 1.1 p.u.

$$\frac{1}{X_c} = 2\pi fC = 1.1 \text{ p.u.}$$

$$C = \frac{1.1}{2\pi \times 50}$$

$$C/\text{km} = \frac{1.1}{2\pi \times 50 \times l}$$

$$\text{Velocity of propagation} = \frac{1}{\sqrt{LC}}$$

$$30 \times 10^5 = \frac{l}{\sqrt{\frac{0.04}{2\pi \times 50} \times \frac{1.1}{2\pi \times 50}}}$$

$$l = 2124.5 \text{ km}$$

Example 37: A single-phase 50 Hz generator supplies an inductive load of 5 MW at a power factor of 0.7 lagging by means of an overhead transmission line 20 km long. The line resistance and inductance are 0.019 Ω and 0.62 mH per km. The voltage at receiving end is required to be kept at 10 kV.

Calculate

1. The sending end voltage and voltage regulation of the line.
2. The value of capacitors to be placed in parallel with the load such that the regulation is reduced to 60% of that obtained in the previous problem.
3. Calculate the transmission efficiency in case (i) and (ii).

Solution: Given transmission line is 20 km, so it is considered as a short line.

$$\therefore I = I_R = I_S \text{ given by}$$

$$I_R = \frac{5 \times 10^3}{10 \times 0.7} = \frac{P}{V \cos \phi} = 714 \text{ A}$$

Source voltage

$$|V_s| = |V_R| + |I|(R \cos \phi_R + X \sin \phi_R)$$

$$= 10,000 + 714(0.38 \times 0.7 + 3.89 \times 0.714)$$

Sending end voltage = 12.173 kV

$$\text{Voltage regulation} = \frac{12.173 - 10}{10} \times 100$$

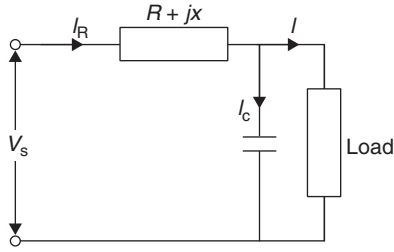
$$= 21.73\%$$

(ii) Desired voltage regulation = 60% of 21.73

$$= 0.6 \times 21.73 = 13.03\%$$

$$\text{Regulation} \frac{V_s - 10}{10} = 0.13$$

$$V_s = 11.3 \text{ kV}$$



Source voltage in the above circuit after placing capacitor to improve voltage regulation

$$V_s = V_r + I_R(R \cos \phi_r + X \sin \phi_r)$$

$$V_s - V_r = 0.13 \text{ kV} = I_R(R \cos \phi_r + X \sin \phi_r)$$

Since capacitance does not draw any real power

$$|I_R| = \frac{5000}{10 \times \cos \phi_r}$$

From the above two equations, $\cos \phi_r = 0.8685$, $I_R = 575.7 \text{ A}$

Now capacitor current $I_c = I_R - I$

$$= 575(0.8685 - j0.495) - 714(0.7 - j0.714)$$

$$= 0.195 + j225.17$$

Capacitor current $I_c = j225.17 \text{ A}$

$$X_c = \frac{1}{\omega C} = \frac{|V_R|}{|I_c|} = \frac{10 \times 1000}{225.17}$$

$$C = \frac{225.17}{10^4 \times \omega} = \frac{225.17}{10^4 \times 314} \Rightarrow C = 71.71 \mu\text{F}$$

(iii) Efficiency of transmission

$$\eta = \frac{\text{Output}}{\text{Output} + \text{Losses}}$$

$$\text{Case (i): } \eta = \frac{5000}{5000 + 714^2 \times 0.38 \times 10^{-3}} = 0.962\%$$

$$\text{Case (ii): } \eta = \frac{5000}{500 + 575^2 \times 0.38 \times 10^{-3}} = 97.5\%$$

Example 38: A 200 km, 3- ϕ , 50 Hz transmission line delivering 25 MVA at 0.85 lagging power factor to a balanced load at 132 kV. The line conductors are spaced equilaterally 2.5 m apart. The conductor resistance is 0.1 Ω /km and its effective diameter is 1.5 cm. Calculate the sending end voltage and voltage regulation.

Solution: Transmission line parameters

$$L = 0.461 \log \frac{D}{r'} = 0.461 \cdot \log \frac{250}{0.778 \times 0.75} = 1.21 \text{ mH/km}$$

$$C = \frac{0.0242}{\log \frac{D}{r}} = \frac{0.0242}{\log \frac{250}{0.75}} = 0.0096 \mu\text{F/km}$$

$$R = 0.1 \times 200 = 20 \Omega$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 1.21 \times 10^{-3} \times 200 = 76.02 \Omega$$

$$Z = R + jX_L = 20 + j76.02 = 78.6 \angle 75.2^\circ$$

$$Y = j\omega C = j.314 \times 0.0096 \times 10^{-6} \times 200$$

$$Y = 6.02 \times 10^{-4} \angle 90^\circ$$

$$I_R = \frac{25 \times 1000}{\sqrt{3} \times 132} \angle 31.78 = 109.3 \angle -31.78$$

$$V_R (\text{per phase}) = \frac{132}{\sqrt{2}} \angle 0^\circ = 76.2 \angle 0^\circ \text{ kV}$$

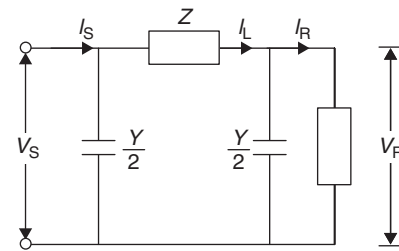


Figure 15 Medium line π -representation

Source voltage

$$\begin{aligned} V_s &= \left(1 + \frac{1}{2}YZ\right)V_R + ZI_R \\ &= \left[1 + \frac{1}{2} \times 6.02 \times 10^{-4} \angle 90^\circ \times 78.6 \angle 75.2^\circ\right] \times 76.2 \\ &\quad + 78.6 \angle 75.2^\circ \times 109.3 \angle -31.78 \times 10^{-3} \\ &= 76.2 + 1.8 \angle 166.2 + 8.59 \angle 43.42 \\ V_s &= 80.69 + 6.3j = 80.93 \angle 4.48^\circ \end{aligned}$$

$$|V_s|(\text{line}) = 80.93 \times \sqrt{3} = 140.17 \text{ kV}$$

$$\begin{aligned} 1 + \frac{1}{2}YZ &= 1 + \frac{1}{2} \times 6.02 \times 10^{-4} \angle 90^\circ \times 78.6 \angle 75.2^\circ \\ &= 1 + 0.023 \angle 165.2 = 0.977 + j0.006 \end{aligned}$$

$$|V_{R_0}|(\text{line no load}) = \frac{140.17}{0.977} = 143.47 \text{ kV}$$

$$\text{Voltage regulation} = \frac{143.47 - 140.17}{140.17} = 2.4\%$$

Example 39: A 50 Hz, 250 km long transmission line has a total series impedance of $30 + j100 \Omega$ and a total shunt admittance of $10^{-3} \Omega$. The receiving end load is 40 MW at 220 kV with 0.8 lagging power factor. Calculate the sending end voltage, current, power and power factor using.

(i) Short line approximation

(ii) Nominal π -method

Solution: Impedance of the line $Z = 30 + j100 \Omega$
 $= 104.4 \angle 73.3^\circ$
 $Y = 10^{-3} \angle 90^\circ \therefore$

The receiving end load is 40 MW at 220 kV, 0.8 p.f. lagging

$$\therefore I_R = \frac{40}{\sqrt{3} \times 220 \times 0.8} \angle -36.9^\circ$$

$$= 0.1312 \angle -36.9^\circ \text{ kA} \quad V_R = \frac{220}{\sqrt{3}} \angle 0^\circ = 127 \angle 0^\circ \text{ kV}$$

(i) Short line approximation

$$\begin{aligned} V_s &= V_R + I_R Z = 127 + (0.164 \angle -36.9^\circ \times 104.4 \angle 73.3^\circ) \\ &= 127 + 17.12 \angle 36.4^\circ \\ &= 141.14 \angle 4.127^\circ \\ &= 140.7 + 10.15j \end{aligned}$$

$$|V_s|_{\text{line}} = 244.46^\circ \text{ kV}$$

$$I_s = I_R = 0.1312 \angle -36.9^\circ$$

Sending end power factor

$$= \cos(4.127 + 36.9) = 0.754 \text{ lagging}$$

$$\begin{aligned} \text{Sending end power} &= \sqrt{3} \times 244.4 \times 0.1312 \times 0.754 \\ &= 41.87 \text{ MW} \end{aligned}$$

(ii) Nominal- π method

$$\begin{aligned} A = D &= 1 + \frac{1}{2} YZ = 1 + \frac{1}{2} \times 10^{-3} \angle 90^\circ \times 104.4 \angle 73.3^\circ \\ &= 1 + 0.0522 \angle 163.3^\circ = 0.95 \angle 0.90^\circ \end{aligned}$$

$$B = Z = 104.4 \angle 73.3^\circ$$

$$C = Y \left(1 + \frac{1}{4} YZ \right) = Y \left(1 + \frac{1}{4} \times 10^{-3} \angle 90^\circ \times 104.4 \angle 73.3^\circ \right)$$

$$= 9.75 \times 10^{-4} \angle 90.4^\circ$$

$$\begin{aligned} V_s &= AV_R + BI_R = 0.95 \angle 0.90^\circ \times 127 + \\ &\quad 104.4 \angle 73.3^\circ \times 0.13 \angle -36.9^\circ \\ &= 120.6 \angle 0.9^\circ + 13.57 \angle 36.4^\circ \end{aligned}$$

$$V_s = 131.88 \angle 4.32^\circ$$

$$|V_s|_{\text{line}} = 228.4 \text{ kV}$$

$$I_s = CV_R + DI_R$$

$$= 9.75 \times 10^{-4} \angle 90.4^\circ \times 127 + 0.95 \angle 0.90^\circ \times 0.13 \angle -36.9^\circ$$

$$= 0.123 \angle 90.4^\circ + 0.1235 \angle -36^\circ$$

$$I_s = 0.111 \angle 26.96^\circ$$

$$\text{Sending power factor} = \cos(26.96 - 4.32) = \cos 22.64 = 0.922 \text{ leading}$$

Sending end power

$$= \sqrt{3} \times 228.4 \times 0.111 \times 0.922 = 40.5 \text{ MW}$$

UNDERGROUND CABLES

Underground transmission has more advantages over overhead transmission like low maintenance, cost, good looking, free from damage due to lightning and thunder storms and has no interference with communication circuit. The underground transmission is done by underground cables.

The cable used in underground transmission generally has three parts.

1. The core (or) conductor.
2. The insulating material.
3. A metallic sheath of lead is provided around the insulation to provide external protection.

The characteristics of insulating material used in cables are

- (i) High dielectric strength.
- (ii) Good mechanical properties and the insulating material should not be affected by chemicals.
- (iii) High insulation resistance.

Depend upon the operating voltage and temperature, different types of insulating materials used in cables those are listed as follows.

- (i) Polyvinyl chloride (PVC)
- (ii) Impregnated paper
- (iii) Vulcanized rubber
- (iv) Polythene

Classification of Cables

The underground cables are classified depending upon the operating voltages are

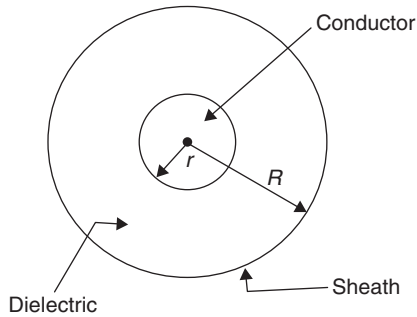
1. Low-tension cable (LV-cable): These cables are generally used for an operating voltage below 1 kV.
2. High-tension cable (HV-cable): These cables are used for an operating voltage below 11 kV.
3. Super tension cable (ST-cable): These cables are used for an operating voltage up to 33 kV.
4. Extra high-tension cable (EHT-cable): These cables are used for an operating voltage of 66 kV.
5. Extra super tension cable: These cables are used for an operating voltage of above 132 kV.

Dielectric Stress in Cable

In a cable the current carries in a core (or) conductor of the cable. So, the stress is more on the surface of the core and it decreases towards the surface of the cable (or) sheath of the cable.

Maximum gradient at the surface of the conductor is

$$g_{\text{max}} = \frac{V}{r \ln \left(\frac{R}{r} \right)}$$



Minimum gradient at the sheath of the cable

$$g_{\min} = \frac{V}{R \ln \left(\frac{R}{r} \right)}$$

Here, V is the voltage applied to the cable
 r – Radius of the conductor (or) core
 R – Radius of the cable

Grading of Cable

Grading of cable is done according to the stress on the dielectric material of the cable. There are two methods of grading.

1. Capacitance grading
2. Inter sheath grading

Capacitance Grading

In this method, different dielectric materials having different permittivities are used and these materials are used in cables by two ways.

1. The factor of safety for all the materials is same, i.e., the material which has highest product of dielectric strength and permittivity should be placed near to the core of the cable.
2. The same working stress for different materials but the material which has highest permittivity placed near to the conductor of the cable.

Intersheath Grading

In this method, different metallic sheaths are used in between the core and sheath of the cable for the same dielectric material by applying different voltages to different metallic sheath from operating voltage to earth potential. So, the stress on the same material also changed corresponding to the voltage.

Laying and Heating of Cables

Generally there are three methods of laying of cables

1. Draw-in-system
2. Direct laying
3. Solid system

In cable if the rate of heat generation is more than the rate of heat dissipation, then the temperature of the cable increases. The sources of heat generation in cables are

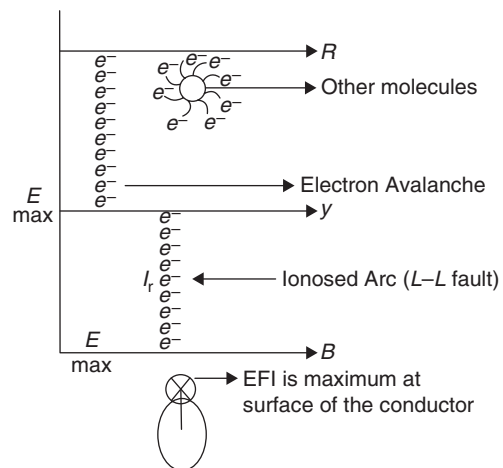
1. Dielectric loss
2. Copper loss in the core
3. Sheath losses

CORONA

The ionization of air surrounding the power conductor is corona. There are free electrons normally present in the free space due to radioactivity and cosmic rays. The gradient around the surface of the conductor increases with increases in the potential between the conductors. The free electrons present in the air will move with certain velocity depending on the field strength. These moving electrons collide with the molecules of the air and some of the fast moving electrons may also dislodge electrons from the air molecules, thereby increasing the number of electrons. Hence the process of ionization is thus cumulative and may lead to electron avalanche. Hence the air surrounding the conductor may be ionized.

When a voltage higher than the critical voltage is applied between two parallel polished wires, the glow is quite obvious. After it is operated for some time, reddish heads or tufts form along the wire, bluish white glow is appeared around the surface of the wire.

It is noticed that the reddish tufts or beads are formed when the conductor is negative, and a smoother bluish white glow is formed when the conductor is positive. The corona due to AC is same as the corona due to DC when viewed through stroboscope. Due to corona, a hissing noise is observed and ozone gas is formed which is detected by its characteristic odour.

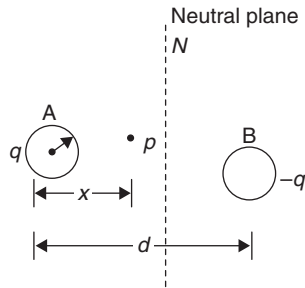


Overhead transmission lines are used for all economic reasons than underground cables. If line-to-line operating voltage ≥ 275 kV, then there will be abnormal electric field intensity (EFI) at the surface of the conductor.

EFI (increase or decrease a potential difference a mobility of electrons).

Critical Disruptive Voltage

Consider a single-phase transmission line. Let 'r' be the radius of each conductor and 'd' be the distance of separation such that 'd' is far greater than 'r'. Let 'q' be the charge per unit length of the conductor on one of the conductors and hence '-q' is on the other. If the operating voltage is 'V', the potential of conductor 'A' with respect to neutral plane 'N' will be $\frac{V}{2}$ and that of 'B' will be $\frac{-V}{2}$. Consider a point 'p' at a distance 'x' where we want to find the electric field intensity.



The field due to 'A' will be repulsive and due to 'B' would be attractive by bringing a unit positive charge from infinity to that point 'p'. Hence the EFI's due to A and B at 'p' would be additive

$$E = \frac{q}{2\pi\epsilon_0 x} + \frac{q}{2\pi\epsilon_0 (d-x)} = \frac{q}{2\pi\epsilon_0} \left[\frac{1}{x} + \frac{1}{d-x} \right]$$

The potential difference between the conductors,

$$\begin{aligned} V &= - \int_{d-r}^r E dx = - \int_{d-r}^r \frac{q}{2\pi\epsilon_0} \left[\frac{1}{x} + \frac{1}{d-x} \right] dx \\ &= \frac{q}{2\pi\epsilon_0} [\ln x - \ln(d-x)] r^{d-r} \\ &= \frac{q}{2\pi\epsilon_0} [\ln(d-r) - \ln(d-(d-r)) - \ln r + \ln(d-r)] \\ &= \frac{q}{2\pi\epsilon_0} \times 2 \ln \left[\frac{d-r}{r} \right] = \frac{q}{\pi\epsilon_0} \ln \left[\frac{d-r}{r} \right] \end{aligned}$$

as $d \gg r$, $d-r \approx d$,

$$\Rightarrow V = \frac{q}{\pi\epsilon_0} \ln \frac{d}{r}$$

Now the EFI at the point, which is 'x' distance from the centre of the conductor 'A' is given by,

$$E = \frac{q}{2\pi\epsilon_0} \left[\frac{1}{x} + \frac{1}{d-x} \right] = \frac{q}{2\pi\epsilon_0} \cdot \frac{d}{x(d-x)}$$

From (1), substitute 'q' in above equation.

$$q = \frac{\pi\epsilon_0 V}{\ln \frac{d}{r}}$$

$$\Rightarrow E = \frac{\pi\epsilon_0 V}{\ln \frac{d}{r}} \times \frac{1}{2\pi\epsilon_0} \times \frac{d}{x(d-x)} = \frac{V}{2 \ln \frac{d}{r}} \times \frac{d}{x(d-x)}$$

Let $V' = \frac{V}{2}$ which is line to neutral voltage. In case of three-phase system, $V' = \frac{V}{\sqrt{3}}$, where 'V' is line-to-line voltage

$$\Rightarrow E = \frac{V'd}{x(d-x) \ln \frac{d}{r}}$$

From the above expression, it is clear the EFI is inversely proportional to 'x', i.e. as 'x' decreases, EFI increases and it is maximum when $x = r$, i.e., at the surface of the conductor and this value is given by

$$E_{\max} = \frac{V'd}{(d-r)r \ln \frac{d}{r}} \approx \frac{V'}{r \ln \frac{d}{r}} \Rightarrow V' = E_{\max} r \ln \frac{d}{r}$$

Critical disruptive voltage is hence defined as the voltage at which the complete disruption of dielectric occurs. This voltage gives the gradient at the surface of the conductor equal to the breakdown strength of the air. This dielectric strength is normally denoted by g_0 and is equal to 30 kV/cm peak at normal temperature and pressure, i.e. 25°C and 76 cm of Hg. At any other temperature and pressure,

$$g_0' = g_0 \times \delta$$

where 'δ' is the air density correction and is given by

$$\delta = \frac{3.92 b}{273 + t}$$

where 'b' is the barometric pressure in cm of Hg and 't' the temperature in °C.

Hence, the critical disruptive voltage is given by

$$V' = r g_0' \delta \ln \frac{d}{r} \text{ kV.}$$

where the gradient 'g' is nothing but electric field intensity (EFI).

The above expression is derived based on the consideration that the conductor is solid and the surface is smooth. ACSR conductors are used for higher voltages. The potential gradient for ACSR conductors will be greater than the potential gradient of an equivalent smooth conductor, and hence the breakdown voltage of a stranded conductor will be comparatively less than a smooth and solid conductor. The irregularities on the surface of such a conductor are increased further due to the deposition of dust and dirt on its surface and hence the breakdown voltage is further reduced.

Hence, the final expression for the critical disruptive voltage after considering atmospheric conditions and the surface of the conductor is

$$V' = r g_0' \delta m_0 \ln \frac{d}{r} \text{ kV.}$$

When the voltage applied corresponds to the critical disruptive voltage, corona creates but may not be visible as the charged ions in the air must receive some finite energy to cause further ionization by collision. For a radial field, it must reach a gradient g_v at the surface of the conductor to cause a gradient g_0 , a finite distance away from the surface of the conductor. ' g_v ' is not constant as ' g_0 ' is, and is a function of the size of the conductor.

$$g_v = g_0 \delta \left(1 - \frac{0.3}{\sqrt{r\delta}} \right) \text{ kV/cm}$$

If ' V_v ' is the critical visual disruptive voltage, then

$$\Rightarrow g_v = \frac{V_v}{r \ln \frac{d}{r}} = g_0 \delta \left(1 + \frac{0.3}{\sqrt{r\delta}} \right)$$

$$\Rightarrow V_v = r g_0 \delta \left[1 + \frac{0.3}{\sqrt{r\delta}} \right] \ln \frac{d}{r} \text{ kV}$$

In case the irregularity factor is taken into account,

$$\begin{aligned} V_v &= g_0 m_v \delta r \left[1 + \frac{0.3}{\sqrt{r\delta}} \right] \ln \frac{d}{r} \\ &= 21.1 m_v \delta r \left[1 + \frac{0.3}{\sqrt{r\delta}} \right] \ln \frac{d}{r} \text{ kV RMS} \end{aligned}$$

where ' r ' is the radius in cm. The irregularity factor m_v has the following values:

$m_v = 1.0$ for polished wires

= 0.98 to 0.93 for rough conductor exposed to atmospheric conditions

= 0.72 for local corona on stranded conductors

Since the surface of the conductor is irregular, the corona does not start immediately on the whole surface but it takes place at difference points of the conductor which are pointed and this is known as local corona.

Example 40: A three-phase line has conductors 4 cm in diameter spaced equally 2 m apart. If the dielectric strength of the air is 30 kV (max) per cm, find the disruptive critical voltage for the line. Take air density factor, $d = 0.951$ and irregularity factor = 0.9.

Solution: Conductor radius = $\frac{4}{2} = 2$ cm

Conductor spacing = 2 m = 200 cm

Dielectric strength of the air,

$$g_0 = 30 \text{ kV/cm (max)}$$

$$= 21.2 \text{ kV (RMS) per cm}$$

Disruptive critical voltage

$$V' = r g_0 \delta m_0 \ln \left(\frac{d}{r} \right) \text{ kV/ph}$$

$$= 2 \times 21.2 \times 0.951 \times 0.9 \times \ln \left(\frac{200}{2} \right)$$

$$= 167.28 \text{ kV/ph}$$

$$\begin{aligned} \therefore \text{Line voltage (RMS)} &= \sqrt{3} \times 167.28 \\ &= 289.72 \text{ kV} \end{aligned}$$

Example 41: A 132 kV line with 2 cm diameter conductors is built so that corona takes place if the line voltage exceeds 220 kV (RMS). If the value of potential gradient at which ionization occurs can be taken as 30 kV per cm, find the spacing between the conductors.

Solution: If the line is 3- ϕ , Radius of the conductor, $r = \frac{2}{2} = 1$ cm. Dielectric strength of the air, $g_0 = \frac{30}{\sqrt{2}} = 21.2$ kV (RMS). Disruptive voltage/phase = $\frac{220 \text{ kV}}{\sqrt{3}} = 127.020$

kV. If the conductors are assumed to be smooth conductors, then irregularity factor $m_0 = 1$. At standard temperature and pressure, air density factor $d = 1$

$$\therefore V' = r g_0 \delta m_0 \ln \left(\frac{d}{r} \right) \text{ kV}$$

where ' d ' is spacing between the conductors.

$$\begin{aligned} 127.020 &= 1 \times 21.2 \times 1 \times 1 \times \ln \left(\frac{d}{1} \right) \\ \Rightarrow \ln \left(\frac{d}{1} \right) &= 5.9915 \Rightarrow d = 400.014 \text{ cm} \end{aligned}$$

Corona Loss

The electric field produces ions in the space which further leads to the production of space charge which move around the conductor. If there is a continuous transmission of energy, there is a continuous motion for these charges. Hence, it can indirectly be said that the energy required for the charges to remain in motion is derived from the supply system. In order to maintain the flow of energy over the conductor in the field where in this additional energy would have been otherwise absent, it is necessary to supply this additional loss from the supply system. This additional power is referred as corona loss.

An empirical relation was derived by Peek by conducting a number of experiments on corona loss.

$$P = 241 \times \frac{(f + 25)}{\delta} \sqrt{\frac{r}{d}} (V_p - V_c)^2 \times 10^{-5} \text{ kW/km/phase}$$

where ' f ' is the supply of frequency,

' δ ' – Air density correction factor,

V_p – The operating voltage in kV

V_c – Critical disruptive voltage

$$\text{and } \delta = \frac{3.92 b}{273 + t},$$

where b = Atmospheric pressure

t = Temperature.

The above equation is for a fair weather condition. For a foul weather condition, the power loss can be obtained by taking V_0 as 0.8 times the fair weather value. The above empirical formula has certain limitations and is applicable only if the frequency is in between 25 and 120 Hz, the conductor radius is greater than 0.25 cm and the ratio of $\frac{V_p}{V_0} > 1.8$.

Example 42: A 3- ϕ 220 kV, 50 Hz transmission line consists of 3 cm diameter conductor spaced 4 m apart in an equilateral triangle form. If the temperature is 40°C, atmospheric pressure is 76 cm, calculate the corona loss per km of the line. Take $m_0 = 0.9$.

Solution: We know that

$$P = 241 \times 10^{-5} \frac{(f+25)}{\delta} \sqrt{\frac{r}{d}} (V_p - V_0)^2 \text{ kW/km/phase}$$

$$\delta = \frac{3.92 \times b}{273 + t} = \frac{3.92 \times 76}{273 + 40} = 0.952$$

$$g_0 = 21.2 \text{ kV/cm (RMS)}$$

Critical disruptive voltage per phase

$$V_0 = m_0 g_0 \delta r \log_e \frac{d}{r} \text{ kV}$$

$$V_0 = 0.9 \times 21.2 \times 0.952 \times 1.5 \times \log_e \frac{200}{1.5}$$

$$= 27.24 \times 4.80 = 133.281 \text{ kV}$$

Supply voltage per phase,

$$V_p = \frac{220}{\sqrt{3}} = 127 \text{ kV}$$

$$\therefore P = 241 \times 10^{-5} \times \frac{(50+25)}{0.952} \times \sqrt{\frac{1.5}{200}} \times (127 - 133.281)^2$$

$$\text{kW/km/phase}$$

$$= 0.01644 \times 39.450 = 0.648 \text{ kW/km/phase}$$

$\therefore$ Total corona loss per km for the three phases

$$= 3 \times 0.648 = 1.944 \text{ kW.}$$

Example 43: Which of the following will minimize the corona loss?

- (A) Reduction in conductor size
- (B) Reduction in smoothness of the conductor
- (C) Sharp points are provided in the line hardware
- (D) Current density in conductor is reduced.

Solution: (D)

Example 44: The corona loss on a particular system at 50 Hz is 1.2 kW/km/phase. The corona loss at 60 Hz would be

- (A) 1.2 kW/km/ph
- (B) 1.36 kW/km/ph
- (C) 1.14 kW/km/ph
- (D) 1.42 kW/km/ph

Solution: (B)

Corona loss $\propto (f+25)$

$$\frac{P_2}{P_1} = \frac{f_2 + 25}{f_1 + 25}$$

$$P_2 = \frac{60 + 25}{50 + 25} \times P_1 = 1.36 \text{ kW/km/ph}$$

Factors Affecting Corona Loss

The factors that affect corona loss on overhead transmission lines:

1. Electrical factors
2. Atmospheric factors and
3. Factors with respect to conductors

Electrical Factors

According to the above power loss expressions corona loss is a function of frequency. Hence if the frequency increases, the corona loss also increases proportionately. Hence we can infer that DC corona loss is comparatively less than AC corona loss. Hence if in case of AC supply, there may be harmonic which may further increase the corona loss.

Field Around the Conductor

The field around the conductor is a function of the voltage. Along with voltage, field also depends on the configuration of the conductors. The height of the conductors from the ground also effects the corona loss. The smaller the height, the greater will be the corona loss.

Atmospheric Factors

Pressure and Temperature Effect

From the expression for power loss, we can observe that it is a function of air density correction factor ' δ ' and even critical disruptive voltage is a function of ' δ '. The lower the value of ' δ ', the higher is the loss. For lower values of ' δ ', the pressure should be low and the temperature should be high. Hence the corona loss is low in plain areas and high in hilly areas.

Dust, Rain, Snow and Storm Effect

The particles of dust deposit on the conductor, thereby reducing the critical disruptive voltage for local corona and also the rains, snow and hail storm reduce the critical disruptive voltage and the corona loss increases.

Factors with Respect to Conductor

Diameter of the Conductor

From the expression, we can see that the corona loss is directly proportional to the radius of the conductor and also we know that loss $\propto (V_p - V_0)^2$, where V_0 is the critical disruptive voltage which is inversely proportional to the radius of the conductors. Hence, the larger the size of the conductor, the lower is the corona loss.

Number of Conductors/phase

If the operating voltage ≥ 380 kV, it is found that using one conductor per phase produces large coroner loss and also large radio interference in the communication lines running parallel to power lines. Hence, using '2' or more conductors per phase reduces the corona loss, i.e. using bundled conductors. By using bundled conductors, the self-GMD of the conductor reduces increases the critical disruptive voltage and thereby the corona loss reduces.

Heating of the Conductor by Load Current

The current passing through the conductor produces I^2R losses which would help to reduce the corona loss. If there is no such heat produced, then the conductor's temperature would be slightly less than the surrounding temperature. In the absence of heating, due to high humidity at times, additional corona is introduced.

Methods of Reducing Corona Loss

The losses can be reduced by

1. Large diameter conductors
2. Hollow conductors
3. Bundled conductors
4. By increasing the conductor's spacing

Advantages and Disadvantages of Corona

Advantages

1. Due to corona formation, the air surrounding the conductor also becomes conducting and therefore virtually the diameter of the conductor increases. This increased diameter reduces electrostatic stresses between the conductors.
2. Corona reduces the effects of transients produced by surges by dissipating them as corona loss.

Disadvantages

1. Corona reduces the energy to be transmitted thereby reducing the transmission efficiency of the transmission line.
2. The ozone produced during the corona may cause corrosion of the conductor.
3. The current drawn by the conductor during the time of corona is non-sinusoidal and produces non-sinusoidal voltage drop which may cause inductive interference with neighbouring communication lines.
4. Due to corona, the effective capacitance of the conductors is increased as the effective diameter of the conductor is increases. The charging current flowing also increases.

Example 45: A 3- ϕ line is operating at 220 kV and 50 Hz. The line has a conductor of 1.3 cm diameter arranged in a 3 m delta connection. Assume air density factor of 1.05 and

the dielectric strength of air to be 21.1 kV/cm, the corona losses will be ($m = 0.75$)

- (A) 5 kW/km (B) 10 kW/km
(C) 15 kW/km (D) 20 kW/km

Solution: (C)

$$\text{Phase voltage} = \frac{220}{\sqrt{3}} \text{ kV} = 127 \text{ kV}$$

Critical disruptive voltage

$$\begin{aligned} (E_0) &= mg_0 \delta \left(\ln \frac{d}{r} \right) \text{ kV/phase} \\ &= 0.75 \times 1.05 \times 21.1 \ln \left(\frac{300}{0.65} \right) \\ &= 101.93 \approx 102 \text{ kV} \end{aligned}$$

Critical disruptive voltage $E_0(L - L)$

$$= \sqrt{3} \times 102 = 176.6 \text{ kV}$$

Corona loss in the system is given by

$$\begin{aligned} P &= 241 \times 10^{-5} \left(\frac{f + 25}{8} \right) \sqrt{\frac{r}{D}} (E - E_0)^2 \text{ kW/phase/km} \\ &= 241 \times 10^{-5} \times \frac{75}{8} \times 29.09 \\ &= 5.007 \text{ kW/phase/km} \\ &= 15.02 \text{ kW/km} \end{aligned}$$

SKIN EFFECT

When a current-carrying conductor is carrying direct current (DC), the current is uniformly distributed over the whole cross section of the conductor. But, if an alternating current is flowing through the conductor, the current does not distribute uniformly, instead, it is concentrated more on the surface of the conductor compared to the current flowing through the centre of the conductor. This is known as skin effect.

The tendency of the alternating current to concentrate near the surface of a conductor is known as skin effect.

Due to skin effect, the effective area of cross section of the conductor through which the current flows is reduced. Hence, the resistance of the conductor increases while carrying an alternating current. The conductor consisting of a large number of strands each carrying part of the total current. The inner strands carrying currents give rise to flux which link only the inner strands whereas the strands outside produce flux which links both inner and outside strands. The flux linkages per ampere are more for the inner strands than for the outer strands. Thus the inductance as well as impedance of the inner strands is more than that of

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the outer strands, resulting in more current to flow in outer strands. Skin effect depends on the following factors.

1. Nature of the material
2. Radius of the conductor
3. Frequency of supply
4. Shape of wire

Skin effect is more if the frequency of supply is more and is proportional to the size of the conductor.

Proximity Effect

The alternating magnetic flux produced in the conductor due to current flowing through the neighbouring conductor gives rise to circulating currents which apparently increases the resistance of the conductor. This phenomenon is called proximity effect. Consider a two-wire system in which the lines of flux link with inner and outer elements of the conductor but most of them link with the elements farther apart compared to the inner elements. Hence the inductance of the farthest elements are more compared to the nearer elements, thereby impedance of the farther elements are more compared to nearer elements. This gives rise to non-uniform distribution of current. The effective resistance is thereby increased due to non-uniform distribution of current. The proximity effect is more in case of underground cables, where the distance between overhead line transmission lines is more. Proximity effect depends on conductor's size, frequency, distance between conductors and permeability of conductor material.

Ferranti Effect

A long-transmission line operated under no load or lightly loaded condition, the receiving end voltage is more than the sending end voltage. This phenomenon is known as Ferranti effect. This is explained as follows.

We know the equation of voltage

$$V = \frac{V_r + I_r Z_c}{2} e^{\alpha X} e^{j\beta X} + \frac{V_r + I_r Z_c}{2} e^{-\alpha X} e^{-j\beta X}$$

under no-load condition,

The above equation reduces to

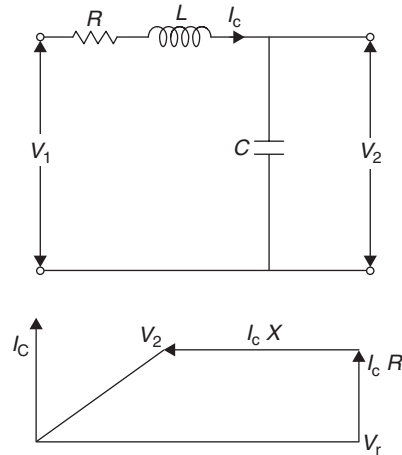
When $x = L, I_r = 0$

$$V_s = \frac{V_r}{2} e^{\alpha L} e^{j\beta L} + \frac{V_r}{2} e^{-\alpha L} e^{-j\beta L}$$

At $L = 0, V_r = \frac{V_s}{2} + \frac{V_s}{2}$

As ' L ' increases, the incident component of sending end voltage increases exponentially and turns the vector anticlockwise through an angle βL whereas the reflected sending end voltage decreases by the same amount and is rotated clockwise through the same angle βL . The sum of these two components of sending end voltage gives a voltage which is smaller than the receiving end voltage V_r .

Ferranti effect can be explained by approximating the distributed parameters of the long transmission line by lumped impedance as shown below.



As the capacitive reactance of the line is more than the inductive reactance, under no load condition the line current is of leading power factor. The charging current produces drop in the reactance of the line which is in phase opposition to the receiving end voltage and therefore the sending end voltage becomes smaller than the receiving end voltage. This is shown in the above phasor diagram.

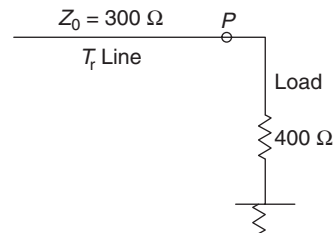
Example 46: The surge impedance of a 200 km long overhead line is 200Ω . For a 100 km length of the same line, the surge impedance will be

- (A) 200Ω (B) 100Ω
(C) 400Ω (D) 150Ω

Solution: (A)

Surge impedance is independent of length of the line

Example 47: The reflection co-efficient for the transmission line shown in figure at 'P' is



- (A) 0.25 (B) 0.142
(C) 0.33 (D) 7

Solution: (B)

$$\begin{aligned} \text{Reflection co-efficient} &= \frac{Z_L - Z_s}{Z_L + Z_s} \\ &= \frac{400 - 300}{400 + 300} = 0.142 \end{aligned}$$

Example 48: An overhead line having a surge impedance of 400Ω is connected in series with an underground cable having a surge impedance of 80Ω . If surge of 40 kV travels from the line towards the cable junction, the value of the transmitted voltage wave at the junction is

- (A) 160 kV (B) 40 kV
(C) 80 kV (D) 13.3 kV

Solution: (D)

$$\begin{aligned} \text{The transmitted voltage} &= \frac{2Z_L}{Z_L + Z_s} \times V \\ &= \frac{2 \times 80}{80 + 400} \times 40 \times 10^3 \\ &= 13.33 \text{ kV} \end{aligned}$$

Consider a power system network as shown in the figure.

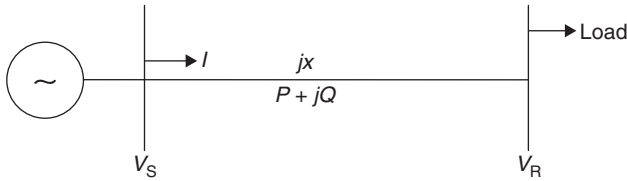


Figure 16 Power system single-line diagram

Generally the load is Inductive load. The complex power at the sending end is

$$S = V_s^* I = P - jQ$$

$$I = \frac{P - jQ}{V_s^*}$$

The receiving end voltage $V_R = V_s - j \left(\frac{P - jQ}{V_s} \right) X$
($\because V_s$ - reference voltage)

$$V_R = \left[V_s - \frac{Q}{V_s} X \right] - j \frac{P}{V_s} X$$

From the above equation V_R is not considerable affect by the variation of real power then by the variation reactive power of the load, Bus voltage is varied. So, by controlling the reactive power the voltage is maintained within the specified limits.

METHODS OF VOLTAGE CONTROL

The different voltage control methods are

1. Shunt reactors
2. Shunt capacitors
3. Tap-changing transformers
4. Synchronous condensers
5. Series capacitors
6. Static VAR system

Shunt Reactors

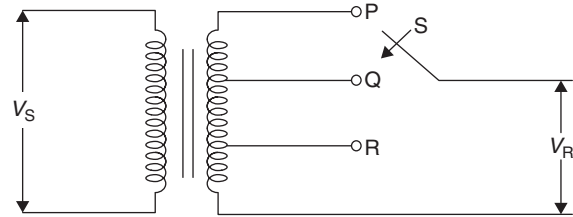
In this method, inductors are connected across the load. Shunt reactors generally used when the bus voltage more than the specified voltage, i.e., under light load condition ((or) Ferranti effect).

Shunt Capacitor

In this method, capacitors connected in parallel with the load. Shunt capacitors are generally used when the bus voltage less than the specified voltage, i.e., under voltage condition. By connecting shunt capacitors, the VARS power required by the load decreases so that the reactive power drawn by the load from the source decreases and the power factor, voltage of the bus Increases.

Tap-changing Transformer

In this method, the voltage of the line changes by changing the position of the winding connection as shown in the figure.

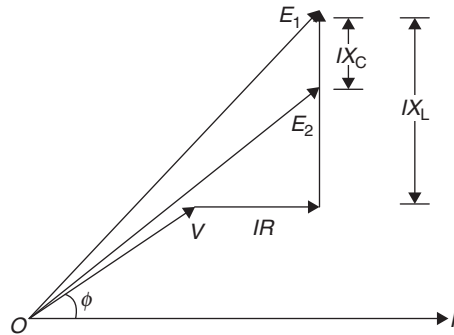


The above figure shows that the secondary voltage of the transformer changes by changing the corresponding switch position.

Series Capacitors

In this method, a capacitor is connected in series with the line. So, the resultant reactance of the line decreases then the corresponding voltage drop of the line also decreases.

The change in voltage by connecting series capacitor is $\Delta V = IX_c \sin \phi$.



In the above phasor diagram E_1 represent the sending end voltage before connecting the series capacitor and E_2 represent the sending end voltage after connecting the series capacitor.

Therefore by connecting series capacitor the voltage drop decreases by IX_c factor.

POWER FACTOR IMPROVEMENT

As the electrical energy is generated, transmitted and distributed in the form of alternating wave, hence power factor is considered. As most of the loads are inductive in nature, therefore they have low lagging power factor. Low power factor causes the increase active power loss. To ensure the most favourable conditions for a supply system from engineering and economical standpoint, it is important to have power factor close to unity.

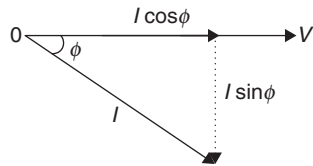
Power Factor

The cosine of angle between voltage and current in an AC circuit is known as power factor.

In an AC circuit, there is generally phase difference ' ϕ ' between voltage and current. The term $\cos\phi$ is called the power factor of the circuit.

If the circuit is inductive, the current lags the voltage and power factor is said to be lagging. In case of capacitive circuit, current leads the voltage and the power factor is said to be leading.

Consider an inductive circuit taking a lagging current ' I ' from the supply voltage ' V ' the lagging angle being equal to ' ϕ '. The vector diagram for the above circuit is shown below.



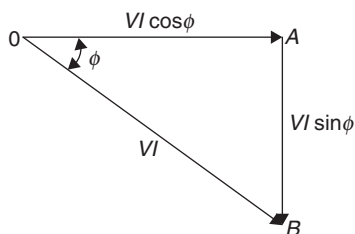
The current ' I ' is resolved into two perpendicular components,

1. $I \cos\phi$ in phase with ' V '
2. $I \sin\phi$, 90° out of phase with V . (for L and C)

$I \cos\phi$ is known as active (or) wattful component while the component $I \sin\phi$ is called Reactive (or) wattless component. If the reactive component is small, then the phase angle ' ϕ ' is small and hence the power factor $\cos\phi$ will be high. Hence the circuit having small reactive current ($I \sin\phi$) will have high power factor and vice versa. The value of power factor can never be more than unity.

Power Triangle

Power factor can also be analysed by the power drawn by the AC circuit. If each side of the current triangle is multiplied by voltage V , then we get the power triangle.



$OA = VI \cos\phi$ which represents the active power in watts (or) kW.

$AB = VI \sin\phi$ represents the reactive power in VAR or kVAR.

$OB = VI$ represents the apparent power in VA or kVA

From the power triangle, we can infer that,

1. The apparent power in AC circuit has two components, i.e. active and reactive power at right angles to each other.

$$OB^2 = OA^2 + AB^2$$

$$\Rightarrow (\text{Apparent power})^2 = (\text{Active power})^2 + (\text{Reactive power})^2$$

$$2. \text{ Power factor } \cos\phi = \frac{OA}{OB} = \frac{\text{Active power}}{\text{Apparent power}}$$

3. For leading currents, the power triangle should be reversed. To improve the power factor of the load, the device taking leading reactive power (e.g. capacitor) is connected in parallel with the load, neutralizing partly the lagging reactive power.

Disadvantages of Low Power Factor

We know that $P_L = V_L I_L \cos\phi$

$$\Rightarrow I_L = \frac{P}{V_L \cos\phi} \text{ (for single-phase)}$$

$$P = \sqrt{3} V_L I_L \cos\phi$$

$$\Rightarrow I_L = \frac{P}{\sqrt{3} V_L \cos\phi} \text{ (for three-phase)}$$

From the above expression, we can conclude that for the fixed power and voltage, the load current is inversely proportional to the power factor.

Due to the low power factor, the current drawn from the source is large resulting in the following disadvantages.

1. The electrical machinery is always rated in kVA. While designing the machine, the power factor of the load is not known.

$$\text{kVA} = \frac{\text{kW}}{\cos\phi}$$

From the above formula, kVA rating of the Equipment is inversely proportional to the power factor. Hence, lower the power factor, the kVA of the machinery should be increased.

2. The large current drawn at low power factor causes more copper losses ($I^2 R$) in all the elements of the supply system. Hence the efficiency is reduced.

Economics of Power Factor Improvement

By using power factor improvement devices the power factor of the load increases. Therefore net maximum demand power drawn from the supply decreases then the corresponding total current and transmission losses decreases

which leads to the net saving on the economic cost. But by considering the cost and maintain of power factor correction equipment the total saving is decreases. Generally P is the expenditure per kVA per annum of the power factor correction equipment and q is the rate per kVA of maximum demand per annum. Therefore the corresponding economic power factor improvement is $\cos \phi$, i.e.,

$$\cos \phi = \sqrt{1 - \left(\frac{P}{q}\right)^2}$$

The power factor is improved by different methods.

1. Static VAR compensator
2. Synchronous condenser
3. STATCOM
4. Static capacitor

Static VAR Compensator

In this method, a capacitor bank fixed or switched (or) fixed, capacitor and switch reactor bank are used in parallel. These type of compensations are generally used when reactive power leading and lagging by corresponding switching ON and OFF of the capacitors and reactors into the circuit. By changing the reactive power, the power factor of the circuit also changes.

Static Capacitor

In the above method both capacitors and reactors are connected but In these method only capacitors are used in parallel with the load then there is net improvement of the power factor and also the reactive power drawn from the supply also reduces.

Synchronous Condenser

It is a synchronous motor operating under no-load and over excitation condition. When these motor is connected in parallel with the load then the motor supplying reactive power to the load. There for the corresponding power factor of the load Increases.

Example 49: A shunt reactor of 100 MVAR is operated at 98% of its rated voltage and 95% of its rated frequency the reactive power absorbed by the reactor is

- | | |
|----------------|-----------------|
| (A) 98.92 MVAR | (B) 101.09 MVAR |
| (C) 95.32 MVAR | (D) 104.32 MVAR |

Solution: (B)

Reactive power absorbed by reactor = $\frac{V^2}{X_L}$

$$Q_1 = \frac{V_1^2}{2\pi f_1 L} = 100 \text{ MVAR}$$

$$V_2 = 0.98 V_1$$

$$f_2 = 0.95 f_1$$

New case reactive power absorbed = $\frac{V_2^2}{2\pi f_2 L}$

$$Q_2 = \frac{(0.98 V_1)^2}{2\pi (0.95 f_1) L}$$

$$Q_2 = \frac{0.98^2}{0.95} Q_1 = 1.0109 Q_1$$

$$Q_2 = 101.09 \text{ MVAR}$$

Example 50: A balanced delta connected load of $(8 + j6) \Omega$ per phase is connected to a 400 V, 60 Hz, 3- ϕ supply lines. If the input power factor is to be improved to 0.9 by connecting a bank of star connected capacitor, the required kVAR of the bank is.

- | | |
|---------------|---------------|
| (A) 28.8 kVAR | (B) 18.5 kVAR |
| (C) 10.7 kVAR | (D) 20.4 kVAR |

Solution: (C)

Per-phase load current = $\frac{400}{8 + j6} = 40 \angle -36.86^\circ$

Load power factor = $\cos 36.86 = 0.8$ lagging

Capacitor placement affects the reactive component of load but real power component of the load remains unchanged

$$\therefore I_1 \cos \phi_1 = I_2 \cos \phi_2$$

$$40 \times 0.8 = I_2 \times 0.9 \Rightarrow I_2 = 35.55 \text{ A}$$

$$I_2 = I_1 + I_c \text{ from the system configuration}$$

$$I_c = I_2 - I_1 [\because \phi_2 = \cos^{-1} 0.9]$$

$$= 35.55 \angle -25.84 - 40 \angle -36.86$$

$$I_c = 8.5 \angle 90^\circ$$

Reactive power of capacitor bank

$$= Q \text{ at } 0.8 \text{ power factor} - Q \text{ at } 0.9 \text{ power factor}$$

$$= \sqrt{3} V_{L_1} I_{L_1} \sin \phi_1 - \sqrt{3} V_{L_2} I_{L_2} \sin \phi_2$$

$$= \sqrt{3} \times 400 \times \sqrt{3} \times 40 \times 0.6 - \sqrt{3} \times 400 \times \sqrt{3} \times 35.5 \times \sin 25$$

$$= 10.7 \text{ kVAR}$$

Example 51: At an industrial substation with a 4 MW load, a capacitor of 2.5 MVAR is installed to maintain the load factor at 0.97 lagging. If the capacitor goes out of service the load power factor is

- | | |
|---------|-----------|
| (A) 0.9 | (B) 0.75 |
| (C) 0.8 | (D) 0.842 |

Solution: (B)

Initial phase difference $\phi_1 = \cos^{-1} 0.97$
 $= 14.07^\circ$

kVAR supplied by capacitor bank = $P(\tan \phi_1 - \tan \phi_2)$

$$4 \times 10^6 (\tan \phi_1 - \tan 14.07) = 2.5 \times 10^6$$

$$\phi_1 = 41.2^\circ$$

Power factor before adding capacitor bank

$$\cos \phi_1 = 0.752$$

Example 52: A loss less transmission line having surge impedance loading of 2000 MW. A series capacitive compensation of 30% the surge impedance loading of the compensated transmission line will be

- (A) 1673 MW (B) 2390 MW
(C) 600 MW (D) 2600 MW

Solution: (B)

$$\text{Surge impedance loading (SIL)} = \frac{V^2}{Z_C}$$

$$\text{Surge impedance loading} \propto \frac{1}{Z_C}$$

When the capacitive compensation of 30% is provided, the series impedance of the line is reduced to 70% of its original value which in turn reduced the characteristic impedance.

Characteristic impedance $\propto \sqrt{Z_{\text{series}}}$

$$\frac{Z_{C_1}}{Z_{C_2}} = \frac{\sqrt{Z_{\text{series}}}}{\sqrt{0.7 \times Z_{\text{series}}}}$$

Characteristic impedance after compensation

$$Z_{C_2} = 0.836 Z_{C_1}$$

Since surge impedance loading $\alpha = \frac{1}{Z_C}$

$$\frac{\text{SIL}_1}{\text{SIL}_2} = \frac{Z_{C_2}}{Z_{C_1}}$$

$$\begin{aligned} \text{SIL}_2 &= \text{SIL}_1 \times \frac{Z_{C_1}}{Z_{C_2}} \\ &= 1.195 \times 2000 = 2390 \text{ MW} \end{aligned}$$

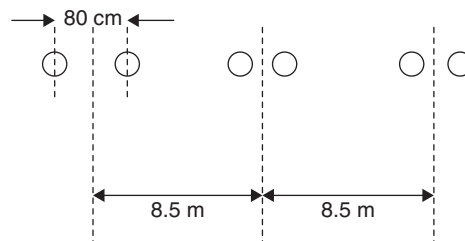
EXERCISES

Practice Problems I

Directions for questions 1 to 27: Select the correct alternative from the given choices.

- A generating station has a maximum demand of 60 MW. The plant capacity factor is 0.5 and the load factor is 0.6. The plant running as per schedule is fully loaded. The daily energy produced will be
(A) 720 MWh (B) 964 MWh
(C) 864 MWh (D) 820 MWh
- The inductance in mH/km of a single-phase overhead transmission line of diameter 1 cm and distance of separation 60 cm is
(A) $0.05 + 0.2 \ln 60$
(B) $0.05 + 0.2 \ln \left(\frac{60}{0.5} \right)$
(C) $0.2 \ln \left(\frac{60}{0.5} \right)$
(D) $0.2 \ln 60$
- A 50 km transmission line when excited with a source of frequency 500 Hz, it will be modelled as
(A) Short line (B) Medium line
(C) Long line (D) Data insufficient
- An existing AC system has a string efficiency of 80%. The same line, if used for DC transmission, then string efficiency will be
(A) 100% (B) >80%
(C) <80% (D) 0%

- The figure shows a single circuit 460 kV line using two bundled conductors per phase. If diameter of each conductor is 5 cm, the inductance/km/phase is



- (A) 0.24 mH/km/phase
(B) 0.76 mH/km/phase
(C) 0.89 mH/km/phase
(D) 0.90 mH/km/phase
- A conductor is composed of seven identical strands of copper, each having a radius r . The self-GMD of the conductor will be
(A) $2.17 r$ (B) $2.645 r$
(C) $2.141 r$ (D) $1.21 r$
 - Determine the capacitance for a 2-conductor 1- ϕ line operating at 50 Hz, the conductors being of 2 cm in diameter and spaced 3 m apart.
(A) 9.74×10^{-9} F/km (B) 4.87×10^{-9} F/km
(C) 3.05×10^{-9} F/km (D) None of these
 - A 1- ϕ transmission line having $R = 1.4 \Omega$ and $X = 0.8 \Omega$ is delivering a load of 200 kVA at 2500 V. The voltage at sending end when p.f. of the load is 0.8 p.f. lag is
(A) 2.069 kV (B) 2.840 kV
(C) 2.627 kV (D) 3.54 kV

9. The parameters of a single circuit 3- ϕ , 50 Hz transmission line are

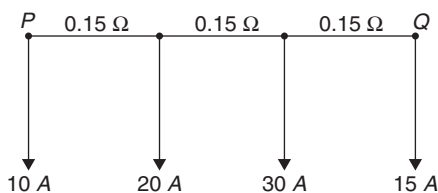
$$R = 0.2 \, \Omega/\text{km}, L = 1.3 \, \text{mH}/\text{km}$$

$$C = 0.01 \, \mu\text{F}/\text{km}, L = 120 \, \text{km}$$

The efficiency of the line if it delivers 40 MW at 132 kV and 0.8 p.f. lag is

- (A) 85.4% (B) 90.0%
(C) 98% (D) 92.6%

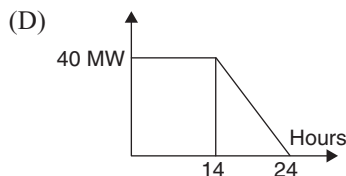
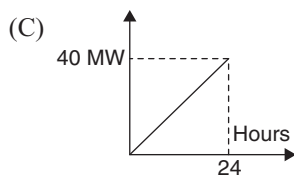
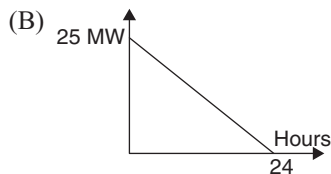
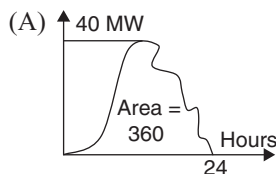
10. A DC distribution system is shown with load currents as marked. The two ends of the feeder P and Q are fed such that $V_P - V_Q = 3 \, \text{V}$. The value of voltage V_P for a minimum voltage of 220 V at any point along the feeder is



- (A) 220 V (B) 223 V
(C) 226 V (D) 229 V

11. An underground cable with a surge impedance of $100 \, \Omega$ is connected in series with an overhead line having a surge impedance of $400 \, \Omega$. If a source of 50 kV travels from the line towards the line cable junctions, the value of the transmitted voltage wave at the junction is
- (A) 20 kV (B) 30 kV
(C) -30 kV (D) 80 kV

12. For the figures given below, which curve has the least load factor?

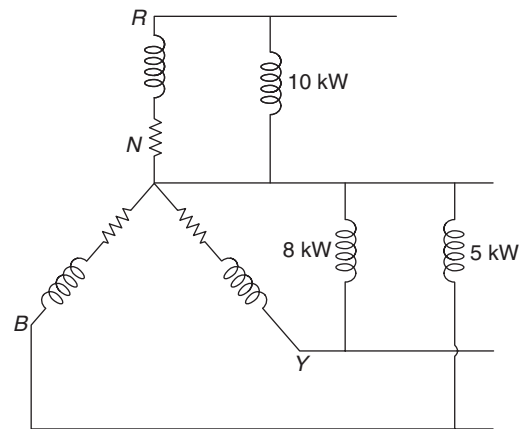


- (A) Curve A (B) Curve B
(C) Curve C (D) Curve D

13. A 275 kV transmission line has the following line constants. $A = 0.85 \angle 5^\circ$, $B = 200 \angle 75^\circ$. The power that can be received at unity p.f. if voltage profile at each end is to be maintained at 275 kV is

- (A) 118 MW (B) 218 MW
(C) 98 MW (D) None of these

14. In a 3- ϕ , 4-wire system, the line voltage is 400 V and now inductive loads of 10, 8 and 5 kW are connected between the three line conductors and the neutral as shown in figure. Calculate the current in the neutral conductor.



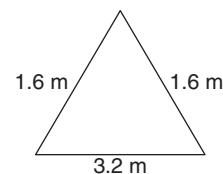
- (A) 15.8 A (B) 16.8 A
(C) 17.8 A (D) 18.8 A

15. A 3- ϕ transmission line consists of 1 cm radius conductors spaced symmetrically 4 m apart. Dielectric strength of air is 30 kV/cm. The line voltage for corona to commence is

- (A) 110 kV (B) 220 kV
(C) 330 kV (D) 400 kV

Common Data for Questions 16 and 17:

The conductors of a 3- ϕ transmission line are arranged as shown, radius of each line = 0.4 cm



The operating voltage is 132 kV.

16. The capacitance per phase is

- (A) $8.9 \times 10^{-9} \, \text{F}/\text{km}$
(B) $9.2 \times 10^{-6} \, \text{F}/\text{km}$
(C) $11.5 \times 10^{-9} \, \text{F}/\text{km}$
(D) $7.5 \times 10^{-7} \, \text{F}/\text{km}$

17. The charging current when the transmission line is operated at 264 kV is

- (A) 0.312 A/km
- (B) 0.257 A/km
- (C) 0.427 A/km
- (D) 0.214 A/km

Common Data for Questions 18 and 19:

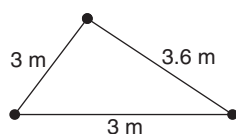
A 3- ϕ 50 Hz transmission line is 400 km long. The voltage at the sending end is 220 kV. The line parameters are $r = 0.125 \Omega/\text{km}$, $x = 0.4 \Omega/\text{km}$, $y = 2.8 \times 10^{-6} \text{ S}/\text{km}$

18. The sending end current and receiving end voltage are.
 - (A) 76 A, 121 kV
 - (B) 152 A, 242 kV
 - (C) 152 A, 226 kV
 - (D) 76 A, 242 kV
19. The maximum permissible line length of receiving end no load voltage is not to exceed 235 kV
 - (A) 138 km
 - (B) 238 km
 - (C) 338 km
 - (D) 438 km
20. A 100 MVA, 11 kV alternator has synchronous reactance 0.3 p.u. The p.u. synchronous reactance on a 50 MVA, 33 kV base is
 - (A) 1.67 p.u.
 - (B) 0.45 p.u.
 - (C) 0.016 p.u.
 - (D) 0.167 p.u.
21. A 3- ϕ induction motor rated at 400 V, 50 Hz is running at a p.f. of 0.707 lag. The input to the motor is 8 kVA. It is proposed to improve the p.f. of the system to 0.9 by a Δ connected bank of capacitors. The value of capacitance/phase required is
 - (A) 0.58 μF
 - (B) 58 μF
 - (C) 0.19 μF
 - (D) 19 μF
22. A shunt capacitor rated 100 MVAR is operated at 99% of rated voltage and 98% of rated frequency. The reactive power absorbed would be approximately
 - (A) 100 MVAR
 - (B) 98 MVAR
 - (C) 96 MVAR
 - (D) 95 MVAR
23. The per-unit impedance of a circuit element is 0.30. If the base kV and base MVA are halved, then the new value of per unit impedance is
 - (A) 0.9
 - (B) 0.6
 - (C) 0.3
 - (D) 0.15
24. Two buses 1 and 2 are connected by an impedance $(0 + j10) \Omega$. The bus 1 voltage is $400\angle 45^\circ$ and bus 2 voltage is $400\angle 0^\circ$ V. The real and reactive powers supplied by bus 1, respectively, are
 - (A) +11314 W, +4686 VAR
 - (B) +11314 W, -4686 VAR
 - (C) -11314 W, -2687 VAR
 - (D) -5656 W, +2687 VAR
25. A load is drawing 500 kW at 0.707 lag. A synchronous motor improves the p.f. of the load to 0.96. The motor carries a load of 300 kW. The leading reactive power supplied by the motor is
 - (A) 500 kVAR
 - (B) 233.33 kVAR
 - (C) 266.67 kVAR
 - (D) 250 kVAR
26. The voltage phasor of a circuit is $20\angle 15^\circ$ V and current phasor is $5\angle -45^\circ$ A. The active and reactive powers in the circuit are
 - (A) 50 W and -86.67 VAR
 - (B) 50 W and 86.67 VAR
 - (C) 86.67 W and 50 VAR
 - (D) 86.67 W and -50 VAR
27. A 240 V single-phase AC source is connected to a load with an impedance of $20\angle 45^\circ$. A capacitor is connected in parallel with the load. If the capacitor supplies 2400 VAR, the real power supplied by source is
 - (A) 3394 W
 - (B) 2400 W
 - (C) 994 W
 - (D) None of these

Practice Problems 2

Directions for questions 1 to 40: Select the correct alternative from the given choices.

1. A 1- ϕ two conductor circuit has a spacing of 3 m between them and each conductor has a diameter of 10 mm. The inductance of each circuit per km is
 - (A) 1.24 mH/km
 - (B) 1.19 mH/km
 - (C) 2.38 mH/km
 - (D) 0.59 mH/km
2. The conductors of a 3- ϕ 50 Hz circuit are each of diameter 21 mm. The spacing is as shown. The inductance is



- (A) 10.5×10^{-4} H/km
- (B) 8.2×10^{-4} H/km
- (C) 5.6×10^{-3} H/km
- (D) None of these

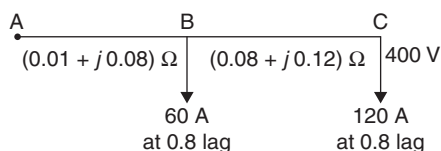
3. A two conductor 1- ϕ line operating at 50 Hz has a diameter of 20 mm and spacing between the conductors is 3 m. Height of conductor above the ground is 6 m. The capacitance of line to neutral will be
 - (A) 9.7 pF/km
 - (B) 10.2 pF/km
 - (C) 8.7 $\mu\text{F}/\text{km}$
 - (D) 2.4 $\mu\text{F}/\text{km}$
4. The power losses in a 3- ϕ 20 km line delivering 10 MW at 11 kV, 50 Hz, 0.8 p.f. lagging is 10% of power delivered. The resistance of each conductor is
 - (A) 0.774 Ω
 - (B) 2.14 Ω
 - (C) 1.54 Ω
 - (D) 0.86 Ω
5. If bundled conductors are used instead of single conductors, then
 - (A) Inductance and capacitance increase
 - (B) Inductance and capacitance decrease
 - (C) Inductance increases and capacitance decreases
 - (D) Inductance decreases and capacitance increases

6. A 400 V, 3- ϕ 4-wire system supplies a Y-connected load. Resistance of each line is $0.2\ \Omega$ and of neutral is $0.4\ \Omega$. The load impedances are

$$Z_R = 6 + j10, Z_Y = 10\ \Omega$$

$$Z_B = 8 - j10\ \Omega. \text{ Current in neutral is}$$

- (A) 16.56 A (B) 17.56 A
(C) 18.56 A (D) 19.56 A
7. The non-uniform distribution of voltage across the units in a string of suspension-type insulator is due to
- (A) Non-uniform distance between the cross arm and the units
(B) Existence of the stray capacitance between the metallic junctions of the units and tower body
(C) Non-uniform distance of separation of units from tower body
(D) Unequal self-capacitance of the units
8. A long overhead transmission line is terminated by its characteristic impedance. The ratio of voltage to current at different points along the line
- (A) Progressively increases from receiving end to the sending end
(B) Progressively decreases from receiving end to the sending end
(C) Remains the same at the two ends being maximum at the centre of the line
(D) Remains the same at all points
9. A 1- ϕ , AC distributor supplies two single-phase loads as shown in the figure.



The voltage drop from A to C is

- (A) 21 V (B) 26.7 V
(C) 41 V (D) 51 V
10. The use of hollow conductors reduce corona loss as
- (A) Provides better ventilation of the conductor
(B) Eddy currents in the conductor are eliminated
(C) Current density is reduced
(D) For a given cross section radius of conductor is increases
11. An insulator string has three units. The capacitance of the link between two units to earth is one eighth of the capacitance of each insulator. Each unit has a safe working voltage of 18.5 kV. The system voltage is
- (A) 55.5 kV (B) 46.7 kV
(C) 37 kV (D) 36.7 kV
12. The increase in resistance of a conductor carrying alternating current, due to non-uniform distribution of current is called
- (A) Hall effect (B) Proximity effect
(C) Skin effect (D) Ferranti effect

13. The transmission capacity of a line is related to its inductance varies in
- (A) Direct proportion
(B) Inverse proportion
(C) Square
(D) Independent of inductance
14. A lossless radial transmission line with surge impedance loading
- (A) Has sending end voltage higher than receiving end voltage
(B) Has flat voltage profile and unity p.f. at all points along it
(C) Takes positive VAR at sending end and zero VAR at receiving end
(D) Takes negative VAR at sending end and zero VAR at receiving end
15. The surge impedance of a 400 km line is $500\ \Omega$. For a 200 km length of same line, the impedance will be
- (A) $250\ \Omega$ (B) $1000\ \Omega$
(C) $500\ \Omega$ (D) $750\ \Omega$
16. In a string of suspension insulators, the voltage distribution across the different units could be made uniform by using grading ring because it
- (A) Decreases the capacitances of upper insulators units causing equal voltage drop
(B) Increases capacitances of lower insulator units to cause equal voltage drop
(C) Forms capacitances with link pins which helps to cancel the charging current from link pins
(D) Forms capacitances with link pins to carry the charging current from the link pins
17. The cause of Ferranti effect in transmission lines is
- (A) Line inductance
(B) Line capacitance
(C) Line capacitance and inductance
(D) Line resistances and capacitance
18. Series capacitors are used in a transmission line to
- (A) To increase the phase angles at sending end and receiving ends under heavy load conditions
(B) Reduce the phase angle between sending end and receiving ends under heavy load conditions
(C) To reduce the capacitive current
(D) None of these
19. The load factor and capacity factor of a plant are equal if
- (A) Peak load is equal to capacity of plant
(B) Average load is half the capacity of the plant
(C) Average load is same as peak load
(D) Group diversity factor is equal to peak diversity factor
20. The voltage across the units in a two unit suspension insulator is 40% and 60%. The ratio of capacitance of insulator to that of earth will be

- (A) 0.4 (B) 0.45
(C) 0.5 (D) 0.55
21. The receiving end voltage of a long transmission line at no load condition is
(A) Less than sending end voltage
(B) Greater than sending end voltage
(C) Equal to sending end voltage
(D) Zero voltage
22. The level of insulation of a 400 kV extra high voltage line is decided based on
(A) Lightning over voltage
(B) Switching over voltage
(C) Corona inception voltage
(D) Radio and TV interference
23. The transmission line feeding power on either side of main transmission line is
(A) Primary distribution
(B) Primary transmission
(C) Secondary transmission
(D) Secondary distribution
24. Suppose the conductors are dead ended and there is a change in the direction of transmission line, insulators used are
(A) Strain type (B) Pin type
(C) Shackle type (D) Suspension type
25. For a fixed inductive shunt compensation, series capacitive compensation
(A) Increases virtual surge impedance loading
(B) Decreases vertical surge impedance loading
(C) Increases effective length of line
(D) None of these
30. Shunt reactors are connected in transmission lines to
(A) Limit the fault current
(B) Improve stability limit
(C) Absorb the excess reactive power
(D) Generate requisite reactive power
31. The excitation of a synchronous generator connected to an infinite bus bar delivering power at a lagging power factor is increased. Then
(A) Current delivered increases
(B) Terminal voltage increases
(C) Voltage angle δ increases
(D) Both A and C
32. The voltage of an infinite bus bar is $1\angle 0^\circ$ and that of an alternator connected to it is $0.95\angle 0^\circ$. The alternator acts as a
(A) Shunt coil
(B) Shunt capacitor
(C) Shunt resistance
(D) None of these
33. The power dispatch through a transmission line can be increased by installing
(A) Shunt capacitance
(B) Series capacitance
(C) Shunt reactance
(D) Series reactance
34. The combined frequency regulation of machines in an area of capacity 3000 MW and operating at a normal frequency of 60 Hz is 0.15 p.u. The regulation in Hz/MW is
(A) $\frac{3}{1000}$ (B) $\frac{0.3}{1000}$
(C) $\frac{1}{200}$ (D) $\frac{1}{100}$

Common Data for Questions 26 and 27:

A 50 Hz, 3- ϕ , transmission line 30 km long has a total series impedance of $(40 + j125) \Omega$ and shunt admittance of 10^{-3} mho. The load is 50 MW at 220 kV with 0.8 lagging p.f.

26. Sending end voltage using nominal π is
(A) 216 kV (B) 204 kV
(C) 232 kV (D) 238 kV
27. The sending end power factor is
(A) 0.8 lag (B) 0.9 lag
(C) 0.98 lag (D) 1
28. A pure inductance absorbs reactive power from the AC line when
(A) Both applied voltage and current decreases
(B) Both applied voltage and current increases
(C) Applied voltage increases and current decreases
(D) Applied voltage decreases and current increases
29. The power factor of the load is corrected by connecting a static capacitor in parallel with the load. The power taken by the load
(A) Increases (B) Decreases
(C) Remains same (D) Is uncertain
35. Which amongst the following is not true?
(A) Synchronous phase modifiers are installed at sending end
(B) Synchronous phase modifiers are installed at load end
(C) Synchronous phase modifiers are specially designed synchronous motors
(D) Synchronous phase modifiers do not carry load
36. In EHV lines, series capacitance compensation is used to
(A) Improve the stability of the system
(B) Reduce the loading on the line
(C) Improve the protection of the line
(D) Reduce the voltage profile

II. Question Nos.: 37 and 38

A line operating at 60 Hz is 81% series compensated.

37. The resonating frequency is
(A) 60 Hz (B) 66.6 Hz
(C) 50 Hz (D) 54 Hz

38. Now if the line is 100% series compensated. Resonant frequency is
 (A) 60 Hz (B) 66.6 Hz
 (C) 50 Hz (D) 54 Hz.
39. The p.u. value of impedance when referred to the LV side is
 (A) $2.13 \angle 32^\circ$ (B) $0.0085 \angle 62^\circ$
 (C) $0.085 \angle 62^\circ$ (D) $0.213 \angle 32^\circ$
40. If the transformer has an exciting current of 10 A on HV side, p.u. value of current referred to LV side is
 (A) 1 p.u. (B) 2 p.u.
 (C) 0.5 p.u. (D) 0.25 p.u.

III. Question Nos.: 39 and 40

A20 kVA, 2000/200 V single-phase transformer has an equivalent impedance of $Z = 8 + j15$ when referred to the HV side.

PREVIOUS YEARS' QUESTIONS

- The concept of an electrically short, medium, and long line is primarily based on the [2006]
 (A) nominal voltage of the line
 (B) physical length of the line
 (C) wavelength of the line
 (D) power transmitted over the line
 - A 400 V, 50 Hz, three-phase balanced source supplies power to a star connected load whose rating is $12\sqrt{3}$ kVA, 0.8 p.f. (lag). The rating (in kVAR) of the delta connected (capacitive) reactive power bank necessary to bring the p.f. to unity is [2006]
 (A) 28.78 (B) 21.60
 (C) 16.60 (D) 12.47
 - The A, B, C, D constants of a 220 kV line are:
 $A = D = 0.94 \angle 1^\circ$, $B = 130 \angle 73^\circ$, $C = 0.001 \angle 90^\circ$. If the sending end voltage of the line for a given load delivered at nominal voltage is 240 kV, then % voltage regulation of the line is [2006]
 (A) 5 (B) 9
 (C) 16 (D) 21
 - Consider a bundled conductor of an overhead line, consisting of three identical sub-conductors placed at the corners of an equilateral triangle as shown in the figure. If we neglect the charges on the other phase conductors and ground, and assume that spacing between sub-conductors is much larger than their radius, the maximum electric field intensity is experienced at [2007]
-
- Point X (B) Point Y
 (C) Point Z (D) Point W
- The total reactance and total susceptance of a lossless overhead EHV line, operating at 50 Hz, are given by 0.045 p.u. and 1.2 p.u., respectively. If the velocity of wave propagation is 3×10^5 km/s, then the approximate length of the line is [2007]
 (A) 122 km (B) 172 km
 (C) 222 km (D) 272 km
 - Out of the following plant categories
 (i) Nuclear (ii) Run-of-river
 (iii) Pump storage (iv) Diesel
 The base load power plants are [2009]
 (A) (i) and (ii) (B) (ii) and (iii)
 (C) (i), (ii) and (iv) (D) (i), (iii) and (iv)
 - For a fixed value of complex power flow in a transmission line having a sending end voltage V , the real power loss will be proportional to [2009]
 (A) V (B) V^2
 (C) $\frac{1}{V}$ (D) $1/V$
 - Match the items in List-I with the items in List-II and select the correct answer using the codes given below the lists. [2009]
- | List I
To | List II
Use |
|------------------------------------|---------------------|
| a. Improve power factor | 1. Shunt reactor |
| b. Reduce the current ripples | 2. Shunt capacitor |
| c. Increase the power flow in line | 3. Series capacitor |
| d. Reduce the Ferranti effect | 4. Series reactor |
- $a \rightarrow 2$ $b \rightarrow 3$ $c \rightarrow 4$ $d \rightarrow 1$
 (B) $a \rightarrow 2$ $b \rightarrow 4$ $c \rightarrow 3$ $d \rightarrow 1$
 (C) $a \rightarrow 4$ $b \rightarrow 3$ $c \rightarrow 1$ $d \rightarrow 2$
 (D) $a \rightarrow 4$ $b \rightarrow 1$ $c \rightarrow 3$ $d \rightarrow 2$
- Match the items in List-I with the items in List-II and select the correct answer using the codes given below the lists. [2009]
- | List I
Type of transmission line | List II
Type of distance relay preferred |
|-------------------------------------|---|
| a. Short line | 1. Ohm relay |
| b. Medium line | 2. Reactance relay |
| c. Long line | 3. Mho relay |

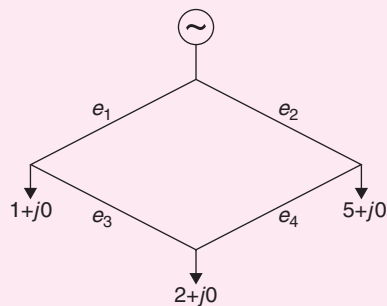
- (A) $a \rightarrow 2 \rightarrow b \rightarrow 1 \rightarrow c \rightarrow 3$
 (B) $a \rightarrow 3 \rightarrow b \rightarrow 2 \rightarrow c \rightarrow 1$
 (C) $a \rightarrow 1 \rightarrow b \rightarrow 2 \rightarrow c \rightarrow 3$
 (D) $a \rightarrow 1 \rightarrow b \rightarrow 3 \rightarrow c \rightarrow 2$

10. Three generators are feeding a load of 100 MW. The details of the generators are [2009]

	Rating (MW)	Efficiency (%)	Regulation (p.u.) on 100 MVA base
Generator-1	100	20	0.02
Generator-2	100	30	0.04
Generator-3	100	40	0.03

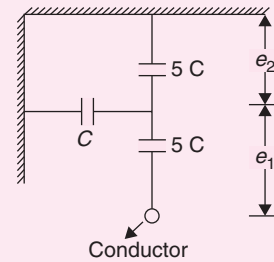
In the event of increased load power demand, which of the following will happen?

- (A) All the generators will share equal power
 (B) Generator-3 will share more power compared to Generator-1
 (C) Generator-1 will share more power compared to Generator-2
 (D) Generator-2 will share more power compared to Generator-3
11. An extra high voltage transmission line of length 300 km can be approximated by a lossless line having propagation constant $\beta = 0.00127$ radians per km. Then the percentage ratio of line length to wavelength will be given by [2008]
- (A) 24.24% (B) 12.12%
 (C) 19.05% (D) 6.06%
12. Single line diagram of a 4-bus single source distribution system is shown below, branches e_1, e_2, e_3 and e_4 have equal impedances. The load current values indicated in the figure are in per unit. Distribution company's policy requires radial system operation with minimum loss. This can be achieved by opening of the branch [2008]

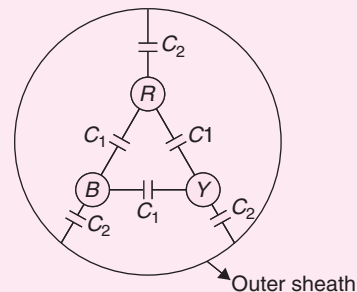


- (A) e_1 (B) e_2
 (C) e_3 (D) e_4

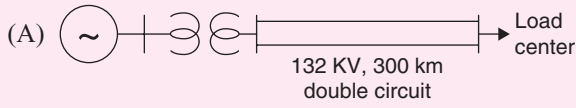
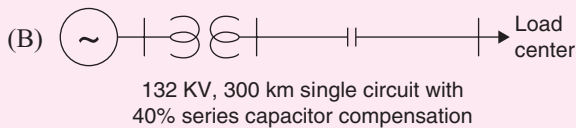
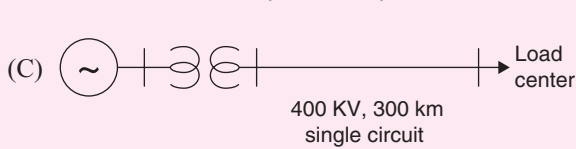
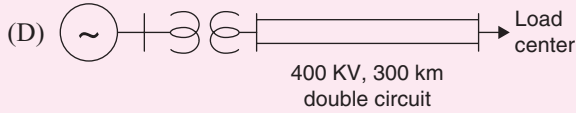
13. Consider a three-phase, 50 Hz, 11 kV distribution system. Each of the conductors is suspended by an insulator string having two identical porcelain insulators. The self-capacitance of the insulator is 5 times the shunt capacitance between the link and the ground, as shown in the figure. The voltage across the two insulators is [2010]



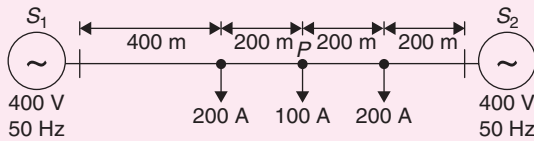
- (A) $e_1 = 3.74$ kV, $e_2 = 2.61$ kV
 (B) $e_1 = 3.46$ kV, $e_2 = 2.89$ kV
 (C) $e_1 = 6.0$ kV, $e_2 = 4.23$ kV
 (D) $e_1 = 5.5$ kV, $e_2 = 5.5$ kV
14. Consider a three-core, three-phase, 50 Hz, 11 kV cable whose conductors are denoted as R, Y and B in the figure. The inter-phase capacitance (C_1) between each pair of conductors is 0.2 mF and the capacitance between each line conductor and the sheath is 0.4 mF. The per-phase charging current is [2010]



- (A) 2.0 A (B) 2.4 A
 (C) 2.7 A (D) 3.5 A
15. For enhancing the power transmission in a long EHV transmission line, the most preferred method is to connect a [2011]
- (A) Series inductive compensator in the line.
 (B) Shunt inductive compensator at the receiving end.
 (C) Series capacitive compensator in the line.
 (D) Shunt capacitive compensator at the sending end.
16. A nuclear power station of 500 MW capacity is located at 300 km away from a load centre. So the most suitable power evacuation transmission configuration among the following options. [2011]

- (A)  132 KV, 300 km double circuit
- (B)  132 KV, 300 km single circuit with 40% series capacitor compensation
- (C)  400 KV, 300 km single circuit
- (D)  400 KV, 300 km double circuit

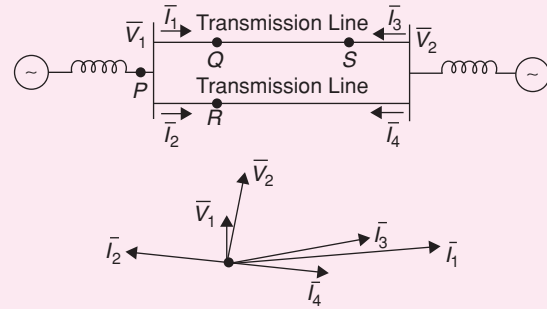
17. A distribution feeder of 1 km length having resistance, but negligible reactance, is fed from both the ends by 400 V, 50 Hz balanced sources. Both voltage sources S_1 and S_2 are in phase. The feeder supplies concentrated loads of unity power factor as shown in the figure. [2014]



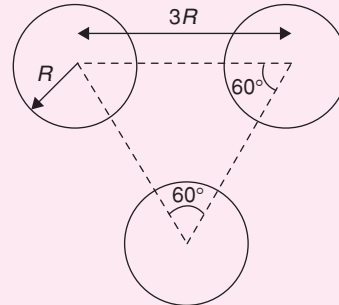
The contributions of S_1 and S_2 in 100 A current supplied at location P, respectively, are

- (A) 75 A and 25 A
 (B) 50 A and 50 A
 (C) 25 A and 75 A
 (D) 0 A and 100 A
18. Shunt reactors are sometimes used in high-voltage transmission systems to [2004]
 (A) limit the short-circuit current through the line
 (B) compensate for the series reactance of the line under heavily loaded condition.
 (C) limit overvoltages at the load side under lightly loaded condition.
 (D) compensate for the voltage drop in the line under heavily loaded condition.
19. Base load power plants are [2015]
 P: wind farms.
 Q: run-of-river plants.
 R: nuclear power plants.
 S: diesel power plants.
 (A) P, Q and S only
 (B) P, R and S only
 (C) P, Q and R only
 (D) Q and R only
20. A sustained three-phase fault occurs in the power system shown in the figure. The current and voltage

phasors during the fault (on a common reference), after the natural transients have died down, are also shown. Where is the fault located? [2015]



- (A) Location P
 (B) Location Q
 (C) Location R
 (D) Location S
21. A composite conductor consists of three conductors of radius R each. The conductors are arranged as shown below. The geometric mean radius (GMR) (in cm) of the composite conductor is kR . The value of k is _____. [2015]



22. A three-phase cable is supplying 800 kW and 600 kVAR to an inductive load. It is intended to supply an additional resistive load of 100 kW through the same cable without increasing the heat dissipation in the cable, by providing a three-phase bank of capacitors connected in star across the load. Given the line voltage is 3.3 kV, 50 Hz, the capacitance per phase of the bank, expressed in microfarads, is _____. [2016]
23. Single-phase transmission line has two conductors each of 10 mm radius. These are fixed at a center-to-center distance of 1 m in a horizontal plane. This is now converted to a three-phase transmission line by introducing a third conductor of the same radius. This conductor is fixed at an equal distance D from the two single-phase conductors. The three-phase line is fully transposed. The positive sequence inductance per phase of the three-phase system is to be 5% more than that of the inductance per conductor of the single-phase system. The distance D , in meters, is _____. [2016]

24. The inductance and capacitance of a 400kV, three-phases 50HZ lossless transmission line are 1.6mH/km/phase and 10nF/km/phase respectively. The sending end voltage is maintained at 400kV. To maintain a voltage of 400kV at the receiving end, when the line is delivering 300MW load, the shunt compensation required is [2016]
 (A) Capacitive (B) Inductive
 (C) Resistive (D) Zero
25. At no load condition, a 3-phase, 50Hz, lossless power transmission line has sending-end and receiving-end voltages of 400KV and 420KV respectively. Assuming the velocity of travelling wave to be the velocity of light, the length of the line, in km, is _____. [2016]
26. The power consumption of an industry is 500kVA, at 0.8 p.f lagging. A synchronous motor is added to raise the power factor of the industry to unity. If the power intake of the motor is 100kW, the p. f. of the motor is _____. [2016]

ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. C | 4. A | 5. C | 6. A | 7. B | 8. C | 9. D | 10. C |
| 11. A | 12. A | 13. A | 14. D | 15. B | 16. A | 17. C | 18. B | 19. C | 20. C |
| 21. D | 22. C | 23. B | 24. A | 25. C | 26. B | 27. A | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. A | 4. A | 5. D | 6. C | 7. B | 8. D | 9. B | 10. D |
| 11. B | 12. C | 13. B | 14. B | 15. C | 16. C | 17. C | 18. B | 19. A | 20. C |
| 21. B | 22. B | 23. C | 24. A | 25. D | 26. D | 27. C | 28. A | 29. C | 30. C |
| 31. D | 32. A | 33. B | 34. A | 35. A | 36. A | 37. D | 38. A | 39. C | 40. A |

Previous Years' Questions

- | | | | | | | | | | |
|------------------|-----------|----------|-------|---------------|-----------|-------|-------|-------|-------|
| 1. B | 2. D | 3. C | 4. B | 5. C | 6. A | 7. C | 8. B | 9. A | 10. C |
| 11. D | 12. D | 13. B | 14. A | 15. C | 16. D | 17. D | 18. C | 19. C | 20. B |
| 21. 1.85 to 1.95 | 22. 48.23 | 23. 1.42 | 24. B | 25. 294.847Km | 26. 0.316 | | | | |

Chapter 2

Fault Analysis

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Transients in transmission line
- Symmetrical components
- Sequence networks of synchronous generator
- Negative sequence impedance network
- Zero sequence impedance network
- Delta connected load
- Sequence diagrams for transformers
- Unbalanced fault analysis
- Single line to ground fault
- Double line to ground fault

INTRODUCTION

Faults in power systems occur because of various reasons like equipment failure, lightning strikes, falling of branches or trees on the transmission lines, switching surges, insulations failures and other electrical or mechanical causes. All these disturbances are collectively called 'faults' in power systems.

A fault usually causes high current flowing through the lines and if enough protection is not provided it may result in damages for the power apparatus. The determination of the values of such currents enables us to make proper selection of circuit breakers, protective relays and also helps to ensure that the associated apparatus will withstand the forces which arise due to the fault currents during the occurrence of fault prior to the interrupting device during the fault.

Percentage wise, the various causes of faults are

- | | | |
|---------------------------------|---|-----|
| 1. Lightning | → | 6% |
| 2. Equipment failure | → | 10% |
| 3. Switching to a fault | → | 10% |
| 4. Sleet, wind | → | 10% |
| 5. Tree falling, sabotage, etc. | → | 10% |

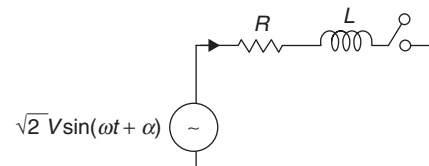
Types of faults and probability of occurrence are given below:

- | | | |
|----------------------------------|---|-----|
| 1. Single-phase to ground faults | – | 70% |
| 2. Phase-to-phase faults | – | 15% |
| 3. Two-phase to ground faults | – | 10% |
| 4. Three-phase fault | – | 5% |

TRANSIENTS IN TRANSMISSION LINE

Assumptions in the analysis

1. Transmission line is fed from a constant voltage source.
2. Short circuit occurred when the line is unloaded.
3. Line capacitance is neglected and the line is represented by a lumped RL series circuit.



The above figure shows the RL circuit supplied by a voltage source according to given assumptions.

The switch is assumed to be closed at $t = 0$ sec. The parameter ' α ' indicates the instant on the voltage source when switch is closed. When the switch is closed, the current flowing through the circuit can be written as

$$i = i_s + i_t$$

where i_s = Steady-state current
 i_t = Transient current

$$i_s = \frac{\sqrt{2}V}{|Z|} \sin(\omega t + \alpha - \theta)$$

$$i_t = \frac{\sqrt{2}V}{|Z|} \sin(\theta - \alpha) e^{-\left(\frac{R}{L}\right)t}$$

where

$$Z = \sqrt{R^2 + \omega^2 L^2} \angle \tan^{-1} \frac{\omega L}{R}; \theta = \tan^{-1} \frac{\omega L}{R}$$

$$\text{Total current, } i = \frac{\sqrt{2}V}{|Z|} \sin(\omega t + \alpha - \theta) + \frac{\sqrt{2}V}{|Z|} \sin(\theta - \alpha) e^{-\left(\frac{R}{L}\right)t}$$

Steady-state sinusoidal current is called 'symmetrical short-circuit current' and the unidirectional transient component is called the 'DC off set current'. First peak in this

transient current is known as 'maximum momentary short-circuit current' and it is expressed as

$$i_{mm} = \frac{\sqrt{2}V}{|Z|} \sin(\theta - \alpha) + \frac{\sqrt{2}V}{|Z|}$$

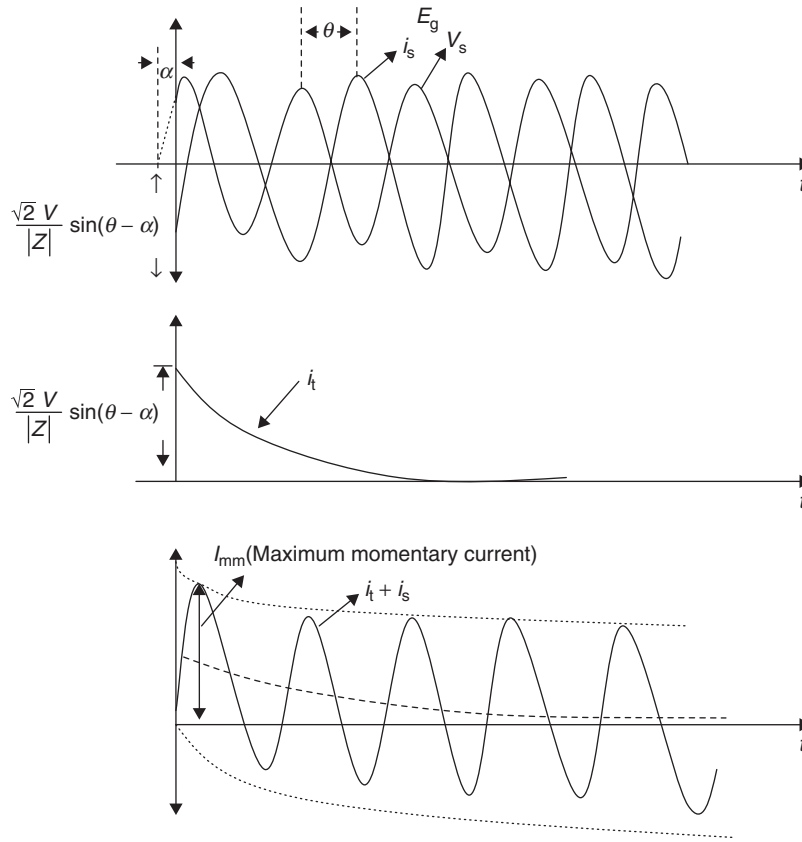
Since the transmission line resistance is small,

$$\theta \cong 90^\circ$$

and i_{mm} has maximum possible value for $\alpha = 0$.

$$i_{mm} = 2 \frac{\sqrt{2}V}{|Z|}$$

$$= 2 \times \text{Maximum symmetrical short-circuit current}$$



Short Circuit of an Unloaded Synchronous Generator

When a three-phase symmetrical short circuit occurs at the terminals of an unloaded synchronous generator, it is assumed that there is no DC off set in the armature current and the magnitude of the current decreases exponentially from a high initial value. The instantaneous expression for the fault current is given by

$$i_f(t) = \sqrt{2}V_t \left[\left(\frac{1}{x_d''} - \frac{1}{x_d'} \right) e^{-t/T_d''} + \left(\frac{1}{x_d'} - \frac{1}{x_d} \right) e^{-t/T_d'} + \frac{1}{x_d} \right] \sin\left(\omega t + \alpha - \frac{\pi}{2}\right)$$

where V_t is the magnitude of the terminal voltage.

' α ' is the phase angle of the voltage.

X_d'' is the direct axis subtransient reactance.

X_d' is the direct axis transient reactance.

X_d is the direct axis synchronous reactance.

With $X_d'' < X_d' < X_d$. The time constants are.

T_d'' is the direct axis sub transient time constant.

T_d' is the direct axis transient time constant.

If the fault occurs in the system at $t = 0$, the RMS value of the current is given as

$$I_f(0) = I_f'' = \frac{V_t}{X_d''}$$

which is called the 'subtransient fault current'. The duration of the sub transient current is dictated by the time constant T_d'' . As the time progresses and $T_d'' < t < T_d'$, the first exponential term of fault current will short decaying and will eventually vanish. However, since 't' is still nearly equal to zero, we have the following RMS value of the current.

$$I_f' = \frac{V_t}{X_d'}$$

This is called the transient fault current. Now as the time progress, further and the second exponential term also decays, we get the following RMS value of the current for the sinusoidal steady state.

$$I_f = \frac{V_t}{X_d}$$

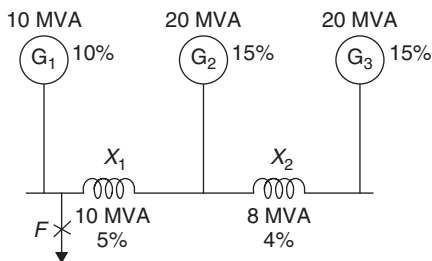
In addition to the AC, the fault currents will also contain the DC offset, the maximum value of the DC off set is given by

$$i_{DC}^{\max} = \sqrt{2} I_f'' e^{-t/T_A}$$

where T_A is the armature time constant.

Solved Examples

Example 1: An interconnected generator-reactor system is shown in figure. The base values for a given per cent reactance are the ratings of the individual pieces of equipment. A three-phase short circuit occurs at point 'F'. Determine the fault current and the fault kVA if the busbar line to line voltage is 11 kV and base MVA is 50 MVA.



Solution: Reactance diagram of the system is to be represented on 50 MVA base. New values of reactance on new base are

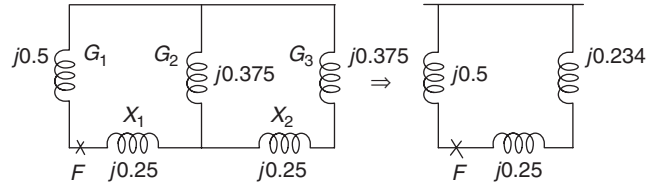
$$X_{G_1} = 0.10 \times \frac{50}{10} = 0.10 \times \frac{\text{New VA}}{\text{Old VA}} = 0.5 \text{ p.u.}$$

$$X_{G_2} = 0.15 \times \frac{50}{20} = 0.375 \text{ p.u.}$$

$$X_{G_3} = 0.15 \times \frac{50}{20} = 0.375 \text{ p.u.}$$

$$X_1 = 0.05 \times \frac{50}{10} = 0.25 \text{ p.u.}$$

$$X_2 = 0.04 \times \frac{50}{8} = 0.25 \text{ p.u.}$$



Total reactance from the fault point to neutral is per unit

$$\text{reactance} = j \frac{0.5(0.2344 + 0.25)}{0.5 + 0.2344 + 0.25} = j0.246$$

$$\text{Fault MVA} = \frac{\text{Base MVA}}{Z_{F \text{ p.u.}}} = \frac{50 \times 10^6}{0.246} = 203.25 \text{ MVA}$$

$$\text{Fault current} = \frac{\text{Fault MVA}}{\sqrt{3} \times 11 \text{ kV}} = \frac{203.25 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 10.66 \text{ kA}$$

Example 2: A 60 Hz alternator is rated 500 MVA, 20 kV, with $X_d = 1.0$ p.u. and $X_d'' = 0.3$ p.u. It supplies a purely resistive load of 400 MVA at 20 kV. The load is directly connected across the generator terminals when a symmetrical fault occurs at the load terminals. The initial RMS current in the generator circuit is

- (A) 4.811 p.u. (B) 11.54 p.u.
(C) 4.16 p.u. (D) 8.33 p.u.

Solution: (C)

Initial fault current depends on sub-transient reactance $X_d'' = 0.3$ p.u.

$$\text{Initial fault current} = \frac{E}{X_{F \text{ actual}}}$$

$$X_{F \text{ actual}} = X_{d \text{ actual}}'' = 0.3 \times \frac{V_{\text{base}}^2}{VA_{\text{base}}} = 0.3 \times \frac{20^2}{500} = 0.24 \Omega$$

Initial fault current

$$= \frac{E}{X_{F \text{ actual}}} = \frac{\left(\frac{20}{\sqrt{3}}\right) \times 10^3}{0.24} = 48.11 \text{ kA}$$

Base current in load circuit

$$= \frac{\text{Load MVA}}{\sqrt{3} \times \text{kV}} = \frac{400 \times 10^6}{\sqrt{3} \times 20 \times 10^3} = 11.547 \text{ kA}$$

$$\text{Per-unit value of fault current } I_F = \frac{I_{F \text{ actual}}}{I_{\text{base}}}$$

$$= \frac{48.11}{11.547} = 4.16 \text{ p.u.}$$

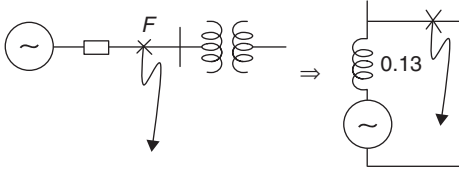
Example 3: A three-phase generator rated at 110 MVA, 11 kV is connected through circuit breakers to a transformer. The generator is having direct axis sub-transient reactance

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$X_d'' = 13\%$ and synchronous reactance $X_d = 130\%$. The generator is operating at no-load and rated voltage when a three-phase short-circuit fault occurs between the breaker and the transformer. The magnitude of initial symmetrical RMS current in the breaker will be

- (A) 76.9 kA (B) 44.4 kA
(C) 38.45 kA (D) 7.69 kA

Solution: (B)



$$\text{Fault current } I_f = \frac{E}{X_d''} = \frac{1}{0.13} = 7.69 \text{ p.u.}$$

$$I_{\text{fault actual}} = I_{\text{base}} \times I_{f \text{ p.u.}} = \frac{110 \times 10^6}{\sqrt{3} \times 11 \times 10^3} \times 7.69 = 44.4 \text{ kA}$$

Example 4: At a 220 kV substation of a power system, it is given that the three-phase fault level is 3500 MVA. Calculate the driving point reactance at the faulted bus

- (A) 4.61 Ω (B) 27.65 Ω
(C) 6.91 Ω (D) 13.82 Ω

Solution: (D)

The short-circuit MVA = $3V_{\text{base}} I_{\text{sc}}$

$$\text{Short-circuit current } (I_{\text{sc}}) = \frac{V_{\text{base}}}{X_f}$$

$$\therefore 3V_{\text{base}} \cdot \frac{V_{\text{base}}}{X_{f_{\text{eq}}}} = 3500$$

$$X_{f_{\text{eq}}} = \frac{3V_{\text{base}}^2}{3500} = \frac{3 \times \left(\frac{220}{\sqrt{3}}\right)^2}{3500}$$

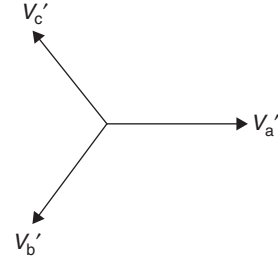
$$X_{f_{\text{eq}}} = 13.82 \Omega$$

SYMMETRICAL COMPONENTS

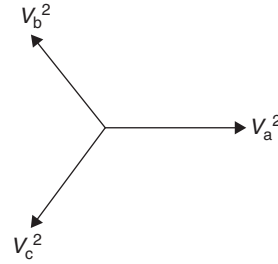
The key idea of symmetrical component analysis is to decompose the unbalanced system into three sequences of balanced networks. The networks are then coupled only at the point of the unbalanced.

The three sequence networks are known as the

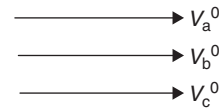
1. Positive-sequence components consisting of three phasors equal in magnitude, displaced from each other by 120° in phase, and having the same phase sequence as the original phasors.



2. Negative-sequence components consisting of three phasors equal in magnitude, displaced from each other by 120° in phase, and having the phase sequence opposite to that of the original phasors.



3. Zero-sequence components consisting of three phasors equal in magnitude and with zero-phase displacements from each other.



Assume three unbalanced voltage phasors, V_A , V_B and V_C having a positive sequence (abc). Using symmetrical components, it is possible to represent each phasor voltage as

$$V_A = V_A^0 + V_A^1 + V_A^2$$

$$V_B = V_B^0 + V_B^1 + V_B^2$$

$$V_C = V_C^0 + V_C^1 + V_C^2$$

Operator 'a': Function of the operator 'a' was introduced, when it is operated upon a phasor that is rotated by $+120^\circ$ without changing the magnitude of it.

$$a = 1 \angle 120^\circ = e^{j120^\circ} = -0.5 + j0.866$$

$$a^2 = 1 \angle 240^\circ = e^{j240^\circ} = -0.5 - j0.866$$

$$a^3 = 1 \angle 360^\circ = 1 \angle 0^\circ = e^{j360^\circ} = 1$$

$$\therefore 1 + a + a^2 = 0$$

$$a^4 = a; a^5 = a^2; a^6 = 1$$

$$a - a^2 = j\sqrt{3}; a^2 - a = -j\sqrt{3}$$

Assuming phase 'a' as reference, the above relation between the unsymmetrical components of phase 'b' and 'c' in terms of symmetrical components of phase 'a' is given by

$$V_a = V_a^0 + V_a^1 + V_a^2$$

$$V_b = V_a^0 + a^2 V_a^1 + a V_a^2$$

$$V_c = V_a^0 + a V_a^1 + a^2 V_a^2$$

Matrix representation of the above relation is given by

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix} = [T] \begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix}$$

Inverse relation is given

$$\begin{bmatrix} V_a^0 \\ V_a^1 \\ V_a^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

The complex power flowing into a three-phase circuit through three lines a , b and c is

$$S_3 \phi = P + jQ = V_a I_a^* + V_b I_b^* + V_c I_c^*$$

where V_a , V_b and V_c are the voltages and I_a , I_b and I_c are the currents flowing into the circuit in the three lines.

$$\begin{aligned} S_3 \phi &= (T V_a^{012})^T (T I_a^{012})^* \\ &= V_a^{012^T} T^T T^* I_a^{012^*} \end{aligned}$$

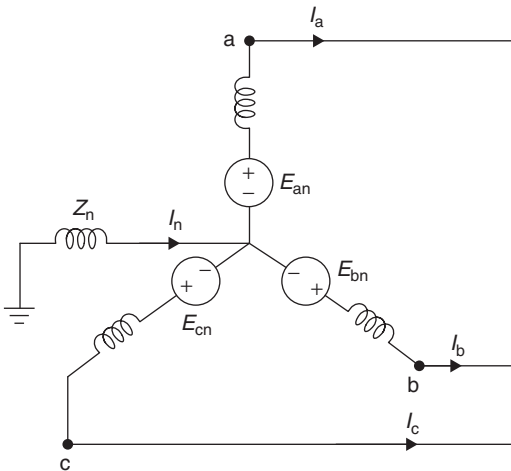
Since $T^T T^* = 3$,

$$\begin{aligned} \therefore S_3 \phi &= 3 (V^{012^T} I^{012^*}) \\ S_3 &= 3 V^0 I^{0*} + 3 V^1 I^{1*} + 3 V^2 I^{2*} \end{aligned}$$

The above equation shows that the total unbalanced three-phase power is equal to the sum of the three symmetrical component powers.

Sequence Networks of Synchronous Generator

A Synchronous generator, grounded through a reactor is shown in figure below.



When the fault takes place at the terminals, the current I_a , I_b and I_c flows in the lines. Whenever the fault involves ground, the total current $I_a + I_b + I_c = I_n$ flows into the neutral of the generator and the line currents can be resolved into symmetrical components.

Positive Sequence Impedance and Network

Synchronous generator induces positive sequence emf's only due to its symmetrical winding designs. When the machine carries positive sequence currents only, this mode of operation is the balanced mode. As the short circuit progresses in time, the machine equivalently offers a direct axis reactance whose value reduces from sub transient reactance (X_d'') to transient reactance (X_d') and finally to steady-state reactance (X_d). If the armature resistance is assumed negligible, the positive sequence impedances of machine is

$$Z_1 = jX_d'' \text{ (sub transient period } \rightarrow 1 \text{ cycle)}$$

$$= jX_d' \text{ (transient period } \rightarrow 3-4 \text{ cycles)}$$

$$= jX_d \text{ (steady-state value)}$$

Since the neutral current $I_n = 0$, the neutral impedance Z_n does not appear in the model. Since the positive-sequence network is a balanced network, it can be represented by the single-phase network model as shown in figure.

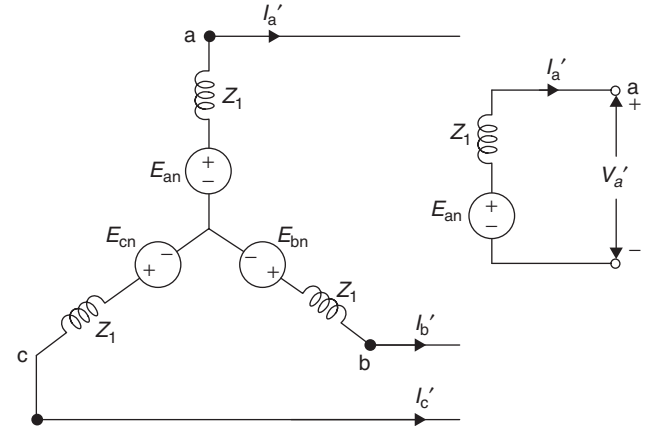


Figure 1 Positive-sequence network and its single-phase equivalent

Negative Sequence Impedance Network

Synchronous machine has no negative sequence induced voltages. With the flow of negative sequence currents in the stator, a rotating field is created which rotates in opposite direction to that of the positive sequence, therefore, at double synchronous speed with respect to rotor. The negative sequence mmf is alternately presented with reluctance of direct and quadrature axes. The negative sequence impedance presented by the machine with consideration given to the damper windings is defined as

$$Z_2 = j \frac{X_q' + X_d''}{2}; |Z_2| < |Z_1|$$

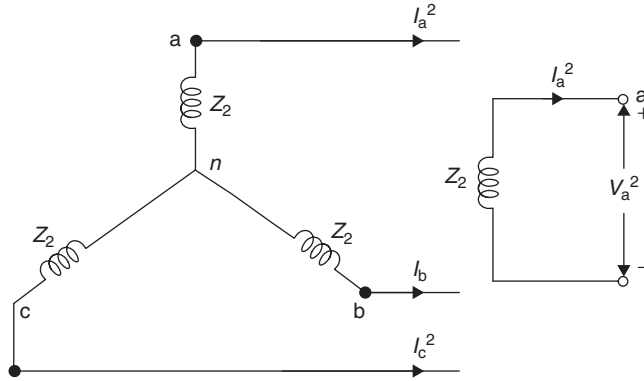


Figure 2 Negative-sequence network and its single-phase equivalent

Zero-sequence Impedance Network

Synchronous machine has no zero-sequence voltages induced, the flow of zero-sequence currents creates three mmf's which are in time phase, but are distributed in space by 120° . The current flowing through the impedance Z_n between neutral to ground is $I_n = 3I_{a_0}$. The zero-sequence voltage drop of zero sequence from point 'a' to ground is

$$V_{a_0} = -(3Z_n + Z_{g_0})I_{a_0}$$

The zero-sequence circuit, which is a single-phase circuit assumed to carry only the zero-sequence current of one phase, must therefore have an impedance of $3Z_n + Z_{g_0}$. The total zero-sequence impedance through which I_a^0 flows is

$$Z_0 = 3Z_n + Z_{g_0}$$

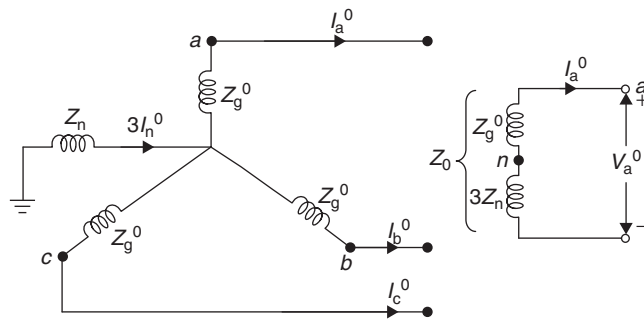
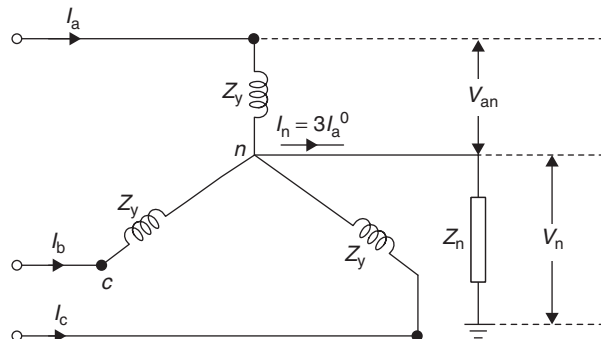


Figure 3 Zero-sequence network and its single-phase equivalent

Sequence Circuit of Y- and Δ-connected Loads

Consider the following Y-connected load.



In the above Y-connected load, sum of the line current is equal to the neutral current I_n and it is given by

$$I_n = I_a + I_b + I_c$$

Voltage of phase 'a' with respect to neutral is given by

$$V_a = V_{an} + V_n,$$

where $V_n = 3I_a^0 Z_n$

We can also write the voltage drops to ground from each of the lines a , b and c as

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \end{bmatrix} + \begin{bmatrix} V_n \\ V_n \\ V_n \end{bmatrix} = Z_Y \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} + 3I_a^0 Z_n \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The $a-b-c$ voltages and currents in the above equation can be replaced by their symmetrical components and the symmetrical component relation is given by

$$\begin{bmatrix} V^0 \\ V^1 \\ V^2 \end{bmatrix} = \begin{bmatrix} Z_Y + 3Z_n & 0 & 0 \\ 0 & Z_Y & 0 \\ 0 & 0 & Z_Y \end{bmatrix} \begin{bmatrix} I^0 \\ I^1 \\ I^2 \end{bmatrix}$$

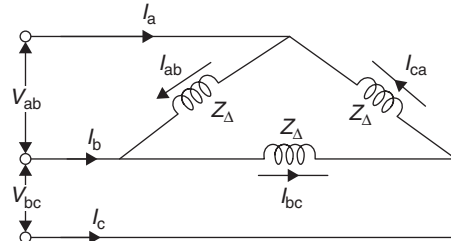
Delta-Connected Load

The delta (Δ) circuit cannot provide a path through neutral. Therefore for a delta-connected load or its equivalent 'Y' connected cannot contain any zero-sequence component.

$$V_{ab} = Z_{\Delta} \cdot I_{ab}$$

$$V_{bc} = Z_{\Delta} \cdot I_{bc}$$

$$V_{ca} = Z_{\Delta} \cdot I_{ca}$$



The sum of line-to-line voltages or phase currents is always zero.

$$\frac{1}{3} (V_{ab} + V_{bc} + V_{ca}) = V_{abc} = 0$$

$$\frac{1}{3} (I_{ab} + I_{bc} + I_{ca}) = I_{abc} = 0$$

Therefore, for a delta-connected load without sources or mutual coupling there will be no zero-sequence currents at the lines (there are some cases where a circulating currents may circulate inside a delta load and not seen at the terminals of the zero sequence circuit).

System parameters relations are decoupled and written as

$$V^0 = (Z_Y + 3Z_n)I^0$$

$$V^1 = Z_Y I^1$$

$$V^2 = Z_Y I^2$$

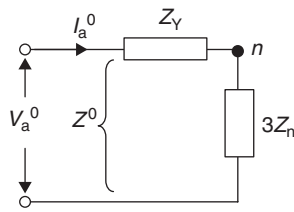


Figure 4 Zero sequence network

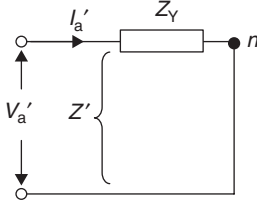


Figure 5 Positive sequence

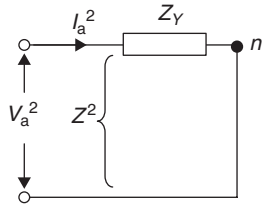


Figure 6 Negative sequence

For ungrounded or Isolated neutral star-connected loads, no zero sequence current can flow and $Z_n = \infty$. Then the positive- and negative-sequence networks remain unchanged but zero-sequence network is as shown in the figure below.

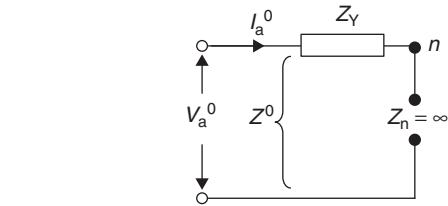


Figure 7 Zero sequence circuit

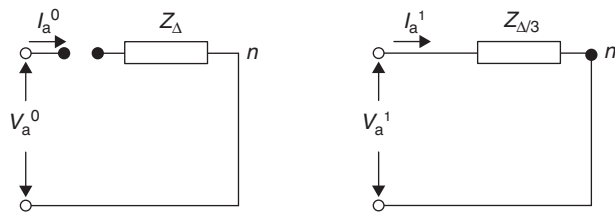


Figure 8 Positive sequence circuit

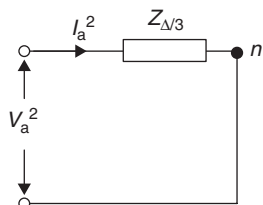
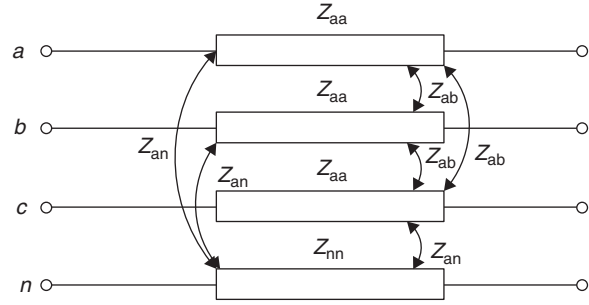


Figure 9 Negative sequence circuit

Sequence Diagrams of Transmission Lines

For transmission lines, assume we have the following, with mutual impedances.



The phase relationships are given by

$$\begin{bmatrix} \Delta V_a \\ \Delta V_b \\ \Delta V_c \end{bmatrix} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

where Z_s = Self-impedance of the phase $= Z_{aa} + Z_{nn} - 2Z_{an}$

Z_m = Mutual impedance between the phases

$$= Z_{ab} + Z_{nn} - 2Z_{an}$$

$$\therefore \Delta V = ZI$$

After converting relationships to sequence representation

$$\Delta V_s = A^{-1} Z A I_s$$

ΔV_s and I_s are symmetrical voltages and currents.

$$A^{-1} Z A = \begin{bmatrix} Z_s - Z_m & 0 & 0 \\ 0 & Z_s - Z_m & 0 \\ 0 & 0 & Z_s + 2Z_m \end{bmatrix}$$

$$\therefore Z^1 = Z_s - Z_m$$

$$Z^2 = Z_s - Z_m$$

$$Z^0 = Z_s + 2Z_m$$

where

$$Z_s = Z_{aa} + Z_{nn} - 2Z_{an}$$

$$Z_m = Z_{ab} + Z_{nn} - 2Z_{an}$$

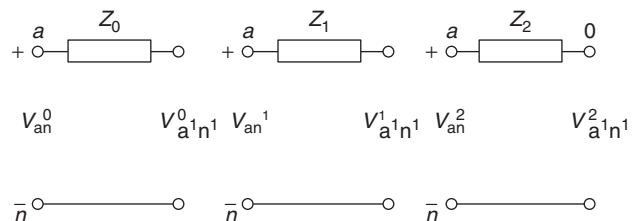


Figure 10 Positive-, negative- and zero-sequence networks of transmission lines

Sequence Diagrams for Transformers

Similar to the transmission lines, the positive and negative sequence diagrams for transformers are same and equal to its leakage impedances.

$$Z^1 = Z^2 = Z_{\text{leakage}}$$

The zero-sequence network of the transformer depends on how the transformer is grounded and its type of connection current flows only through the winding connected in star and the star point in earthed. For delta winding, a star winding or with neutral isolated, the zero-sequence current cannot flow.

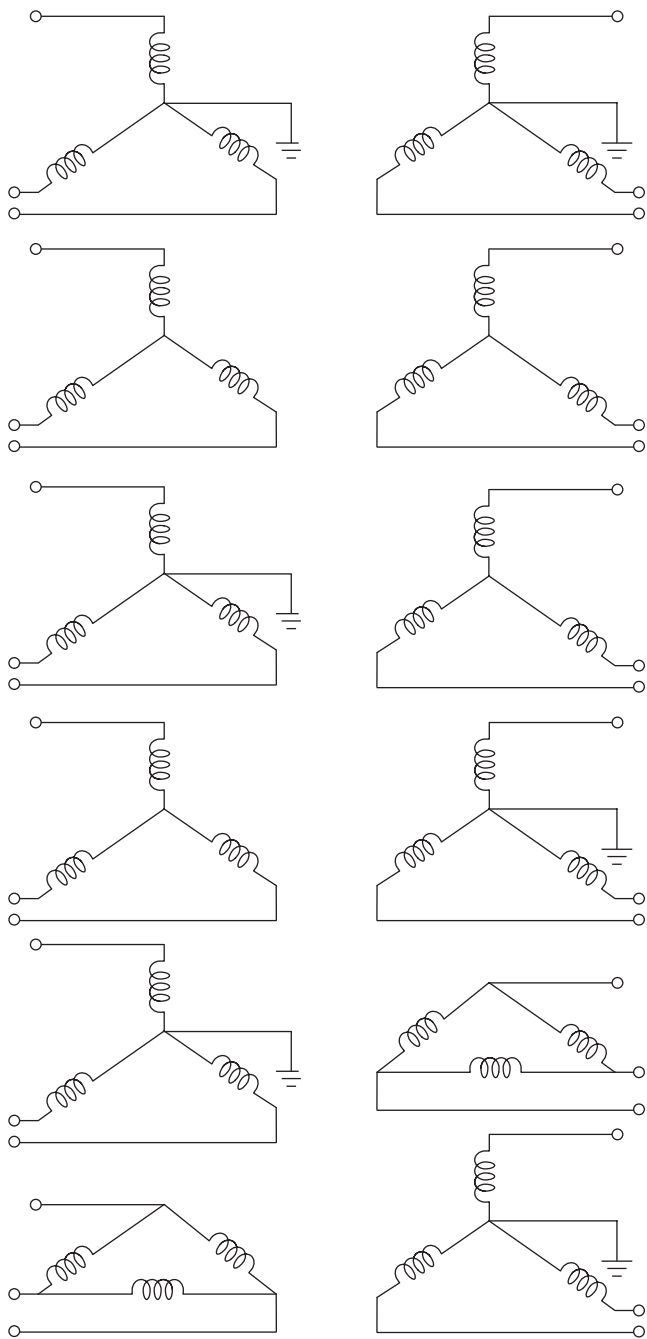


Figure 11 (Continued)

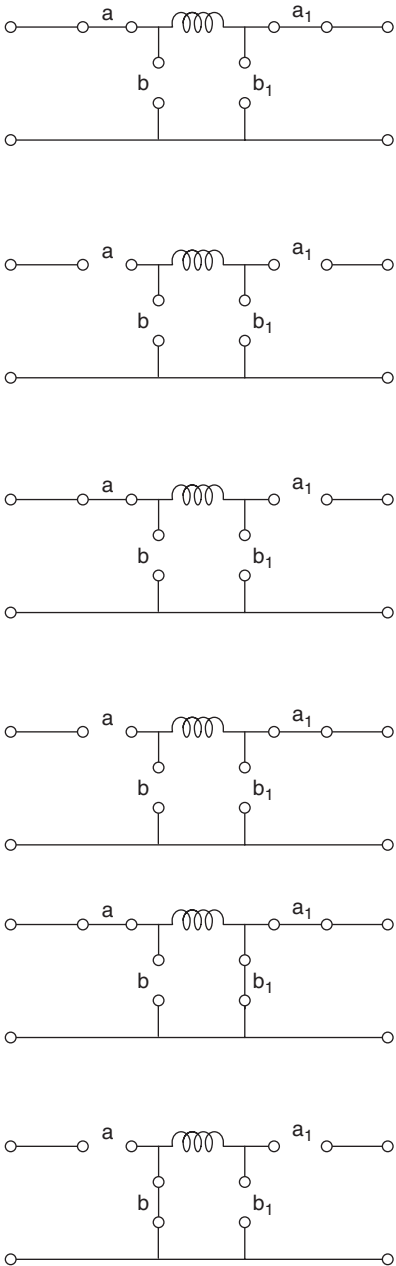


Figure 12 (Continued)

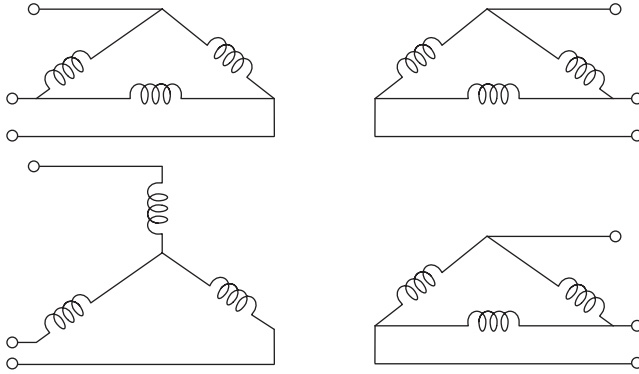


Figure 11 Type of transformer connection

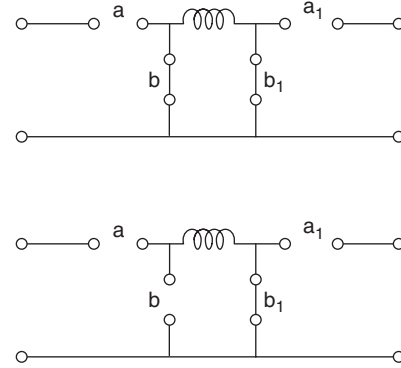


Figure 12 Zero-sequence network

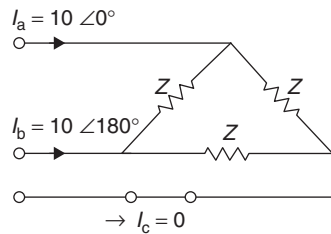
Example 5: One conductor of a 3- ϕ line is open. The current flowing to the Δ -connected load through line 'a' is 10 A with the current in line 'a' as reference and assuming that line 'c' is open, find symmetrical components of line currents.

Solution: Circuit is shown in figure, and the line currents are given by

$$I_a = 10 \angle 0^\circ \text{ A}$$

$$I_b = 10 \angle 180^\circ \text{ A}$$

$$I_c = 0 \text{ A}$$



$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 10 \angle 0^\circ \\ 10 \angle 180^\circ \\ 0 \end{bmatrix}$$

$$I_a^0 = 0 \text{ A}$$

$$I_a^1 = 5.78 \angle -30^\circ \text{ A}$$

$$I_a^2 = 5.78 \angle 30^\circ \text{ A}$$

$$I_b^0 = a^2 I_a^0 = 0 \text{ A}$$

$$I_b^1 = a^2 I_a^1 = 5.78 \angle -50^\circ \text{ A}$$

$$I_b^2 = a^2 I_a^2 = 5.78 \angle 150^\circ \text{ A}$$

$$I_c^0 = a I_a^0 = 0 \text{ A}$$

$$I_c^1 = a I_a^1 = 5.78 \angle 90^\circ \text{ A}$$

$$I_c^2 = a I_a^2 = 5.78 \angle -90^\circ \text{ A}$$

Note: Symmetrical components of phase 'c' are non-zero even though there is no current in that phase but the sum of the corresponding phase symmetrical components is equal to zero.

Example 6: The parameters of transposed overhead transmission line are given as

$$\text{Self-reactance } X_s = 0.45 \text{ } \Omega/\text{km}$$

$$\text{Mutual reactance } X_m = 0.12 \text{ } \Omega/\text{km}$$

The positive-sequence reactance X_1 and zero-sequence reactance X_0 , respectively, in Ω/km are

- (A) 0.69, 0.33 (B) 0.33, 0.69
(C) 0.33, 0.66 (D) 0.66, 0.33

Solution: (B)

$$\text{Positive-sequence reactance} = X_s - X_m$$

$$= 0.45 - 0.12 = 0.33 \text{ } \Omega/\text{km}$$

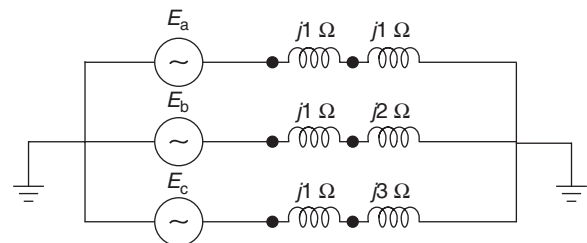
$$\text{Zero-sequence reactance} = X_s + 2X_m$$

$$= 0.45 + 0.24 = 0.69 \text{ } \Omega/\text{km}$$

Example 7: A three-phase alternator generating unbalanced voltages is connected to an unbalanced load through a three-phase transmission line as shown in figure. The neutral of the alternator and the star point of the load are solidly grounded. The phase voltages of the alternator are

$$E_a = 10 \angle 0^\circ \text{ V}, \quad E_b = 10 \angle -90^\circ \text{ V}, \quad E_c = 10 \angle 120^\circ \text{ V}$$

The negative sequence component of the load current is



- (A) 0.33 A (B) 5 A (C) 3.33 A (D) 2.5 A

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Solution: (A)

Since the circuit is grounded on both the sides, each phase can be treated as a single-phase circuit and the phase currents can be written as

$$I_a = \frac{E_a}{X_a} = \frac{10 \angle 0^\circ}{j2} = 5 \angle -90^\circ$$

$$I_b = \frac{E_b}{X_b} = \frac{10 \angle -90^\circ}{j3} = 3.33 \angle -180^\circ$$

$$I_c = \frac{E_c}{X_c} = \frac{10 \angle 120^\circ}{j4} = 2.5 \angle 30^\circ$$

$$\begin{aligned} \text{Negative-sequence currents} &= I_a^2 = \frac{1}{3}(I_a + a^2 I_b + a I_c) \\ &= \frac{1}{3}(5 \angle -90^\circ + 3.33 \angle -300^\circ + 2.5 \angle 150^\circ) \\ &= 0.33 \angle -120^\circ \end{aligned}$$

Example 8: A three-phase, 30 MVA, 6.6 kV alternator having 15% reactance is connected through a 30 MVA, 6.6/33 kV delta–star-connected transformer of 5% reactance to a 33 kV transmission line having a negligible resistance and a reactance of 4 Ω . At the receiving end of the transmission line there is a 30 MVA, 33/6.6 kV delta–star-connected transformer of 5% reactance stepping down the voltage at 6.6 kV, both the transformers have their neutral solidly grounded. Draw the one-line diagram and the positive-, negative- and zero-sequence networks of this system.

Solution: One-line diagram of the system is shown in the figure below

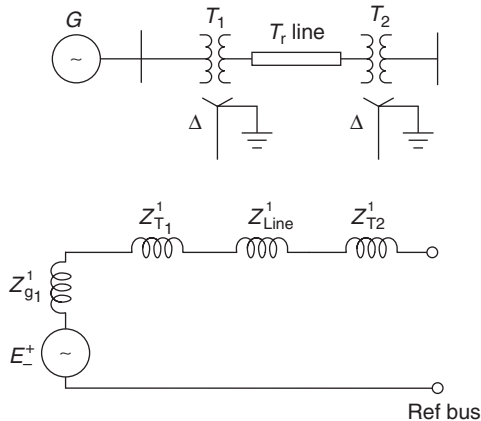


Figure 13 Positive-sequence equivalent network

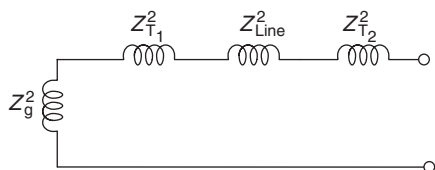


Figure 14 Negative-sequence equivalent network

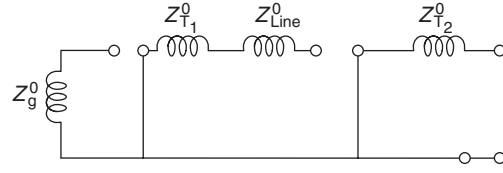
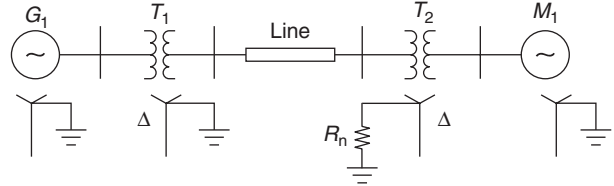


Figure 15 Zero-sequence equivalent network

Example 9: Single-line diagram of a power system is given by

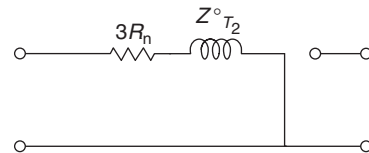


Which one of the following is the zero-sequence equivalent of the above system?

- (A)
- (B)
- (C)
- (D)

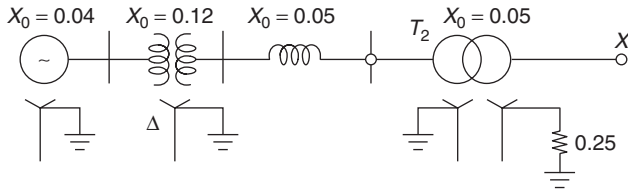
Solution: (D)

For 3- ϕ transformer 2, equivalent circuit is given by



Example 10: A generator is connected to a transformer which feeds another transformer through a short transmission line. The zero-sequence impedances values are expressed in p.u. on a common base and are indicated in

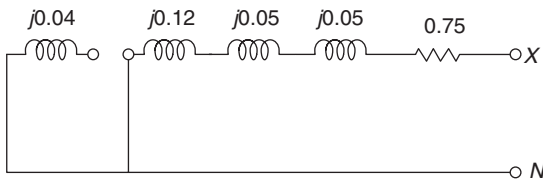
figure. The Thevenin's equivalent zero-sequence impedance at point 'X' is



- (A) $0.75 + j0.22$ (B) $0.75 + j0.26$
(C) $0.25 + j0.26$ (D) $0.25 + j0.22$

Solution: (A)

Zero-sequence equivalent network is given by



Equivalent impedance at point 'X' is, $Z_{X_n} = j0.22 + 0.75$

UNBALANCED FAULT ANALYSIS

Various types of unsymmetrical faults that occurs in power systems are

Shunt type faults

1. Single line to ground (LG) fault
2. Line to line (LL) fault
3. Double line to ground (LLG) fault

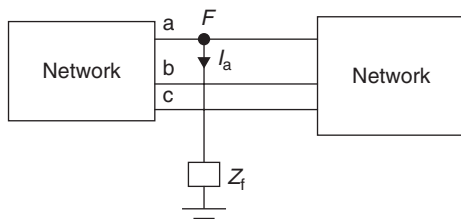
Series type of faults

1. One conductor open fault
2. Two conductor open fault

Symmetrical components are very commonly used in calculating currents arising from unbalanced short circuits.

Single Line to Ground Fault

A single line to ground fault at a point in power system is through a fault impedance as shown in the figure.



At the point of fault occurrence, the currents come out of the network and line to ground voltages are given by

$$\begin{aligned} I_b &= 0 \\ I_c &= 0 \\ V_a &= I_a Z_f \end{aligned}$$

Symmetrical components of the fault currents are,

$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \Rightarrow I_a^1 = I_a^2 = I_a^0 = \frac{1}{3} I_a \quad (1)$$

Similarly symmetrical components of the fault voltages are

$$V_a^1 + V_a^2 + V_a^0 = Z_f I_a = 3Z_f I_a^1 \quad (2)$$

Expression (1) and (2) together describes the sequence networks connection shown in the figure.

Note: Positive-, negative- and zero-sequence components currents of a phase are equal if there is a LG fault in that particular phase.

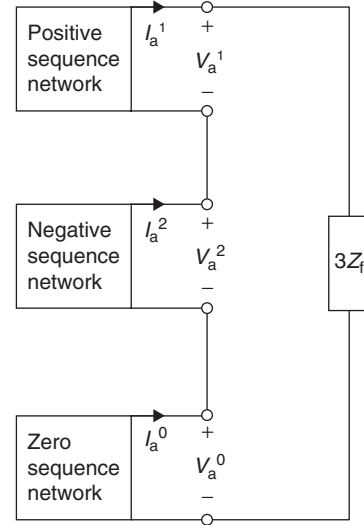
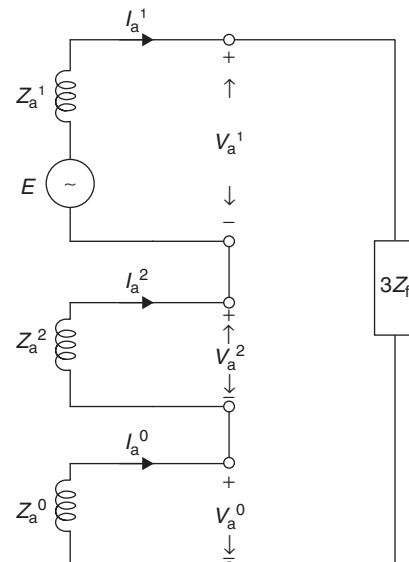


Figure 16 Sequence connection for LG Fault

The Thevenin's equivalent of sequence networks is given by



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From the above equivalent reactance diagram,

$$I_a^1 = \frac{E_a}{(Z_1 + Z_2 + Z_0) + 3Z_f}$$

Fault current in phase 'a'

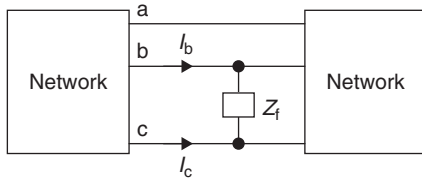
$$I_a = 3I_{a1} = \frac{3E_a}{(Z_1 + Z_2 + Z_0) + 3Z_f}$$

Phase voltage in faulted phase

$$V_a = I_a Z_f = \frac{3E_a}{(Z_1 + Z_2 + Z_0) + 3Z_f} \cdot Z_f$$

Line-to-Line Fault

A line-to-line fault in a power system on phases 'b' and 'c' through a fault impedance Z_f is shown in the following figure.



The current and voltages at the fault can be expressed as

$$I_a = 0$$

$$I_b = -I_c \text{ and } V_b - V_c = I_b Z_f$$

Symmetrical components of fault currents and voltages are given by

$$I_a^0 = 0 \text{ and } I_a^1 = -I_a^2$$

$$V_a^1 = V_a^2 \text{ and } V_a^1 - V_a^2 = Z_f I_a^1$$

The above relations describe the sequence network connections as shown in the figure.

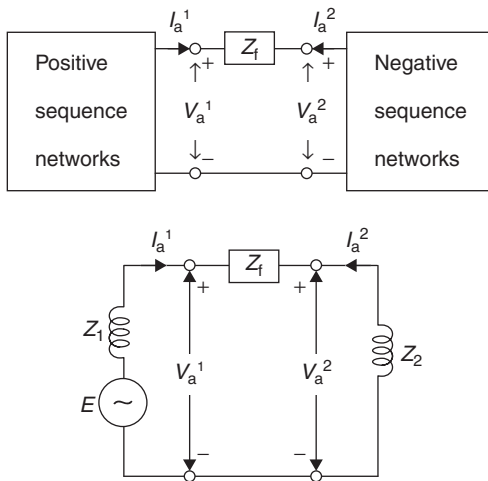


Figure 17 Sequence networks connection and their Thevenin's equivalent network

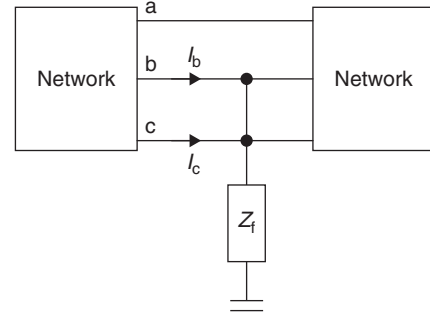
From the equivalent reactance diagram,

$$I_a^1 = \frac{E_a}{Z_1 + Z_2 + Z_f}$$

$$I_f = I_b = -I_c = \frac{-j\sqrt{3} E_a}{Z_1 + Z_2 + Z_f}$$

Double Line to Ground Fault

A double line to ground fault in power system is shown in the figure below.



The voltage and current conditions at the fault are expressed as

$$I_a = 0 \text{ or } I_a^1 + I_a^2 + I_a^0 = 0$$

$$V_b = V_c = Z_f(I_b + I_c) = 3Z_f I_a^0$$

Symmetrical voltages and currents relations are given by

$$V_a^1 = V_a^2 = \frac{1}{3} [V_a + (a + a^2)V_b]$$

$$V_a^0 = \frac{1}{3} (V_a + 2V_b)$$

$$V_a^1 - V_a^0 = \frac{1}{3} (2 - a - a^2)V_b = V_b = 3Z_f I_a^0$$

The above relations describe the sequential network connections as shown in the following figure.

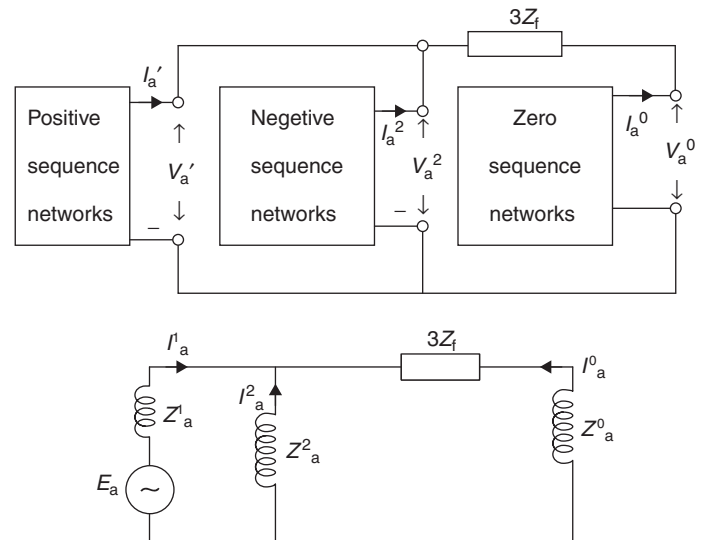


Figure 18 Sequence-network connections and their Thevenin's equivalent network

From the above sequential networks,

$$\Rightarrow I_a^1 = \frac{E_a}{Z_a^1 + \frac{Z_a^2(Z_a^0 + 3Z_f)}{Z_a^0 + Z_a^2 + 3Z_f}}$$

$$I_f = 3I_{a_0} = 3 \left[\frac{E_a}{Z_a^2} - \left(\frac{Z_a^1 + Z_a^2}{Z_a^2} \right) I_a^1 \right]$$

Example 11: The severity of the line to ground and three-phase fault at the terminals of an unloaded synchronous generator is to be same. If the terminal voltage is 'E' and the positive, negative and zero-sequence impedances are Z_1 , Z_2 and Z_0 , respectively, then the required inductive reactance for neutral grounding is

- (A) $2Z_1 - Z_2 - Z_0$ (B) $\frac{1}{3}[2Z_1 - Z_2 - Z_0]$
 (C) $Z_1 + Z_2 + Z_0$ (D) $\frac{1}{3}[Z_1 + Z_2 + Z_3]$

Solution: (B)

For 3- ϕ fault, fault current $I_f = \frac{E}{Z_1}$ p.u.

For fault of line to ground

$$I_{f_{lg}} = \frac{3E}{Z_1 + Z_2 + Z_0 + 3Z_n} \text{ p.u.}$$

If severity of line to ground and 3- ϕ faults are same

$$\frac{E}{Z_1} = \frac{3E}{Z_1 + Z_2 + Z_0 + 3Z_n}$$

$$Z_1 + Z_2 + Z_0 + 3Z_n = 3Z_1$$

$$Z_n = \frac{2Z_1 - Z_2 - Z_0}{3}$$

$$\Rightarrow Z_n = \frac{1}{3}[2Z_1 - Z_2 - Z_0]$$

Example 12: The per-unit values of positive-, negative- and zero-sequence reactance of a network at fault are 0.2, 0.17 and 0.12 p.u., respectively. The fault current for double line to ground fault is

- (A) 4.718 p.u. (B) 2.04 p.u.
 (C) 1.57 p.u. (D) 16.26 p.u.

Solution: (A)

For LLG fault

$$I_f = \frac{3(Z_2 + Z_0)}{Z_1 Z_2 + Z_1 Z_0 + Z_2 Z_0}$$

$$= \frac{3(0.17 + 0.12)}{0.2 \times 0.7 + 0.2 \times 0.12 + 0.17 \times 0.12}$$

$$I_f = 4.718 \text{ p.u.}$$

Example 13: Zero-sequence current of fault current is 100 A, then the SLG fault current is

- (A) 100 A (B) 33.33 A
 (C) 300 A (D) 200 A

Solution: (C)

$$\text{Single-line to ground fault current} = 3I_{a_0}$$

$$= 3 \times 100 \text{ A}$$

$$= 300 \text{ A}$$

Example 14: A 500 MVA, 22 kV generator has $X_d'' = X_1 = X_2 = 15\%$ and $X_0 = 5\%$. Its neutral is grounded through a reactor of 0.3 Ω . The generator is operating at rated voltage, and no load. If it is disconnected from the system when a SLG fault occurs at its terminals, the sub-transient current in the fault phase is

- (A) 13.12 kA (B) 31.5 kA
 (C) 2.4 kA (D) 10.5 kA

Solution: (B)

Fault current for SLG fault $I_f = 3I_{a_1}$

$$I_f = 3I_{a_1} = \frac{3E_a}{Z_1 + Z_2 + Z_0 + 3Z_n}$$

$$= \frac{3}{0.15 + 0.15 + 0.05 + 0.9} = 2.4 \text{ p.u.}$$

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3} V_{\text{base}}} = \frac{500}{\sqrt{3} \times 22} = 13.12 \text{ kA}$$

Fault current = $I_{f \text{ p.u.}} \times I_{\text{base}}$

$$= 2.4 \times 13.12 \text{ kA}$$

$$= 31.49 \text{ kA}$$

Example 15: A 25 MVA, 11 kV alternator has positive, negative and zero sequence reactance of 0.3 p.u., 0.35 p.u. and 0.05 p.u., respectively. What value of resistance must be put in the generator neutral so that the fault current for a line to ground fault of zero fault impedance will not exceed the rated line current?

- (A) 4.84 Ω (B) 4.7 Ω (C) 0.972 Ω (D) 2.35 Ω

Solution: (B)

Rated current in per unit $I_{\text{rated}} = 1$ p.u.

Fault current for SLG fault

$$I_f = \frac{3E_a}{Z_1 + Z_2 + Z_0 + 3Z_n} = 1$$

$$\frac{3 \times 1}{j(0.3 + 0.35 + 0.05) + 3R_n} = 1$$

$$R_n = \frac{1}{3} \sqrt{3^2 - (0.3 + 0.35 + 0.05)^2}$$

$$= \frac{1}{3} \sqrt{9 - 0.49} = 0.972 \text{ p.u.}$$

$$Z_{\text{base}} = \frac{(KV)_{\text{base}}^2}{MVA_{\text{base}}} = \frac{11^2}{25} = 4.84 \Omega$$

Actual $R_n = R_{n \text{ p.u.}} \times Z_{\text{base}}$

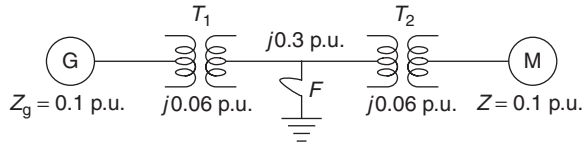
$$= 0.972 \times 4.84 = 4.7 \Omega$$

EXERCISES

Practice Problems I

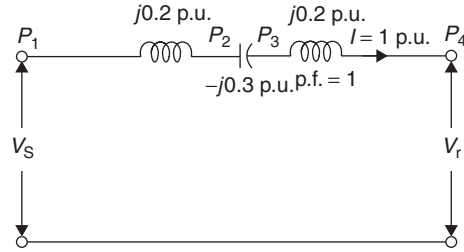
Directions for questions 1 to 20: Select the correct alternative from the given choices.

1. A power system is represented by a single line diagram with all the reactance marked in per unit (p.u.) on the same base

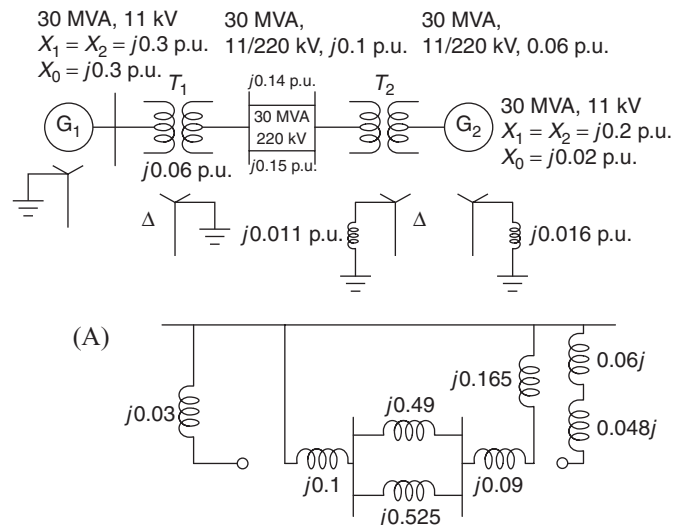


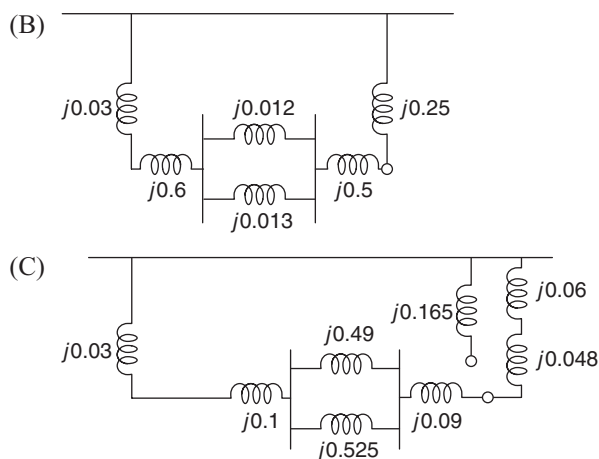
The system is on no load when three-phase fault occurs at F on the high voltage side of the transformer T_2 . The fault current will be

- (A) $-j0.726$ p.u. (B) $-j0.84$ p.u.
(C) $-j7.42$ p.u. (D) $-j8.4$ p.u.
2. The 3- ϕ power system operating at 122 kV with 0.6 p.f. Recovery voltage = 0.8 times full line value, Breaking current is symmetrical, frequency of oscillation of restriking voltage is 20 kHz. Assuming the neutral is grounded and the fault does not involve with the ground, the average rate of rise of restriking voltage is
(A) 20.82 kV/ms (B) 30.76 kV/ms
(C) 56.5 V/ms (D) 30.76 V/s
3. Two 11 kV, 30 MVA, 3 ϕ star-connected generators operate in parallel. The positive-, negative- and zero-sequence reactances of each are $j0.2$ p.u., $j0.35$ p.u., $j0.15$ p.u., respectively. Assume that the generator is grounded solidly. Voltage of the healthy phase for a double line to ground fault at the terminals is (for solid fault, $z_f = 0$)
(A) 0.625 V (B) 0.766 V
(C) 0.52 V (D) 0.25 V
4. A 15 MVA, 13.2 kV alternator has $X_1 = X_2 = 30\%$ and $X_0 = 8\%$ Reactance that must be put in the generator neutral so that the fault current for a line to ground fault of zero fault impedance will not exceed the rated current is
(A) 2.6 W (B) 6.8 W
(C) 9 W (D) 4.8 W
5. In a three-phase, 4-wire system, the line currents under abnormal conditions are given by $(10 + j6)$, $(12 - j8)$ and $(-14 + j12)$. The positive sequence currents in lines will be
(A) $-2 + 4j$ (B) $-1.26 + 2.7j$
(C) $3.4 - 2.4j$ (D) $1.26 - 2.7j$
6. For the transmission line model with a series capacitor at its midpoint, the maximum voltage on the line is at location



- (A) P_1 (B) P_2 (C) P_3 (D) P_4
7. A power system network with the capacity of 150 MVA has a source impedance of 15% at a point. The fault at the point is
(A) 10 MA (B) 150 MVA
(C) 3500 MVA (D) 1000 MVA
8. The positive-, negative- and zero-sequence impedances of a solidly grounded system under steady-state condition follow the relations
(A) $Z_1 > Z_2 > Z_0$
(B) $Z_1 < Z_2 < Z_0$
(C) $Z_0 < Z_1 < Z_2$
(D) None of these
9. A 3- ϕ . 10 MVA, 2.8 kV alternator with a reactance of 6% is connected to a feeder of series impedance of $0.2 + j0.6$ ohm/phase per km. The transformer rated at 5 MVA, 1.8/22 kV has a reactance of 5%. The fault current supplied by the generator operating under no load with a voltage of 3.8 kV, when a 3- ϕ symmetrical fault occurs at a point 10 km along the feeder is
(A) $(1406 \angle 76.288^\circ)$ A
(B) $(436 \angle 76.288^\circ)$ A
(C) $(15 + j80)$ A
(D) $(400 - j83.4)$ A
10. The zero-sequence network of the single-line diagram is shown below.





(D) None of these

11. An earth fault occurs on a busbar to which 3, 20 MVA, 3.3 kV, 3- ϕ alternators are connected in parallel. When only one alternator is solidly grounded and others are isolated, the fault current is [Given $X_d'' = 12\%$, $X_2 = 10\%$ and $X_0 = 13.5\%$]

(A) 8.895×10^4 A (B) 6.85×10^4 A
(C) 0.00789 A (D) 5.65 mA

12. A single-phase load of 100 kVA is connected across line bc of a three-phase supply of 3.3 kV. The symmetrical components of line currents are

(A) 17.5, -17.5 and 0 (B) -17.5, 17.5 and 0
(C) 12.5, -12.5 and 0 (D) -12.5, 12.5 and 0

13. A generator is unloaded and operated at rated voltage when a fault occurs. Neglecting resistance identify the fault. The values of currents and voltages are given as follows

$$I_{a0} = j2.37 \text{ p.u.}$$

$$I_{a1} = -j8.45 \text{ p.u.}$$

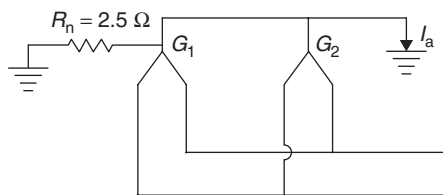
$$I_{a2} = j6.08 \text{ p.u.}$$

$$V_{a0} = V_{a1} = V_{a2} = j0.237 \text{ p.u.}$$

(A) L-G fault (B) L-L fault
(C) L-L-G fault (D) Open conductor fault

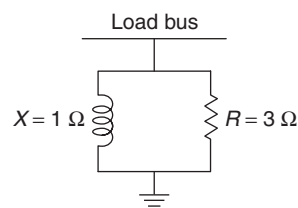
Common data for Questions 14–16:

Two 12.8 kV, 25 MVA, three-phase, star-connected generators operate in parallel as shown in the following figure.



The positive, negative and zero sequence reactances are, respectively, $j0.20$, $j0.16$, $j0.12$ p.u. The star point of one of the generator is isolated and that of the other is earthed through a 2.5Ω resistor. A single line to ground fault occurs at the terminals of one of the generators

14. The fault current will be
(A) $2.55 \angle -15^\circ$ p.u. (B) $1.0675 \angle -15^\circ$ p.u.
(C) $2.93 \angle -30^\circ$ p.u. (D) $3.0675 \angle -30^\circ$ p.u.
15. The current in the grounding resistor is
(A) $3.0675 \angle -30^\circ$ p.u. (B) $2.55 \angle -15^\circ$ p.u.
(C) $1.0675 \angle -15^\circ$ p.u. (D) $2.93 \angle -30^\circ$ p.u.
16. The voltage across the grounding resistor is
(A) $3067.5 \angle -15^\circ$ p.u. (B) $2.9223 \angle -15^\circ$ p.u.
(C) $3.0675 \angle -15^\circ$ p.u. (D) $0.9551 \angle -15^\circ$ p.u.
17. A three-phase, 20 MVA, 13.2 kV 50 Hz generator with solidly earthed neutral has sub transient reactance X_d'' of 25%; direct-axis reactance X_d' of 20% and synchronous reactance X_d of 50%. Negative - sequence reactance $X_2 = 10\%$ and zero-sequence reactance $X_0 = 6\%$. The generator is operated on open circuit when fault occurs. Take $E_a = 1.0$. If the fault is a three-phase short without an impedance, the value of initial symmetrical sub transient, transient and sustained RMS values of the line current under faulty condition will be
(A) $-j4.0, -j5.0, -j2$ p.u.
(B) $-j1.667, -j4.0, -j5.0$ p.u.
(C) $-j15, -j14, -j11.667$ p.u.
(D) $-j11.667, -j14, -j15$ p.u.
18. Three 10 MVA, 13.8 kV, three-phase star-connected alternators are operating in parallel. Each has $X_d'' = 10\%$, $X_2 = 8\%$ and $X_0 = 3\%$. If an earth fault occurs on one bus bar, the value of fault current when all the three alternators are solidly grounded is
(A) $j25,000$ A (B) $-j4062.27$ A
(C) $j5,000$ A (D) $j15,000$ A
19. A 20 MVA, 6.6 kV alternator with solidly grounded neutral has a sub transient reactance of 0.25 p.u. The negative and zero sequence reactances are 0.30 p.u. and 0.15 p.u., respectively. A single phase to ground fault occurs at the terminals of this unloaded generator. The fault impedance is 0.01 p.u. Then the value of three sequent components of voltages as percentage of rated values will be
(A) 65.75%, -41.1% and -20.55%
(B) -65.75%, 41.1% and 20.55%
(C) 41.75%, -36.2% and -15.4%
(D) -41.75%, 36.2% and 15.4%
20. A load of 25 kVA rating is connected to the bus as shown. If the bus voltage is 400 V, the p.u. value of resistance is



(A) 0.469 (B) 0.0469
(C) 0.187 (D) 0.0187

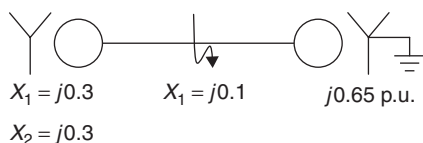
Practice Problems 2

Directions for questions 1 to 17: Select the correct alternative from the given choices.

- Which of the following statements are true?
 - A balanced three-phase system consists of positive sequence components only.
 - The magnitude of zero-sequence component is one third of current in neutral wire.
 - In a delta-connected load, zero-sequence component is equal to neutral current.
 - The current of a single-phase load drawn from a three-phase system comprises equal positive-, negative- and zero-sequence components.

(A) 1, 2 only (B) 2, 3 only
(C) 2, 3, 4 (D) 1, 2, 3, and 4
- A generator of negligible resistance having 1.0 per unit voltage behind transient reactance is subjected to a line to line fault resulting in a fault current of 2.68 p.u. The per unit value of zero-sequence reactance when it is subjected to a line to ground fault resulting in 2.68 p.u. fault current is

(A) 0.473 p.u. (B) 0.265 p.u.
(C) 0.98 p.u. (D) 1 p.u.
- A single line to ground fault occurs in a star-connected three-phase generator. It supplies power to a star-connected inductive load through a transmission line. The star point of the load is grounded and the generator neutral is ungrounded. The single-line diagram is shown below. Fault takes place half way down the line.



Prior to fault, the network is balanced and voltage at fault location is $1 \angle 0^\circ$ p.u. The current through the fault path is

- (A) $-j2.36$ p.u. (B) $j1.45$ p.u.
(C) $j3.26$ p.u. (D) $j3.4$ p.u.
- A 100 kVA, 400 V synchronous generator (0.2 p.u. sub-transient reactance) is supplying a 80 kW load at 0.8 lagging power factor. The initial symmetrical RMS current for a three-phase fault at generator terminals is

(A) 0.08 p.u. (B) 0.06 p.u.
(C) 0.04 p.u. (D) 0.02 p.u.
 - Ratio of L-G fault current to three-phase fault

(A) $\frac{3X_1}{X_1 + X_2 + X_0}$ (B) $\frac{1.732X_1}{X_1 + X_2}$
(C) $\frac{X_1}{3\left(X_1 + X_2 + \frac{X_0}{3}\right)}$ (D) Both (A) and (C)

- The generator emf is 0.5 p.u. and the transient reactance is 0.15. The value of transient current is

(A) $2.5 \angle -90^\circ$ (B) $5 \angle 30^\circ$
(C) $3.33 \angle -90^\circ$ (D) $2.5 \angle 30^\circ$
- $E_g = 0.5$ p.u. and $X_d = 0.2$. The value of short-circuit current is

(A) $2.5 \angle -90^\circ$ (B) $5 \angle -90^\circ$
(C) $2.5 \angle 90^\circ$ (D) $5 \angle 90^\circ$
- Which is true for steady state, transient and sub-transient reactances for a synchronous machine?

(A) $X_d > X'_d > X''_d$ (B) $X''_d > X'_d > X_d$
(C) $X'_d > X''_d > X_d$ (D) $X_d > X''_d > X'_d$
- X_1 , X_2 and X_0 values of two generators operated parallel are given as 0.08, 0.24 and 0.10 per unit, respectively. If the generator is solidly grounded, the voltage of healthy phase for a double line to ground fault at terminals of generator is

(A) 0.603 p.u. (B) 0.969 p.u.
(C) 0.152 p.u. (D) 0.846 p.u.
- The fault which causes a greater electro-magnetic interference between the power line and a nearby communication line

(A) Three-phase fault (B) L-L-G fault
(C) L-L fault (D) L-G fault
- The zero-sequence impedances of an ideal star-delta-connected transformer (star-grounded)

(A) Looking from star side is zero and looking from delta side is infinite
(B) Looking from star side as well as delta side is infinite
(C) Looking from star side as well as delta side is zero
(D) Looking from star side is infinite and looking from delta side is zero.
- For a line-to-line fault analysis using symmetrical components

(A) The positive-, negative- and zero-sequence networks at the fault point are connected in series
(B) The positive-, negative- and zero- sequence networks at the fault point are connected in parallel
(C) The positive- and negative-sequence networks at the fault point are connected in series
(D) The positive- and negative-sequence networks at the fault point are connected in parallel
- The line currents in ampere in phases a, b and c, respectively, are $(300 + j200)$, $(50 - j300)$ and $(-200 + j500)$ referred to the same reference vector. The negative sequence components of currents will be

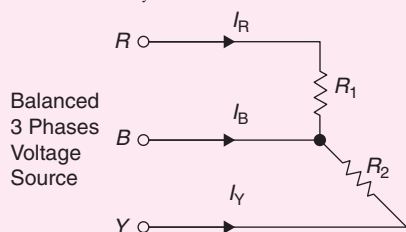
(A) $138.56 \angle 200.14$ (B) $11.82 \angle 100.02$
(C) $112.82 \angle 200.14$ (D) $98.41 \angle 100.02$
- If a positive sequence current passes through a transformer and its phase shift is 60 degrees, the negative sequence current flowing through the transformer will have a phase shift of

- (A) 60 deg (B) -60 deg
(C) 120 deg (D) -120 deg
15. The reason for not considering the load currents in short-circuit calculations are
- Short-circuit currents are much larger than load currents
 - Short-circuit currents are greatly out of phase with load currents
- The correct alternative is
- Both (i) and (ii) are correct
 - Both (i) and (ii) are wrong
 - (i) is correct and (ii) is wrong
 - (i) is wrong and (ii) is correct

16. For a single line to ground fault, the rise in potential on a healthy phase, if we are considering the voltage as V will be
- $\frac{1}{\sqrt{2}}$ V kV
 - $\sqrt{2}$ V kV
 - $\sqrt{3}$ V kV
 - $\frac{1}{\sqrt{3}}$ V kV
17. When a line to ground fault occurs, the current in the phase is 120 A. The zero-sequence current is
- 0 A
 - 40 A
 - 20 A
 - 120 A

PREVIOUS YEARS' QUESTIONS

1. For the three-phase circuit shown in Figure, the ratio of the current $I_R : I_Y : I_B$ is given by [2005]



- 1:1: $\sqrt{3}$
 - 1:1:2
 - 1:1:0
 - 1:1: $\sqrt{\frac{3}{2}}$
2. The parameters of a transposed overhead transmission line are given as:
Self-reactance $x_s = 0.4 \Omega/\text{km}$ and mutual reactance $x_m = 0.1 \Omega/\text{km}$
The positive sequence reactance x_1 and zero sequence reactance x_0 , respectively, in Ω/km are [2005]
- 0.3, 0.2
 - 0.5, 0.2
 - 0.5, 0.6
 - 0.3, 0.6

Common Data for Questions 3 and 4:

At a 220 kV substation of a power system, it is given that the three-phase fault level is 4000 MVA and single-line to ground fault level is 5000 MVA. Neglecting the resistance and the shunt susceptances of the system,

3. The positive-sequence driving point reactance at the bus is: [2005]
- 2.5 Ω
 - 4.033 Ω
 - 5.5 Ω
 - 12.1 Ω
4. And the zero-sequence driving point reactance at the bus is: [2005]
- 2.2 Ω
 - 4.84 Ω
 - 18.18 Ω
 - 22.72 Ω

5. Three identical star-connected resistors of 1.0 p.u. are connected to an unbalanced three-phase supply. The load neutral is isolated. The symmetrical components of the line voltages in p.u. are:

$V_{ab1} = X \angle \theta_1$, $V_{ab2} = Y \angle \theta_2$. If all the p.u. calculations are with the respective base values, the phase-to-neutral sequence voltages are [2006]

- $V_{an1} = X \angle (\theta_1 + 30^\circ)$, $V_{an2} = Y \angle (\theta_2 - 30^\circ)$
- $V_{an1} = X \angle (\theta_1 - 30^\circ)$, $V_{an2} = Y \angle (\theta_2 + 30^\circ)$
- $V_{an1} = \frac{1}{\sqrt{3}} X \angle (\theta_1 - 30^\circ)$,
 $V_{an2} = \frac{1}{\sqrt{3}} Y \angle (\theta_2 + 30^\circ)$
- $V_{an1} = \frac{1}{\sqrt{3}} X \angle (\theta_1 - 60^\circ)$,
 $V_{an2} = \frac{1}{\sqrt{3}} Y \angle (\theta_2 - 60^\circ)$

Common Data for Questions 6 and 7:

For a power system the admittance and impedance matrices for the fault studies are as follows.

$$Y_{\text{bus}} = \begin{bmatrix} -j8.75 & j1.25 & j2.50 \\ j1.25 & -j6.25 & j2.50 \\ j2.50 & j2.50 & -j5.00 \end{bmatrix}$$

$$Z_{\text{bus}} = \begin{bmatrix} j0.16 & j0.08 & j0.12 \\ j0.08 & j0.24 & j0.16 \\ j0.12 & j0.16 & j0.34 \end{bmatrix}$$

The pre-fault voltages are 1.0 p.u. at all the buses. The system was unloaded prior to the fault. A solid three-phase fault takes place at bus 2.

6. The post fault voltages at buses 1 and 3 in per unit, respectively, are [2006]

- (A) 0.24, 0.63 (B) 0.31, 0.76
(C) 0.33, 0.67 (D) 0.67, 0.33
7. The per unit fault feeds from generators connected to buses 1 and 2, respectively, are [2006]
(A) 1.20, 2.51 (B) 1.55, 2.61
(C) 1.66, 2.50 (D) 5.00, 2.50
8. A three-phase balanced star-connected voltage source with frequency ω rad/s is connected to a star-connected balanced load which is purely inductive. The instantaneous line currents and phase to neutral voltages are denoted by (i_a, i_b, i_c) and (v_{an}, v_{bn}, v_{cn}) , respectively, and their RMS values are denoted by V and I .

$$\text{If } R = [v_{an}, v_{bn}, v_{cn}] \begin{bmatrix} 0 & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix},$$

then the magnitude of R is [2007]

- (A) $3VI$ (B) VI (C) $0.7VI$ (D) 0
9. Suppose we define a sequence transformation between 'a-b-c' and 'p-n-o' variables as follows:

$$\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = k \begin{bmatrix} 1 & 1 & 1 \\ \alpha^2 & \alpha & 1 \\ \alpha & \alpha^2 & 1 \end{bmatrix} \begin{bmatrix} f_p \\ f_n \\ f_o \end{bmatrix}, \text{ where } \alpha = e^{j\frac{2\pi}{3}} \text{ and } k \text{ is a constant.}$$

$$\text{Now, if it is given that: } \begin{bmatrix} V_p \\ V_n \\ V_o \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 2.0 \end{bmatrix} \begin{bmatrix} i_p \\ i_n \\ i_o \end{bmatrix}$$

$$\text{and } \begin{bmatrix} V_a \\ V_b \\ -V_c \end{bmatrix} = Z \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \text{ then, [2007]}$$

$$(A) Z = \begin{bmatrix} 1.0 & 0.5 & 0.75 \\ 0.75 & 1.0 & 0.5 \\ 0.5 & 0.75 & 1.0 \end{bmatrix}$$

$$(B) Z = \begin{bmatrix} 1.0 & 0.5 & 0.5 \\ 0.5 & 1.0 & 0.5 \\ 0.5 & 0.5 & 1.0 \end{bmatrix}$$

$$(C) Z = 3k^2 \begin{bmatrix} 1.0 & 0.75 & 0.5 \\ 0.5 & 1.0 & 0.75 \\ 0.75 & 0.5 & 1.0 \end{bmatrix}$$

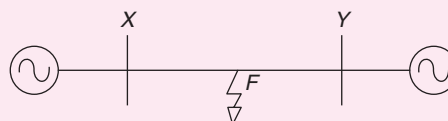
$$(D) Z = \frac{k^2}{3} \begin{bmatrix} 1.0 & -0.5 & -0.5 \\ -0.5 & 1.0 & -0.5 \\ -0.5 & -0.5 & 1.0 \end{bmatrix}$$

10. A 230 V (phase), 50 Hz, three-phase, 4-wire system has a phase sequence ABC. A unity power-factor load of 4 kW is connected between phase A and neutral N. It is desired to achieve zero neutral current through the use of a pure inductor and a pure capacitor in the other two phases. The value of inductor and capacitor is [2007]

- (A) 73.95 mH in phase C and 139.02 mF in phase B
(B) 72.95 mH in phase B and 139.02 mF in phase C
(C) 42.12 mH in phase C and 240.79 mF in phase B
(D) 42.12 mH in phase B and 240.79 mF in phase C

11. A two-machine power system is shown below. Transmission line XY has positive-sequence impedance of $Z_1 \Omega$ and zero-sequence impedance of $Z_0 \Omega$.

An 'a' phase to ground fault with zero fault impedance occurs at the centre of the transmission line. Bus voltage at X and line current from X to F for the phase 'a' are given by $V_a V$ and $I_a A$, respectively. Then, the impedance measured by the ground distance relay located at the terminal X of line XY will be given by [2008]

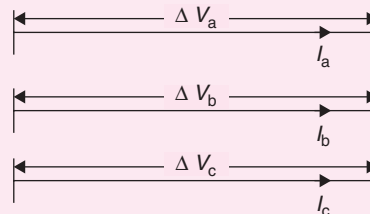


- (A) $\frac{Z_1}{2} \Omega$ (B) $\frac{Z_0}{2} \Omega$
(C) $\frac{(Z_0 + Z_1)}{2} \Omega$ (D) $\frac{V_a}{I_a} \Omega$

12. A three-phase transmission line is shown in the figure voltage drop across the transmission line is given by the following equation

$$\begin{bmatrix} \Delta V_a \\ \Delta V_b \\ \Delta V_c \end{bmatrix} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix} \begin{bmatrix} I_s \\ I_b \\ I_c \end{bmatrix}$$

Shunt capacitance of the line can be neglected. If the line has positive-sequence impedance of 15Ω and zero-sequence impedance of 48Ω , then the values of Z_s and Z_m will be [2008]



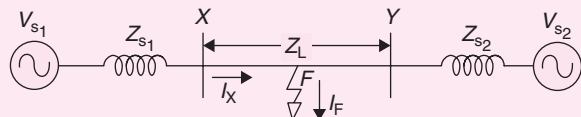
- (A) $Z_s = 31.5 \Omega$; $Z_m = 16.5 \Omega$
(B) $Z_s = 26 \Omega$; $Z_m = 11 \Omega$
(C) $Z_s = 16.5 \Omega$; $Z_m = 31.5 \Omega$
(D) $Z_s = 11 \Omega$; $Z_m = 26 \Omega$

13. A lossless transmission line having surge impedance loading (SIL) of 2280 MW is provided with a uniformly distributed series capacitive compensation of 30%. Then, SIL of the compensated transmission line will be [2008]

(A) 1835 MW (B) 2280 MW
(C) 2725 MW (D) 3257 MW

Common Data for Questions 14 to 16:

Consider a power system shown below:



Given that:

$$V_{s1} = V_{s2} = 1.0 + j0.0 \text{ p.u.};$$

The positive-sequence impedances are $Z_{s1} = Z_{s2} = 0.001 + j0.01 \text{ p.u.}$ and $Z_L = 0.006 + j0.06 \text{ p.u.}$

Three-phase base MVA = 100

Voltage base = 400 kV (Line to line)

Nominal system frequency = 50 Hz

The reference voltage for phase 'a' is defined as $v(t) = V_m \cos(\omega t)$.

A symmetrical three-phase fault occurs at centre of the line, i.e. point 'F' at time t_0 . The positive sequence impedance from source S_1 to point 'F' equals $0.004 + j0.04 \text{ p.u.}$ The waveform corresponding to phase 'a' fault current from bus X reveals that decaying DC offset current is negative and in magnitude at its maximum initial value. Assume that the negative-sequence impedances are equal to positive-sequence impedances, and the zero-sequence impedances are three times positive-sequence impedances.

14. The instant (t_0) of the fault will be [2008]

(A) 4.682 ms (B) 9.667 ms
(C) 14.667 ms (D) 19.667 ms

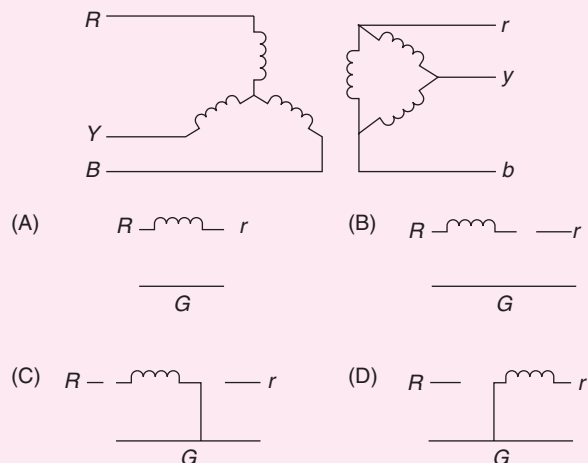
15. The RMS value of the AC component of fault current (I_x) will be [2008]

(A) 3.59 KA
(B) 5.07 KA
(C) 7.18 KA
(D) 10.15 KA

16. Instead of the three-phase fault, if a single line to ground fault occurs on phase 'a' at point 'F' with zero fault impedance, then the RMS value of the AC component of fault current (I_x) for phase 'a' will be [2008]

(A) 4.97 p.u.
(B) 7.0 p.u.
(C) 14.93 p.u.
(D) 29.85 p.u.

17. The zero-sequence circuit of the three-phase transformer shown in the figure is [2010]



18. For the power system shown in the figure below, the specifications of the components are the following:

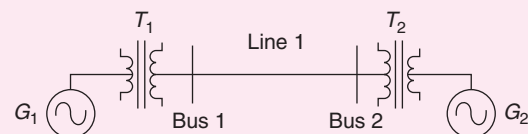
G_1 : 25 kV, 100 MVA, $X = 9\%$

G_2 : 25 kV, 100 MVA, $X = 9\%$

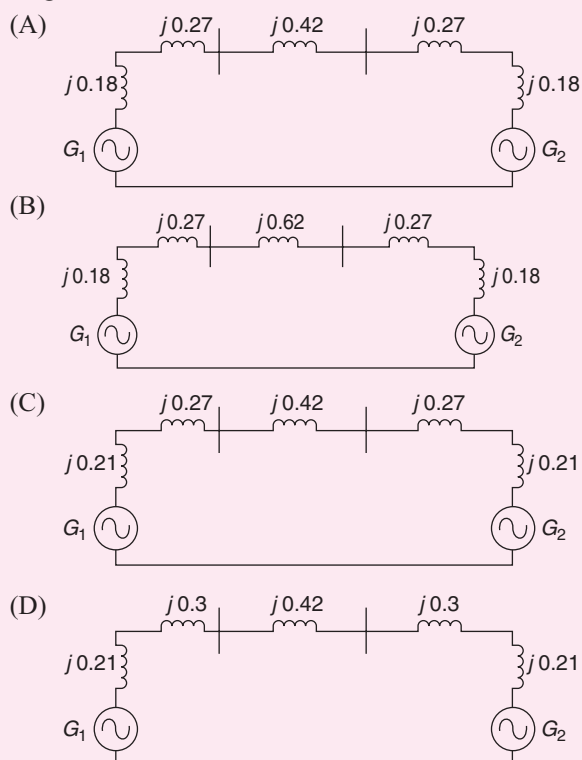
T_1 : 25 kV/220 kV, 90 MVA, $X = 12\%$

T_2 : 220 kV/ 25 kV, 90 MVA, $X = 12\%$

Line1: 220 kV, $X = 150 \Omega$

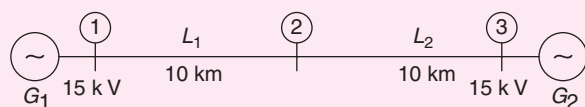


Choose 25 kV as the base voltage at the generator G_1 , and 200 MVA as the MVA base. The impedance diagram is [2010]



Common Data for Questions 19 and 20:

Two generator units G_1 and G_2 are connected by 15 kV line with a bus at the mid-point as shown below

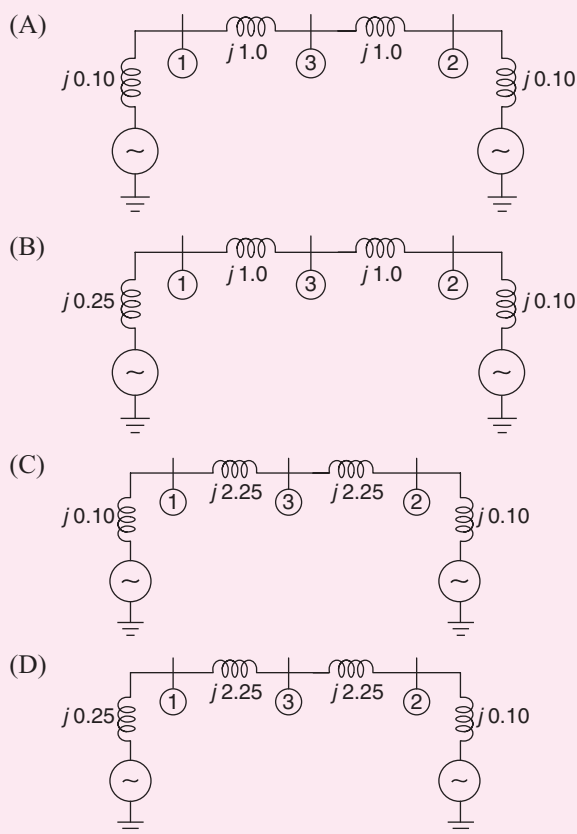


$G_1 = 250$ MVA, 15 kV, positive-sequence reactance $X = 25\%$ on its own base

$G_2 = 100$ MVA, 15 kV, positive-sequence reactance $X = 10\%$ on its own base

L_1 and $L_2 = 10$ km, positive-sequence reactance $X = 0.225 \Omega/\text{km}$

19. For the above system the positive sequence diagram with the p.u. values on the 100 MVA, common base is [2011]



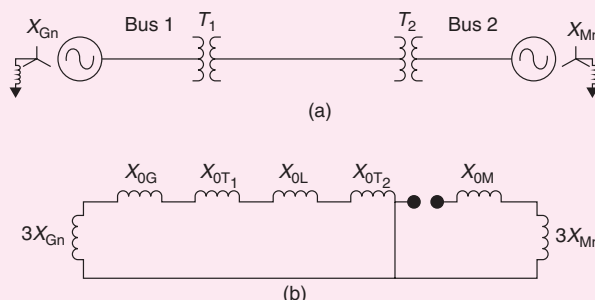
20. In the above system, the three-phase fault MVA at the bus 3 is [2011]

- (A) 82.55 MVA
(B) 85.11 MVA
(C) 170.91 MVA
(D) 181.82 MVA

21. The sequence components of the fault current are as follows: $I_{\text{positive}} = j1.5$ p.u., $I_{\text{negative}} = -j0.5$ p.u., $I_{\text{zero}} = -j1$ p.u. The type of fault in the system is [2012]

- (A) LG
(B) LL
(C) LLG
(D) LLLG

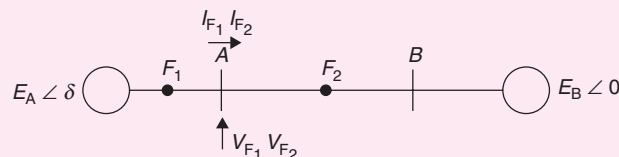
22. A 2-bus system corresponding zero-sequence network is shown in the figure.



- The transformers T_1 and T_2 are connected as [2014]

- (A) Δ and Δ
(B) Δ and Δ
(C) Δ and Δ
(D) Δ and Δ

23. Three-phase to ground fault takes place at locations F_1 and F_2 in the system shown in the following figure.



If the fault takes place at location F_1 , then the voltage and the current at bus A are V_{F_1} and I_{F_1} , respectively. If the fault takes place at location F_2 , then the voltage and the current at bus A are V_{F_2} and I_{F_2} , respectively. The correct statement about voltages and currents during faults at F_1 and F_2 is

- (A) V_{F_1} leads I_{F_1} and V_{F_2} leads I_{F_2}
(B) V_{F_1} leads I_{F_1} and V_{F_2} lags I_{F_2}
(C) V_{F_1} lags I_{F_1} and V_{F_2} leads I_{F_2}
(D) V_{F_1} lags I_{F_1} and V_{F_2} lags I_{F_2}

24. In an unbalanced three-phase system, phase current $I_a = 1 \angle (-90^\circ)$ p.u., negative sequence current $I_{b2} = 4 \angle (-150^\circ)$ p.u., zero sequence current $I_{c0} = 3 \angle 90^\circ$ p.u. The magnitude of phase current I_b in p.u. is [2014]

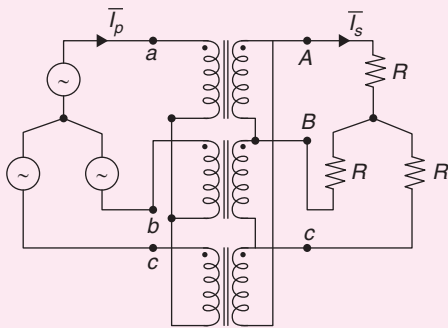
- (A) 1.00
(B) 7.81
(C) 11.53
(D) 13.00

25. A three-phase star-connected load is drawing power at a voltage of 0.9 p.u. and 0.8 power factor lagging. The three-phase base power and base current are

100 MVA and 437.38 A, respectively. The line-to-line load voltage in kV is _____. [2014]

26. At three-phase, 100 MVA, 25 kV generator has solidly grounded neutral. The positive, negative, and the zero sequence reactances of the generator are 0.2 p.u., 0.2 p.u., and 0.05 p.u., respectively, at the machine base quantities. If a bolted single phase to ground fault occurs at the terminal of the unloaded generator, the fault current in amperes immediately after the fault is _____. [2014]

27. A balanced (positive sequence) three-phase AC voltage source is connected to a balanced, star connected load through a star-delta transformer as shown in the figure. The line-to-



line voltage rating is 230 V on the star side, and 115 V on the delta side. If the magnetizing current is neglected and $\vec{I}_s = 100 \angle 0^\circ$ A, then what is the value of $\vec{I}_p$ in Ampere? [2015]

- (A) $50 \angle 30^\circ$ (B) $50 \angle -30^\circ$
(C) $50\sqrt{3} \angle 30^\circ$ (D) $200 \angle 30^\circ$
28. The magnitude of three-phase fault currents at buses A and B of a power system are 10 pu and 8 pu, respectively. Neglect all resistances in the system and consider the pre-fault system to be unloaded. The pre-fault voltage at all buses in the system is 1.0 pu. The voltage magnitude at bus B during a three-phase fault at bus A is 0.8 pu. The voltage magnitude at bus A during a three-phase fault at bus B, in pu, is _____. [2016]

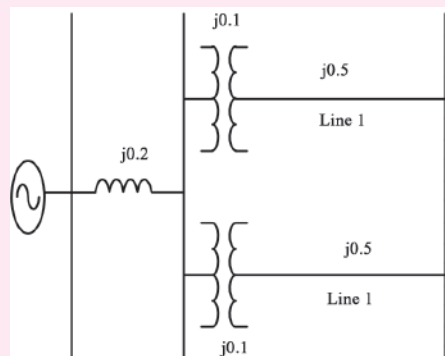
29. A 30 MVA, 3-phase, 50 Hz, 13.8 kV, star-connected synchronous generator has positive, negative and zero sequence reactances, 15%, 15% and 5% respectively. A reactance (X_n) is connected between the neutral of the generator and ground. A double line to ground

fault takes place involving phases 'b' and 'c', with a fault impedance of $j0.1$ p.u. The value of X_n (in p.u.) that will limit the positive sequence generator current to 4270 A is [2016]

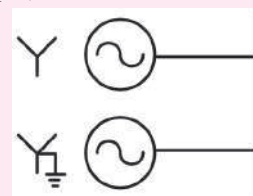
30. A 50MVA, 10KV, 50HZ, star-connected, unloaded three- phase alternator has a synchronous reactance of 1p.u and a sub-transient reactance of 0.2p.u. If a 3-phase short circuit occurs close to the generator terminals, the ratio of initial and final values of the sinusoidal component of the short circuit current is _____. [2016]

31. The single line diagram of a balanced power system is shown in the figure. The voltage magnitude at the generator internal bus is constant and 1.0p.u. the p.u reactances of different components in the system are also shown in the figure. The infinite bus voltage magnitude is 1.0p.u. A three phase fault occurs at the middle of line 2.

The ratio of the maximum real power that can be transferred during the pre - fault condition to the maximum real power that can be transferred under the faulted condition is _____



32. Two identical unloaded generators are connected in parallel as shown in the figure. Both the generators are having positive, negative and zero sequence impedances of $j0.4$ pu., $j0.3$ pu., and $j0.15$ pu., respectively. If the pre- fault voltage is 1pu, for a line - to - ground (L - G) fault at the terminals of the generators, the fault current, in pu., is _____. [2016]



ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. B | 4. C | 5. B | 6. C | 7. D | 8. A | 9. A | 10. A |
| 11. A | 12. A | 13. C | 14. A | 15. B | 16. D | 17. A | 18. B | 19. A | 20. A |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|------|------|-------|
| 1. D | 2. A | 3. A | 4. C | 5. A | 6. C | 7. A | 8. A | 9. A | 10. D |
| 11. A | 12. D | 13. C | 14. B | 15. A | 16. C | 17. B | | | |

Previous Years' Questions

- | | | | | | | | | | |
|---------|-------|-------|-------|-----------|-----------|-------|-----------|----------|-------|
| 1. A | 2. D | 3. D | 4. B | 5. C | 6. D | 7. C | 8. A | 9. B | 10. B |
| 11. D | 12. B | 13. C | 14. A | 15. A | 16. C | 17. C | 18. B | 19. A | 20. D |
| 21. C | 22. B | 23. C | 24. C | 25. 118.8 | 26. 15396 | 27. A | 28. 0.832 | 29. 1.08 | 30. 5 |
| 31. 2.3 | 32. 6 | | | | | | | | |

Chapter 3

Load Flow Analysis and Protection

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Bus admittance matrix
- Sparsity in Y_{bus} matrix
- Load flow analysis
- Formation of power flow equations
- Load bus
- Generator bus
- Load flow algorithm
- Optimal operation of generators
- Power system protection
- Protective zones
- Time multiplier setting

BUS ADMITTANCE MATRIX

Nodal admittance matrix or admittance matrix or Y matrix or Y_{bus} is an $n \times n$ matrix describing a power system with ' n ' buses. It represents the nodal admittance of the buses in a power system.

In realistic systems which contain thousands of buses, the Y_b matrix is quite sparse. Each bus in a real power system is usually connected to only a few other buses through the transmission lines. The Y matrix is also one of the data requirements needed to formulate a power flow study

$$Y_{bus} = \begin{bmatrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \dots & y_{nn} \end{bmatrix}$$

Consider a four-bus power system shown in figure below.

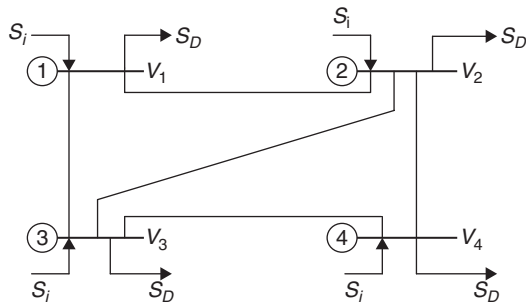


Figure 1 Single-line diagram of a four bus network

Equivalent network of the above single-line diagram is given by

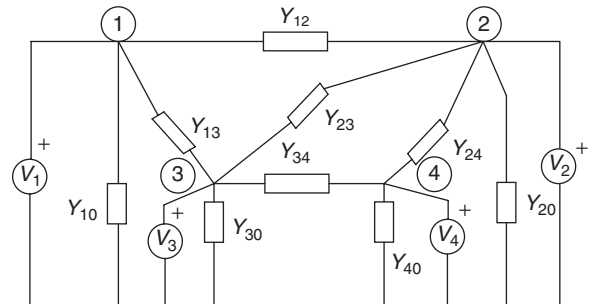


Figure 2 Equivalent circuit of the single-line diagram shown above

Apply Kirchhoff's current law for the above network,

$$I_1 = V_1 y_{10} + (V_1 - V_2) y_{12} + (V_1 - V_3) y_{13}$$

$$I_2 = V_2 y_{20} + (V_2 - V_1) y_{12} + (V_2 - V_3) y_{23} + (V_2 - V_4) y_{24}$$

$$I_3 = V_3 y_{30} + (V_3 - V_1) y_{13} + (V_3 - V_2) y_{23} + (V_3 - V_4) y_{34}$$

$$I_4 = V_4 y_{40} + (V_4 - V_2) y_{24} + (V_4 - V_3) y_{34}$$

Rearrange the equations and rewrite them in matrix form

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} y_{10} + y_{12} & -y_{12} & -Y_{13} & 0 \\ +y_{13} & y_{20} + y_{12} & -y_{23} & -y_{24} \\ -y_{12} & +y_{23} + y_{24} & y_{30} + y_{13} & -y_{34} \\ -y_{13} & -y_{23} & +y_{23} + y_{24} & y_{40} + y_{24} \\ 0 & -y_{24} & -y_{34} & +y_{34} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

Above matrix form can be represented as

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & Y_{14} \\ Y_{21} & Y_{22} & Y_{23} & Y_{24} \\ Y_{31} & Y_{32} & Y_{33} & Y_{34} \\ Y_{41} & Y_{42} & Y_{43} & Y_{44} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

where $Y_{11} = y_{10} + y_{12} + y_{13}$

$$Y_{22} = y_{20} + y_{12} + y_{23} + y_{24}$$

$$Y_{33} = y_{30} + y_{13} + y_{23} + y_{34}$$

$$Y_{44} = y_{40} + y_{24} + y_{34}$$

$$Y_{12} = Y_{21} = -y_{12}$$

$$Y_{13} = Y_{31} = -y_{13}$$

$$Y_{14} = Y_{41} = -y_{14} = 0$$

$$Y_{23} = Y_{32} = -y_{23}$$

$$Y_{24} = Y_{42} = -y_{24}$$

$$Y_{34} = Y_{43} = -y_{34}$$

Each admittance Y_{ii} is called the self-admittance of node 'i' and equal to the algebraic sum of the admittances terminating on the node.

$$Y_{ii} = y_i + \sum_{\substack{K=1 \\ K \neq i}}^N y_{iK}$$

Each off diagonal term Y_{ij} is called the mutual admittance (or transfer admittance) between nodes 'i' and 'j' and is equal to the negative of the sum off all admittances connected directly between those nodes.

$$Y_{ij} = -y_{ij}$$

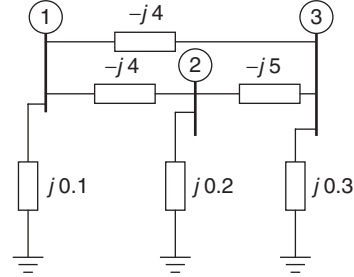
The admittance matrix is symmetric $Y_{ij} = Y_{ji}$.

Relation between bus impedance matrix and admittance matrix is given by

$$[Z_{bus}] = [Y_{bus}]^{-1}$$

Solved Examples

Example 1: Three-bus admittance network of a power system is given by



$$(A) \begin{bmatrix} -j7.9 & -j4 & -j4 \\ -j4 & -j8.8 & -j5 \\ -j4 & -j5 & -j8.7 \end{bmatrix}$$

$$(B) \begin{bmatrix} -j7.9 & +j4 & j4 \\ j4 & -j8.8 & j5 \\ j4 & j5 & -j8.7 \end{bmatrix}$$

$$(C) \begin{bmatrix} -j0.1 & -j4 & -j4 \\ -j4 & -j0.2 & -j5 \\ -j4 & -j5 & -j0.3 \end{bmatrix}$$

$$(D) \begin{bmatrix} j0.1 & -j4 & -j4 \\ -j4 & j0.2 & -j5 \\ -j4 & -j5 & j0.3 \end{bmatrix}$$

Solution: (B)

Admittance matrix order is 3×3 elements of the admittance matrix are

$$Y_{11} = y_{10} + y_{12} + y_{13} = j0.1 - j4 - j4 = -j7.9.$$

$$Y_{22} = y_{20} + y_{21} + y_{23} = j0.2 - j4 - j5 = -j8.8$$

$$Y_{33} = y_{30} + y_{31} + y_{32} = j0.3 - j5 - j4 = -j8.7$$

$$Y_{12} = y_{21} = j4$$

$$Y_{13} = y_{31} = j4$$

$$Y_{23} = y_{32} = j5$$

$$Y_{bus} = \begin{bmatrix} -j7.9 & j4 & j4 \\ j4 & -j8.8 & j5 \\ j4 & j5 & -j8.7 \end{bmatrix}$$

Sparsity in Y_{bus} Matrix

An element in Y_{bus} has a non-zero value, if the two buses are directly connected through a transmission line or a transformer. The non-existence of a direct connection between

two buses means that the off-diagonal transfer admittances of the Y_{bus} matrix will be zero. Normally in a large interconnected power systems, each bus may be connected with a few neighbouring buses. The non-existence of a direct connection between many buses means that the vast majority of a real world Y_{bus} matrix will be zero. Thus, for a large power systems Y_{bus} is highly sparse. In large systems, sparsity may be as high as 97%.

Example 2: For the Y-bus matrix of a 4-bus system given in per unit, the buses having shunt elements are

$$Y_{bus} = j \begin{bmatrix} -7 & 2 & 3 & 0 \\ 2 & -12 & 4 & 2.5 \\ 3 & 4 & -9 & 2 \\ 0 & 2.5 & 2 & -4.5 \end{bmatrix}$$

- (A) 1 and 2 (B) 2 and 3
(C) 3 and 4 (D) 1, 2, and 3

Solution: (A)

In an admittance matrix, sum of elements in a row is non-zero if the corresponding node is having shunt branch.

$$\text{For bus 1} = \sum_{j=1}^4 y_{1j} = y_{11} + y_{12} + y_{13} + y_{14} = -2 \neq 0$$

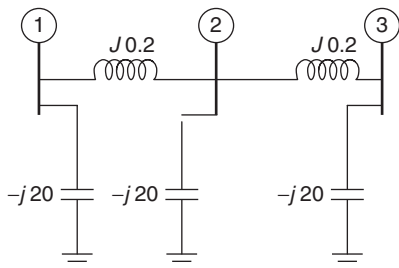
$$\text{For bus 2} = \sum_{j=1}^4 y_{2j} = y_{21} + y_{22} + y_{23} + y_{24} = -3.5 \neq 0$$

$$\text{For bus 3} = \sum_{j=1}^4 y_{3j} = 0$$

$$\text{For bus 4} = \sum_{j=1}^4 y_{4j} = 0$$

Bus 1 and 2 has shunt elements.

Example 3: Impedance diagram of a network is given in the figure in per unit. The diagonal elements Y_{22} of the bus admittance matrix Y_{bus} of the network is



- (A) $-j9.95$ (B) $-j19.6$ (C) $j0.4$ (D) $j20$

Solution: (A)

Diagonal element $Y_{22} = y_{20} + y_{21} + y_{23}$

$$= \frac{-1}{j20} + \frac{1}{j0.2} + \frac{1}{j0.2} = -9.95j$$

Example 4: Sparsity of a 100 bus network is given by 95%.

Number of elements in the network is given by

- (A) 500 (B) 4700 (C) 9500 (D) 200

Solution: (D)

Number of elements in a network

$$= \frac{N - n}{2}$$

where N = Number of non-zero elements in an admittance matrix

= 5% of total elements are non-zeros.

$$= \frac{5}{100} \times 10,000 = 500$$

n = order of the Y_{bus}

$$\text{Number of element} = \frac{500 - 100}{2} = 200 \text{ elements}$$

BUS IMPEDANCE MATRIX

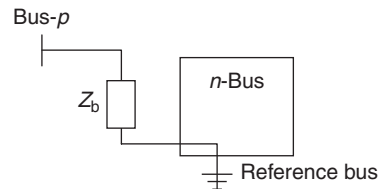
The process of building the bus impedance matrix proceeds by adding impedance one at a time till all the impedances have been included. With each addition of impedance, a new matrix is produced. The order of the matrix may or may not increase depending upon whether the addition of the impedance creates a new bus or not.

Assume that for a partial power system network with ' n ' buses, a reference node 'O' and the bus impedance matrix is $Z_{original}$ is known. When an element with impedance Z_b is added to the partial network, four types of modifications are possible.

1. Addition of a tree branch, with impedance Z_b from a new bus to the reference bus.
2. Addition of a tree branch with impedance Z_b from a new bus to an existing bus.
3. Adding a co-tree link with impedance Z_b between two existing buses.
4. Adding a co-tree link with impedance Z_b between an existing bus to the reference bus.

Modification of an Existing Bus Impedance Matrix

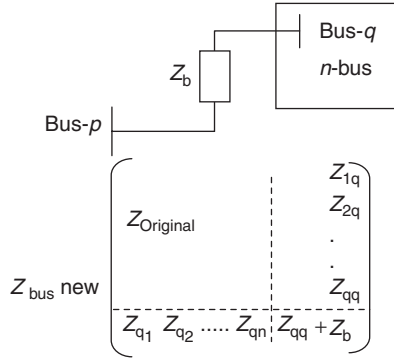
Case I: Adding Z_b from a new bus- p to reference bus



$$Z_{bus \text{ new}} = \begin{bmatrix} Z_{original} & \begin{matrix} 0 \\ 0 \\ \vdots \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & Z_b \end{bmatrix}$$

In this case, the elements of $(n + 1)^{\text{th}}$ column and rows are all zeros except the diagonal. The diagonal element is added to branch impedance Z_b . The elements of the original Z_{bus} matrix are not altered.

Case II: Adding Z_b from a new bus- p to existing bus- q



The elements of the $(n + 1)^{\text{th}}$ column are the elements of the q^{th} column and the elements of the $(n + 1)^{\text{th}}$ row are the elements of the q^{th} row. The diagonal element is given by the sum of Z_{qq} and Z_b .

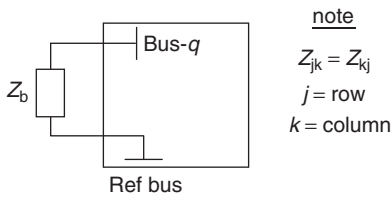
The elements of original Z_{bus} matrix are not altered.

Case III: Adding Z_b from an existing bus- q to reference bus.

It will be an addition as that of case II. The new impedance matrix of order $(n + 1)$ can be formed as that of case II.

So the bus impedance matrix has to be modified by eliminating $(n + 1)^{\text{th}}$ row and $(n + 1)^{\text{th}}$ column. This reduced bus impedance matrix is the actual new bus impedance matrix. Every element of actual new bus impedance matrix can be determined by using the equation

$$Z_{jk \text{ actual}} = Z_{jk} - \frac{Z_{j(n+1)}Z_{(n+1)k}}{Z_{(n+1)(n+1)}}$$



Case IV: Adding Z_b between two existing buses ' h ' and ' q '.

Hence the elements of the $(n + 1)^{\text{th}}$ row is the difference between the elements of row- h and row- q .

Here the elements of $(n + 1)^{\text{th}}$ column is the difference between the elements of column- h and column- q . The elements of $(n + 1)^{\text{th}}$ row is the difference between the elements of row- h and row- q .

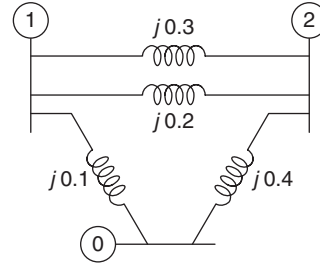
$$Z_{\text{bus new}} = \begin{pmatrix} & & Z_{1h} - Z_{1q} \\ & & Z_{2h} - Z_{2q} \\ & & \vdots \\ & & Z_{qh} - Z_{qn} \\ Z_{\text{original}} & & Z_{nh} - Z_{nq} \\ \hline Z_{h1} - Z_{q1} - Z_{hn} - \dots & & Z_{qn} - Z_{(n+1)(n-1)} \end{pmatrix}$$

$$\text{where } Z_{(n+1)(n+1)} = Z_b + Z_{hh} + Z_{qq} - 2Z_{hq}$$

Since the modification does not involve addition of new bus, the order of new bus impedance matrix has to be reduced to $n \times n$ by eliminating the $(n + 1)^{\text{th}}$ column and $(n + 1)^{\text{th}}$ row. This reduced bus impedance matrix is the actual bus impedance matrix.

$$Z_{jk \text{ actual}} = Z_{jk} - \frac{Z_{j(n+1)}Z_{(n+1)k}}{Z_{(n+1)(n+1)}}$$

Example 5: From the bus impedance matrix for the network shown in the figure.



Solution: Consider a reference bus as 'O', In this case buses 1 and 2 are connected to the reference bus. Connect bus 1 to the reference bus.

$$Z_{\text{bus}}^1 = j0.1$$

Add bus 2 to the bus 1 (with $j0.2$ as impedance)

$$Z_{\text{bus}}^2 = \begin{matrix} \textcircled{1} & \textcircled{2} \\ \downarrow & \downarrow \\ \begin{bmatrix} j0.1 & j0.1 \\ j0.1 & j0.3 \end{bmatrix} \end{matrix}$$

Add a link between bus (2) and reference bus

$$\begin{aligned} Z_{\text{bus}}^{\text{new}} &= Z_{\text{bus}}^2 - \frac{1}{Z_{22} + Z_{\text{new}}} \begin{bmatrix} j0.2 \\ j0.3 \end{bmatrix} \begin{bmatrix} j0.2 & j0.3 \end{bmatrix} \\ &= \begin{bmatrix} j0.1 & j0.1 \\ j0.1 & j0.3 \end{bmatrix} + \frac{1}{j0.3 + j0.4} \begin{bmatrix} 0.04 & 0.06 \\ 0.06 & 0.09 \end{bmatrix} \\ &= j \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.3 \end{bmatrix} - j \begin{bmatrix} 0.057 & 0.085 \\ 0.085 & 0.128 \end{bmatrix} \\ Z_{\text{bus}}^{\text{new}} &= j \begin{bmatrix} 0.043 & 0.015 \\ 0.015 & 0.172 \end{bmatrix} \end{aligned}$$

Add a link between bus (1) and (2)

$$\begin{aligned} Z_{\text{bus}}^{\text{new}} &= Z_{\text{bus}}^{\text{old}} - \frac{1}{Z_{11} + Z_{22} - 2Z_{12} + z_{\text{new}}} \begin{bmatrix} j0.0285 \\ -j0.157 \end{bmatrix} \begin{bmatrix} j0.08 - j0.157 \end{bmatrix} \\ &= Z_{\text{bus}}^{\text{old}} - j \frac{1}{0.043 + 0.172 - 0.03 + 0.3} \begin{bmatrix} 7.8 \times 10^{-4} & -4.4 \times 10^{-3} \\ -4.4 \times 10^{-3} & 0.0246 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
&= Z_{\text{bus}}^{\text{old}} - j \begin{bmatrix} 0.0016 & -0.009 \\ -0.009 & 0.0506 \end{bmatrix} \\
&= j \begin{bmatrix} 0.043 & 0.015 \\ 0.015 & 0.172 \end{bmatrix} - j \begin{bmatrix} 0.0016 & -0.009 \\ -0.009 & 0.0506 \end{bmatrix} \\
Z_{\text{bus}}^{\text{final}} &= j \begin{bmatrix} 0.0414 & 0.024 \\ 0.024 & 0.1214 \end{bmatrix}
\end{aligned}$$

LOAD FLOW ANALYSIS

A three-phase electric power system under normal steady state conditions of operation is expected to deliver the demanded real and reactive power to the consumer terminals at the rated voltage and frequency, within the specified limits of tolerance. To obtain a complete description of the behaviour of the power system, it is essential to know the voltages at various node points or buses and the power flowing through the elements of power system.

Power flow studies provide the required information regarding bus voltage and power flowing through transmission lines, transformers, and other elements of power systems for a specified load demand subjected to the regulating capabilities of generators, condensers, tap changing transformers, phase shifting transformers as well as specified net interchange of power with adjoining power systems.

In the power flow analysis, the load powers are assumed as known constants. A given set of loads on the buses can be served from a given set of generators in an infinite number of power flow configurations. Power flow analysis concerns itself not only with the actual physical mechanism which controls the power flow in the network meshes but also with how a best or optimum flow configuration from among the available possibilities is selected.

1. The sum of real power injected at the generating bus must equal the sum of total system load demand plus system losses at each instant of time.
2. The power transfer capability of a transmission line is limited by the thermal loading limit and the stability limit.
3. Keep the voltage levels of certain buses within close tolerance.
4. The power system must fulfil contractual scheduled interchange of power for neighbouring systems, via its tie lines.

Formation of Power Flow Equations

The general form of network equations for n -bus systems in the bus frame of reference in admittance form is

$$I_{\text{bus}} = Y_{\text{bus}} V_{\text{bus}}$$

where I_{bus} is the bus current vector
 V_{bus} is the bus voltage vector
 Y_{bus} is the bus admittance matrix

In general injected current at any bus ' i ' takes the form

$$I_i = Y_{i1}V_1 + Y_{i2}V_2 + Y_{i3}V_3 + \dots + Y_{in}V_n \text{ for } i = 1, 2, 3, \dots, n.$$

$$I_i = \sum_{m=1}^n Y_{im} V_m$$

In power systems, complex power injected at bus ' i '

$$S_i = P_i + jQ_i = V_i I_i^* \text{ for } i = 1, 2, 3, \dots, n$$

$$\Rightarrow S_i^* = P_i - jQ_i = V_i^* I_i$$

$$S_i^* = (Y_{i1}V_1 + Y_{i2}V_2 + Y_{i3}V_3 + \dots + Y_{in}V_n)V_i^*$$

$$\therefore S_i^* = P_i - jQ_i = \sum_{m=1}^n Y_{im} V_m V_i^* \text{ for } i = 1, 2, 3, \dots, n$$

Above equation gives the general form for static power flow equations.

Let the bus voltages and admittance be written in rectangular and polar form

$$V_i = e_i + jf_i = |V_i| \angle \delta_i$$

$$|V_i| = \sqrt{e_i^2 + f_i^2}; \delta_i = \tan^{-1} \frac{f_i}{e_i}$$

$$Y_{im} = G_{im} + jB_{im} = |Y_{im}| \angle \theta_{im}$$

$$|Y_{im}| = \sqrt{G_{im}^2 + B_{im}^2}; \theta_{im} = \tan^{-1} \frac{B_{im}}{G_{im}}$$

Substitute V_i and Y_{im} in S_i^* equation

$$P_i - jQ_i = \sum_{m=1}^n (G_{im} + jB_{im})(e_m + jf_m)(e_i - jf_i)$$

Equating the real and imaginary parts, we get

$$P_i = e_i \sum_{m=1}^n (G_{im} e_m - B_{im} f_m) + f_i \sum_{m=1}^n (G_{im} f_m + B_{im} e_m)$$

$$Q_i = e_i \sum_{m=1}^n (G_{im} f_m - B_{im} e_m) + f_i \sum_{m=1}^n (G_{im} e_m + B_{im} f_m)$$

for $i = 1, 2, 3, \dots, n$.

If the polar form of bus voltage V_i and admittance Y_{im} are substituted in S_i^* expression.

$$P_i - jQ_i = |V_i| e^{-j\delta_i} \sum_{m=1}^n |Y_{im}| e^{j\theta_{im}} |V_m| e^{j\delta_m}$$

for $i = 1, 2, 3, \dots, n$

$$P_i = |V_i| \sum_{m=1}^n |Y_{im}| |V_m| \cos(\theta_{im} - \delta_i + \delta_m)$$

$$Q_i = |V_i| \sum_{m=1}^n |Y_{im}| |V_m| \sin(\theta_{im} - \delta_i + \delta_m)$$

for $i = 1, 2, 3, \dots, n$.

A typical bus of a power network is shown in the figure. Each bus ' i ' is associated with four variables.

- Voltage magnitude $|V_i|$
- Phase angle δ_i
- Net injected real power P_i
- Net injected reactive power Q_i

For ‘ n ’ bus system there are total number of $4n$ variables. Above mentioned ‘ $2n$ ’ equations in polar or rectangular form are solved for $2n$ variables provided the other ‘ $2n$ ’ variables are specified as input data.

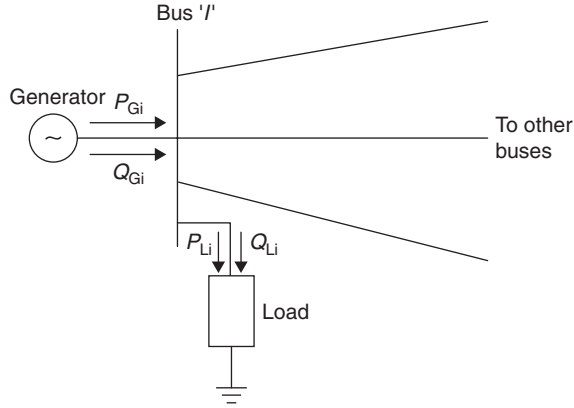


Figure 3 A typical bus of power systems

The study of various methods of solution to power system network is referred as load flow study. The solution provides the voltages at various buses, power flowing in various lines and line losses.

A load flow study of a power system generally requires the following steps

1. Representation of the system by single-line diagram.
2. Obtaining the impedance diagram using the information in single-line diagram.
3. Formulation of network equations
4. Solution of network equations

Types of Buses

The buses of power system can be classified into following three types.

1. Load bus (or PQ-bus)
2. Generator bus (or voltage controlled bus or PV bus)
3. Slack bus (or swing bus or reference bus)

Bus type	Quantities specified	Quantities to be obtained
Load bus	P, Q	$ V , d$
Generator bus	$P, V $	Q, d
Slack bus	$ V , d$	P, Q

Load Bus

The bus is called load bus, when real and reactive components of power are specified for the bus.

The load flow equations can be solved to find the magnitude and phase of bus voltage. In case of load bus, the voltage is allowed to vary within permissible limits.

Generator Bus

The bus is called generator bus, when real power and magnitude of bus voltage are specified for the bus. The load flow equation can be solved to find the reactive power and phase of bus voltage.

Slack Bus

The bus is called slack bus, if the magnitude and phase of bus voltage are specified for the bus.

The slack bus is the reference bus for load flow equation and usually one of the generator bus is selected as the slack bus.

Gauss–Seidel (G–S) Iterative Technique

In this Gauss–Seidel load flow, the load buses and generator busses are treated differently. In both these type of buses we use the complex power equation for updating voltages

$$V_i = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{V_i^*} - \sum_{m=1}^n Y_{im} V_m \right]$$

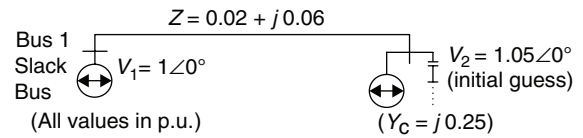
For PV buses, even though the real power is specified, its reactive power is unknown. But to update the voltage of PV buses, we must first estimate the reactive power of this bus. Reactive power expression is given by

$$Q_{i, \text{inj}} = -\text{Im}g \left[V_i^* \sum_{m=1}^n Y_{im} V_m \right]$$

Q_i value for K^{th} iteration is given by

$$Q_i^K = -\text{Im}g \left[V_i^{*(K-1)} \left\{ Y_{i1} V_1 + Y_{i2} V_2^K + Y_{i3} V_3^K + \dots + Y_{in} V_n^{(K-1)} \right\} \right]$$

Example for Gauss–Seidel Method



using Gauss–Seidel method, what should be next iteration value of V_2 ?

$$V_2^{(n+1)} = \frac{1}{Y_{22}} \left(\frac{S_2^{n4}}{V_2^{n4}} - \sum_{\substack{K=1 \\ K \neq 2}}^n Y_{2K} V_K^n \right)$$

$$= \frac{1}{Y_{22}} \left(\frac{S_2^{n4}}{V_2^{n4}} - Y_{21} V_1^n \right)$$

$$V_2^{(1)} = \frac{1}{Y_{22}} \left(\frac{S_2^{04}}{V_2^{04}} - Y_{21} V_1^{(0)} \right) = 1.045 \angle -0.83^\circ$$

Advantages of Gauss–Seidel (G.S) Method

1. Calculations are simple, so the programming task is easier.
2. Memory requirement is less.
3. Useful for small size systems.

Disadvantages of G.S Method

1. Requires large number of iterations to reach convergence.
2. Not suitable for large systems.
3. Converge time increases with size of the system.

Newton–Raphson Method

Let us assume that an n -bus power system contains a total of n_p number of P–Q buses while the number of P–V (Generator) buses be n_g such that $n = n_p + n_g + 1$. Bus 1 is assumed to be the slack bus. We shall further use the mismatch equation ΔP_i and ΔQ_i as specified below.

$$\Delta P_i = P_{i \text{ specified}} - P_{i \text{ calculated}}$$

$$\Delta Q_i = Q_{i \text{ specified}} - Q_{i \text{ calculated}}$$

The approach to Newton–Raphson load flow is similar to that of solving a system of non-linear equations using ‘Newton–Raphson’ method. At each iteration, we have to form a Jacobian matrix and solve for the correction factor.

$$J \begin{bmatrix} \Delta \delta_2 \\ \Delta \delta_3 \\ \vdots \\ \Delta \delta_n \\ \Delta |V_2|/V_2 \\ \vdots \\ \Delta |V_{i+n_p}|/|V_{1+np}| \end{bmatrix} = \begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \vdots \\ \Delta P_n \\ \Delta Q_2 \\ \vdots \\ \Delta Q_{1+n_p} \end{bmatrix}$$

where ‘J’ is the Jacobian matrix and it is divided into sub-matrices as

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

Size of the Jacobian matrix is $(n + n_{p-1}) \times (n + n_{p-1})$

The dimensions of the submatrices are as follows.

$$J_{11} \rightarrow (n-1) \times (n-1)$$

$$J_{12} \rightarrow (n-1) \times n_p$$

$$J_{21} \rightarrow n_p \times (n-1)$$

$$J_{22} \rightarrow n_p \times n_p$$

Sub matrices are given by

$$J_{11} = \begin{bmatrix} \partial P_2 / \partial \delta_2 & \cdot & \cdot & \partial P_2 / \partial \delta_n \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \partial P_n / \partial \delta_2 & \cdot & \cdot & \partial P_n / \partial \delta_n \end{bmatrix} = \begin{bmatrix} L_{22} & \cdot & \cdot & L_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ L_{n2} & \cdot & \cdot & L_{nn} \end{bmatrix}$$

where

$$L_{ik} = \frac{\partial P_i}{\partial \delta_k} = -|Y_{ik} V_i V_k| \sin(\theta_{ik} + \delta_k - \delta_i) \text{ for } i \neq K$$

$$L_{ii} = \frac{\partial P_i}{\partial \delta_i} = \sum_{k=1, k \neq i}^n |Y_{ik} V_i V_k| \sin(\theta_{ik} + \delta_k - \delta_i)$$

$$J_{12} = \begin{bmatrix} |V_2| \frac{\partial P_2}{\partial |V_2|} & |V_{1+np}| \frac{\partial P_2}{\partial |V_{1+np}|} \\ |V_2| \frac{\partial P_n}{\partial |V_2|} & |V_{1+np}| \frac{\partial P_n}{\partial |V_{1+np}|} \end{bmatrix} = \begin{bmatrix} N_{22} & N_{2np+1} \\ N_{n2} & N_{nnp+1} \end{bmatrix}$$

where

$$N_{ik} = |V_k| \cdot \frac{\partial P_i}{\partial |V_k|} = |Y_{ik} V_i V_k| \cos(\theta_{ik} + \delta_k - \delta_i)$$

for $i \neq K$

$$N_{ii} = |V_i| \frac{\partial P_i}{\partial |V_i|} = 2|V_i|^2 G_{ii} + \sum_{k=1, k \neq i}^n |Y_{ik} V_i V_k| \cos(\theta_{ik} + \delta_k - \delta_i)$$

$$J_{21} = \begin{bmatrix} \partial Q_2 / \partial \delta_2 & \cdot & \cdot & \partial Q_2 / \partial \delta_n \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \partial Q_n / \partial \delta_2 & \cdot & \cdot & \partial Q_n / \partial \delta_n \end{bmatrix} = \begin{bmatrix} M_{22} & \cdot & \cdot & M_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ M_{n2} & \cdot & \cdot & M_{nn} \end{bmatrix}$$

where

$$M_{ik} = \frac{\partial Q_i}{\partial \delta_k} = -|Y_{ik} V_i V_k| \cos(\theta_{ik} + \delta_k - \delta_i).$$

$$M_{ii} = \frac{\partial Q_i}{\partial \delta_i} = \sum_{k=1, k \neq i}^n |Y_{ik} V_i V_k| \cos(\theta_{ik} + \delta_k - \delta_i) = P_i - |V_i|^2 G_{ii}$$

$$J_{22} = \begin{bmatrix} |V_2| \frac{\partial Q_2}{\partial |V_2|} & |V_{1+np}| \frac{\partial Q_2}{\partial |V_{1+np}|} \\ |V_2| \frac{\partial Q_{1+np}}{\partial |V_2|} & |V_{1+np}| \frac{\partial Q_{1+np}}{\partial |V_{1+np}|} \end{bmatrix} = \begin{bmatrix} O_{22} & Q_{2np} \\ O_{np2} & O_{np np} \end{bmatrix}$$

where

$$O_{ik} = |V_k| \cdot \frac{\partial Q_i}{\partial |V_k|} = -|V_i| |Y_{ik} V_i V_k| \sin(\theta_{ik} + \delta_k - \delta_i)$$

for $i \neq K$.

$$Q_{ii} = |V_i| \frac{\partial Q_i}{\partial |V_i|} = 2|V_i|^2 B_{ii} - L_{ii}$$

Load Flow Algorithm

The Newton–Raphson procedure is as follows:

Step 1: Choose initial values for the voltage magnitude $|V|^{\circ}$ of all n_p load buses and $(n - 1)$ angles δ° of the voltages of all the buses except slack bus.

Step 2: Use the estimated voltages and phase angles to calculate $(n - 1)$ number of injected real power P°_{calc} and corresponding real power mismatch ΔP° for all buses except slack bus.

Step 3: Use the estimated voltages and phase angles to calculate n_p number of injected reactive power Q°_{calc} and corresponding reactive power mismatch ΔQ° for the PQ buses.

Step 4: Use the estimated $|V|^{\circ}$ and δ° to formulate Jacobian matrix J°

Step 5: Solve the load flow equation for δ° and $\Delta|V|^{\circ}/|V|^{\circ}$

Step 6: Obtain the updates from.

$$\delta' = \delta^{\circ} + \Delta\delta^{\circ}$$

$$|V|^1 = |V|^{\circ} \left[1 + \frac{|\Delta V|^{\circ}}{|V|^{\circ}} \right]$$

Step 7: Check if all the mismatches are below a small number (tolerance). Terminate process if yes. Otherwise go back to step 1 to start the next iteration with the updates given in the step '6'.

Advantages of Newton–Raphson (N.R) Method

1. The N.R method is faster, more reliable and results are more accurate.
2. Requires less number of iterations for convergence.
3. The number of iterations is independent of the size of system.
4. Suitable for large-size systems.

Disadvantages of Newton–Raphson (N.R) Method

1. The programming logic is more complex than G.S method.
2. The memory requirement is more.
3. Number of calculations per iteration is higher than G.S method.

Comparison of Gauss–Seidel and Newton–Raphson Method of Load Flow Solutions

S. No.	Gauss–Seidel method	Newton–Raphson method
1.	Variables are expressed in rectangular co-ordinates.	Variables are expressed in polar co-ordinates.
2.	Computation time per iteration is less.	Computation time per iteration is more.

(Continued)

S. No.	Gauss–Seidel method	Newton–Raphson method
3.	It has linear convergence characteristics.	It has quadratic convergence characteristics.
4.	The number of iterations required for convergence increases with size of system.	The number of iterations is independent of the size of the system.
5.	The choice of slack bus is critical.	The choice of slack bus is arbitrary.

Example 7: In load flow analysis of power system,

(A) One of the bus is taken as slack bus

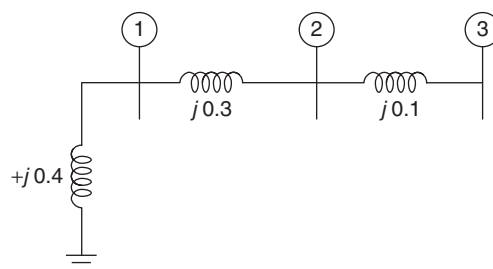
(B) Newton–Raphson method is preferred over Gauss–Seidel method

(C) Bus admittance matrix is a sparse matrix

(D) All the above

Solution: (D)

Example 8: Bus impedance matrix of the network given below is



$$(A) \ j \begin{bmatrix} 0.4 & 0.3 & 0 \\ 0.3 & 0.4 & 0.1 \\ 0 & 0.1 & 0.1 \end{bmatrix}$$

$$(B) \ j \begin{bmatrix} 0.4 & 0.4 & 0.4 \\ 0.4 & 0.7 & 0.7 \\ 0.4 & 0.7 & 0.8 \end{bmatrix}$$

$$(C) \ j \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 0.7 & 0 \\ 0 & 0 & 0.8 \end{bmatrix}$$

$$(D) \ j \begin{bmatrix} 0.4 & 0.4 & 0.4 \\ 0.4 & 0.7 & 0.4 \\ 0.4 & 0.4 & 0.8 \end{bmatrix}$$

Solution: (B)

Formation of bus 1

$$Z^1_{bus} = \overset{\textcircled{1}}{[j0.4]}$$

Formation of bus 2

$$Z^2_{bus} = \overset{\textcircled{1}}{[j0.4 \ j0.4]} \overset{\textcircled{2}}{[j0.4 \ j0.7]}$$

Formation of bus 3

$$Z^3_{bus} = \overset{\textcircled{1}}{[j0.4 \ j0.4 \ j0.4]} \overset{\textcircled{2}}{[j0.4 \ j0.7 \ j0.7]} \overset{\textcircled{3}}{[j0.4 \ j0.7 \ j0.8]}$$

Example 9: A power system consist of 40 bus with 9 voltage controlled buses, the size of the Jacobian matrix is

- (A) 70×70 (B) 80×80
(C) 62×62 (D) 79×79

Solution: (A)

Total number of buses (n) = 40

Voltage-controlled buses or generator buses = 9

Order of the Jacobian matrix

$$\begin{aligned} &= (2 \times \text{no. of load buses}) + \\ &\quad (\text{Number of generator buses} - 1) \\ &= 2(40 - 9) + (9 - 1) \\ &= 70 \times 70 \end{aligned}$$

Example 10: Voltage of a bus can be controlled by controlling

- (A) Active power of the bus
(B) Reactive power of the bus
(C) Both A and B
(D) Phase angle

Solution: (B)

Example 11: The bus impedance matrix of a 4-bus power system is given by

$$Z_{\text{bus}} = \begin{bmatrix} j0.3435 & j0.2860 & j0.2723 & j0.2277 \\ j0.2860 & j0.3408 & j0.2586 & j0.2414 \\ j0.2723 & j0.2586 & j0.2791 & j0.2209 \\ j0.2277 & j0.2414 & j0.2209 & j0.2791 \end{bmatrix}$$

A branch having an impedance of $j0.2 \Omega$ is connected between bus 2 and the reference. Then the values of $Z_{22\text{new}}$ and $Z_{23\text{new}}$ of the bus impedance matrix of the modified network are, respectively?

Solution: $Z_{\text{Bus}}(\text{new}) = Z_{\text{Bus}}(\text{old})$

$$= \frac{1}{Z_{jj} + Z_b} \begin{bmatrix} Z_{1j} \\ \cdot \\ \cdot \\ Z_{nj} \end{bmatrix} [Z_{j1} \dots Z_{jn}]$$

New elements (Z_b) is connected between j^{th} and reference bus

Here $j = 2, n = 4$

$$\begin{aligned} &= \frac{1}{j0.3408 + j0.2} \begin{bmatrix} j0.2860 \\ j0.3408 \\ j0.2586 \\ j0.2414 \end{bmatrix} \\ &\quad \times [j0.2860 \ j0.3408 \ j0.2586 \ j0.2414] \end{aligned}$$

We require only changes in Z_{22} and Z_{23} .

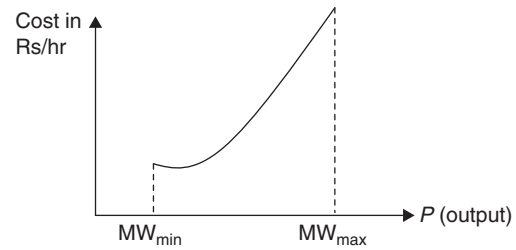
$$= \begin{bmatrix} - & - & - & - \\ - & j0.2147 & j0.16 & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix}$$

$$Z_{22\text{ new}} = Z_{22}(\text{old}) - j0.2147 = j0.1260$$

$$Z_{23\text{ new}} = Z_{23}(\text{old}) - j0.16296 = j0.956$$

OPTIMAL OPERATION OF GENERATORS

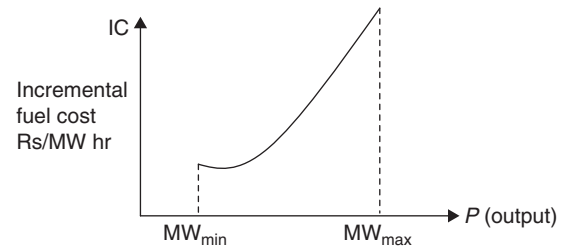
Generator operating cost: The major component of generator operating cost is the fuel input/hour, while the maintenance contributes only to a small extent. The fuel cost is most significant in case of thermal and nuclear stations. Typical cost-power output characteristics of a steam power plant are given below.



In the above characteristics, $MW_{\min}$ is the minimum loading limit below which it is uneconomical to operate the unit and $MW_{\max}$ is the maximum output limit. An analytical expression for operating cost can be written as

$$C_i = \frac{1}{2} a_i P_{Gi}^2 + b_i P_{Gi} + C_i \text{ Rs/hour}$$

The slope of the curve, i.e., $\frac{dC_i}{dP_{Gi}}$ is called the 'incremental fuel cost' (IC) and is expressed in units of rupees per megawatt hour (Rs/MW hr). A typical plot of incremental fuel cost versus power output is given below.



When there are ' n ' no. of generators present in the power system, total cost of generation is expressed as

$$C = \sum_{i=1}^n C_i(P_{Gi})$$

Total cost ' C ' is to be minimized under the equality constraint that $\sum_{i=1}^n P_{Gi} - P_{\text{Demand}} = 0$

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The problem can be solved by Lagrange multipliers.

$$L = \sum_{i=1}^n C_i(P_{Gi}) - \lambda \left[\sum_{i=1}^n P_{Gi} - P_D \right]$$

where λ is the Lagrangian multiplier

Minimization of 'C' is achieved by the condition

$$\frac{\partial L}{\partial P_{Gi}} = 0 \Rightarrow \frac{\partial C_i}{\partial P_{Gi}} = \lambda \text{ for } i = 1, 2, 3, \dots, n$$

where $\frac{\partial C_i}{\partial P_{Gi}}$ is the incremental cost of the i^{th} generator

$$\therefore \frac{\partial C_i}{\partial P_{Gi}} = \frac{\partial C_2}{\partial P_{G2}} = \frac{\partial C_3}{\partial P_{G3}} = \dots = \frac{\partial C_n}{\partial P_{Gn}} = \lambda$$

$\therefore$ Optimal operation of generators corresponds to the equal incremental cost point of all generators.

If a particular generator loading P_{Gi} reaches the limit $P_{Gi_{\max}}$ or $P_{Gi_{\min}}$, its loading from now on is held fixed at this and the remaining load is then shared between the remaining generators on equal incremental cost basis.

When line losses are accounted for optimal scheduling, the equality constraint of meeting the load demand is modified as

$$\sum_{i=1}^n P_{Gi} - P_D - P_L = 0$$

where P_L = Total system transmission loss

Lagrangian is written as

$$L = \sum_{i=1}^n C_i(P_{Gi}) - \lambda \left[\sum_{i=1}^n P_{Gi} - P_D - P_L \right]$$

where the transmission losses are function of active power generation at all buses.

$$P_L = P_L(P_{G1}, P_{G2}, P_{G3}, \dots, P_{Gn})$$

For optimal power dispatch

$$\frac{\partial L}{\partial P_{Gi}} = \frac{\partial C_i}{\partial P_{Gi}} - \lambda + \lambda \frac{\partial P_L}{\partial P_{Gi}} = 0, i = 1, 2, \dots, k$$

$$\frac{\partial C_i}{\partial P_{Gi}} = \lambda \left(1 - \frac{\partial P_L}{\partial P_{Gi}} \right)$$

$$\frac{\partial C_i / \partial P_{Gi}}{(1 - \partial P_L / \partial P_{Gi})} = \lambda \text{ For } i = 1, 2, 3, \dots, n$$

where $L_i = \frac{1}{\left(1 - \frac{\partial P_L}{\partial P_{Gi}} \right)}$ is called penalty factor of i^{th} plant.

The partial derivative $\frac{\partial P_L}{\partial P_{Gi}}$ is referred as the incremental transmission loss. (ITL)_i, associated with i^{th} generator. So, the condition for optimal generation can be given by

$$IC_i = \lambda [1 - ITL_i] \text{ for } i = 1, 2, 3, \dots, n$$

So far optimal scheduling of plants, it is necessary to compute ITL of each plant, the functional dependence of transmission losses on real power of generating plants is given by

$$P_L = \sum_{p=1}^n \sum_{q=1}^n P_{Gp} B_{pq} P_{Gq}$$

where P_{Gp}, P_{Gq} = Real power generation at p^{th} and q^{th} plants

B_{mn} = Loss coefficients which are constants under certain assumed operating conditions.

Example 12: The fuel costs of a two-unit plant are given by

$$C_1 = 100 + 2P_1 + 0.005P_1^2$$

$$C_2 = 200 + 2P_2 + 0.01P_1^2$$

where P_1 and P_2 are in MW. The plant supplies a load of 400 MW. The economic load scheduling of two units will be?

$$\begin{aligned} \text{Solution: } \frac{dC_1}{dP_1} &= 2 + 0.005 \times 2P_1 \\ &= 2 + 0.01P_1 \end{aligned}$$

$$\frac{dC_2}{dP_2} = 2 + 0.02P_2$$

For optimum load division, the two incremental costs should be equal.

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2}$$

$$2 + 0.01P_1 = 2 + 0.02P_2$$

$$P_1 = 2P_2$$

Total load is 400 MW

$$\text{i.e., } P_1 + P_2 = 400 \text{ MW}$$

$$2P_2 + P_2 = 400 \text{ MW}$$

$$3P_2 = 400 \text{ MW}$$

$$\therefore P_2 = 133.4 \text{ MW}$$

$$P_1 = 2P_2 = 266.6 \text{ MW}$$

Example 13: The power system has two generating plants and the power is being dispatched economically with $P_1 = 125$ MW, $P_2 = 250$ MW. The cost coefficients are

$$B_{11} = 0.1 \times 10^{-2} \text{ MW}^{-1}$$

$$B_{12} = -0.01 \times 10^{-2} \text{ MW}^{-1}$$

$$B_{22} = 0.13 \times 10^{-2} \text{ MW}^{-1}$$

to raise the total load on the system by 1 MW will cost an additional Rs 200 per hour. The penalty factor is _____?

$$\text{Solution: } P_L = P_1^2 B_{11} + 2P_1 P_2 B_{12} + P_2^2 B_{22}$$

$$\frac{\partial P_L}{\partial P_1} = 2P_1 B_{11} + 2P_2 B_{12}$$

$$= 2 \times 125 \times 0.1 \times 10^{-2} + 2 \times 250 \times (-0.01 \times 10^{-2}) = 0.2$$

Penalty factor for plant 1

$$L_1 = \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} = \frac{1}{1 - 0.2} = 1.25$$

Example 14: A generating station has two units. The incremental operating costs of the two units are as follows, G_1 : $\lambda_1 = 0.1P_1 + 20$, G_2 : $\lambda_2 = 0.2P_2 + 10$, where P_1, P_2 are in MW. Total loading is 200 MW. The economic loading will be
(A) $P_1 = 150, P_2 = 50$ (B) $P_1 = 100, P_2 = 100$
(C) $P_1 = 175, P_2 = 25$ (D) $P_1 = 50, P_2 = 150$

Solution: (B)

Given $P_1 + P_2 = 200$ (1)
and we know that $l_1 = l_2$, for economic load scheduling

$$0.1P_1 + 20 = 0.2P_2 + 10$$

$$\Rightarrow 0.1P_1 - 0.2P_2 = -10 \quad (2)$$

Solving (1) and (2), we get

$$P_1 = 100 \text{ MW}, P_2 = 100 \text{ MW}$$

Example 15: A power system has two generating plants and the power is being dispatched economically with $P_1 = 100$ MW and $P_2 = 200$ MW. The loss coefficients are $B_{11} = 0.2 \times 10^{-2} \text{ MW}^{-1}$, $B_{12} = -0.02 \times 10^{-2} \text{ MW}^{-1}$, $B_{22} = 0.09 \times 10^{-2} \text{ MW}^{-1}$. The penalty factor for plant-1 will be
(A) 1.35 (B) 1.21 (C) 1.47 (D) 1.63

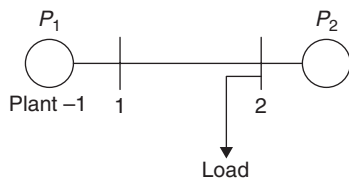
Solution: (C)

$$P_L = P_1^2 B_{11} + 2P_1 P_2 B_{12} + P_2^2 B_{22}$$

$$\begin{aligned} \frac{\partial P_L}{\partial P_1} &= 2P_1 B_{11} + 2P_2 B_{12} \\ &= [2(100)(0.2) \times 10^{-2}] + [2 \times 200 \times (-0.02) \times 10^{-2}] \\ &= 0.4 - 0.08 = 0.32 \end{aligned}$$

$$\begin{aligned} \text{Penalty factor, } L_1 &= \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} \\ &= \frac{1}{1 - 0.32} = 1.47 \end{aligned}$$

Example 16: A system consists of two plants connected by a transmission line as shown in the fig. The load is at plant-2. If a load of 200 MW is transmitted from plant-1 to the load, there is a loss of 20 MW. Determine the loss factor B_{11} .



- (A) $5 \times 10^{-4} \text{ MW}^{-1}$ (B) $10 \times 10^{-4} \text{ MW}^{-1}$
(C) $15 \times 10^{-4} \text{ MW}^{-1}$ (D) $20 \times 10^{-4} \text{ MW}^{-1}$

Solution: (A)

$P_L = P_1^2 B_{11} + 2P_1 P_2 B_{12} + P_2^2 B_{22}$
 $\therefore$ The load is at plant-2 bus

$$B_{12} = B_{21} = 0 \text{ and } B_{22} = 0$$

$$\therefore P_L = P_1^2 B_{11}$$

$$\Rightarrow B_{11} = \frac{P_L}{P_1^2} = \frac{20}{(200)^2}$$

$$\therefore B_{11} = 5 \times 10^{-4} \text{ MW}^{-1}$$

Example 17: A power system has two generating plants and the power is being dispatched economically with $P_1 = 100$ MW and $P_2 = 200$ MW. The loss coefficients are $B_{11} = 0.5 \times 10^{-2} \text{ MW}^{-1}$, $B_{12} = -0.02 \times 10^{-2} \text{ MW}^{-1}$, $B_{22} = 0.15 \times 10^{-2} \text{ MW}^{-1}$. To raise the total load on the system by 1 MW will cost an additional Rs 300 per hour. The penalty factor for plant-1 will be
(A) 10 (B) 11.5 (C) 12.5 (D) 15

Solution: (C)

$$P_L = P_1^2 B_{11} + 2P_1 P_2 B_{12} + P_2^2 B_{22}$$

$$\frac{\partial P_L}{\partial P_1} = 2P_1 B_{11} + 2P_2 B_{12}$$

$$\begin{aligned} &= 2(100)(0.5 \times 10^{-2}) + 2(200)(-0.02 \times 10^{-2}) \\ &= 1 - 0.08 = 0.92 \end{aligned}$$

$$\begin{aligned} \text{Penalty factor } L_1 &= \frac{1}{1 - \frac{\partial P_L}{\partial P_1}} \\ &= \frac{1}{1 - 0.92} = 12.5 \end{aligned}$$

Example 18: The fuel costs of two unit plants are given by

$$C_1 = 50 + 5P_1 + 0.5P_1^2,$$

$$C_2 = 10 + 15P_2 + P_2^2$$

where P_1, P_2 are in MW. The plant supplies load of 100 MW. The economic load scheduling of the two units will be

- (A) $P_1 = 70, P_2 = 30$ (B) $P_1 = 30, P_2 = 70$
(C) $P_1 = 50, P_2 = 50$ (D) $P_1 = 60, P_2 = 40$

Solution: (A)

$$\frac{dC_1}{dP_1} = \frac{dC}{dP_2}$$

$$5 + P_1 = 15 + 2P_2 \quad (3)$$

$$P_1 - 2P_2 = 10 \quad (3)$$

$$\text{Given, } P_1 + P_2 = 100 \quad (4)$$

Solving (3) and (4), we get

$$P_1 = 70 \text{ MW}, P_2 = 30 \text{ MW}$$

POWER SYSTEM PROTECTION

Undesirable effects of faults in power system are given by

1. Power system components carrying abnormal currents get overheated.
2. Operating voltages can go above or below their acceptable values.
3. Power flow is restricted or completely blocked due to short circuit.
4. Power system areas can lose synchronism.
5. Arcing faults can vaporize equipment in the vicinity leading to possibility of fire and explosion.
6. Open-circuit faults will lead to abnormal system operation and danger to personnel.

So, the faults should be instantly detected from the faulty section. Isolated from the rest of the system in the shortest possible time. Faults are detected automatically by means of relay and the fault section isolated by means of circuit breakers. The functional characteristics of the protection system are enumerated as follows.

1. **Reliability:** Reliability of a relay is a basic requirement. It must operate when it is required. Every component and circuit which is involved in operation of relay must be found in healthy operating condition when called upon to act.
2. **Speed:** A protective relay must operate at the required speed. It should neither be too slow which may result in damage to the equipment nor should it be too fast which may result in undesired operation during transient faults. The time is in the order of 30–100 ms depending upon the fault level of the section involved.
3. **Selectivity:** Relaying equipment must clearly discriminate between normal and abnormal system conditions, so it never operates unnecessarily.
4. **Sensitivity:** A relay should be sufficiently sensitive so that it operates reliably when required under the actual conditions in the system which produces the least tendency for operation.

Basic Definitions in Protection

Pick-up Level

The value of the actuating quantity which is on the threshold above which the relay operates.

Reset Level

The value of the actuating quantity below which a relay opens its contacts and comes to original position.

Reset Time

The time which elapses between the instant when the actuating quantity becomes less than the reset value to the instant when the relay contact returns to its normal position.

Operating Time

The time which elapses between the instant when the actuating quantity exceeds the pick-up value to the instant when the relay contacts close.

Primary Relays

The relays which are directly connected in the circuit to be protected.

Secondary Relays

The relays which are connected in the circuit to be protected through current and potential transformers.

Auxiliary Relays

Relays which operate in response to the opening or closing of its operating circuit to assist another relay.

Reach

A distance relay operates whenever the impedance seen by the relay is less than a pre-specified value. The impedance or the corresponding distance is known as the reach of the relay.

Over Reach

The tendency of the relay to operate at impedance more than its setting.

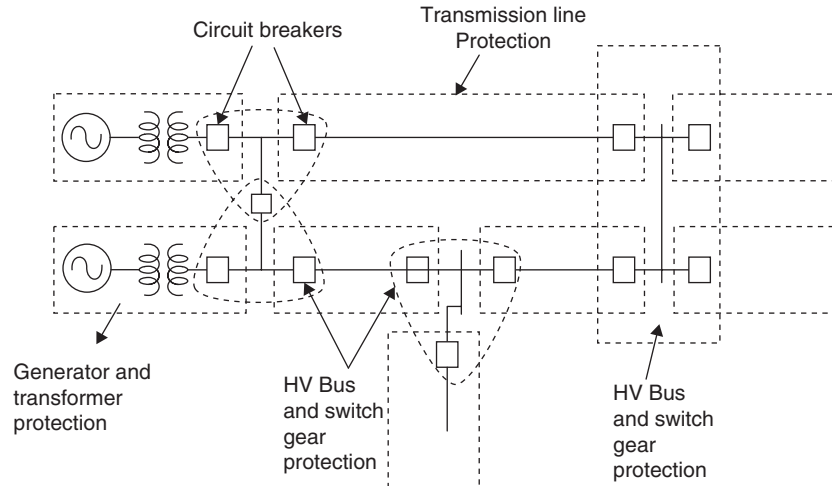
Under Reach

The tendency of the relay to operate at impedance less than its setting.

Protective Zones

In order to limit the number of elements disconnected by the protective system during a fault, the protective system is divided into number of zones. Protective zones of a power system are as shown in the figure below.

Each protective zone has the primary responsibility to disconnect the element or elements in the zone in the event of fault. The protection provided by each zone to its elements is known as 'primary protection' in the event of primary protection failure, backup protection is provided.



CLASSIFICATION OF RELAYS

Based on the Time of Operation

1. Instantaneous relay
2. Definite minimum time relay
3. Inverse time relay
4. Inverse definite minimum time relay.

Based on the Construction

1. Electromagnetic relay
2. Induction relay
3. Static relay

Based upon Application

1. Overcurrent relay
2. Overvoltage relay
3. Differential relay
4. Distance relay

Essential Characteristics of Relay

1. Sensitivity
2. Selectivity
3. Speed
4. Reliability
5. Economy
6. Simplicity

Pick-up Value

The minimum value of operating quantity at which the relay is at wedge of operation is called 'pickup value', i.e. ready to operate.

Reset Value

The maximum value of operating quantity at which the relay is at wedge of reset is called 'reset value'.

Time Multiplier Setting (TMS)

$$TMS = \frac{\text{Actual time of operation required}}{\text{Required time of operation when TMS} = 1}$$

Plug Setting Multiplier (PSM)

It is the ratio of CT secondary fault current to relay operating current.

$$\text{Relay operating current} = \frac{\text{Current setting} \times \text{CT secondary rated current}}{\text{CT secondary faulty current}}$$

$$PSM = \frac{\text{CT secondary faulty current}}{\text{Current setting} \times \text{CT secondary rated current}}$$

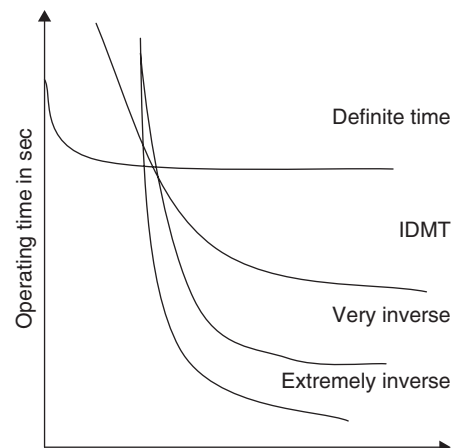
(or)

$$PSM = \frac{\text{Fault current (primary)}}{\text{Current setting} \times \text{CT secondary current} \times \text{CT ratio}}$$

Overcurrent Relays

Depending upon the time of operation, the relays are categorized as follows.

Characteristics of Various Overcurrent Relays



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Instantaneous Overcurrent Relay

- In which no intentional time delay is provided for the operation.
- The time of operation of such relay is approximately 0.1 s.

Inverse Time-current Relay

- In which the operating time reduces as the activating quantity increases in magnitude.

Inverse Definite Minimum Time Current Relay (IDMT)

- In which the operating time is approximately inversely proportional to the fault current near pick-up value and becomes substantially constant slightly above the pickup value of the relay.

Very Inverse Relay

- In which the saturation of the core occurs at a later stage.

Extremely Inverse Relay

- In which the saturation of the core occurs at a still later stage.

Overcurrent (OC) Relay

The relay operates when current through it satisfies the condition

$$Q = K_1 |I|^2 - K_4 > 0$$

$$I > \sqrt{\frac{K_4}{K_1}} = |I_p|$$

where I_p = pick up value of the relay, K_1 and K_4 are constants.

The above condition for relay operation can also be written as.

$$|I| > |I_p| \text{ trip (Relay trips CB)}$$

$$|I| < |I_p| \text{ block (Relay does not trip CB)}$$

The value of the pickup current can be easily adjusted by altering the constant K_1 by means of plug setting.

For co-ordination of overcurrent relays, it is essential that these must have inverse-time characteristics where operating time decreases as relay current increases these relays are called as 'Inverse definite minimum time lag relays'. Characteristics of an IDMTL relay are expressed mathematically as

$$T_{op} = f(|I_{p.u.}| - 1) \text{ for } |I_{p.u.}| > 1$$

$$|I_{p.u.}| = |I|/|I_p|$$

- The inverse time current relays are non-directional relays and are used for the protection of feeders, transmission lines, transformers, machines and other numerous applications.

Directional Relays

At a certain points in power system, current is flowing in one direction (specified direction), if due to any changes in

the circuit condition, the current flows in opposite direction, the relay will develop positive torque and will operate.

In a directional relay

$$Q = K_3 |V| |I| \cos(\theta - \tau) - K_4$$

where θ = Angle by which $|I|$ leads $|V|$

τ = Adjustable angle parameter

The relay operator when

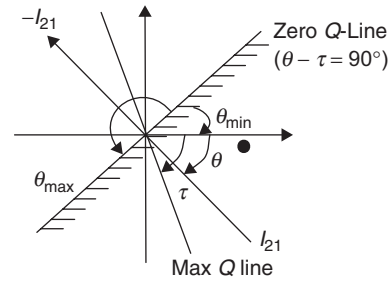
$$Q = K_3 |V| |I| \cos(\theta - \tau) > 0$$

or

$$\cos(\theta - \tau) > 0 \text{ trip}$$

$$\cos(\theta - \tau) < 0 \text{ block}$$

The operating characteristics of directional relay are shown in figure below.



In a directional relay, V is taken as the reference phasor, under fault, it undergoes only a small change in phase angle, This phase is called as the 'polarizing quantity' of the relay. The relay characteristics also can be expressed as

$$\theta_{min} > \theta > \theta_{max} \text{ trip}$$

$$\theta_{min} < \theta < \theta_{max} \text{ block}$$

Impedance Relay

From the general relay equation we have a special case

$$Q = K_1 |I|^2 + K_2 |V|^2 - K_4$$

If effect of K_4 is neglected, the relay operates when

$$K_1 |I|^2 + K_2 |V|^2 > 0$$

Or

$$\frac{|V|}{|I|} = |Z| < \left(\frac{K_1}{-K_2} \right)^{1/2} = |Z_{rs}|$$

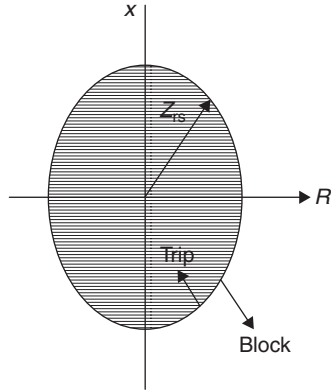
If K_2 is negative, such a relay senses impedance magnitude, and operates if the magnitude of impedance seen from its location is less than a specified value. Since $|V|$ prevents relay operation and $|I|$ tries to operate it, voltage is the restraining quantity and current is the operating quantity of the relay.

The characteristic impedance can be generally expressed as

$$|Z| < |Z_{rs}| \text{ trip}$$

$$|Z| > |Z_{rs}| \text{ Block}$$

where $|Z_{rs}|$ = impedance setting of the relay. RX -plane of the impedance characteristics is given by



Mho Relay

A modified impedance relay called 'Mho relay' in case of Mho relay

$$Q = K_3 |V| |I| \cos(\theta - \tau) - (-K_1) |V|^2 - K_4$$

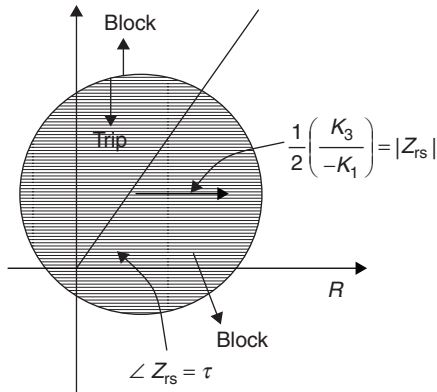
For relay operation neglecting K_4

$$K_3 |V| |I| \cos(\theta - \tau) - (-K_1) |V|^2 > 0$$

$$\frac{|V|}{|I|} = |Z| < \frac{K_3}{-K_1} \cos(\theta - \tau)$$

$$|Z - Z_{rs}| < |Z_{rs}| \text{ trip}$$

$$|Z - Z_{rs}| > |Z_{rs}| \text{ block}$$



Rx -diagram of a Mho relay

Reactance Relay

A reactance relay is an overcurrent relay with directional restraint the direction element is arranged to yield maximum Q -contribution when its current lags its voltage by 90°

$$Q = K_1 |I|^2 \cos(\theta - \tau) - (-K_1) |V|^2 - K_4$$

For relay operation neglecting K_4

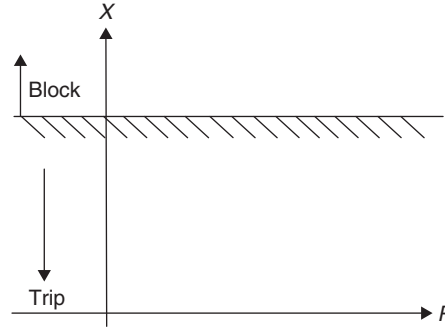
$$K_1 |I|^2 \cos(\theta - \tau) - (-K_1) |V|^2 > 0$$

$$\frac{|V|}{|I|} \sin \theta < \frac{K_1}{-K_2}$$

$$|Z| \sin \theta < \frac{K_1}{-K_2}$$

$$x < \frac{K_1}{-K_2}$$

These characteristics are represented in R - x -plane as shown below.



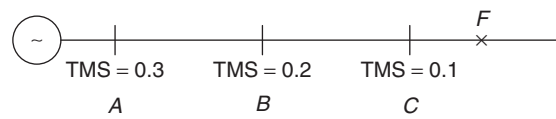
Protection of Feeders

Radial Feeders

Overcurrent protection for radial feeder is done by the following methods.

Time-graded System

The time of operation of the relays at various locations is so adjusted that the relay farthest from the source will have minimum time of operation and as it is approached towards the source the operating time increases.



Current Graded Method

In this method for the relay farthest from the source the current setting (cs) is kept minimum. As we go towards the source, the current setting gradually increases.

~	TMS = 0.3	TMS = 0.2	TMS = 0.1
	CS = 150%	CS = 125%	CS = 100%
	PSM = 1	PSM = 2	PSM = 3

$$\text{Current setting} \propto \frac{1}{\text{PSM}}$$

$$\text{PSM} \propto \frac{1}{\text{TMS}}$$

Time Current Graded System

According to this system the TMS and current setting for the relay farther from the source is kept minimum and as we go towards the source both these values increase.

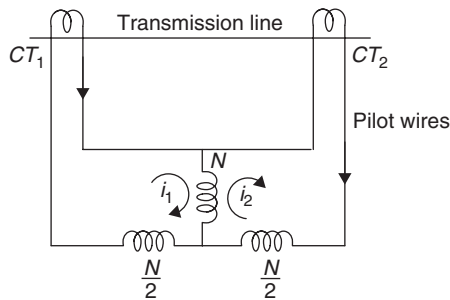
Parallel Feeders

- In this protection scheme, directional as well as non-directional overcurrent relays are used.
- Non directional relays are provided near the source and directional relays are provided near the load.

Ring Main Feeders

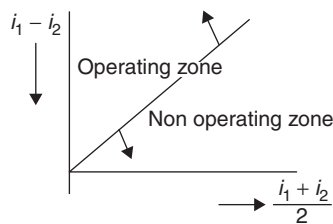
- The ring main considered two radial feeders.
- Non-directional relays must be placed near the source.
- Directional relays are placed near the load.

Differential Relays



$$\text{Operating coil} \propto (i_1 - i_2) N$$

$$\text{Restraining force} \propto \frac{i_1 N}{2} + \frac{i_2 N}{2} \\ \propto \frac{(i_1 + i_2) N}{2}$$



- Operating zone and non-operating zone is fixed by percentage differential relay.
- A differential protection using percentage differential relay is called as merz-price circulating current coil.
- A differential relay will not operate for inter turn fault in the generator winding. For inter turn fault protection of generator winding, split phase relay is used.
- For short circuits, normally percentage differential protection is recommended for transformers rated more than 1 MVA. For low rating, overcurrent relaying is used.

Neutral Grounding

The process of connecting neutral point of three-phase system to Earth (i.e. soil), either directly or through some circuit element (E.g. resistance, reactance, etc.) is called neutral grounding.

Methods of Neutral Grounding

Solid or Effective Grounding

Neutral point of a 3- ϕ system is directly connected to earth through a wire of negligible resistance and reactance.

This type of grounding is used for voltages up to 33 kV with total power capacity not exceeding 5000 kVA.

Resistance Grounding

When the neutral point of a three-phase system is connected to earth through a resistor, it is called resistance grounding.

It is used for voltages between 2.2 kV and 33 kV when power capacity is more than 5000 kVA.

Reactance Grounding

- A reactance is inserted between the neutral and ground.
- High transient voltage appears under fault condition.

Resonant Grounding

Capacitance currents are responsible for producing arcing grounds. These capacitive currents flow because capacitance exists between each line and Earth.

When the value of 'L' for arc suppression coil is such that the fault current, I_f exactly balances the capacitive current I_c , it is called resonant grounding (or Peterson coil grounding).

For resonant grounding,

$$I_L = I_C$$

$$\frac{V_{ph}}{X_L} = \frac{3V_{ph}}{X_C}$$

$$X_L = \frac{X_C}{3}$$

$$\omega L = \frac{1}{3\omega C}$$

$$\therefore L = \frac{1}{3\omega^2 C}$$

Solid State Relay

A solid state relay (SSR) is an electronic switching device in which a small control signal controls a larger load current or voltage. It comprises a voltage or current sensor which responds to appropriate input (control signal), a solid state electronic switching device of some kind which switches power to the load circuitry either on or off, and some coupling mechanism to control signal to activate this switch without mechanical parts. It serves same functions as an electromechanical relay, but has no moving parts.

Many SSRs use optical coupling. The control voltage energizes an LED which illuminates and switches on a photo sensitive diode (photo-voltaic). The diode current turns on a back to back thyristor, silicon control rectifier, or MOSFET to switch the load.

Advantages of SSRs

- SSRs are faster
- Increased life time
- Clean, bounce less operation
- No sparking, silent operation
- Identically smaller than electromagnetic relay

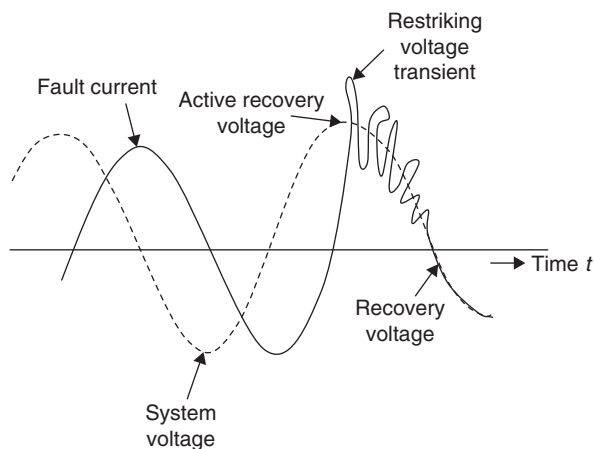
Disadvantages of SSRs

- Isolated bias supply is required for gate charge circuit.
- SSRs have higher transient reverse recovery time due to the presence of diode.

CIRCUIT BREAKER

Circuit breakers can be used for opening and closing of electrical circuit under normal and abnormal conditions. Generally circuit breaker consist of both moving and fixed contacts, when the circuit under normal (or) healthy conditions the contacts are closed and when the circuit under abnormal (or) fault condition the contacts are separated.

Consider a circuit operated under normal conditions and after some time interval fault occurs on the system. Then the variation of voltage and currents is as shown below.



When the circuit is interrupted by a circuit breaker under fault condition an arc is formed between the two contacts of the circuit breaker. This arc can be extinguished by two methods.

1. Low resistance (or) current zero method
2. High resistance method

Arc Voltage

The voltage appeared across the contacts of the circuit breaker during arcing period is known as arc voltage.

Restriking Voltage

The transient voltage appears across the circuit breaker contacts at current zero point is known as restriking voltage.

Average Rate of Rise of Restriking Voltage (RRRV)

It is the ratio of peak value of restriking voltage to the time taken to reach peak value voltage.

The average RRRV

$$= \frac{\text{Peak value of restriking voltage}}{\text{Time taken in attaining peak value}}$$

$$= \frac{2V_{\max}}{\pi\sqrt{LC}}$$

Insertion P₆

Ratings of Circuit Breaker

The circuit Breaker performance is analysed by the following terms.

1. Breaking capacity
2. Making capacity
3. Short time capacity

Breaking Capacity

It is the product of rated breaking current in terms of kA and rated voltage in terms of kV.

For three-phase, the Breaking capacity = $\sqrt{3}VI$

Making Capacity

The making capacity of circuit breaker is the ability of the circuit breaker to withstand the electromagnetic force under short-circuit condition. Generally making capacity is 2.55 times of breaking capacity.

Making capacity = $2.55 \times \text{Breaking capacity}$.

Short Time Capacity

It is the capacity of the circuit Breaker which can safely applied to the circuit Breaker under its normal condition for 3 seconds time period.

Types of Circuit Breakers

1. Air-break circuit breakers
2. Oil circuit breakers (OCB)
3. Vacuum circuit breakers (VCB)
4. Air-blast circuit breakers (ABCB)
5. SF₆ circuit Breakers

Air Break Circuit Breakers

Air Break circuit Breakers are generally indoor type. These are used for DC circuit up to 12 kV and for medium- and low-voltage AC circuits up to 6.6 kV, 400–200 A.

The constructional details of an ABCB are shown in figure. The arc is lengthened in the breakers by the magnetic

field and are runners and is finally extinguished by are splitters. There are two sets of contacts:

- Main contacts
- Arcing contacts

The main contacts are first to dislodge, while the arcing contacts are still closed under spring pressure. Thus the min contacts do not open any current and have long life. Arcing contacts made of hard copper alloy are easily replicable.

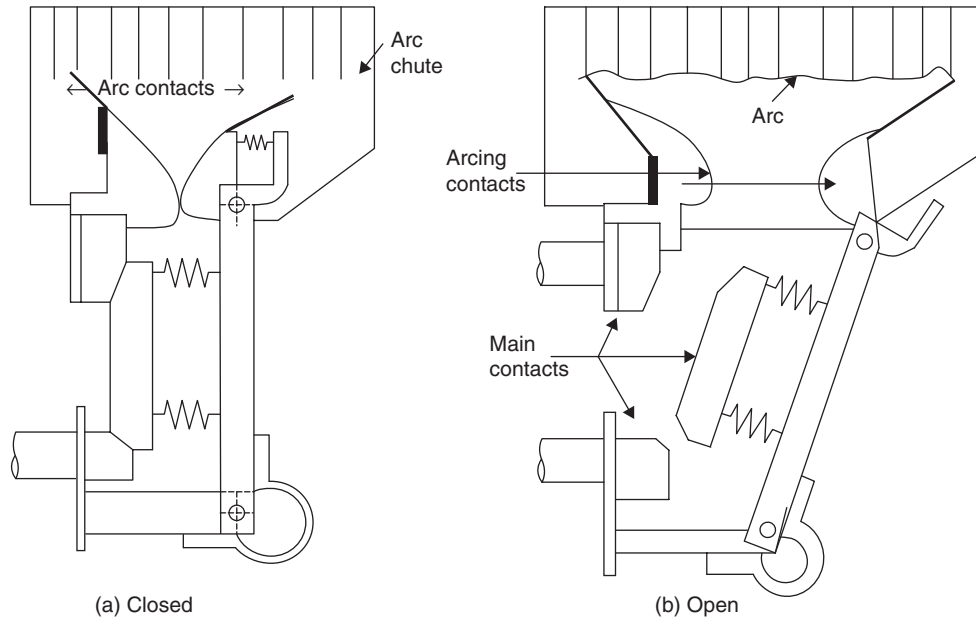
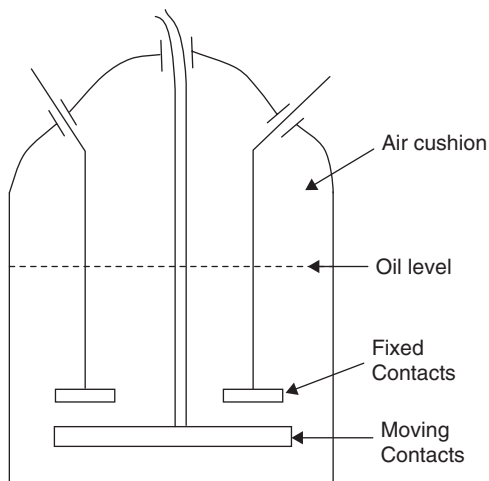


Figure 4 Air break circuit breakers

Oil Circuit Breakers (OCB)

Oil is used as the arc-quenching medium. These CB's have been developed for the voltage levels of 3.6 kV, 7.2 kV, 12 kV, 36 kV, 72.5 kV, 145 kV and 245 kV. The arc control devices are based on axial-flow and/or cross-flow principle. For higher current ratings, cross flow principle is preferred. The technique of arc quenching in oil circuit breakers is shown below.



In oil circuit breakers, as the contacts separate, the heat of the arc causes the oil to decompose in to hydrogen (70%)

acetylene, etc. The gaseous formation causes the pressure inside the arc control device to rise and as a result, the arc is pushed across the side vents, there by elongating it. As the contacts move farther apart lengthening of arc occurs and hence the arc will be extinguished. For a specific design and speed of contracts, the gas pressure generated is a function of arc current and arcing time.

Disadvantages of OCB:

1. Arc products are inflammable.
2. Oil is hygroscopic, and must be sealed air-tight in chamber.
3. Dielectric strength of oil is reduced by carbonization during the arcing process.

Vacuum Circuit Breakers

The separation of current carrying contacts comes the metal vapour to the released from the contacts giving rise to plasma – electrons and positive arc. The vapour density depends up on the current in the arc. In the decreasing phase of the current the rate of release of metal vapour quickly disperses and the medium regains its dielectric strength the arc is thus extinguished in just half a cycle. The contact separation needed is of the order of a few millimetres—least among all circuit Breaker types. VCBs have been employed at voltages up to 72.5 kV.

Air Blast Circuit Breakers (ABCB)

Blast of air at high speeds directed at the arc is very effective in cooling it, and in removing the products of ionization after current zero with consequent arc extinction within a cycle. The high speed pressure is produced by externally generated pressure. The breaker is designed to direct a jet of air derived from the high pressure source to the contact space at the instant of contact separation. Various possible alternative arrangements for achieving this are shown below.

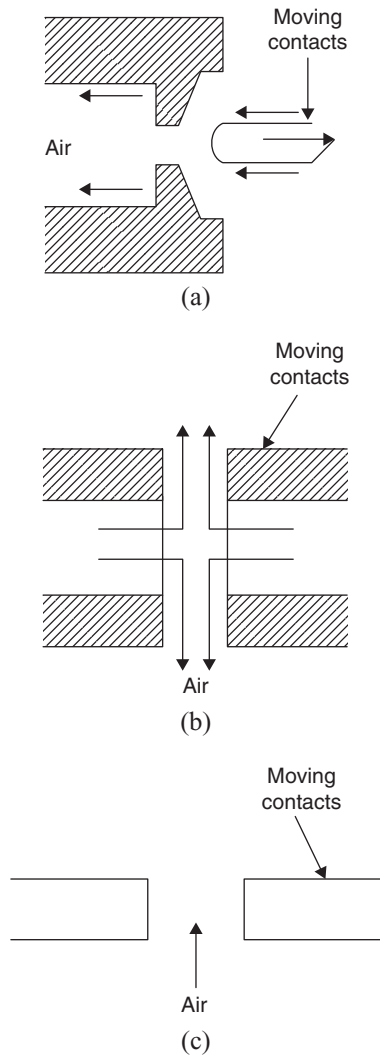


Figure 5 (a) Axial blast with hollow fixed contact; (b) Radial blast; (c) Cross blast

Advantages

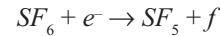
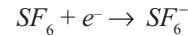
- Reliable because an external source is used for arc extinction.
- clean, non-compassable, and non-inflammable.
- Fresh medium is used every time.
- Faster contact travel.
- Small contact travel is involved.

Disadvantages

- High air noise while operating
- Chopping possibility
- Independent pressure system is needed for ABCB.

Sulphur Hexafluoride (SF_6)CB

The attachment of the electron with the natural gas molecules may occur in two ways



The negative ions formed are relatively heavier than free electrons and therefore under a given field the ions do not attain sufficient energy to lead cumulative ionization in the gas. This property gives rise to very high dielectric strength of SF_6 . The property of SF_6 makes it very effective in quenching arcs.

Advantages

- Current-chopping tendency is minimized.
- There is no exhaust of high-pressure gas to atmosphere.
- There is short time arc, low-contact erosion and no contact replacement.
- No carbon deposition takes place.
- The smaller size of conductors and clearances lead to small overall breaker size.
- They are non-inflammable.

Example 19: Which of the following protection relay is inherently directional?

- (A) Reactance relay (B) Impedance relay
(C) OHM relay (D) MHO relay

Solution: (D)

Example 20: Purpose of harmonic restraint in protection transformers is to protect against

- (A) Lightning
(B) Unbalanced operation
(C) Switching overvoltages
(D) Magnetizing inrush current

Solution: (D)

Example 21: Reactance relay is mainly used for

- (A) Earth faults (B) Open-circuit faults
(C) Phase faults (D) None of the above

Solution: (A)

Example 22: A negative sequence relay is commonly used to protect

- (A) A transformer (B) An alternator
(C) A transmission line (D) A bus bar

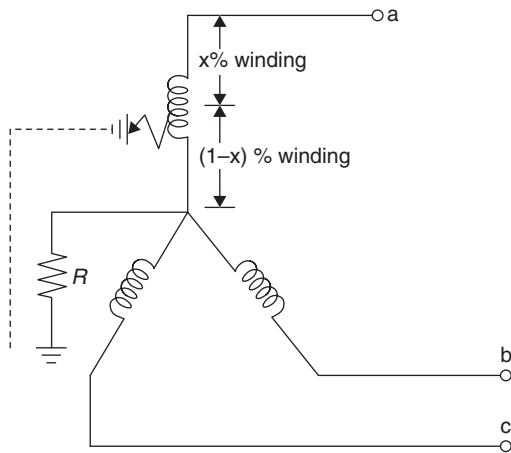
Solution: (B)

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Example 23: The neutral of 10 MVA, 11 kV, alternator is earthed through a resistance of $5\ \Omega$. The earth fault relay is set to operate at 0.65 A. The CTs have a ratio of 1000/5. What percentage of the alternator winding is protected?

- (A) 89.7% (B) 98.3%
(C) 10.3% (D) 1.7%

Solution: Fault current (I_f) = $(1-x) \frac{V_{ph}}{R}$



$\therefore$ Primary fault setting current (I_p) = $(1-x) \frac{V_{ph}}{R}$

Earthing resistor current setting (I_s) = $\frac{V_{ph}}{R}$

$$\therefore I_p = (1-x)I_s$$

$$\Rightarrow x = \left(1 - \frac{I_p}{I_s}\right)$$

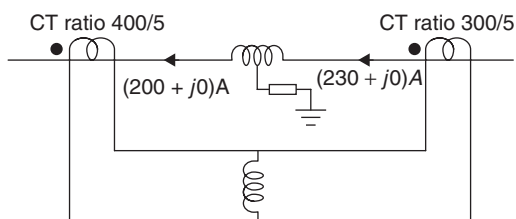
$\therefore$ Given that the earth fault relay set current = 0.65 A.

Primary fault setting current (I_p) = $0.65 \times \frac{1000}{5} = 130\text{ A}$
Earth resistor current setting I_s

$$= \frac{V_{ph}}{R} = \frac{11000}{\sqrt{3}} \times \frac{1}{5} = 1270.17\text{ A}$$

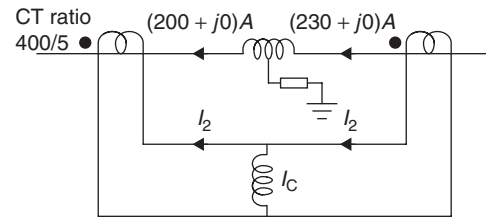
$$\% \text{ of winding protected} = \left(1 - \frac{130}{1270.17}\right) \times 100 = 89.7\%$$

Example 24: Consider a stator winding of an alternator with an internal high resistance ground fault. The currents under the fault condition are shown in the figure the winding is protected using differential current scheme with current transformer of ratio 300/5 A as shown. The current through the operating coil is



- (A) 3.83 A
(B) 3.33 A
(C) 1.33 A
(D) 0.369 A

Solution: (C)



$$I_1 = 200 \times \frac{5}{400} = 2.5\text{ A}$$

$$I_2 = 230 \times \frac{5}{300} = 3.83\text{ A}$$

$$I_c = I_2 - I_1 = 1.33\text{ A}$$

Example 25: A 3- ϕ , 33 kV oil-circuit breaker is rated 1100 A, 2000 MVA, 2.5 s. The symmetrical braking current is

- (A) 35 kA
(B) 3600 A
(C) 1200 A
(D) 105 A

Solution: (A)

$$\text{Symmetrical breaking current} = \frac{P}{\sqrt{3} V_L}$$

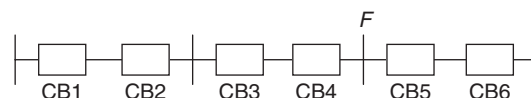
$$= \frac{2000 \times 10^6}{\sqrt{3} \times 33 \times 10^3} = 35\text{ kA}$$

Example 26: The interrupting time of a circuit breaker is the time period between

- (A) Initiation of short circuit and the parting of primary arc contacts.
(B) Energizing of the trip circuit and the parting of primary arc contacts.
(C) Energizing of the trip circuit and the arc extinction on an opening operation.
(D) Initiation of short circuit and the arc extinction on an opening operation.

Solution: (C)

Example 27: Three sections of a feeder are provided with circuit breaker: CB1, CB2, CB3, CB4, CB5, CB6, for a fault 'F' as indicated in the following figure.



- (A) CB5 must be set to trip before CB1, CB2, CB3 and CB4 trip.
 (B) CB5 must be set to trip after CB1 trip
 (C) CB5 must be set to trip after CB2 trips
 (D) CB5 must be set to trip after CB2, and CB4 trips

Solution: (A)

Example 28: Resistance switching is normally employed in

- (A) Bulk oil breakers
 (B) Minimum oil breakers
 (C) Air blast circuit breakers
 (D) All breakers

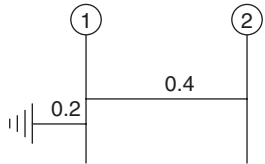
Solution: (D)

EXERCISES

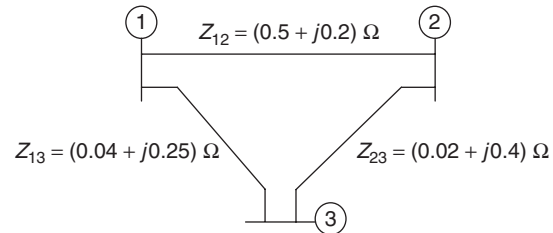
Practice Problems I

Directions for questions 1 to 22: Select the correct alternative from the given choices.

- In power flow analysis, a load connected at a bus acts as
 - Constant impedance connected at the bus
 - Voltage and frequency dependent source at the bus.
 - Constant real and reactive power drawn from the bus
 - Constant current drawn from the bus
- A system has two buses marked 1 and 2. The per unit impedances are also marked. The bus admittance matrix is



- $\begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.4 \end{bmatrix}$
 - $\begin{bmatrix} 7.5 & 2.5 \\ 2.5 & 2.5 \end{bmatrix}$
 - $\begin{bmatrix} 0.6 & -0.4 \\ -0.4 & 0.4 \end{bmatrix}$
 - $\begin{bmatrix} 7.5 & -2.5 \\ -2.5 & 7.5 \end{bmatrix}$
- The load flow studies is performed using Y bus matrix rather than Z bus matrix. This is because
 - Y bus is a square matrix
 - Y bus is a sparse matrix
 - Z bus is a sparse matrix
 - Z bus is a singular matrix
 - The number of iterations required for solving load flow problems in Gauss Seidel and Newton–Raphson methods is
 - Both directly proportional to number of buses
 - Directly proportional to number of buses and independent of number of buses, respectively
 - Independent of number of buses and directly proportional to number of buses, respectively
 - both independent of the number of system
 - The value of Y_{11} of the bus admittance matrix of the network is



- $1.8 - j8.6$
 - $1.3 - j7.2$
 - $1.8 + j8.6$
 - $1.3 + j7.2$
- Two alternators in parallel supplying a total load of 75 MW have the following data:
 Machine 1: 50 MVA with 4% speed regulation
 Machine 2: 70 MVA with 5% speed regulation
 The load shared between machines 1 and 2 is
 - $P_1 = 50$ MW, $P_2 = 25$ MW
 - $P_1 = 25$ MW, $P_2 = 50$ MW
 - $P_1 = 40$ MW, $P_2 = 35$ MW
 - $P_1 = 35$ MW, $P_2 = 40$ MW
 - Two generators delivering a total of 220 MW have their incremental cost characteristics as follows

$$\frac{dF_1}{dP_1} = 1 + 0.15P_1$$

$$\frac{dF_2}{dP_2} = 4.0 + 0.1P_2$$
 For economic operation P_1 and P_2 should be
 - $P_1 = 120$ MW, $P_2 = 100$ MW
 - $P_1 = 100$ MW, $P_2 = 120$ MW
 - $P_1 = 80$ MW, $P_2 = 140$ MW
 - $P_1 = 140$ MW, $P_2 = 80$ MW
 - The most inappropriate among the disadvantages of Gauss–Seidel method of load flow solution is
 - Slow convergence
 - Unreliable convergence
 - A good initial guess is essential for convergence.
 - Choice of slack bus affects convergence
 - For the Y bus matrix of a 4-bus system given in per unit, the buses having shunt elements are,

$$Y_{\text{BUS}} = j \begin{bmatrix} -5 & 2 & 2.5 & 0 \\ 2 & -10 & 2.5 & 4 \\ 2.5 & 2.5 & -9 & 4 \\ 0 & 4 & 4 & -8 \end{bmatrix}$$

- (A) 3 and 4 (B) 2 and 3
(C) 1 and 2 (D) 1, 2 and 4
10. If the equation $x^2 - 500 = 0$ is to be solved using Newton–Raphson method, the iterative steps involved is
- (A) $x_{R+1} = x_R + \frac{500}{x_R}$ (B) $x_{R+1} = \frac{1}{2} \left(x_R + \frac{500}{x_R} \right)$
(C) $x_{R+1} = \frac{1}{2} \left(x_R - \frac{500}{x_R} \right)$ (D) $x_{R+1} = x_R - \frac{500}{x_R}$
11. The equation $e^x - 1 = 0$ is to be solved using Newton–Raphson method with initial guess $x_0 = 1$. After one step, estimate x of solution is given by
(A) 0.36788 (B) 0.71828
(C) 0.20587 (D) 0.0000
12. The specified variables in a slack bus are
(A) Active power demand and reactive power demand at the bus
(B) Active power demand and magnitude of voltage at the bus
(C) Reactive power demand and phase angle of the voltage
(D) Magnitude of the voltage and phase angle of the voltage

Common Data for Questions 13 and 14:

The fuel cost of a system having two plants are given by

$$C_1 = (100 + 2P_1 + 0.005P_1^2) \text{ Rupees}$$

$$C_2 = (100 + 2P_2 + 0.01P_2^2) \text{ Rupees}$$

where P_1 and P_2 are in MW.

The loss coefficients are

$$B_{11} = 0.1 \times 10^{-2} \text{ MW}^{-1}$$

$$B_{12} = -0.09 \times 10^{-2} \text{ MW}^{-1}$$

$$B_{22} = 0.09 \times 10^{-2} \text{ MW}^{-1}$$

13. If the system supplies a load of 600 MW, the scheduling of two plants will be
(A) $P_1 = 266.7 \text{ MW}$, $P_2 = 333.3 \text{ MW}$
(B) $P_1 = 333.3 \text{ MW}$, $P_2 = 266.7 \text{ MW}$
(C) $P_1 = 400 \text{ MW}$, $P_2 = 200 \text{ MW}$
(D) $P_1 = 200 \text{ MW}$, $P_2 = 400 \text{ MW}$
14. The penalty factor of plant 1 is
(A) 1.79 (B) 1.92 (C) 1.25 (D) 1.36
15. The instantaneous value of current interrupted is 8 A. If the inductance and capacitance of a system are 2.0 H and 0.025 μF , voltage across the breaker contacts will be
(A) 71.55 V (B) 71.55 kV
(C) 5.155 kV (D) 715.5 kV
16. A 10 A, 3 second overcurrent relay having a current setting of 125% and a time setting multiplier of 0.4 is

connected to supply circuit through a 400/10 current transformer. When the circuit carries a fault current of 4000 A, operating time is?

Given

PSM	4	6	8	1	0
TSM	2.8	3.2	3.5	1.2	0.8

- (A) 1.4 s (B) 2.2 s (C) 3.6 s (D) 4.25 s
17. A percentage differential relay is employed for protecting a 80 MVA, 110/66 kV, Δ/Y , three-phase power transformer.
The ratios of primary and secondary currents in the CT's located in delta and wye side of power transformer are 400/10 A and 1100/10 A, respectively. The relay current at full load is
(A) -0.5219 A (B) 0.12 A
(C) 0.68 A (D) 6.5 A
18. The bus bars of a power station are in two sections P and Q separated by a reactor. Section P has a 20 MVA generator of 10% reactance and Q has two generators; one is 25MVA with 13% reactance and other 10 MVA with 8% reactance. Feeders are connected to bus bars through transformers, each rated 10 MVA and 6% reactance. Maximum short-circuit MVA with which the circuit breakers on the outgoing side of the transformers have to deal [The reactor is rated at 15MVA and 16% reactance] is
(A) 150 MVA (B) 163 MVA
(C) 180 MVA (D) 100 MVA
19. For a 132 kV system, the inductive reactance and capacitance up to the location of circuit breaker is 10 Ω and 0.06 μF .
A resistance of 300 Ω is connected across the contacts. The value of resistance which will give damped frequency of oscillation of $1/4^{\text{th}}$ natural frequency
(A) 128.65 Ω (B) 363.37 Ω
(C) 280.56 Ω (D) 500.08 Ω

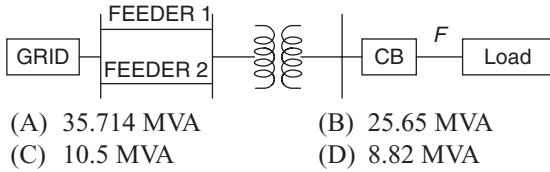
20. A 50 Hz three-phase alternator with grounded neutral has an inductance of 1.4 mH per phase and is connected to the bus bars through a circuit breaker. The capacitance to earth of the circuit between the alternator and CB is 0.00165 μF per phase. Time to attain maximum RRRV is
(A) 1.85 μs (B) 2.38 μs
(C) 1.75 μs (D) 3.55 μs
21. Consider grid as infinite bus. Choose 10 MVA as base Transformer: Three-phase 66/33 kV, 10 MVA,

$$0.02 + j0.06 \text{ p.u. impedance}$$

Load: Three-phase, 11 kV, 6000 kVA, 0.8 lag, $j0.6 \text{ p.u. impedance}$.

Impedance of each feeder = $8 + j16 \Omega$.

The required MVA rating of CB is



22. In a short-circuit test on a circuit breaker, time to reach the peak restriking voltage is 85 ms and the peak restriking voltage is 150 kV. The average rate of rise of restriking voltage is

(A) 1.76 kV/ μ S (B) 2.08 kV/ μ S
(C) 0.82 kV/ μ S (D) 5.04 kV/ μ S

Practice Problems 2

Directions for questions 1 to 15: Select the correct alternative from the given choices.

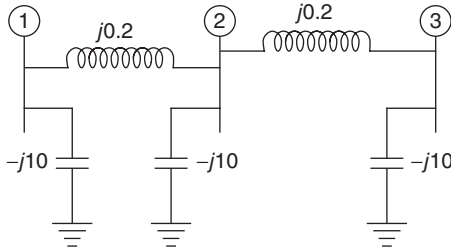
1. The power generated by two plants are $P_1 = 80$ MW and $P_2 = 100$ MW

If the loss coefficients are $B_{11} = 0.002$, $B_{22} = 0.004$ and $B_{12} = -0.0008$

The power loss is

(A) 12.0 MW (B) 16.4 MW
(C) 15.4 MW (D) 10.0 MW

2. The network shown in the figure has impedances in p.u. as indicated. The diagonal element Y_{22} of the bus admittance matrix Y_{bus} of the network is

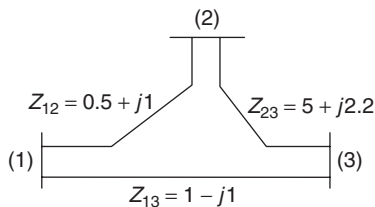


(A) $-j9.6$ (B) $+j0.104$
(C) $+j9.9$ (D) $-j9.9$

3. If X is the reactance of a line and R is the resistance of the line, the power transferred is maximum when
(A) $X = R$ (B) $X = 2R$
(C) $X = \sqrt{3}R$ (D) $X = \sqrt{2}R$

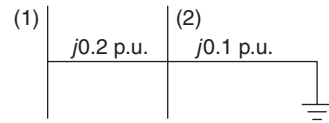
4. The fuel cost of two or more plants operating in parallel is minimum when
(A) The penalty factor of each plant is the same.
(B) Incremental fuel cost of each plant is the same.
(C) The incremental fuel cost multiplied by its penalty factor for each plant is the same.
(D) Ratio of incremental fuel cost to penalty factor of each plant is the same.

5. The value of Y_{11} of the bus admittance matrix for the network shown is



(A) $0.9 - j0.3$ (B) $0.1 - j0.3$
(C) $1 - j0.3$ (D) $0.9 - j0.8$

6. The bus admittance matrix for the network shown below is



(A) $\begin{bmatrix} j5 & j5 \\ j5 & -j13 \end{bmatrix}^{-1}$ (B) $\begin{bmatrix} j5 & j5 \\ j5 & -j15 \end{bmatrix}$
(C) $\begin{bmatrix} j0.2 & j0.1 \\ -j0.2 & -j0.1 \end{bmatrix}$ (D) $\begin{bmatrix} j10 & -j15 \\ j5 & j15 \end{bmatrix}$

7. For a generator rated at 30 MVA, 22 kV, 3-cycle circuit breaker is employed. The interrupting current is [Take $X'_d = 30\%$, $E_g = 1$ p.u.]

(A) 4.8 kA (B) 3.15 kA
(C) 5.6 A (D) None of these

8. For a relay, the operating speed depends upon,

(A) Spring tension
(B) Armature core air gap
(C) Rate of flux build-up
(D) All of these

9. For a system, the normal line current is 120 A. The circuit is designed to operate at 125% of normal line current and the trip coil is connected through a CT of ratio of 100:1. The trip mechanism should be set to operate at

(A) 1.5 A (B) 2.5 A
(C) 1.2 A (D) 2 A

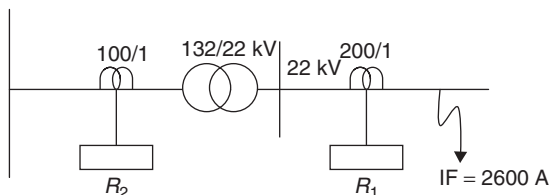
10. For universal relays, the torque equation is

(A) $K_3 VI \cos(\theta - \delta)$ (B) $K_1 I^2 - K_2 V^2$
(C) $\frac{V}{I} = \sqrt{\frac{K_1}{K_2}}$ (D) None of these

11. The relay which is operated depending on the ratio of applied voltage to current is

(A) Distance relay
(B) Impedance relay
(C) Induction type overcurrent relay
(D) Both (B) and (A)

12. The shunt resistance connected across the breaker contacts
- (1) Modifies the oscillatory restriking voltage to a periodic wave.
 - (2) Should have a value of $\frac{1}{2} \sqrt{\frac{L}{C}}$ for critical damping.
 - (3) RRRV will be increased
 - (4) Results in increase in rupturing capacity
- Which of the statements is correct?
- (A) 1, 2 (B) 1, 2 and 3
(C) 1, 2 and 4 (D) 1, 2, 3 and 4
13. Maximum value of restriking voltage for a 132 kV system with 3Ω reactance is
- (A) 126.5 V (B) 180.5 kV
(C) 215.55 kV (D) 208.56 kV
14. Fault current level at 22 kV side is 2600 A; CT ratio at 22 kV side is 200:1 and 132 kV side is 100:1. Two relays shown in the figure are set at 125% plug setting. The time of operation of relay R_1 when the time grading margin is 0.6 s and TMs for relay 1150.15 is



Given: relay characteristic curve

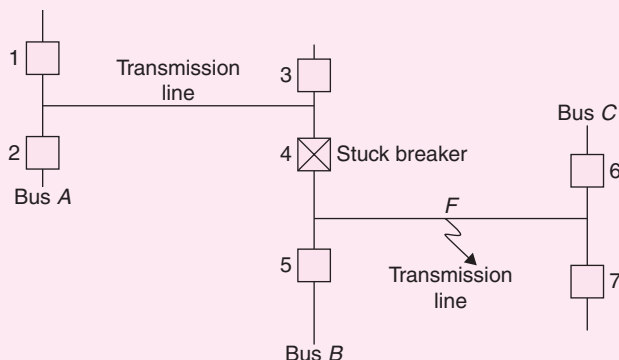
PSM:	1.3	2	4	8	10	20
TSM:	30	10	5	3.3	3	2.2
(A)	0.68 s				(B)	0.25 s
(C)	1.8 s				(D)	0.39 s

15. Which of the following are correct?

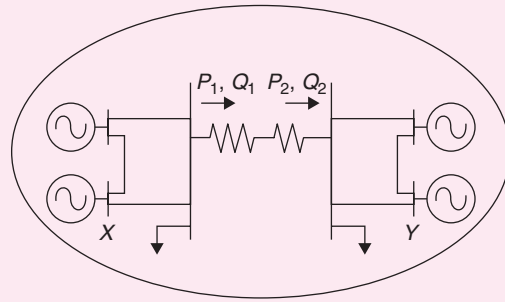
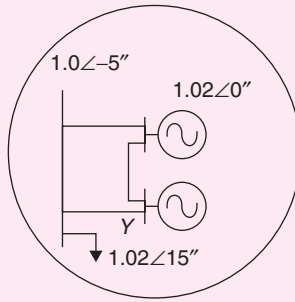
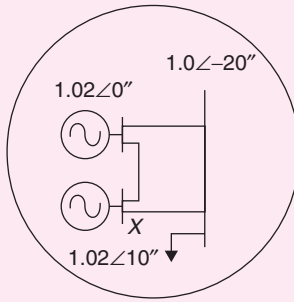
- (1) Relay time is the interval between the occurrence of the fault and up to the activated signal given to circuit breaker.
 - (2) Time interval between closure of trip circuit and final arc extinction is called breaker time.
 - (3) Time elapsed between the instant of occurrence of fault and final arc extinction is called fault clearing time.
 - (4) Time during which stored energy is dissipated after the characteristic quantity has been suddenly restored is reset time
- (A) 1, 2 (B) 1, 2 and 3
(C) 1 and 3 (D) 1, 2 and 4

PREVIOUS YEARS' QUESTIONS

1. Consider the protection system shown in the figure below. The circuit breakers, numbered from 1 to 7 are of identical type. A single line to ground fault with zero fault impedance occurs at the midpoint of the line (at point F), but circuit breaker 4 fails to operate ('stuck breaker'). If the relays are coordinated correctly, a valid sequence of circuit breaker operations is [2007]



- (A) 1, 2, 6, 7, 3, 5
(B) 1, 2, 5, 6, 7, 3
(C) 5, 6, 7, 3, 1, 2
(D) 5, 1, 2, 3, 6, 7
2. Consider the two power systems shown in figure A below, which are initially not interconnected, and are operating in steady state at the same frequency. Separate load flow solutions are computed individually for the two systems, corresponding to the scenario. The bus voltage phasors so obtained are indicated on figure A. These two isolated systems are now interconnected by a short transmission line as shown in figure B, and it is found that $P_1 = P_2 = Q_1 = Q_2 = 0$: [2007]



The bus voltage phase angular difference between generator bus X and generator bus Y after the inter-connection is

- (A) 10° (B) 25° (C) -30° (D) 30°

3. A lossless power system has to serve a load of 250 MW. There are two generators (G_1 and G_2) in the system with cost curves C_1 and C_2 , respectively, defined as follows

$$C_1(P_{G1}) = P_{G1} + 0.055 \times P_{G1}^2$$

$$C_2(P_{G2}) = 3P_{G2} + 0.03 \times P_{G2}^2$$

where P_{G1} and P_{G2} are the MW injections from generator G_1 and G_2 , respectively. Then the minimum cost dispatch will be

[2008]

- (A) $P_{G1} = 250$ MW; $P_{G2} = 0$ MW
 (B) $P_{G1} = 150$ MW; $P_{G2} = 100$ MW
 (C) $P_{G1} = 100$ MW; $P_{G2} = 150$ MW
 (D) $P_{G1} = 0$ MW; $P_{G2} = 250$ MW

4. For the Y -bus matrix of a 4-bus system given in per unit, the buses having shunt elements are $Y_{BUS} =$

$$j \begin{bmatrix} -5 & 2 & 2.5 & 0 \\ 2 & -10 & 2.5 & 4 \\ 2.5 & 2.5 & -9 & 4 \\ 0 & 4 & 4 & -8 \end{bmatrix}$$

[2009]

- (A) 3 and 4 (B) 2 and 3
 (C) 1 and 2 (D) 1, 2 and 4
5. A 500 MW, 21 kV, 50 Hz, three-phase, 2-pole synchronous generator having a rated p.f. = 0.9, has a moment of inertia of 27.5×10^3 kg-m². The inertia constant (H) will be

[2009]

- (A) 2.44 s (B) 2.71 s
 (C) 4.88 s (D) 5.42 s

6. Voltage phasors at the two terminals of a transmission line of length 70 km have a magnitude of 1.0 per unit but are 180 degrees out of phase. Assuming that the maximum load current in the line is $\frac{1}{5}$ th of minimum three-phase fault current, which one of the following transmission line protection schemes will NOT pick up for this condition?

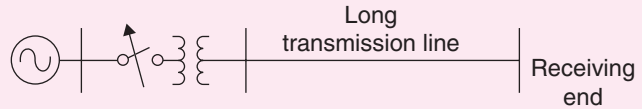
[2008]

- (A) Distance protection using mho relays with zone-1 set to 80% of the line impedance

- (B) Directional overcurrent protection set to pick up at 1.25 times the maximum load current
 (C) Pilot relaying system with directional comparison scheme
 (D) Pilot relaying system with segregated phase comparison scheme

7. A 50 Hz synchronous generator is initially connected to a long lossless transmission line which is open-circuited at the receiving end. With the field voltage held constant, the generator is disconnected from the transmission line. Which of the following may be said about the steady state terminal voltage and field current of the generator?

[2010]



- (A) The magnitude of terminal voltage decreases, and the field current does not change.
 (B) The magnitude of terminal voltage increases, and the field current does not change.
 (C) The magnitude of terminal voltage increases, and the field current increases.
 (D) The magnitude of terminal voltage does not change, and the field current decreases.

8. Consider two buses connected by an impedance of $(0 + j5) \Omega$. The bus 1 voltage is $100\angle 30^\circ$ V, and bus 2 voltage is $100\angle 0^\circ$ V. The real and reactive powers supplied by bus 1, respectively, are

[2010]

- (A) 1000 W, 268 VAr (B) -1000 W, -134 VAr
 (C) 276.9 W, -56.7 VAr (D) -276.9 W, 56.7 VAr

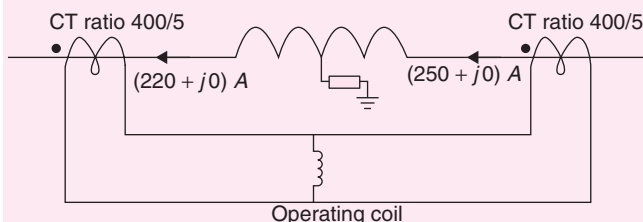
9. A three-phase, 33 kV oil-circuit breaker is rated 1200 A, 2000 MVA, 3 s. The symmetrical breaking current is

[2010]

- (A) 1200 A (B) 3600 A
 (C) 35 kA (D) 104.8 kA

10. Consider a stator winding of an alternator with an internal high-resistance ground fault. The currents under the fault condition are as shown in the figure. The winding is protected using a differential current scheme with current transformers of ratio 400/5 A as shown. The current through the operating coil is

[2010]

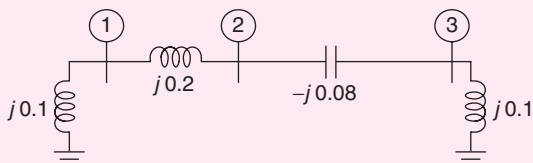


- (A) 0.17875 A (B) 0.2 A
(C) 0.375 A (D) 60 KA

11. A negative sequence relay is commonly used to protect [2011]

- (A) An alternator (B) A transformer
(C) A transmission line (D) A busbar

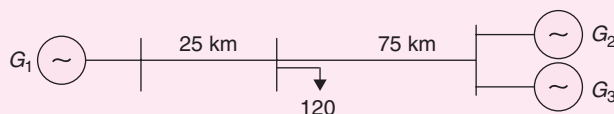
12. A three-bus network is shown in the figure below indicating the p.u. impedances of each element. [2011]



The bus admittance matrix, Y -bus of the network is

- (A) $j \begin{bmatrix} 0.3 & 0.2 & 0 \\ -0.2 & 0.12 & 0.08 \\ 0 & 0.08 & 0.02 \end{bmatrix}$
(B) $j \begin{bmatrix} -15 & 5 & 0 \\ 5 & 7.5 & -12.5 \\ 0 & 12.5 & 2.5 \end{bmatrix}$
(C) $j \begin{bmatrix} 0.1 & 0.2 & 0 \\ 0.2 & 0.12 & -0.08 \\ 0 & -0.08 & 0.10 \end{bmatrix}$
(D) $j \begin{bmatrix} 10 & 5 & 0 \\ 5 & 7.5 & 12.5 \\ 0 & 12.5 & -10 \end{bmatrix}$

13. A load centre of 120 MW derives power from two power stations connected by 220 kV transmission lines of 25 km and 75 km as shown in the figure below. The three generators G_1 , G_2 and G_3 are of 100 MW capacity each and have identical fuel cost characteristics. The minimum loss generation schedule for supplying the 120 MW load is [2011]



- (A) $P_1 = 80 \text{ MW} + \text{losses}$
 $P_2 = 20 \text{ MW}$
 $P_3 = 20 \text{ MW}$
(B) $P_1 = 60 \text{ MW}$
 $P_2 = 30 \text{ MW} + \text{losses}$

- $P_3 = 30 \text{ MW}$
(C) $P_1 = 40 \text{ MW}$
 $P_2 = 40 \text{ MW}$
 $P_3 = 40 \text{ MW} + \text{losses}$
(D) $P_1 = 30 \text{ MW} + \text{losses}$
 $P_2 = 45 \text{ MW}$
 $P_3 = 45 \text{ MW}$

14. The figure shows a two-generator system supplying a load of $P_D = 40 \text{ MW}$, connected at bus 2.



The fuel cost of generators G_1 and G_2 are:

$C_1(P_{G1}) = 10,000 \text{ Rs/MWh}$ and $C_2(P_{G2}) = 12,500 \text{ Rs/MWh}$ and the loss in the line is $P_{\text{loss(p.u.)}} = 0.5 P_{G1}^2$, where the loss coefficient is specified in p.u. on a 100 MVA base. The most economic power generation schedule in MW is [2012]

- (A) $P_{G1} = 20, P_{G2} = 22$
(B) $P_{G1} = 22, P_{G2} = 20$
(C) $P_{G1} = 20, P_{G2} = 20$
(D) $P_{G1} = 0, P_{G2} = 40$

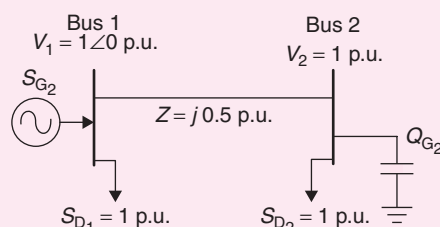
15. The bus admittance matrix of a three-bus-line system is

$$Y = j \begin{bmatrix} -13 & 10 & 5 \\ 10 & -18 & 10 \\ 5 & 10 & -13 \end{bmatrix}$$

If each transmission line between the two buses is represented by an equivalent π -network, the magnitude of the shunt susceptance of the line connecting bus 1 and 2 is [2012]

- (A) 4 (B) 2
(C) 1 (D) 0

16. For the system shown below, S_{D1} and S_{D2} are complex power demands at bus 1 and bus 2, respectively. If $|V_2| = 1 \text{ p.u.}$, the VAR rating of the capacitor (Q_{G2}) connected at bus 2 is [2012]

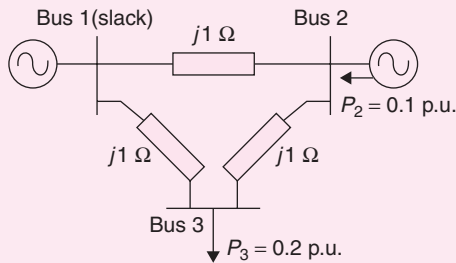


- (A) 0.2 p.u. (B) 0.268 p.u.
(C) 0.312 p.u. (D) 0.4 p.u.

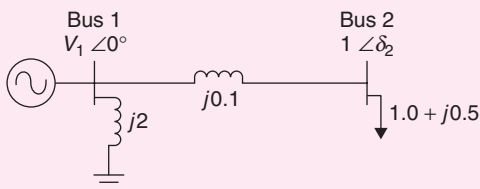
17. For a power system network with n nodes, Z_{33} of its bus impedance matrix is $j0.5$ per unit. The voltage at node 3 is $1.3 \angle -10^\circ$ per unit. If a capacitor having reactance of $-j3.5$ per unit is now added to the network between node 3 and the reference node, the current drawn by the capacitor per unit is [2013]
- (A) $0.325 \angle -100^\circ$ (B) $0.325 \angle 80^\circ$
 (C) $0.371 \angle -100^\circ$ (D) $0.433 \angle 80^\circ$

Common Data for Questions 18 and 19:

In the following network, the voltage magnitudes at all buses are equal to 1 p.u., the voltage phase angles are very small, and the line resistance are negligible. All the line reactances are equal to $j1 \Omega$.

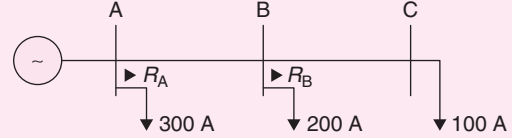


18. The voltage-phase angles in radians at buses 2 and 3 are [2013]
- (A) $\theta_2 = -0.1, \theta_3 = -0.2$ (B) $\theta_2 = 0, \theta_3 = -0.1$
 (C) $\theta_2 = 0.1, \theta_3 = 0.1$ (D) $\theta_2 = 0.1, \theta_3 = 0.2$
19. If the base impedance and the line-to-line base voltage are 100 Ω and 100 kV, respectively, then the real power in MW delivered by the generator connected at the slack bus is [2013]
- (A) -10 (B) 0 (C) 10 (D) 20
20. The fuel cost functions of two power plants are
 Plant P_1 : $C_1 = 0.05 P_{g1}^2 + AP_{g1} + B$
 Plant P_2 : $C_2 = 0.10 P_{g2}^2 + 3AP_{g2} + 2B$
 where P_{g1} and P_{g2} are the generated powers of two plants, and A and B are the constants. If the two plants optimally share 1000 MW load at incremental fuel cost of 100 Rs/MWh, the ratio of load shared by plants P_1 and P_2 is [2014]
- (A) 1:4 (B) 2:3
 (C) 3:2 (D) 4:1
21. A two-bus power system shown in the figure supplies load of $1.0 + j0.5$ p.u.



The values of V_1 in p.u. and δ_2 , respectively, are [2014]

- (A) 0.95 and 6.00° (B) 1.05 and -5.44°
 (C) 1.1 and -6.00° (D) 1.1 and -27.12°
22. The overcurrent relays for the line protection and loads connected at the buses are shown in the figure.



The relays are IDMT in nature having the characteristic

$$t_{op} = \frac{0.14 \times \text{Time Multiplier Setting}}{(\text{Plug Setting Multiplier})^{0.02} - 1}$$

The maximum and minimum fault currents at bus B are 2000 A and 500 A, respectively. Assuming the time multiplier setting and plug setting for relay R_B to be 0.1 and 5 A, respectively, the operating time of R_B (in seconds) is [2014]

23. A 183-bus power system has 150 PQ buses and 32 PV buses. In the general case, to obtain the load flow solution using Newton-Raphson method in polar coordinates, the minimum number of simultaneous equations to be solved is [2014]
24. Consider the economic dispatch problem for a power plant having two generating units. The fuel costs in Rs/MWh along with the generation limits for the two units are given below: [2015]
- $C_1(P_1) = 0.01 P_1^2 + 30P_1 + 10$; $100 \text{ MW} \leq P_1 \leq 150 \text{ MW}$
 $C_2(P_2) = 0.05 P_2^2 + 10P_2 + 10$; $100 \text{ MW} \leq P_2 \leq 180 \text{ mW}$
- The incremental cost (in Rs/MWh) of the power plant when it supplies 200 MW is [2015]
24. Determine the correctness or otherwise of the following Assertion [a] and the Reason [r].
 Assertion: Fast decouples load flow method gives approximate load flow solution because it uses several assumptions.
 Reason: Accuracy depends on the power mismatch vector tolerance. [2015]
- (A) Both [a] and [r] are true and [r] is the correct reason for [a].
 (B) Both [a] and [r] are true but [r] is not the correct reason for [a].
 (C) Both [a] and [r] are false.
 (D) [a] is false and [r] is true.

26. A 3-bus power system network consists of 3 transmission lines. The bus admittance matrix of the uncompensated system is

$$\begin{bmatrix} -j6 & j3 & j4 \\ j3 & -j7 & j5 \\ j4 & j5 & -j8 \end{bmatrix} \text{ pu.}$$

If the shunt capacitance of all transmission lines is 50% compensated, the imaginary part of the 3rd row 3rd column element (in pu) of the bus admittance matrix after compensation is [2015]

- (A) $-j7.0$ (B) $-j8.5$
(C) $-j7.5$ (D) $-j9.0$
27. The incremental costs (in Rupees/MWh) of operating two generating units are functions of their positive powers P_1 and P_2 in MW, and are given by

$$\frac{dC_1}{dP_1} = 0.2P_1 + 50$$

$$\frac{dC_2}{dP_2} = 0.24P_2 + 40$$

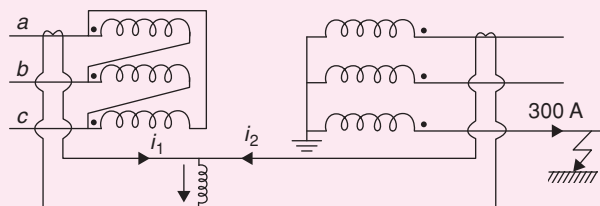
Where

$$20 \text{ MW} \leq P_1 \leq 150 \text{ MW}$$

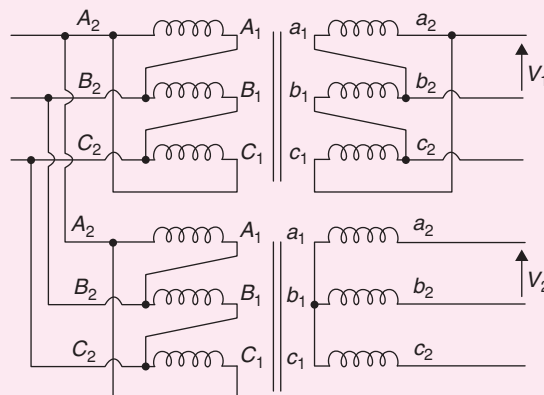
$$20 \text{ MW} \leq P_2 \leq 150 \text{ MW.}$$

For a certain load demand, P_1 and P_2 have been chosen such that $dC_1/dP_1 = 76$ Rs/MWh and $dC_2/dP_2 = 68.8$ ₹/MWh. If the generations are rescheduled to minimize the total cost, then P_2 is _____. [2015]

28. A 3-phase transformer rated for 33 kV/11 kV is connected in delta/star as shown in figure. The current transformers (CTs) on low and high voltage sides have a ratio of 500/5. Find the currents i_1 and i_2 , if the fault current is 300 A as shown in figure. [2015]

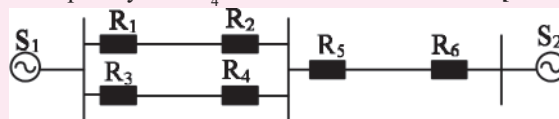


- (A) $i_1 = 1/\sqrt{3}$ A, $i_2 = 0$ A
(B) $i_1 = 0$ A, $i_2 = 0$ A
(C) $i_1 = 0$ A, $i_2 = 1/\sqrt{3}$ A
(D) $i_1 = 1/\sqrt{3}$ A, $i_2 = 1/\sqrt{3}$ A
29. Two three-phase transformers are realized using single-phase transformers as shown in the figure.



The phase difference (in degree) between voltages V_1 and V_2 is _____. [2015]

30. In a 100 bus power system, there are 10 generators. In a particular iteration of Newton Raphson load flow technique (in polar coordinates), two of the PV buses are converted to PQ type. In this iteration, [2016]
- (A) the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes increases by two.
(B) the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes increases by two.
(C) the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes decreases by two.
(D) the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes decreases by two.
31. A power system with two generators is shown in the figure below. The system (generators, buses and transmission lines) is protected by six over current relays R_1 to R_6 . Assuming a mix of directional and non directional relays at appropriate locations, the remote backup relays for R_4 are [2016]



- (A) R_1, R_2 (B) R_2, R_6
(C) R_2, R_5 (D) R_1, R_6
32. A power system has 100 buses including 10 generator buses, for the load flow analysis using Newton – Raphson, method in polar coordinates, the size of the Jacobian is [2016]
- (A) 189×189 (B) 100×100
(C) 90×90 (D) 180×180

ANSWER KEYS**EXERCISES****Practice Problems 1**

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. B | 4. B | 5. A | 6. B | 7. B | 8. C | 9. C | 10. B |
| 11. A | 12. D | 13. C | 14. A | 15. B | 16. A | 17. A | 18. B | 19. B | 20. B |
| 21. A | 22. A | | | | | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|------|------|------|------|-------|
| 1. D | 2. D | 3. C | 4. C | 5. A | 6. B | 7. B | 8. D | 9. A | 10. A |
| 11. D | 12. C | 13. C | 14. C | 15. B | | | | | |

Previous Years' Questions

- | | | | | | | | | | |
|-------|----------|---------|--------|-------|-------|----------------|-------|-------|--------|
| 1. C | 2. A | 3. C | 4. C | 5. A | 6. A | 7. B | 8. A | 9. C | 10. C |
| 11. A | 12. B | 13. A | 14. A | 15. B | 16. B | 17. D | 18. B | 19. C | 20. D |
| 21. B | 22. 0.23 | 23. 332 | 24. 20 | 25. D | 26. B | 27. 135 to 137 | | 28. A | 29. 30 |
| 30. B | 31. D | 32. A | | | | | | | |

Chapter 4

Power Systems Stability

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- Steady state stability
- The swing equation
- Multi-machine systems
- Steady state stability analysis
- Transient stability analysis
- Equal area criterion
- Principal type of transient disturbances
- Sudden change in mechanical input
- Switching operation
- 3-phase

INTRODUCTION

The stability of a system refers to the ability of a system to return back to its steady state when subjected to a disturbance. Power is generated by synchronous generators that operate in synchronism with the rest of the system. A generator is synchronized with a bus when both of them have same frequency, voltage and phase sequence. So, the stability is also defined as the ability of the power system to return to steady state without losing synchronism. The power system stability is categorized as follows:

1. Steady-state stability
2. Transient stability
3. Dynamic stability

Steady-state Stability

Steady-state stability is restricted to small and gradual changes in the system operating conditions. This stability basically concentrates on restricting the bus voltage close to their nominal values. Steady-state stability ensures that phase angle between two buses are not too large and check for the overloading of the power equipment and transmission lines.

Transient Stability

Transient stability involves study of the power system following a major disturbance. Following a large disturbance the synchronous alternator power angle changes due to sudden acceleration of the rotor shaft.

Dynamic Stability

The ability of a power system to maintain stability under continuous small disturbances is investigated under the name of dynamic stability. It is also known as small signal stability.

Power-angle Relationship

Consider the single machine infinite bus system shown in the figure. In this the reactance ' X ' includes the reactance of the transmission line and the synchronous reactance or the transient reactance of the generator. The sending end voltage is then the internal emf of the generator. Let the sending end and receiving end voltages be given by

$$V_s = V_1 \angle \delta, V_R = V_2 \angle 0^\circ$$

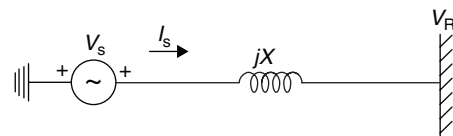


Figure 1 An SMIB system

We then have

$$I_s = \frac{V_1 \angle \delta - V_2}{jX} = \frac{V_1 \cos \delta - V_2 + jV_1 \sin \delta}{jX}$$

The sending end real power and reactive power are given by

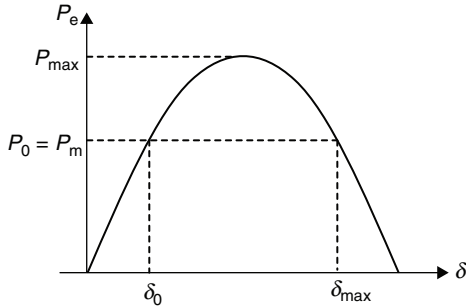
$$P_s + jQ_s = V_s I_s^* = V_1 (\cos \delta + j \sin \delta) \frac{V_1 \cos \delta - V_2 + jV_1 \sin \delta}{jX}$$

$$P_s + jQ_s = \frac{V_1 V_2 \sin \delta + j(V_1^2 - V_1 V_2 \cos \delta)}{X}$$

Since the line is lossless, the real power dispatched from the sending end is equal to the real power received at the receiving end. We can therefore write.

$$P_e = P_s = P_R = \frac{V_1 V_2}{X} \sin \delta = P_{\max} \sin \delta$$

where $P_{\max} = \frac{V_1 V_2}{X}$ is the maximum power that can be transmitted over the transmission line. The power angle curve is shown in figure below



From the figure, we can see that for a given power ' P_0 ' there are two possible values of the angle $\delta \rightarrow \delta_0$ and $\delta_{\max}$.

The angles are given by

$$\delta_0 = \sin^{-1} \left(\frac{P_0}{P_{\max}} \right)$$

$$\delta_{\max} = 180^\circ - \delta_0$$

The Swing Equation

Figure shows the torque, speed and flow of mechanical and electrical powers in a synchronous machine. It is assumed that the windage, friction and iron-loss are neglected. The differential equation governing the rotor dynamics can be written as

$$J \cdot \frac{d^2 \theta_m}{dt^2} = T_m - T_e$$

where θ_m = Angle in rad (mechanical)

T_m = Turbine torque in Nm

T_e = Electromagnetic torque developed in Nm

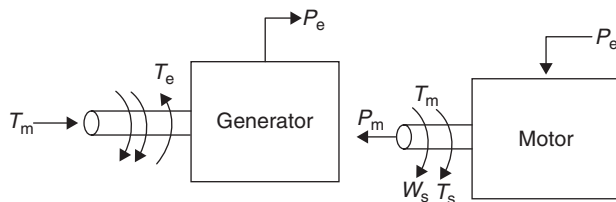


Figure 2 Flow of mechanical and electrical power in a synchronous machine

$$J \omega_{sm} \cdot \frac{d^2 \theta_m}{dt^2} = P_m - P_e$$

ω_{sm} = Synchronous speed

P_m = Mechanical power input in MW

P_e = Electrical power output in MW

$$J \left(\frac{Z}{P} \right)^2 \times \omega_s \cdot \frac{d^2 \theta_e}{dt^2} = P_m - P_e$$

$$M \cdot \frac{d^2 \theta_e}{dt^2} = P_m - P_e$$

where M = Moment of inertia = $J \cdot \left(\frac{2}{P} \right)^2 \omega_s$

If the angular position of the rotor is with respect to the synchronous rotating reference frame

$$\text{Torque angle } (\delta) = \theta_e - \omega_s t$$

$$\frac{d^2 \delta}{dt^2} = \frac{d^2 \theta_e}{dt^2}$$

$$\therefore M \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e$$

$$\frac{GH}{\pi f} \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e$$

$$\frac{H}{\pi f} \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e \text{ in p.u. of machine base ratings}$$

$$M \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e,$$

$$\text{where } M_{\text{p.u.}} = \frac{H}{\pi f}$$

The above equation is called the swing equation.

Multi-machine Systems

In a stability study of a power system with many synchronous machines only one MVA base common to all parts of the system can be chosen. This is accomplished by converting ' H ' for each machine based on its own individual rating to a value determined by the system base

$$H_{\text{system}} = H_{\text{mech}} \cdot \frac{S_{\text{mech}}}{S_{\text{system}}}$$

Consider a power plant with two generators connected to the same bus which is electrically remote from the network disturbances. The swing equations on the common base are

$$\frac{2H_1}{\omega_s} \cdot \frac{d^2 \delta_1}{dt^2} = P_{m_1} - P_{e_1} \text{ per unit}$$

$$\frac{2H_2}{\omega_s} \cdot \frac{d^2 \delta_2}{dt^2} = P_{m_2} - P_{e_2} \text{ per unit}$$

If both the machines swing together, adding the two equations and denoting δ_1 and δ_2 by ' δ '

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$$\frac{2H}{\omega_s} \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e \text{ per unit}$$

where $H = (H_1 + H_2)$, $P_m = P_{m_1} + P_{m_2}$,

$$P_e = P_{e_1} + P_{e_2}$$

For any pair of non-coherent machines in a system, the resultant two machine swing equation can be written as

$$\frac{d^2 \delta_1}{dt^2} - \frac{d^2 \delta_2}{dt^2} = \frac{\omega_s}{2} \left(\frac{P_{m_1} - P_e}{H_1} - \frac{P_{m_2} - P_{e_2}}{H_2} \right)$$

$$\frac{2}{\omega_s} \left(\frac{H_1 H_2}{H_1 + H_2} \right) \frac{d^2 (\delta_1 - \delta_2)}{dt^2} = \frac{P_{m_1} H_2 - P_{m_2} H_1}{H_1 + H_2} - \frac{P_{e_1} H_2 - P_{e_2} H_1}{H_1 + H_2}$$

which can be written as

$$\frac{2}{\omega_s} \cdot H_{12} \cdot \frac{d^2 \delta_{12}}{dt^2} = P_{m_{12}} - P_{e_{12}}$$

where $\delta_{12} = \delta_1 - \delta_2$

$$H_{12} = \frac{H_1 H_2}{H_1 + H_2}$$

$$P_{m_{12}} = \frac{P_{m_1} H_2 - P_{m_2} H_1}{H_1 + H_2}$$

$$P_{e_{12}} = \frac{P_{e_1} H_2 - P_{e_2} H_1}{H_1 + H_2}$$

STEADY-STATE STABILITY ANALYSIS

Assumptions made in steady-state stability analysis:

1. The study considers small amplitude, long duration disturbances.
2. The damping term in the characteristic equation is absent, because of assumption of a lossless system and neglecting the effect of damper windings.
3. Non-linearities are ignored, and hence the linearized form of the swing equation can be used.
4. The response of the governor and the exciter are ignored. This results in the mechanical power and electrical power to be constant throughout the transient period.

In a single machine infinite bus system, dynamics of a synchronous machine are described by the swing equation.

$$M \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e$$

where $M = \frac{H}{\pi f}$ and $P_e = \frac{|E||V|}{X_{12}} \sin \delta = P_{\max} \sin \delta$

Let the system be operating initially at equilibrium. Then the steady-state power transfer $P_{e_0} = P_m$ with losses neglected and a rotor angle of δ_0 . Assume a small increment ΔP in the

electric power output with the mechanical input P_m as fixed. This change causes the torque angle to change to $(\delta_0 + \Delta \delta)$. From the small disturbance, linearized equation can be written as

$$\Delta P_e = \left(\frac{\partial P_e}{\partial \delta_0} \right) \Delta \delta$$

Swing equation can be written as

$$M \cdot \frac{d^2 \Delta \delta}{dt^2} = P_m - (P_{e_0} + \Delta P_e) = -\Delta P_e$$

$$M \frac{d^2 \Delta \delta}{dt^2} + \left[\frac{\partial P_e}{\partial \delta} \right] \Delta \delta = 0$$

Characteristic equation is given by

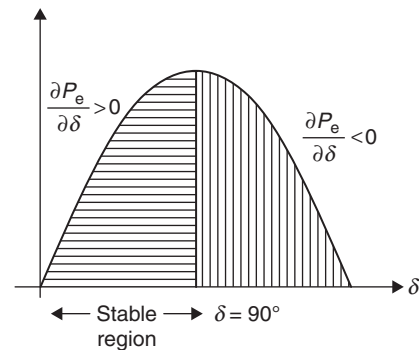
$$MD^2 + \left(\frac{\partial P_e}{\partial \delta_0} \right) = 0$$

Roots of the characteristic equation are

$$D = \pm \left[\frac{-\left(\frac{\partial P_e}{\partial \delta} \right)_0}{M} \right]^{\frac{1}{2}}$$

When $\frac{\partial P_e}{\partial \delta}$ is positive, the roots of the characteristic equation lie on the $j\omega'$ (imaginary) axis, and the system rotor angle behaviour is oscillating about ' δ_0 '. The line resistance and damper winding of machine which have been ignored in the above modelling cause the system oscillations to decrease. The system is therefore stable for a small increment in electrical power.

If $\frac{\partial P_e}{\partial \delta}$ is negative, the roots are real with one of them being positive and other negative but of equal magnitude which is 'unstable and causes rotor angle to increase without bound upon occurrence of a small power increment.



The term $\left(\frac{\partial P_e}{\partial \delta} \right)$ is known as 'synchronizing power coefficient'. This is also called stiffness of the synchronous machine.

The system is unstable if $\frac{\partial P_e}{\partial \delta} < 0$ which implies $\delta > 90^\circ$, so the system is unstable if $\delta > 90^\circ$. The maximum power

that can be transferred without loss of stability (steady state) occurs for $\delta = 90^\circ$ and it is given by

$$P_{\max} = \frac{EV}{X_{12}}$$

$P_{\max}$ is known as steady-state stability limit.

Methods for Improving Steady-state Stability Limit

1. Reduce the reactance of the transmission lines.
 - By adding parallel lines which also increases the reliability of the system.
 - Series capacitors which also provides better voltage regulation.
2. Increasing either or both $|E|$ and $|V|$ by means of excitation control.

Solved Examples

Example 1: A sending end bus transfer power P_0 through a transmission line of p.u. impedance 0.1. If the steady-state stability margin is 40% and the bus voltages at both the ends is 1.0 p.u. The operating power angle and magnitude of P_0 are

- (A) 36.87° , 6 p.u. (B) 36.87° , 4 p.u.
(C) 23.57° , 0.6 p.u. (D) 23.57° , 0.4 p.u.

Solution: (A)

Steady-state stability margin

$$= \frac{P_{\max} - P_0}{P_{\max}} \times 100$$

$$40 = \left(1 - \frac{P_0}{P_{\max}} \right) \times 100$$

$$0.4P_{\max} - P_{\max} = -P_0$$

$$P_0 = 0.6P_{\max} = P_{\max} \sin \delta$$

$$\delta = \sin^{-1}(0.6) = 36.87^\circ$$

$$\text{Magnitude of } P_0 = \frac{V_1 V_2}{X_{12}} \sin \delta_0$$

$$= \frac{1}{0.1} \sin \delta_0 = 6 \text{ p.u.}$$

Example 2: The sending end and receiving end voltages of a transmission line at 100 MW load are equal at 230 kV, per phase line impedance is $(4 + j8)$. Steady-state stability limit of the system is

- (A) 1090 MW/ph (B) 363.3 MW/ph
(C) 3,269.8 MW/ph (D) 1,887.8 MW/ph

Solution: (A)

When the line resistance is considered

$$P_{\max} = \frac{V_S V_R}{|Z|} - \frac{R|V_R|^2}{|Z|^2}$$

where

$$V_S = V_R = \frac{230}{\sqrt{3}} = 132.8 \text{ kV}$$

$$P_{\max} = \frac{(132.8)^2}{\sqrt{4^2 + 8^2}} - \frac{4 \times (132.8)^2}{(\sqrt{8^2 + 4^2})^2}$$

$$= 1971.74 - 881.8$$

$$= 1089.948 \cong 1090 \text{ MW/Ph}$$

Example 3: A 500 MVA, 11 kV, 50 Hz, 4-pole turbo-generator has an inertia constant of $H = 7.5$ MJ/MVA. If the mechanical power input is 552 MW and the electrical power output of 400 MW and stator copper loss is assumed to be negligible, then the angular acceleration is

- (A) 182.4 rpm/s²
(B) 364.8 rpm/s²
(C) 25.33 rpm/s²
(D) 50.66 rpm/s²

Solution: (C)

From the swing equation

$$\frac{H}{\pi f} \cdot \frac{d^2 \delta}{dt^2} = P_m - P_e \text{ in p.u.}$$

$$\frac{7.5}{180 \times 50} \frac{d^2 \delta}{dt^2} = \frac{552 - 400}{500}$$

$$\frac{d^2 \delta}{dt^2} = 364.8 \text{ elec degree/s}^2$$

For a four pole machine

$$\frac{d^2 \delta}{dt^2} = \frac{364.8}{2} = 182.4 \text{ mech.dec/s}^2$$

$$\text{Angular acceleration} = \frac{182.4}{360} \times 50$$

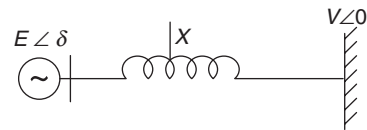
$$= 25.33 \text{ rpm/s}^2$$

Example 4: An alternator having a reactance of 1.3 p.u. is connected to an infinite bus as shown in the figure. It delivers 1.0 p.u. current at 0.8 p.f. lagging at $V = 1.0$ p.u. The steady-state stability limit of the system is

- (A) 0.58 (B) 1.584
(C) 1.3 (D) 1.7

Solution: (B)

System configuration given in the problem is given by



For the given power factor of 0.8, current delivered from the source is $I = (0.8 - j0.6) = 1 \angle -36.86^\circ$

$$\text{Sending end voltage } E = V + jIX_d$$

$$= 1.0 + j(0.8 - j0.6) \times 1.3$$

$$= 2.06 \angle 30.3^\circ$$

$$\text{Steady-state stability limit } P_{\max} = \frac{EV}{X_d} = \frac{2.06 \times 1}{1.3}$$

$$P_{\max} = 1.584 \text{ p.u.}$$

Example 5: A 250 MVA synchronous generator having inertia constant 5 MJ/MVA is connected in parallel with a 200 MVA generator having inertia constant of 4 MJ/MVA. Both the generators are swinging coherently. The equivalent inertia constant of the machine on a base of 100 MVA is

- (A) 20.5 MJ/MVA (B) 12.5 MJ/MVA
(C) 8 MJ/MVA (D) 9 MJ/MVA

Solution: (A)

Both the inertia constants are to be represented with same base

$$H_{\text{p.u.}} = \frac{H_{\text{p.u. old}}}{P_{\text{base new}}} \times P_{\text{base new}}$$

$$H_{\text{p.u.}} = \frac{5 \times 250}{100} = 12.5 \text{ MJ/MVA}$$

$$H_{2\text{p.u.}} = \frac{4 \times 200}{100} = 8 \text{ MJ/MVA}$$

$$H_{\text{eq}} = H_{1\text{p.u.}} + H_{2\text{p.u.}} = 12.5 + 8$$

$$= 20.5 \text{ MJ/MVA}$$

Example 6: A synchronous generator is connected to a 11 kV infinite bus through a transmission line. The reactances of the generator and transmission line are 1.1 and 0.7, respectively. The terminal voltage of the synchronous generator is 15 kV. If the generator delivers 75 MW power to the infinite bus, the load angle (δ) for stable operation is
(A) 55° (B) 125°
(C) Both A and B (D) 63.4°

Solution: (A)

Power angle equation is given by

$$P_e = \frac{|E||V|}{X_{12}} \sin \delta$$

$$75 = \frac{11 \times 15}{(1.1 + 0.7)} \sin \delta$$

$$\delta = 55^\circ$$

TRANSIENT STABILITY ANALYSIS

If the disturbance is large, changes in angular difference may be well enough to cause the loss of synchronism of the machines. The types of disturbances are mostly the faults, sudden. Large load changes may also cause instability of the machines. This type of instability is known as 'transient instability'.

Assumptions in Transient Stability Analysis

1. Resistance of the transmission lines and synchronous machine are neglected.
2. Synchronous machine damper winding is ignored which contributes damping term.
3. Rotor speed is assumed to be synchronous even though it varies insignificantly during the stability transients.
4. Mechanical input to machine is assumed to remain constant the transients.
5. Voltage behind transient reactance is assumed to remain constant.
6. Shunt capacitors ignored in stability study which does not cause any significant error.
7. Loads are modelled as constant admittances.

Equal Area Criterion

To examine the stability of a two machine system without solving the swing equation, a direct approach is given as follows.

Consider a swing equation.

$$H \frac{d^2 \delta}{dt^2} = P_m - P_e = P_a = \text{Accelerating power}$$

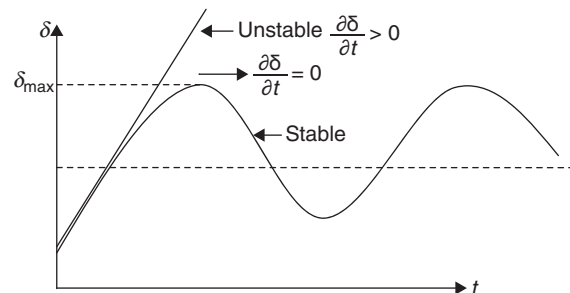
If the system is stable, $\delta(t)$ performs oscillations, whose aptitude decreases in actual practice because of damping terms (Which are neglected in the swing equation) on the other hand, if the system is unstable ' δ ' continues to increase indefinitely with time and the machine loses synchronism. So when the system is stable, then $\delta(t)$ will go to maximum and start to reduce. This fact can be stated as stability criterion, that the system is stable if at some time

$$\frac{d\delta}{dt} = 0$$

and system is unstable if

$$\frac{\partial \delta}{\partial t} > 0 \text{ for sufficiently long time.}$$

The stability criterion stated above can be converted into a simple and easily applicable form for a single machine infinite bus system.



Multiply both sides of swing equation by $\left(2 \cdot \frac{d\delta}{dt}\right)$, we get

$$2 \cdot \frac{d\delta}{dt} \cdot \frac{d^2\delta}{dt^2} = \frac{2P_a}{M} \cdot \frac{d\delta}{dt}$$

Integrate on both sides,

$$\frac{d\delta^2}{dt} = \frac{2}{M} \int_{\delta_0}^{\delta} P_a \cdot d\delta$$

$$\frac{d\delta}{dt} = \left(\frac{2}{M} \int_{\delta_0}^{\delta} P_a \cdot d\delta \right)^{1/2}$$

where ' δ_0 ' is the initial rotor angle before it begins to swing due to disturbance.

The condition for stability can be written as

$$\left(\frac{2}{M} \int_{\delta_0}^{\delta} P_a \cdot d\delta \right)^{1/2} = 0$$

$$\Rightarrow \int_{\delta_0}^{\delta} P_a \cdot d\delta = 0$$

The condition for stability is stated as:

- The system is stable if the area under accelerating power (P_a) – δ curve reduces to zero at some value of δ .
(or)
- Positive area under P_a – δ curve must equal the negative area and hence the name 'equal area' criterion of stability.

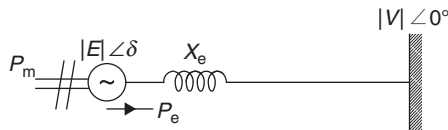
Principal Type of Transient Disturbances

For transient stability studies we have to consider the effect of

1. Sudden change in mechanical input
2. Switching operations
3. Fault with subsequent circuit isolation of the system

Sudden Change in Mechanical Input

Let us consider a synchronous machine connected to infinite bus as shown in the following figure.



Electric power transmitted is given by

$$P_e = \frac{|E||V|}{X_e} \sin \delta = P_{\max} \sin \delta$$

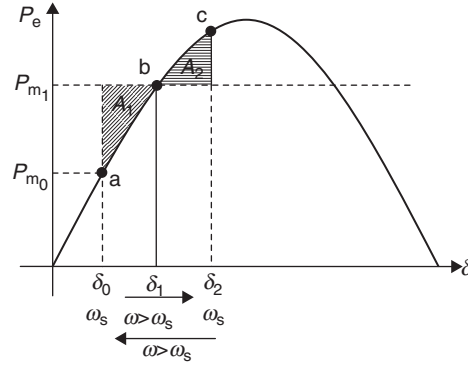
Under steady operating conditions

$$P_m = P_{eo} = P_{\max} \sin \delta_0.$$

Let the mechanical input to the rotor be suddenly increased to P_{m1} by increasing steam input. The accelerating power $P_a = P_{m1} - P_e$ causes the rotor speed to increase and so does

the rotor angle. At angle δ_1 , $P_{m1} - P_e = P_a = 0$ (At point b), but the rotor angle continues to increase as $\omega > \omega_s$. P_a now becomes negative, the rotor speed begins to reduce but the angle continues to increase till at angle δ_2 , $\omega = \omega_s$ once again (at point 'c'), the decelerating area A_2 equals the accelerating

area A_1 i.e., $\int_{\delta_0}^{\delta_2} P_a d\delta = 0$.



As the oscillation decay out because of damping, the system settles to the new steady state when

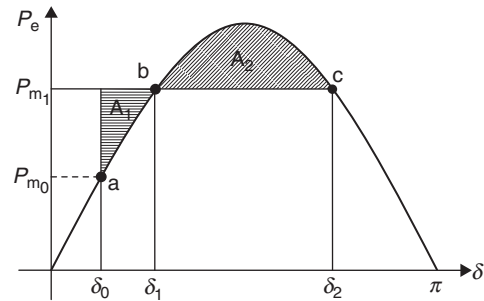
$$P_{m1} = P_e = P_{\max} \sin \delta_1,$$

From the power angle diagram, the expression for A_1 and A_2 can be written as.

$$A_1 = \int_{\delta_0}^{\delta_1} (P_{m1} - P_e) d\delta$$

$$A_2 = \int_{\delta_1}^{\delta_2} (P_e - P_{m1}) d\delta$$

For the system to be stable, it should be possible to find ' δ_2 ' such that $A_1 = A_2$. As P_{m1} increases, a limiting condition is finally reached when A_1 equals the area above the P_{m1} line as shown in the figure below.

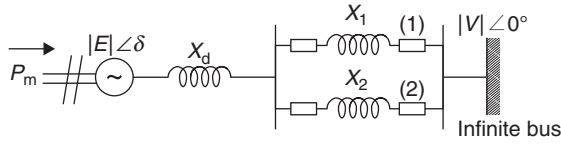


Under these conditions, δ_2 acquired the maximum value such that

$$\delta_2 = \delta_{\max} = \pi - \delta_1 = \pi - \sin^{-1} \left(\frac{P_{m1}}{P_{\max}} \right)$$

Switching Operation

Consider a single machine connected to infinite bus through two parallel lines as shown in the figure.



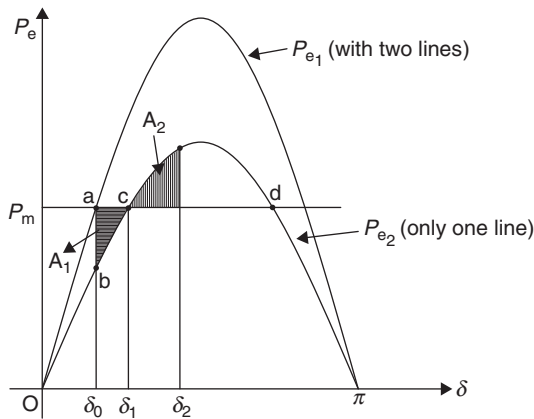
For stability analysis, consider the case when one of the lines is suddenly switched off with the system operating at a steady load. Before switching off the line, power transfer expression is given by

$$P_{e_1} = \frac{|E||V|}{X_d + (X_1 // X_2)} \cdot \sin \delta = P_{\max} \sin \delta.$$

Immediately after the line '2' is switched off

$$P_{e_2} = \frac{|E||V|}{X_d + X_1} \sin \delta = P_{\max_2} \sin \delta.$$

Since $(X_d + X_1) > (X_d + (X_1 // X_2))$, the relation between the maximum powers in two cases becomes $P_{\max_2} < P_{\max_1}$. The system is operating initially with a steady power transfer $P_e = P_m$ at a torque angle δ_0 on curve 1.



Immediately after switching off line 2, operating point on curve 1 shifts to curve 2 (From point 'a' to point 'b') thereafter, operation of the system will be exactly same as sudden change in mechanical input case.

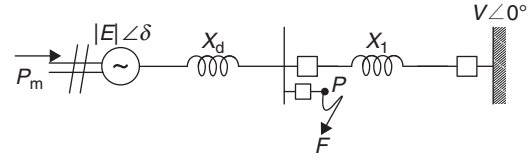
When an area A_2 corresponding to the decelerating energy is equal to an area A_1 corresponding to accelerating area, the system will be stable and finally operates at 'C' corresponding to a new rotor angle $\delta_1 > \delta_0$. This is because a single line after larger reactance and larger rotor angle is needed to transfer the same steady power.

For the limiting case of stability, δ_1 has a maximum value given by

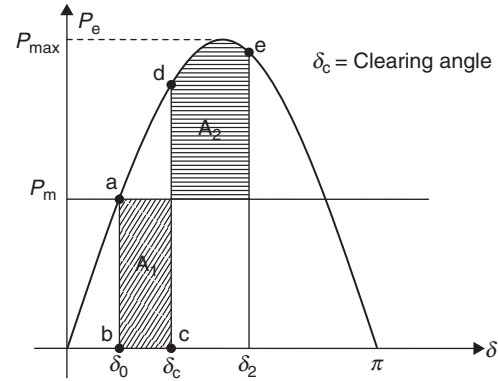
$$\delta_1 = \delta_{\max} = \pi - \delta_c$$

Three-phase Fault on Radial Line

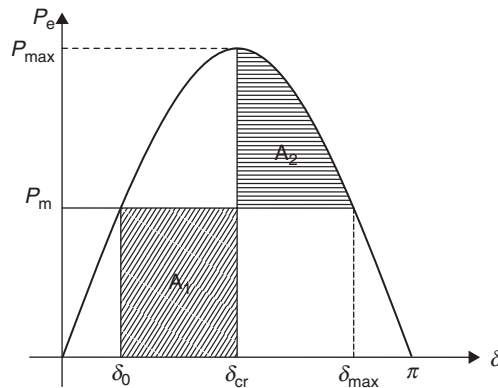
Consider a system operating in steady state and configured as shown in figure below.



If the fault is at point 'p' on the radial line, the electrical output of the generator (power transferred to the load) will reduce to zero. (state point is 'b') The rotor angle of the system starts increasing with the accelerating area and state point moves along path BC. If the fault is cleared by opening circuit breakers at time t_c corresponding to angle δ_c . Value of t_c and δ_c are known as clearing time and clearing angle, respectively. The system once again becomes healthy and transmits $P_0 = P_{\max} \sin \delta$. The rotor now decelerates along de. If an angle δ_1 can be found such that $A_1 = A_2$, the system is found to be stable. The system finally settles down to the steady operating point 'a' in an oscillatory manner because of inherent damping.



As the clearing of the fault line is delayed, A_1 increases and does δ_1 to find $A_2 = A_1$ till $\delta_1 = \delta_{\max}$ as shown in figure below. For a clearing time larger than this value, the system would be unstable as $A_2 < A_1$ the maximum allowable value of clearing time angle for the system to remain stable are, respectively, known as critical clearing time and angle.



From the above figure $\delta_{\max} = \pi - \delta_0$

For a system to be stable $A_1 = A_2$

$$\Rightarrow \int_{\delta_0}^{\delta_{cr}} (P_m - 0) d\delta = \int_{\delta_{cr}}^{\delta_{\max}} (P_{\max} \sin \delta - P_m) d\delta$$

$$\Rightarrow \delta_{cr} = \cos^{-1} [(\pi - 2\delta_0) \sin \delta_0 - \cos \delta_0]$$

where δ_{cr} = critical clearing angle.

From the swing equation

$$\frac{d^2 \delta}{dt^2} = \frac{\pi f}{H} P_m [\because P_e = 0]$$

Integrating twice, we get

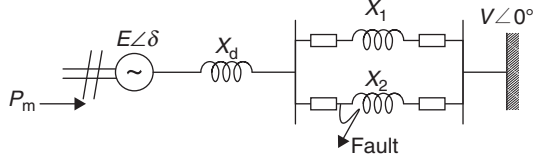
$$\delta_{cr} = \frac{\pi f}{2H} P_m t_{cr}^2 + \delta_0$$

$$t_{cr} = \sqrt{\frac{2H(\delta_{cr} - \delta_0)}{\pi f P_m}},$$

t_{cr} = Critical clearing time

Fault at One End of Parallel Lines

If fault occurs at one of the parallel lines connected between synchronous machine and infinite bus.



Before occurrence of a fault, power transfer expression is given by

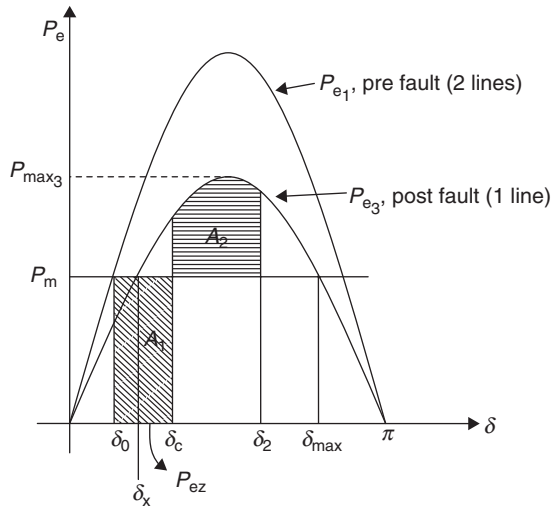
$$P_{e_1} = \frac{|E||V|}{X_d + (X_1 \parallel X_2)} \sin \delta = P_{\max_1} \sin \delta$$

During the period of fault, $P_{e_2} = 0$

After fault is cleared,

$$P_{e_3} = \frac{|E||V|}{X_d + X_1} \sin \delta = P_{\max_3} \sin \delta.$$

Since $[X_d + (X_1 \parallel X_2)] < [X_d + X_1]$, the relation between the maximum power transfer in both the cases is related as $P_{\max_3} < P_{\max_1}$



For stable operation

$$\delta_{\max} = \pi - \delta_x = \pi - \sin^{-1} \left(\frac{P_m}{P_{\max_3}} \right)$$

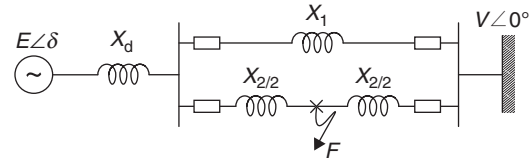
$$A_1 = A_2$$

$$\int_{\delta_0}^{\delta_2} (P_m - 0) d\delta = \int_{\delta_c}^{\delta_2} (P_{\max_3} \sin \delta - P_m) \sin \delta$$

For stable operation of the system, clearing time must be less than critical clearing time at which δ_2 is equal to $\delta_{\max}$.

Fault Away From One of the Parallel Lines

When the fault occurs away from line end, there is some power flow during the fault though considerably reduces, as different from previous case, where $P_e = 0$. Circuit model of the system during fault is shown below.



Power angle curve before fault

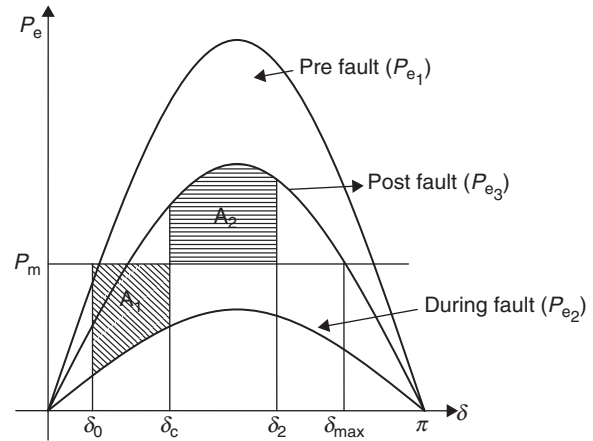
$$P_{e_1} = \frac{|E||V|}{X_d + (X_1/X_2)} \sin \delta$$

Power angle curve during fault

$$P_{e_2} = \frac{|E||V|}{X_F} \sin \delta.$$

If the fault is cleared by removing faulted line after fault is cleared,

$$P_{e_3} = \frac{|E||V|}{X_d + X_1} \sin \delta$$



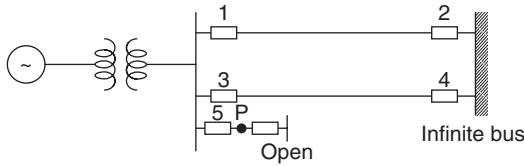
Critical clearing angle

$$\cos \delta_{cr} = \frac{\frac{\pi}{180} P_m (\delta_{\max} - \delta_0) - P_{\max_2} \cos \delta_0 + P_{\max_3} \cos \delta_{\max}}{P_{\max_3} - P_{\max_2}}$$

Design methods for improving transient stability

1. Improved steady-state stability; achieved by:
 - (a) Higher system voltage
 - (b) Additional transmission lines
 - (c) Smaller reactances in transmission lines and transformers
 - (d) Series capacitive compensation
 - (e) Using FACTS
2. High-speed fault clearing
3. High-speed reclosure of circuit breakers
4. Larger machine inertia
5. Fast valving

Example 7: For single-line diagram of three-phase power system given below, the generator is delivering 1.0 p.u. power to infinite bus. The pre fault power angle equation is $P_e = 2.1 \sin \delta$.



Calculate the critical clearing angle and critical clearing time when the system is subjected to a 3-F fault at point 'P'. The fault is cleared by breaker 5. The inertia constant $H = 50$ MJ/MVA.

Solution: Before the fault occurrence

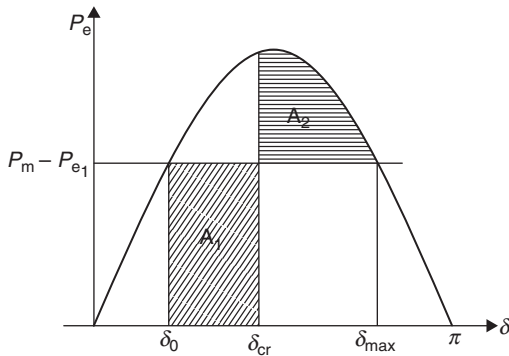
$$P_e = P_{\max} \sin \delta = 2.1 \sin \delta$$

Power transferred $P_e = P_m = 1 = 2.1 \sin \delta$

$$\delta_0 = 28.43^\circ$$

$$\therefore \delta_{\max} = \pi - \delta_0 = 151.56^\circ$$

When the fault occurs, power angle curve is as shown in the figure.



For the system to be stable at critical clearing angle,

$$A_1 = A_2$$

$$\Rightarrow \int_{\delta_0}^{\delta_{cr}} P_m d\delta = \int_{\delta_{cr}}^{\delta_{\max}} (P_{\max} \sin \delta - P_m) d\delta$$

$$\int_{\delta_0}^{\delta_{cr}} P_m d\delta = \int_{\delta_{cr}}^{\delta_{\max}} (2.1 \sin \delta - P_m) d\delta$$

$$P_m(\delta_{cr} - \delta_0) = 2.1(\cos \delta_{cr} - \cos \delta_{\max}) - P_m(\delta_{\max} - \delta_{cr})$$

$$\Rightarrow \cos \delta_{cr} = 0.144$$

$$\text{Critical clearing } \delta_{cr} = 81.72^\circ$$

$$t_{cr} = \left[\frac{2H(\delta_{cr} - \delta_0)}{\pi f P_m} \right]^{\frac{1}{4}} t_{cr} = 0.243 \text{ sec}$$

Example 8: The transient stability of the power system can be effectively improved by

- (A) Excitation improved
- (B) Phase-shifting transformer
- (C) Single-pole switching of circuit breakers
- (D) Increase the turbine value opening

Solution: (C)

Example 9: A loss less single machine infinite bus power systems is shown below.



The synchronous generator transfers 1.0 p.u. of power to the infinite bus. The critical clearing time of the circuit breaker is 0.4 sec. If another identical generator is connected in parallel to the existing generator and each generator is scheduled to supply 0.5 p.u. of power, then the critical clearing time of the circuit breaker will be

- (A) Reduce to 0.2 s
- (B) Reduce but will be more than 0.2 s
- (C) Remains constant at 0.4 s
- (D) Increase beyond 0.4 s

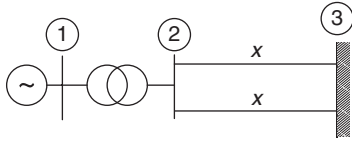
Solution: (D)

$$\text{Critical clearing time } t_{cr} = \sqrt{\frac{2H(\delta_{cr} - \delta_0)}{\pi f P_i}}$$

$$\therefore t_{cr} \propto \sqrt{H}$$

When two identical generators are added in parallel, resultant inertia will be $2H$. Then critical clearing time will increase.

Example 10: A generator with constant 1.0 p.u. terminal voltage supplies power through a step-up transformer of 0.14 p.u. reactance and a double circuit line to an infinite bus bar as shown in the figure. The infinite bus voltage is maintained at 1.0 p.u. Neglecting the resistance and susceptance of the system, the steady-state stability power limit of the system is 6.5 p.u. If one of the double-circuit is tripped, then resulting steady-state stability power limit in p.u. will be



- (A) 6.5 p.u.
 (B) 3.25 p.u.
 (C) 5.96 p.u.
 (D) 2.98 p.u.

Solution: (C)

$$\text{Steady-state stability limit} = \frac{V_1 V_2}{X_{12}} = 6.5$$

$$\frac{1.0 \times 1.0}{\left(0.14 + \frac{X}{2}\right)} = 6.5$$

$$X = 2 \left(\frac{1}{6.5} - 0.14 \right)$$

$$\Rightarrow X = 0.0277.$$

$$\begin{aligned} \text{New steady-state stability limit} &= \frac{V_1 V_2}{0.14 + X} \\ &= \frac{1}{0.14 + 0.0277} = \frac{1}{0.1677} = 5.96 \text{ p.u.} \end{aligned}$$

Example 11: A synchronous generator delivers 0.5 p.u. power in the steady state to an infinite bus through a trans-

mission line of reactance 0.5 p.u. The generator no load voltage is 1.5 p.u. and infinite bus voltage is 1 p.u. The inertial constant of the generator is 5 MW-s/MVA and the generator reactance is 1 p.u. The critical clearing angle, in degrees, for a 3-Ø dead short-circuit fault at the generator terminal is

- (A) 79.4° (B) 30°
 (C) 150° (D) 100.6°

Solution: (A)

Power transferred before fault

$$P_e = \frac{V_1 V_2}{X_{12}} \sin \delta = 0.5 \text{ p.u.}$$

Maximum power can be transferred

$$= \frac{V_1 V_2}{X_{12}} = \frac{1 \times 1.5}{1 + 0.5} = 1 \text{ p.u.}$$

$$\sin \delta_0 = \frac{0.5}{1} = 0.5$$

$$\Rightarrow \delta_0 = \sin^{-1}(0.5) = 30^\circ$$

$$\delta_{\max} = \pi - 30^\circ = 150^\circ.$$

Critical clearing angle (δ_c)

$$\begin{aligned} &= \cos^{-1} \left[\frac{P_e (\delta_{\max} - \delta_0) + P_{m_3} \cos \delta_{\max}}{P_{m_3}} \right] \\ &= 79.45^\circ \end{aligned}$$

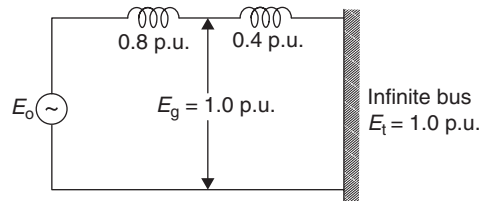
EXERCISES

Practice Problems I

Directions for questions 1 to 15: Select the correct alternative from the given choices.

- A 60 MW, 11 kV, 0.85 lag p.f. water wheel generator has an inertia constant of 4 MJ/MVA. The energy stored in the rotor at synchronous speed is
 (A) 36.5 MJ
 (B) 180 kJ
 (C) 282.3 MJ
 (D) 282.3 kJ
- Which of the following statement is true
 (A) Steady-state stability limit can be increased by increasing its reactance
 (B) Steady-state stability limit is equal to transient stability limit
 (C) Steady-state stability of a power system is improved by increasing generator inertia
 (D) Steady-state stability can be improved by using double circuit line instead of single circuit line
- A generator of equivalent reactance of 0.8 p.u. is connected to an infinite bus through a series reactance

of 0.4 p.u. The terminal voltage of the generator (E_g) is 1.0 p.u. and voltage of infinite bus is 1.0 p.u. The steady-state stability limit is



- (A) 2.17 p.u. (B) 1.6 p.u.
 (C) 3.98 p.u. (D) 1.0 p.u.
- If the mechanical input of a 4-pole, 50 Hz, 20 MVA turbo generator is suddenly raised to 75 MW for an electrical load of 30 MW, the rotor acceleration is (Neglect mechanical and electrical losses and take Inertia constant as 6.0 MJ/MVA)
 (A) 4.82 electrical degrees/s²
 (B) 337.5 electrical degrees/s²
 (C) 3375 electrical degrees/s²
 (D) 280 electrical/rad/s²

5. A 60 MW, 0.85 lag p.f. synchronous generator operates on full load at a frequency of 50 Hz. The load is suddenly reduced to 30 MW. The steam valve begins to close after 0.3 s due to time lag in governor system. The change in frequency that occurs in this time is [Given $H = 4$ MJ/MVA]
- (A) 50.79 Hz (B) 0.79 Hz
(C) 0.69 Hz (D) 0.59 Hz
6. To an infinite bus operating at a voltage of 1 p.u., a 50 Hz generator with a synchronous reactance of 1.2 p.u. is connected. The generator no load voltage is 1.0 p.u. and the inertia constant is 4 MJ/MVA. The frequency of the resulting natural oscillations of the generator when it is suddenly loaded to 60 per cent of its maximum power limit is
- (A) 1.03 Hz (B) 2.89 Hz
(C) 60.42 Hz (D) 50 Hz
7. A 100 MVA synchronous machine has $H_1 = 4.2$ MJ/MVA and a 1200 MVA machine has $H_2 = 3.2$ MJ/MVA. The two machines operate in parallel. The equivalent H constant for the two in 200 MVA base is
- (A) 2.4 MJ/MVA (B) 11.6 MJ/MVA
(C) 21.3 MJ/MVA (D) None of these
8. A 50 Hz, 4-pole turbo generator is rated 200 MVA, 2.2 kV and has an inertia constant of 7.5. Assume the generator is synchronized with a large power system and has a zero accelerating power while delivering a power of 450 MW. Suddenly its input power is changed to 475 MW. The speed of generator in rpm at the end of a period of 10 cycles is
- (A) 1400 rpm (B) 1500 rpm
(C) 1600 rpm (D) 1508.178 rpm
9. For an HVDC valve, the value of output voltage, when the firing angle is 30° and maximum value of line voltage is 200 V is
- (A) 233.83 V (B) 150 V
(C) 100 V (D) 220 V
10. The maximum additional load that can be suddenly applied on a transmission line inter connector carrying 100 MW if the power angle diagram is given by $P = 120 \sin \alpha$
 $P \rightarrow$ Power transmitted in MW
 $\alpha \Rightarrow$ Displacement between voltage phasors at the two ends
- (A) 4.87 MW (B) 3.87 MW
(C) 2.87 MW (D) 1.87 MW
11. It is preferred to transmit bulk power over long distances using high-voltage DC system. This is due to
- (A) Reduced harmonics
(B) Protection system being simple
(C) Low cost of HVDC terminals
(D) Minimum line power losses
12. A 250 MVA, 50 Hz 3- ϕ turbo alternator produces power at 11 kV. The alternator is Y-connected and the neutral

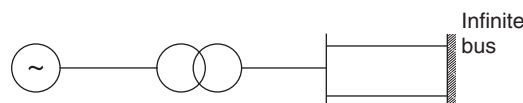
is solidly grounded. The sequence reactance are $X_0 = 0.05$ p.u., $X_1 = 0.2$ p.u.

$X_2 = 0.2$ p.u. The system is running on no load at rated voltage. The magnitude of positive sequence line current for a single line to ground fault at generator terminal would be

- (A) 2.5 p.u. (B) 2.22 p.u.
(C) 5 p.u. (D) 4.44 p.u.

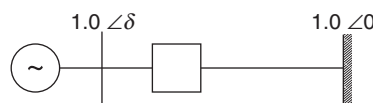
13. A 400 kV transmission line has a maximum power transfer capability $P_{\max}$ at 400 kV. If the line voltage is increased to 800 kV, with series reactance unchanged, the maximum power transfer capability is approximately
- (A) $4P$ (B) $2P$ (C) P (D) $P/4$

14. An alternator with terminal voltage of 1 p.u. supplies power through a transformer of 0.10 p.u. reactance and a double circuit line to an infinite bus bar. The infinite bus voltage is maintained at 1 p.u. Susceptances and resistance are neglected. The steady-state stability power limit of the system is 8 p.u. If one of the lines of the double circuit is tripped, the resulting the steady-state stability power limit will be



- (A) 8 p.u. (B) 3.33 p.u.
(C) 6.67 p.u. (D) 4 p.u.

15. A single machine infinite bus power system is as shown



The alternator transfers 1 p.u. of power to infinite bus. The critical clearing time of circuit breaker is 0.4 s. If another alternator is connected in parallel to the existing alternator and each alternator is scheduled to supply 0.5 p.u. of power, the critical clearing time of breaker

- (A) Reduces to 0.2 s
(B) Reduces but will be greater than 0.2 seconds
(C) Remains at 0.4 s
(D) Increases beyond 0.4 s

Directions for questions 16 to 19: Consists of two statements: one is Assertion (A) and the other is Reason (R). You have to examine these two statements and select the answer using the code given below.

- (A) Both A and R are individually correct and R is the correct explanation of A
(B) Both A and R are individually correct, but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

16. **A:** The most extensively used form of differential relay is the percentage differential or biased beam relay.
R: Percentage differential relay is more sensitive in comparison to differentially connected over current relay.
17. **A:** In comparison to the making capacity of a circuit breaker, breaking capacity is normally higher.
R: The breaking capacity of a circuit breaker is expressed as $\sqrt{3} \times V \times I \times 10^{-6}$ MVA, where V is the rated service.

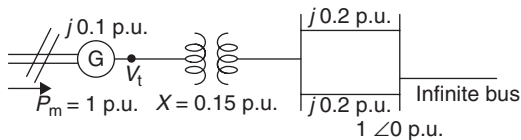
Voltage in volts and I is the RMS value of symmetrical breaking current in amperes.

18. **A:** Extinction of DC arc is much difficult than that of an AC arc.
R: In an AC circuit, current wave passes through zero point twice during each cycle but in dc circuit full current has to be broken.
19. **A:** It is not desirable to operate a high voltage 3- ϕ system of considerable capacitance with an isolated neutral.
R: There is no zero-sequence current in an isolated neutral power system.

Practice Problems 2

Directions for questions 1 to 15: Select the correct alternative from the given choices.

- Swing equation relates
 - Power with load angle
 - Load angle and time
 - Relative motion of rotor with respect to the stator field as a function of time
 - Both (B) and (C)
- A 4-pole 50 Hz, 40 MVA, 13.2 kV turbo alternator has an inertia constant of $H = 6$ kW sec/kVA. The kinetic energy stored in the rotor is
 - 240 MJ
 - 380 kJ
 - 220 MJ
 - 260 kJ
- An alternator is connected to an infinite bus. It delivers 1.6 p.u. current at 0.9 p.f. lag and a voltage of 1.0 p.u. The reactance is 1.1 p.u. Keeping the active power fixed, the excitation is reduced. The critical excitation emf corresponding to operation at stability limit is
 - 1.089 p.u.
 - 2.45 p.u.
 - 1 p.u.
 - None of these
- The maximum power that can be transferred from the generator connected to infinite bus is [Given $V_c = 1.2$ p.u.]



- 1 p.u.
 - 5 p.u.
 - 0.5 p.u.
 - 3 p.u.
- For a 2-pole 50 Hz, 40 MVA turbo alternator, inertia constant is given by 4 MJ/MVA. The moment of inertia is
 - 3.24 kg-m²
 - 0.5095 kg-m²
 - 286.5×10^3 kg-m²
 - 2×10^3 kg-m²
 - The inertia constants of two groups of machines which do not swing together are M_1 and M_2 . The equivalent inertial constant of the system is:

$$(A) M_1 + M_2 \quad (B) M_1 - M_2 \text{ if } M_1 > M_2$$

$$(C) \frac{M_1 + M_2}{M_1 M_2} \quad (D) \frac{M_1 M_2}{M_1 + M_2}$$

- The inertia constant of a 500 MVA alternator is 1 p.u. The value corresponding to 1000 MVA is
 - 2 p.u.
 - 1 p.u.
 - 0.5 p.u.
 - 0.25 p.u.
- Two groups of machines swinging together have their inertia constants M_1 and M_2 . The inertia constant of the system is
 - $M_1 + M_2$
 - $\frac{M_1 + M_2}{M_1 M_2}$
 - $M_1 - M_2, M_1 > M_2$
 - $\frac{M_1 M_2}{M_1 + M_2}$
- A high-voltage DC transmission system, reactive power is needed for rectifier at sending end and inverter at receiving end. During the operation of such a DC link,
 - Inverter supplies leading reactive power and rectifier receives lagging reactive power.
 - Inverter supplies lagging reactive power and rectifier receives leading reactive power.
 - Inverter supplies lagging reactive power and the rectifier receives lagging reactive power.
 - Inverter supplies leading reactive power and rectifier receives leading reactive power.

Data for Linked Questions

I. Question Nos.: 10 and 11

A 100 km, 3-phase, 50 Hz transmission line has the line constants $A = D = 0.853 \angle 1.8^\circ$, $B = 126.5 \angle 50.4^\circ$, $C = 0.002 \angle 90^\circ$. The sending end voltage is 400 kV.

- The receiving end voltage when the load is disconnected will be
 - $2 + j60$ kV
 - $210 \angle 90^\circ$ kV
 - $1.51 \angle 0^\circ$ kV
 - $270.7 \angle -1.8^\circ$ kV
- The sending end current is
 - $541.4 \angle 88.2^\circ$ A
 - $541.4 \angle 60^\circ$ A
 - $17.006 + j541.13$ A
 - Both (A) and (C)

II. Question Nos.: 12 and 13

A 50 Hz, 4-pole turbo generator of rating 30 MVA, 13.2 kV has inertia constant of 8 kW sec/kVA.

12. Angular momentum of rotor in MJ-sec/electrical deg is

- (A) $\frac{1}{150}$ (B) $\frac{4}{150}$ (C) $\frac{4}{6}$ (D) $\frac{40}{600}$

13. If the input rotational losses is 25,100 HP and the electric power developed is 10 MW, the acceleration developed will be

- (A) 2.8 rad/s² (B) 5.72 rad/s²
(C) 3.4 rad/s² (D) 4 rad/s²

III. Question Nos.: 14 and 15

A 50 Hz, 4-pole 100 MVA, 11 kV turbo alternator has an inertia constant of 4.5 kW s/kVA. The input to the alternator is 134102 HP and the electrical power developed is 96 MW.

14. The angular momentum of the rotor is MJ-s/elect deg is

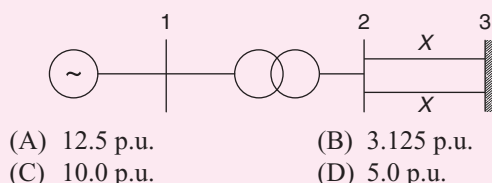
- (A) $\frac{1}{50}$ (B) $\frac{1}{20}$ (C) $\frac{1}{32}$ (D) $\frac{1}{48}$

15. The acceleration of the rotor is

- (A) 1.39 rad/s² (B) 0.39 rad/s²
(C) 2.39 rad/s² (D) 3.39 rad/s²

PREVIOUS YEARS' QUESTIONS

1. A generator with constant 1.0 p.u. terminal voltage supplies power through a step-up transformer of 0.12 p.u. reactance and a double-circuit line to an infinite bus as shown in Figure. The infinite bus voltage is maintained at 1.0 p.u. Neglecting the resistances and susceptances of the system, the steady-state stability power limit of the system is 6.25 p.u. If one of the double-circuit is tripped, the resulting steady-state stability power limit in p.u. will be [2005]



- (A) 12.5 p.u. (B) 3.125 p.u.
(C) 10.0 p.u. (D) 5.0 p.u.

Common Data for Questions 2 and 3:

A generator feeds power to an infinite bus through a double circuit transmission line. A three-phase fault occurs at the middle point of one of the lines. The infinite bus voltage is 1 p.u., the transient internal voltage of the generator is 1.1 p.u. and the equivalent transfer admittance during fault is 0.8 p.u. The 100 MVA generator has an inertia constant of 5 MJ/MVA and it was delivering 1.0 p.u. power prior of the fault with rotor power angle of 30°. The system frequency is 50 Hz.

2. The initial accelerating power (in p.u.) will be

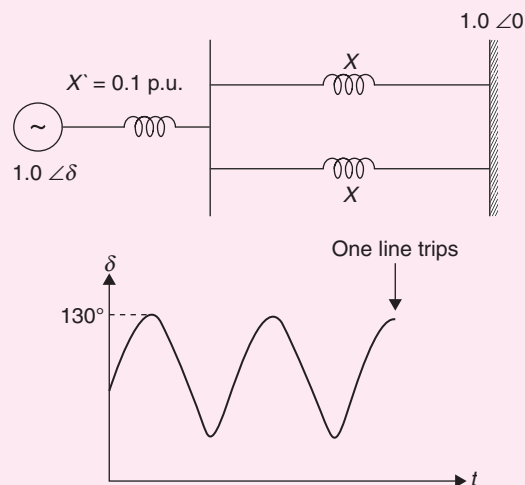
- (A) 1.0 (B) 0.6 (C) 0.56 (D) 0.4 [2006]

3. If the initial accelerating power is X p.u., the initial acceleration in elect deg/s², and the inertia constant in MJ-s/elect deg, respectively, will be [2006]

- (A) $31.4X$, 18 (B) $1800X$, 0.056
(C) $X/1800$, 0.056 (D) $X/31.4$, 18

4. Consider a synchronous generator connected to an infinite bus by two identical parallel transmission lines. The transient reactance x of the generator is 0.1 p.u. and the mechanical power input to it is constant

at 1.0 p.u. Due to some previous disturbance, the rotor angle (δ) is undergoing an undamped oscillation, with the maximum value of $\delta(t)$ equal to 130°. One of the parallel lines trips due to relay mal-operation at an instant when $\delta(t) = 130^\circ$ as shown in the figure. The maximum value of the per unit line reactance, x , such that the system does not lose synchronism subsequent to this tripping is [2007]



- (A) 0.87 (B) 0.74 (C) 0.67 (D) 0.54

5. An isolated 50 Hz synchronous generator is rated at 15 MW which is also the maximum continuous power limit of its prime mover. It is equipped with a speed governor with 5% droop. Initially, the generator is feeding three loads of 4 MW each at 50 Hz. One of these loads is programmed to trip permanently if the frequency falls below 48 Hz. If an additional load of 3.5 MW is connected then the frequency will settle down to [2007]

- (A) 49.417 Hz (B) 49.917 Hz
(C) 50.083 Hz (D) 50.583 Hz

6. A lossless single machine infinite bus power system is shown below

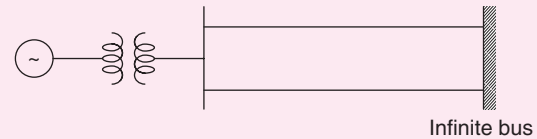


The synchronous generator transfers 1.0 per unit of power to the infinite bus. Critical clearing time of circuit breaker is 0.28 s. If another identical synchronous generator is connected in parallel to the existing generator and each generator is scheduled to supply 0.5 per unit of power. Then the critical clearing time of the circuit breaker will [2008]

- (A) Reduce to 0.14 s
(B) Reduce but will be more than 0.14 s
(C) Remain constant at 0.28 s
(D) Increase beyond 0.28 s
7. A cylindrical rotor generator delivers 0.5 p.u. power in the steady state to an infinite bus through a transmission line of reactance 0.5 p.u. The generator no-load voltage is 1.5 p.u. and the infinite bus voltage is 1 p.u. The inertia constant of the generator is 5 MW-s/MVA and the generator reactance is 1 p.u. The critical clearing angle, in degrees, for a three-phase dead short-circuit fault at the generator terminal is [2012]
- (A) 53.5 (B) 60.2 (C) 70.8 (D) 79.6
8. The angle δ in the swing equation of a synchronous generator is the [2013]
- (A) Angle between stator voltage and current
(B) Angular displacement of the rotor with respect to the stator.
(C) Angular displacement of the stator mmf with respect to a synchronously rotating axis.
(D) Angular displacement of an axis fixed to the rotor with respect to a synchronously rotating axis.
9. A synchronous generator is connected to an infinite bus with excitation voltage $E_f = 1.3$ p.u. The generator has a synchronous reactance of 1.1 p.u. and is delivering real power (P) of 0.6 p.u. to the bus. Assume the infinite bus voltage to be 1.0 p.u. Neglect stator resistance. The reactive power (Q) in p.u. supplied by the generator to the bus under this condition is _____. [2014]
10. There are two generators in a power system. No-load frequencies of the generators are 51.5 Hz and 51 Hz, respectively, and both are having droop constant of 1 Hz/MW. Total load in the system is 2.5 MW. Assuming that the generators are operating under their respective droop characteristics, the frequency of the power system in Hz in the steady state is _____. [2014]
11. A non-salient pole synchronous generator having synchronous reactance of 0.8 p.u. is supplying 1 p.u. power to a unity power factor load at a terminal

voltage of 1.1 p.u. Neglecting the armature resistance, the angle of the voltage behind the synchronous reactance with respect to the angle of the terminal voltage in degrees is _____. [2014]

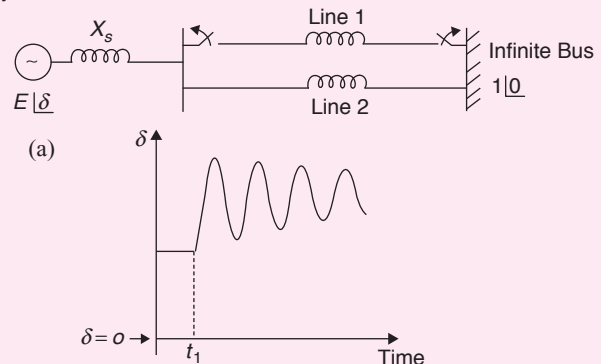
12. The figure shows the single line diagram of a single machine infinite bus system.

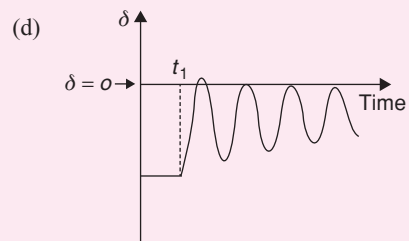
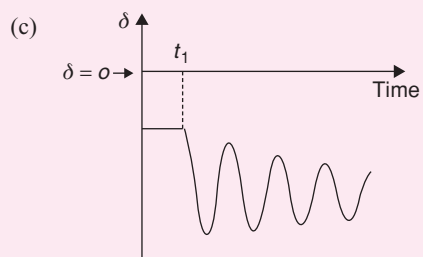
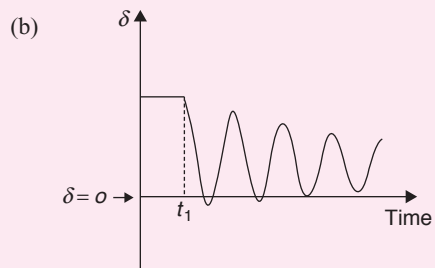


The inertia constant of the synchronous generator $H = 5$ MW-s/MVA. Frequency is 50 Hz. Mechanical power is 1 p.u. The system is operating at the stable equilibrium point with rotor angle δ equal to 30° . A three-phase short circuit fault occurs at a certain location on one of the circuits of the double-circuit transmission line. During fault, electrical power in p.u. is $P_{\max} \sin \delta$. If the values of δ and $d\delta/dt$ at the instant of fault clearing are 45° and 3.762 radian/s, respectively, then $P_{\max}$ (in p.u.) is _____. [2014]

13. A 50 Hz generating unit has H-constant of 2 MJ/MVA. The machine is initially operating in steady state at synchronous speed, and producing 1 pu of real power. The initial value of the rotor angle δ is 5° , when a bolted three phase to ground short circuit fault occurs at the terminal of the generator. Assuming the input mechanical power to remain at 1 pu, the value of δ in degrees, 0.02 second after the fault is _____. [2015]
14. The synchronous generator shown in the figure is supplying active power to an infinite bus via two short, lossless transmission lines, and is initially in steady state. The mechanical power input to the generator and the voltage magnitude E are constant. If one line is tripped at time t_1 by opening the circuit breakers at the two ends (although there is no fault), then it is seen that the generator undergoes a stable transient. Which one of the following waveforms of the rotor angle δ shows the transient correctly? [2015]

Synchronous Generator





ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. A | 4. C | 5. B | 6. A | 7. C | 8. D | 9. A | 10. D |
| 11. D | 12. B | 13. A | 14. C | 15. B | 16. C | 17. D | 18. A | 19. B | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|------|------|------|------|-------|
| 1. D | 2. A | 3. A | 4. B | 5. A | 6. D | 7. C | 8. A | 9. A | 10. D |
| 11. D | 12. B | 13. B | 14. B | 15. A | | | | | |

Previous Years' Questions

- | | | | | | | | | | |
|-----------|----------|----------|-------|------|------|------|------|----------|--------|
| 1. D | 2. C | 3. B | 4. C | 5. A | 6. D | 7. D | 8. D | 9. 0.109 | 10. 50 |
| 11. 33.61 | 12. 0.24 | 13. 5.90 | 14. D | | | | | | |

Chapter 5

HVDC and FACTS

LEARNING OBJECTIVES

After reading this chapter, you will be able to understand:

- HVDC
- Monopolar link
- Bipolar link
- Homopolar link
- FACTS
- Series connected controllers
- Shunt connected controllers
- Combined shunt and series connected controllers
- Other controllers

HIGH-VOLTAGE DC TRANSMISSION

The limitations for use of high-voltage AC transmission of bulk power over long distance leads to high-voltage DC transmission (HVDC). The advantages of HVDC transmission are.

1. Power transmitted per conductor is more for DC and DC line is cheaper as it requires two conductors instead of three and hence costs on insulation and towers are less.
2. The line construction is simpler and ground return is possibly by using single conductor in HVDC transmission.
3. There is no charging current in DC transmission system, No dielectric losses.
4. Skin effect is low in DC systems and hence current density is also high.
5. Corona loss and radio interference are low as compared to AC systems.
6. HVDC does not have any stability problem and hence asynchronous operation of transmission link among the connected machines is possible.
7. Line losses are less in HVDC transmission.
8. Voltage regulation problems are less serious in HVDC since the reactance drop does not exist in DC systems.
9. HVDC system is economical for transmitting bulk power over long distances say above 550 km.
10. Low short-circuit current is required on HVDC lines.

11. Grid control of the converters can drastically reduce the fault current in HVDC.
12. Compensation of lines does not require in DC transmission.

However, HVDC transmission system has a few disadvantages as follows.

1. Application of HVDC is restricted to the distance above 500 km due to its high initial cost which is uneconomical for distances below 500 km.
2. Circuit breaking for protection of HVDC is difficult and expensive.
3. Converter stations require considerable reactive power.
4. Harmonic generation and subsequent filtration are problems in HVDC.
5. Reactive power required by the load is to be supplied locally as no reactive power can be transmitted through HVDC lines.
6. Overload capacity of HVDC converters is low.
7. Maintenance of insulators in HVDC transmission is low.

Principle of HVDC System Operation

A typical HVDC transmission system consists of one rectifier station at the sending end and one inverter station at the receiving end. The two stations are interconnected by a DC transmission line. The rectifier station converts AC to DC while the inverter

station converts DC to AC. By varying the firing angle of the thyristor in the convertor, the DC output voltage magnitude is controlled.

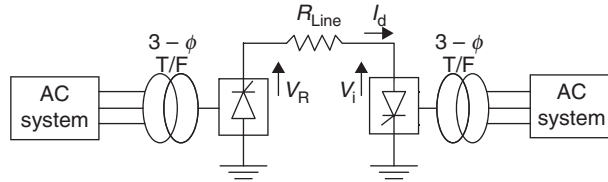


Figure 1 Schematic of a HVDC system

Rectified firing angle is $0^\circ < \alpha < 90^\circ$

Inverter firing angle is $90^\circ < \alpha < 180^\circ$

The current flows from higher voltage to lower voltage by proper setting of the rectifier and inverter. In the DC transmission system the line current (I_d) is given by

$$I_d = \frac{V_R - V_i}{R}$$

where V_R is the DC output voltage at the rectifier and V_i is the DC output voltage at the inverter end.

$$V_R = \left[\left(\frac{3\sqrt{2}}{\pi} \right) V_{AC_r} \cos \alpha - \frac{3X_{cr}}{\pi} I_d \right]$$

$$V_i = \left[\left(\frac{3\sqrt{2}}{\pi} \right) V_{AC_i} \cos \gamma - \frac{3X_{ci}}{\pi} I_d \right]$$

where V_{AC_r} = AC side, line-to-line RMS voltage at rectifier side

V_{AC_i} = AC side, line to line RMS voltage at inverter side

α = Firing angle of rectifier

γ = Extinction angle of inverter

X_{cr} or X_{ci} = Communication reactance at the rectifier and inverter.

The power transfer is given by

$$P = V_i I_d = \left(\frac{V_R - V_i}{R} \right) V_i \text{ W}$$

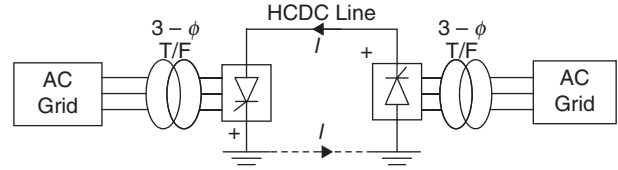
Types of DC Links

HVDC links are classified as

1. Monopolar link
2. Bipolar link
3. Homopolar link

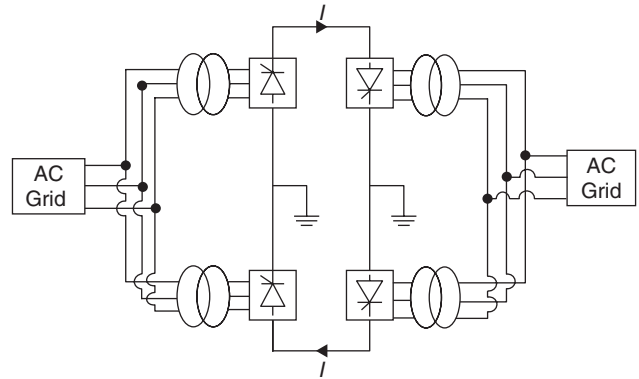
Monopolar Link

This configuration uses only one conductor and ground or sea water is used as return conductor. Negative polarity is used as the transmission conductor due to comparatively lesser ratio interference.



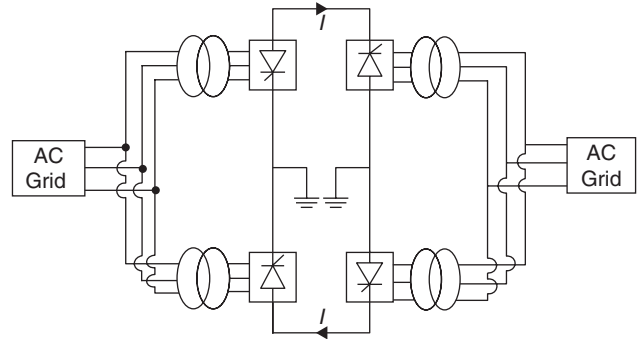
Bipolar Link

This configuration uses two conductors, one positive and another negative. In each terminal, two converters of equal rated voltages are connected in series, neutral points being grounded. Two poles can operate independently when both the terminals are grounded. When the currents in the two conductors are equal, the ground current is zero. In case of fault in one conductor, the other conductor can supply 50% of the rated load along with the ground return.



Homopolar Link

This configuration has two conductors. But having same polarity, ground is used as return path. In case of fault in one conductor, the other conductor can supply 50% of the rated load along with the ground return.



FACTS

Flexible alternating current transmission systems (FACTS) is a technology which provides a methodology for the utilities to effectively utilize their assets, enhance transmission capability by loading lines to their full transmission capability and therefore, minimize the gap between the stability and

the thermal limits, and improve grid reliability. The FACTS technology is based on the use of reliable high-speed power electronics, advanced control technology, advanced micro computers, and powerful analytical tools.

Consider a transmission line of reactance X_{pu} (negligible resistance) connecting two buses in a power system. The system voltage magnitudes in per unit and phase angles are indicated in the diagram.

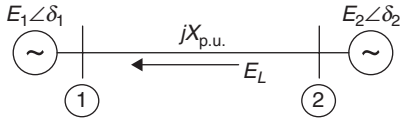


Figure 2 Single-line diagram of AC transmission connecting two buses

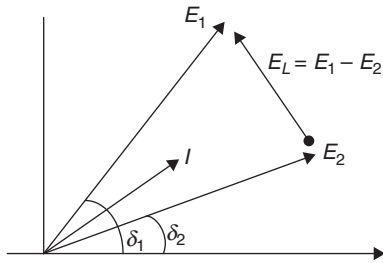


Figure 3 Phasor diagram of the two bus power systems

Line current lags the line voltage drop $E_L = (E_1 - E_2)$ by 90° and is given by

$$I = \frac{E_L}{X} = \frac{E_1 - E_2}{X}$$

Active power flow along the line is given by

$$P = \frac{|E_1||E_2|}{X} \sin \delta$$

where $\delta = \delta_1 - \delta_2$

Reactive power flows Q_1 and Q_2 are expressed as

$$Q_1 = \frac{1}{X} (|E_1|^2 - |E_1||E_2| \cos \delta)$$

$$Q_2 = \frac{1}{X} (|E_2|^2 - |E_1||E_2| \cos \delta)$$

From the expression of active and reactive power it can be observed that the active and reactive power on transmission line are depended on line reactance X , voltage magnitudes $|E_1|$ and $|E_2|$, line voltage drop E_L and phase angle difference between the two voltages $\delta = \delta_1 - \delta_2$.

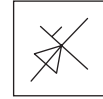
FACTS technology utilizes thyristor-controlled fast-acting controllers to control the following:

1. Line reactance
2. Magnitude of terminal voltages
3. Magnitude of voltage drop (E_L)
4. Phase angle difference

Control action	Parameter affected
(1) Line reactance control	Line current and active power control
(2) Phase angle difference control	Line current and active power control
(3) Voltage injection in series with the line and perpendicular to the current	Real power control through the control of line current
(4) Voltage injection in series with the line and any phase angle with respect to the driving voltage	Accurate control of real and reactive power flow in the line

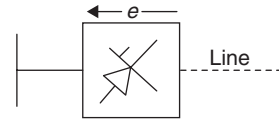
Basic Types of FACTS Controllers

FACTS devices are power electronic based systems and other static equipment that provide control of one or more transmission system parameters. System of FACTS controller is given by

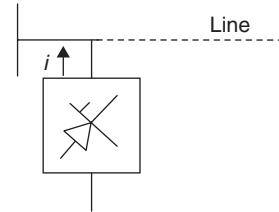


The four basic types of FACTS controllers are given by

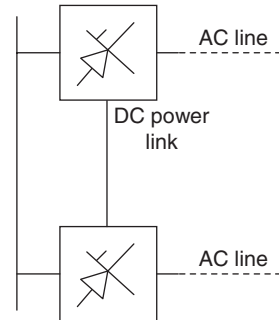
1. Series controllers



2. Shunt controllers



3. Combined series-series controllers



4. Combined series-shunt controllers

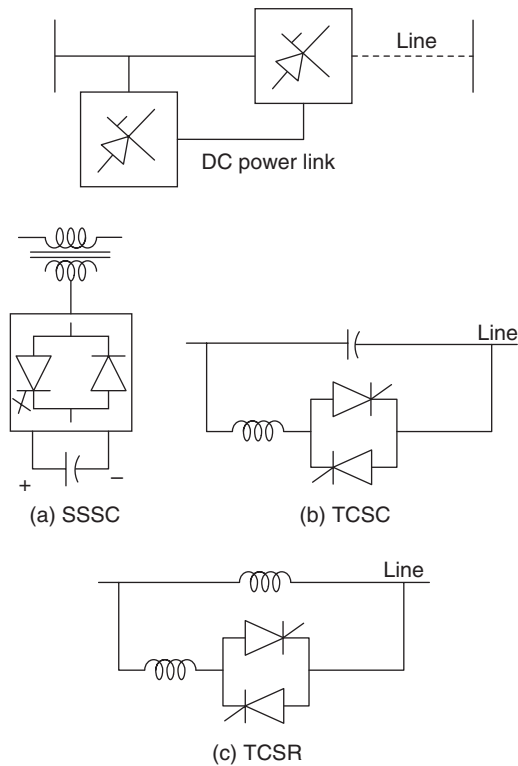


Figure 4 Series-connected FACTS controllers

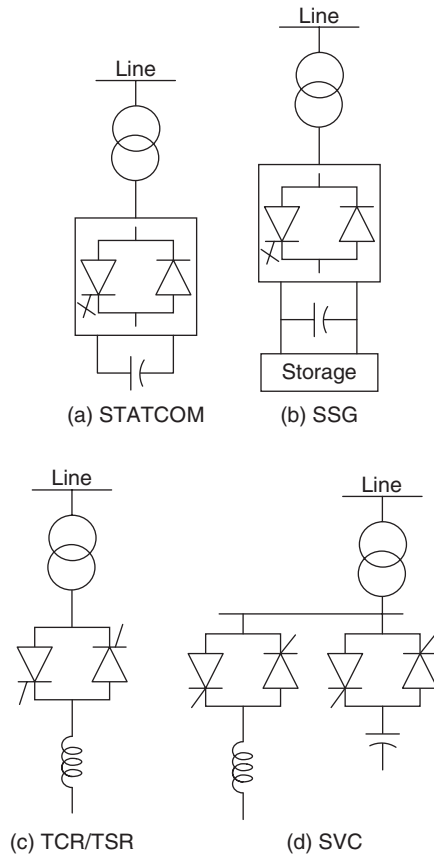


Figure 5 Shunt FACTS controllers

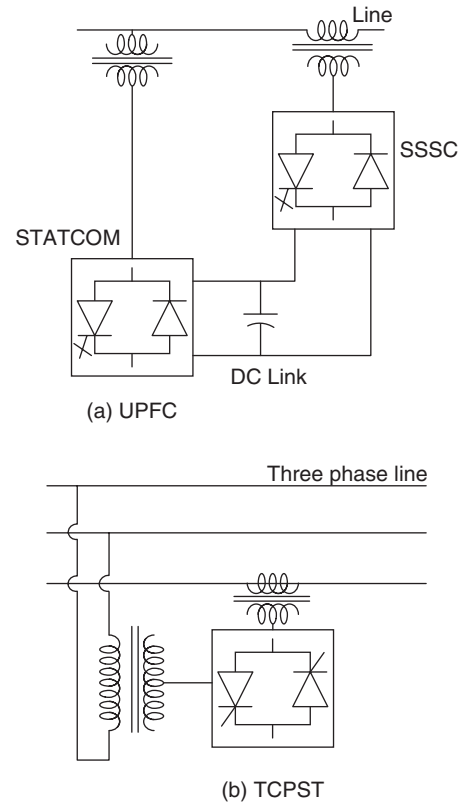


Figure 6 Combined series-shunt controllers

Series-connected Controllers

Static Synchronous Series Compensator (SSSC)

A static synchronous operates as a series compensator whose output voltage is in quadrature with, and controllable independent of, the line current for the purpose of increasing the overall reactive voltage drop across the line and thereby by controlling the transmitted electrical power.

Inter Power Flow Controller (IPFC)

It is a combination of SSSC's, which are controlled to provide independent reactive compensation for the Adjustment of real power flow in each line.

Thyristor-controlled Services Capacitor (TCSC)

This is a capacitive reactive compensator which consists of a series capacitor bank shunted by a thyristor-controlled reactor in order to provide a smoothly variable series capacitive reactance.

Thyristor-controlled Series Reactor (TCSR)

It consists of a series reactor shunted by a thyristor-controlled reactor in order to provide a smoothly variable series inductive reactance.

Shunt-connected Controllers

Static Synchronous Compensator (STATCOM)

It is a shunt-connected static VAR compensator whose capacitive or inductive output current can be controlled independent of the AC system voltage.

Static Synchronous Generator (SSG)

It produces a set of adjustable multiphase output voltages, which may be coupled to an AC power system for the purpose of exchanging independently controllable real and reactive power.

Static VAR Compensator (SVC)

Its output is adjusted to exchange capacitive or inductive current so as to maintain or control specific parameters of the electrical power system.

Thyristor-controlled Reactor (TCR)

Its effective reactance is varied in a continuous manner by partial conduction control of the thyristor value.

Combined Shunt- and Series-connected Controllers**Unified Power Flow Controller (UPFC)**

It is a combination of STATCOM and SSSC to provide real and reactive series line compensation. The UPFC may also provide independently controllable shunt reactive compensation.

Thyristor-controlled Phase-Shifting Transformer (TCPST)

It is adjusted by thyristor switches to provide a rapidly variable phase angle

Interphase Power Controller (IPC)

The active and reactive power can be set independently by adjusting the phase shifts and/or the branch impedances, using mechanical or electrical switches.

Other Controllers**Thyristor-controlled Voltage Limiter (TCVL)**

A thyristor-switched metal-oxide varistor (MOV) used to limit the voltage across its terminals during transient conditions.

Thyristor-controlled Voltage Regulator (TCVR)

Thyristor-controlled transformer is one which can provide variable in-phase voltage with continuous control.

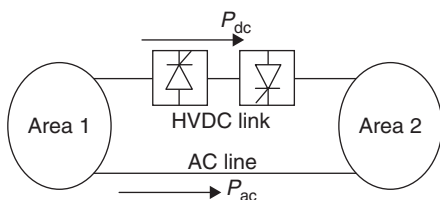
- When the inverter fundamental output voltage is higher than the system line voltage, the STATCOM works as a capacitor, however when the inverter voltage is lower than the system line voltage, the STATCOM acts as an inductor.

Advantages of STATCOM are as follows:

1. Steady-state load ability of lines is improved.
2. The voltage rise due to capacitance switching is substantially reduced both in magnitude and duration.
3. Voltage variation due to customer's loading is reduced.
 - In SVC devices, the current injected into the system depends upon the system voltage.
 - TCSC provides dynamic control of the series compensated lines, which would increase the transfer capability.
 - A thyristor control phase angle regulator (TCPAR) is equivalent to a mechanically phase shifting transformer but, unlike a UPFC, it does not provide a controlled reactive power generation.
 - STATCOM mainly provides dynamic reactive power to the system but it does not directly controls the flow of real power on a transmission line.

EXERCISES**Practice Problems I**

1. In a two area power system both the areas are interconnected by an AC line and a HVDC link as shown in figure. Which one of the following statement is true in steady state.



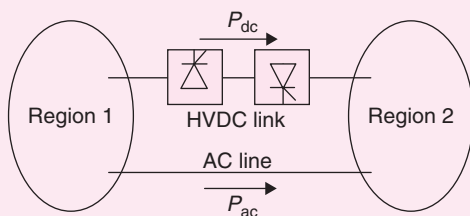
- (A) The total power flow between the regions ($P_{AC} + P_{DC}$) can be changed by controlling the HVDC converters alone.
- (B) Both regions need not have the same frequency.
- (C) The directions of power flow in the HVDC link (P_{DC}) can be reversed.

- (D) The power sharing between the AC line and the HVDC link can be changed by controlling the HVDC converters alone.

2. Which one of the following is an advantage of HVDC transmission over EHV-AC?
 - (A) System stability can be improved
 - (B) VAR compensation is not required
 - (C) Harmonic problem is avoided
 - (D) HVDC terminal equipment inexpensive
3. Main application of HVDC transmission is
 - (A) Bulk power transmission over long distances.
 - (B) Interconnecting two systems with the same nominal frequency.
 - (C) Minimizing harmonics at the converter station.
 - (D) Eliminating reactive power requirement in the operation.

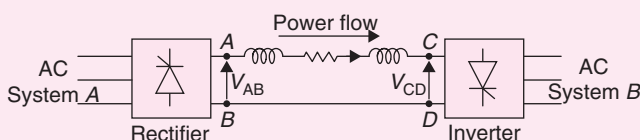
PREVIOUS YEARS' QUESTIONS

- HVDC transmission is mainly used for [2005]
 - Bulk power transmission over very long distances
 - Inter-connecting two systems with the same nominal frequency
 - Eliminating reactive power requirement in the operation
 - Minimizing harmonics at the converter stations
- An HVDC link consists of rectifier, inverter transmission line and other equipments. Which one of the following is true for this link? [2006]
 - The transmission line produces/supplies reactive power
 - The rectifier consumes reactive power and the inverter supplies reactive power from/to the respective connected AC systems
 - Rectifier supplies reactive power and the inverted consumes reactive power to/from the respective connected AC systems
 - Both the converters (rectifier and inverter) consume reactive power from the respective connected AC systems
- Two regional systems, each having several synchronous generators and loads are interconnected by an AC line and a HVDC link as shown in the figure. Which of the following statements is true in the steady state?

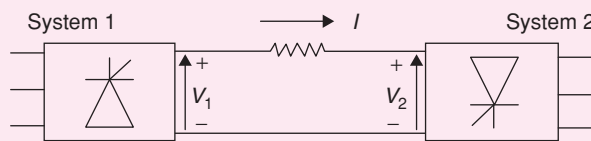


- Both regions need not have the same frequency [2007]
- The total power flow between the regions ($P_{AC} + P_{DC}$) can be changed by controlling the HVDC converters alone

- The power sharing between the AC line and the HVDC link can be changed by controlling the HVDC converters alone.
 - The direction of power flow in the HVDC link (P_{DC}) cannot be reversed.
4. Power is transferred from system A to system B by an HVDC link as shown in the figure. If the voltages V_{AB} and V_{CD} are as indicated in the figure, and $I > 0$, then [2010]



- $V_{AB} < 0, V_{CD} < 0, V_{AB} > V_{CD}$
 - $V_{AB} > 0, V_{CD} > 0, V_{AB} < V_{CD}$
 - $V_{AB} > 0, V_{CD} > 0, V_{AB} > V_{CD}$
 - $V_{AB} > 0, V_{CD} < 0$
5. Consider a HVDC link which uses thyristor based line-commutated converters as shown in the figure. For a power flow of 750 MW from system 1 to System 2, the voltages at the two ends, and the current, are given by: $V_1 = 500$ kV, $V_2 = 485$ kV and $I = 1.5$ kA. If the direction of power flow is to be reversed (that is, from System 2 to System 1) without changing the electrical connections, then which one of the following combinations is feasible? [2015]



- $V_1 = -500$ kV, $V_2 = -485$ kV and $I = 1.5$ kA
- $V_1 = -485$ kV, $V_2 = -500$ kV and $I = 1.5$ kA
- $V_1 = 500$ kV, $V_2 = 485$ kV and $I = -1.5$ kA
- $V_1 = -500$ kV, $V_2 = -485$ kV and $I = -1.5$ kA

ANSWER KEYS

EXERCISES

Practice Problems I

- D
- A
- A

Previous Years' Questions

- A
- B
- C
- C
- B

TEST Power Systems

Time: 60 min.

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. A Non conventional source of energy is
 - (A) Water
 - (B) Radio active substances
 - (C) Fossil fuels
 - (D) Geothermal, ocean tides and waves
2. The function of steel wire in an ACSR conductor is
 - (A) Reducing conductivity
 - (B) Compensating skin effect
 - (C) Providing additional mechanical strength
 - (D) Purely of economical consideration
3. The string efficiency of insulators can be increased by
 - (A) Changing the orientation of strings
 - (B) Increasing the number of strings
 - (C) Correct grading of insulators of various capacitances
 - (D) Reducing the number of strings
4. In a DC transmission line
 - (A) It is necessary for the sending end and receiving end to be operated in synchronism
 - (B) There are no effects due to inductive and capacitive reactances
 - (C) Power transfer capability is limited by stability considerations
 - (D) The effects of inductive and capacitive reactance are greater
5. RRRV depends upon the
 - (A) Inductance of the system only
 - (B) Capacitance of the system only
 - (C) Inductance and capacitance of the system
 - (D) Type of circuit breaker
6. The largest size of thermal generating unit in India
 - (A) 265 MW
 - (B) 310 MW
 - (C) 660 MW
 - (D) 500 MW
7. Base Impedance of a power system is given as
 - (A) $(\text{Base KV})^2/\text{Base MVA}$
 - (B) $\text{Base KV}/(\text{Base MVA})^2$
 - (C) $\text{Base KV}/\text{Base MVA}$
 - (D) $(\text{Base KV})^2/(\text{Base MVA})^2$
8. The values of current and voltage for an unloaded synchronous generator which has a fault occurring at the terminals is given as I_{a_0} , I_{a_1} and I_{a_2} as $j1.37$ p.u., $j2.05$ p.u. and $j0.68$ p.u. respectively $V_{a_0} = V_{a_1} = V_{a_2} = j1.37$ p.u. The fault that has occurred is
 - (A) $L-G$ fault
 - (B) $L-L-G$ fault
 - (C) $L-L$ fault
 - (D) None of the above
9. An overhead transmission line having surge impedance ' Z_1 ' is terminated to an underground cable of surge impedance ' Z_2 '. The reflection coefficient for the travelling wave at the junction of the line and cable is
 - (A) $\frac{Z_2}{Z_1 + Z_2}$
 - (B) $\frac{-Z_2}{Z_1 + Z_2}$
 - (C) $\frac{Z_2 - Z_1}{Z_1 + Z_2}$
 - (D) $\frac{Z_1 - Z_2}{Z_1 + Z_2}$
10. The insulation resistance of a cable of length 20 km is 1 MW, its resistance for 100 km length will be
 - (A) 0.2 MW
 - (B) 1 MW
 - (C) 5 MW
 - (D) None of the above
11. If a generator of 250 MVA has an inertia constant of 20 MJ/MVA, its inertia constant on 500 MVA base is
 - (A) 5 MJ/MVA
 - (B) 10 MJ/MVA
 - (C) 15 MJ/MVA
 - (D) 20 MJ/MVA
12. A power station is to supply three regions of loads whose Peak values are 40 MW, 25 MW and 50 MW. The diversity factor and annual load factor is given as 2.2 and 0.25 respectively. The maximum demand on the station and average load respectively will be
 - (A) 52.27, 13.07
 - (B) 45.08, 15.08
 - (C) 72.28, 16.08
 - (D) 60.04, 10.02
13. A transmission line conductor having a diameter of 18.2 mm, weights 0.80 kg/m. The span is 270 m. The wind pressure is 35 kg/m² of projected area with ice coating of 12 mm. The ultimate strength of the conductor is 6500 kg. If the factor of safety is 2 and ice weights 890 kg/m³; the value of maximum sag will be
 - (A) 6.95 m
 - (B) 5.36 m
 - (C) 4.82 m
 - (D) 5.20 m
14. A 2 km of a three phase metal sheathed belted cable gave a measured capacitance of 1.2 μF between one conductor and the other two conductors joined together with the earth sheath and 1.5 μF measured between all the three conductors joined and sheath. The value of charging current when the cable is connected at 11 KV, 50 Hz supply is
 - (A) 1.42 A/ph
 - (B) 3.09 A/ph
 - (C) 1.59 A/ph
 - (D) 0.68 A/ph
15. In the DC distribution system shown below, the two ends of the feeder are fed by voltage source such that $V_p - V_q = 3$ V. The value of the voltage V_p for a minimum voltage of 220 V at any point along the feeder.

 - (A) 220 V
 - (B) 228.58 V
 - (C) 225.89 V
 - (D) 222.89 V

16. The velocity of wave propagation of a lossless overhead EHV line operating at 50 Hz is 3×10^5 Km/s. If the total reactance and susceptance are given by 0.035 p.u. and 1.3 p.u. respectively, approximate length of the line is
(A) 158 Km
(B) 140 Km
(C) 180 Km
(D) 175 Km
17. A power system network consists of three elements 0–1, 1–2, and 2–0, of per unit impedances 0.2, 0.4 and 0.4 respectively. Its bus impedance matrix is given by _____.
- (A) $\frac{1}{2} \begin{bmatrix} 7.5 & -2.5 \\ -2.5 & 5.0 \end{bmatrix}$ (B) $\frac{1}{2} \begin{bmatrix} 0.16 & 0.08 \\ 0.08 & 0.24 \end{bmatrix}$
- (C) $\frac{1}{2} \begin{bmatrix} 0.16 & -0.18 \\ -0.08 & 0.24 \end{bmatrix}$ (D) $\frac{1}{2} \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.8 \end{bmatrix}$
18. The incremental fuel cost of a power plant consisting of three generating units are
 $IC_1 = 20 + 0.3P_1$
 $IC_2 = 30 + 0.4P_2$
 $IC_3 = 30$
 where P_i is the power in MW generated by unit i , for $i = 1, 2$, and 3. Assume that all three units are operating all the time. Minimum and maximum load on each unit are 50 MW and 300 MW respectively. If the plant supplies a total power demand of 700 MW power generated by each unit is
 (A) $P_1 = 157.14$ MW $P_2 = 242.86$ MW $P_3 = 300$ MW
 (B) $P_1 = 242.86$ MW $P_2 = 157.14$ MW $P_3 = 300$ MW
 (C) $P_1 = 300$ MW $P_2 = 300$ MW $P_3 = 100$ MW
 (D) $P_1 = 233.3$ MW $P_2 = 233.3$ MW $P_3 = 233.4$ MW
19. The leakage reactance referred to lv side for a 3 kVA 11/0.4 KV single phase transformer is given as 0.90 Ω . Then the leakage reactance in per unit is
 (A) 0.001 p.u. (B) 0.017 p.u.
 (C) 0.584 p.u. (D) 1.23 p.u.
20. The maximum allowable tension in an overhead line having a span of 150 m is given as 1200 kg. If the line conductor weights 0.5 kg per meter, then the maximum sag will be
 (A) 1.17 (B) 3.5
 (C) 0.15 (D) 4.8
21. For a step up transformer the line to ground voltage on phases a, b and c are given as 80 KV, 33 KV and 42 KV respectively. The voltage relations are given like phase a lags that of phase b by 150° and leads that of phase c by 80° . Then the positive and negative components of phase voltage are
 (A) $(13.51 - 6.21i)$, $(46.91 + 14.49j)$
 (B) $(13.51 + 15.78j)$, $(46.91 - 14.49j)$
 (C) $(13.51 - 15.78j)$, $(15.63 + 14.49j)$
 (D) $(13.51 + 5.26j)$, $(15.63 - 14.49j)$
22. The losses and other mechanical load for a synchronous motor which improves the p.f. from 0.5 to 0.7 lagging is given as $(150 + 100)$ kW. If the load is given as 1200 kW, then the value of the input to the synchronous motor in kVA
 (A) 649 (B) 550
 (C) 700 (D) 585
23. For a DC three wire system the same supply and load voltages are given to neutral and balanced conditions. If this system is converted into a 3-phase four wire AC system which of the following expression is correct
 (A) $P_2 = P_1$ (B) $P_2 = 1.5 P_1$
 (C) $P_1 = -P_2$ (D) $P_1 = 1.5 P_2$
- Statement for Linked Answer Questions 24 and 25:**
 The maximum demand at a power station is 60 MW, at a load factor of 0.7 and plant capacity factor of 0.6.
24. The annual energy generated in kWh is
 (A) 367.920×10^6 (B) 183.96×10^6
 (C) 735.84×10^6 (D) 147.17×10^6
25. The reserve capacity of the plant is
 (A) 5 MW (B) 10 MW
 (C) 15 MW (D) 20 MW

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. C | 3. C | 4. B | 5. C | 6. C | 7. A | 8. B | 9. C | 10. A |
| 11. B | 12. A | 13. A | 14. B | 15. B | 16. A | 17. D | 18. B | 19. B | 20. A |
| 21. A | 22. A | 23. B | 24. A | 25. B | | | | | |

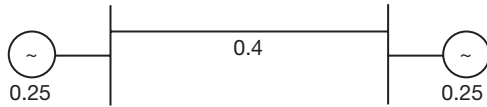
POWER SYSTEMS TEST I

Number of Questions: 35

Section Marks: 90

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. The value of z_{22} for the figure shown below is _____. (All impedances are in pu and ground as reference).



2. Which of the following equation is correct for the minimum voltage value, when the Distributor is fed at both ends with equal voltages.

(A) $V - \frac{IR}{2}$ (B) $V - \frac{IR}{4}$
(C) $V - \frac{IR}{8}$ (D) $V - \frac{IR}{12}$

3. A 3- ϕ transmission line 300km long has the following constants

Resistance/phase/km = 0.20Ω

Reactance/phase/km = 0.5Ω

Shunt admittance/phase/km = $3 \times 10^{-6}S$.

The characteristic impedance of the line is

- (A) $161.55 \angle 68.19$
(B) $0.145 \angle 158.19$
(C) $423.67 \angle -10.90$
(D) $179.5 \times 10^3 \angle -21.81$

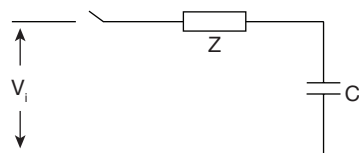
4. The p.u impedance in a power system network is X , if the MVA is increased by two times and voltage becomes half of the previous value. Then the corresponding new p.u impedance is

- (A) $8X$ (B) $4X$
(C) $2X$ (D) X

5. A sending end bus transfers power P_o through a transmission line with per unit impedance 0.4 pu. If the stability margin is 40%, the value of P_o (in pu) for the voltages at sending end and receiving end are 1.2 and 1 pu is _____.

6. Which of the following turbine is Impulse turbine
(A) Pelton wheel turbine (B) Francis turbine
(C) Propeller turbine (D) Kaplan turbine

7. The refracted voltage equation for the following circuit is



- (A) $V(1 - e^{-t/zc})$ (B) $2V(1 - e^{-t/zc})$
(C) $2V(1 - e^{-Zt/c})$ (D) $V(1 - e^{-Zt/c})$

8. The Quantities to be calculated in a voltage controlled bus are

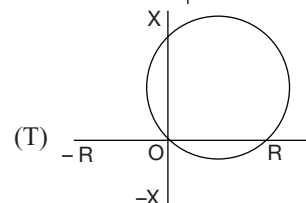
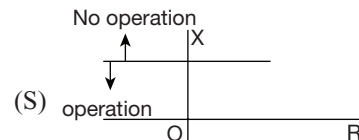
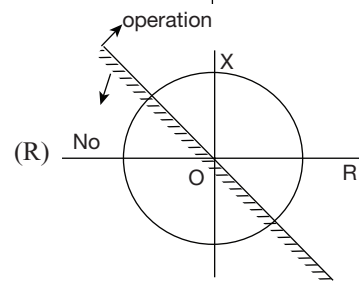
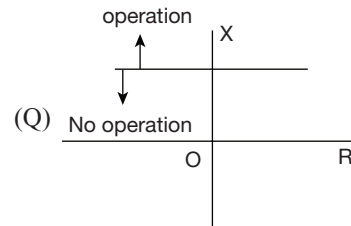
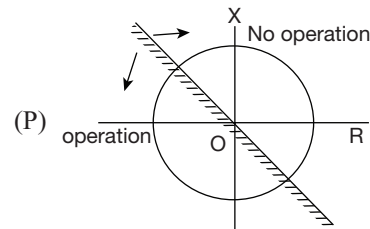
- (A) $|V|, \delta$ (B) P, Q
(C) Q, δ (D) P, V

9. The restraining coil Torque is

- (A) directly proportional to $\frac{(i_1 + i_2) n_r}{2}$
(B) directly proportional to $(i_1 + i_2) n_r$
(C) inversely proportional to $(i_1 + i_2) n_r$
(D) inversely proportional to $\left(\frac{i_1 + i_2}{2}\right) n_r$

10. Match the following Relays.

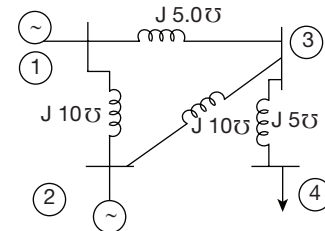
- (1) Impedance Relay with directional unit
(2) Reactance Relay
(3) Mho Relay Characteristics



- (A) 1 – P, 2 – Q, 3 – T (B) 1 – P, 2 – S, 3 – T
(C) 1 – R, 2 – S, 3 – T (D) 1 – R, 2 – Q, 3 – T
11. A 5A IDMT relay has connected to CT of 400/5 ratio with a relay setting of 125%. calculate PSM of the relay with a fault current of 5000A.
(A) 10 (B) 8
(C) 6 (D) 7
12. Isolating switch is used under
(A) No load condition (B) Full load condition
(C) Normal load condition (D) All of the above
13. The efficiency of Thermal Plant is low due to
(A) Condenser (B) Turbine
(C) Super heater (D) Alternator
14. A circuit breaker has a time to reach the peak re-striking voltage of 200 kv at 100 μ sec. The average RRRV of the circuit breaker is
(A) 2 (B) 2×10^6
(C) $200\sqrt{2}$ (D) $200\sqrt{\frac{2}{3}}$
15. A circuit breaker is rated as 2000A, 1500 MVA, 33 kV, 5-second, 3-phase oil circuit breaker. The rated making current of circuit breaker is
(A) 26.24 KA (B) 45.45 KA
(C) 115.90 KA (D) 66.92 KA
16. A single core cable 10 km long has an insulation resistance of 0.5 M Ω . The core diameter is 30 mm and the diameter of the cable over the insulation is 80 mm. The resistivity of the insulating material is
(A) $3.08 \times 10^{10} \Omega\text{m}$ (B) $3.08 \times 10^9 \Omega\text{m}$
(C) $3.08 \times 10^6 \Omega\text{m}$ (D) $3.08 \times 10^3 \Omega\text{m}$
17. The transmission line shown in below has a resistance of 0.205 Ω /km with a loss in the transmission of 7.5% of $\frac{1}{2}$ full load. The line length (in km) is
- 
18. A 1-phase transmission line having a copper conductor of 0.8cm² cross section over which 300 kW at unity power factor and 3,000 V to be delivered. The efficiency of transmission line is 80% and specific resistance as 1.8 $\mu\Omega$ /cm. The maximum length of the line is
(A) 3.75 km (B) 1.67 km
(C) 16.67 km (D) 166.67 km
19. A 3-phase line has a conductor of 5 cm in diameter spaced equilaterally 2 m apart. If the dielectric strength of air is 30 kV(max) per cm. The critical disruptive voltage of the line is ($m_o = 0.9$, $\delta = 0.98$)
(A) 141.91 kV (B) 78.93 kV
(C) 91.04 kV (D) 83.6 kV

20. The most economical size of a single core cable working on 66kV, 3-phase system, if a dielectric stress of 50 kV/cm is
(A) 2.15 cm (B) 5.84 cm
(C) 4.30 cm (D) 11.68 cm
21. The insulation resistance of a single core cable is 500 M Ω per km. If the core diameter is 3cm and resistivity of insulation $5 \times 10^{12} \Omega\text{-m}$. The insulation thickness of the cable is
(A) 2.81cm (B) 1.5cm
(C) 1.8cm (D) 1.31cm
22. A 3-phase, 10 kW induction motor has a $p.f$ of 0.8 lagging. A bank of capacitors is connected in delta across the supply terminals and $p.f$ raised to 0.95 lagging. The KVAR rating of the capacitors connected in each phase is
(A) 4.21 KVAR (B) 1.40 KVAR
(C) 12.63 KVAR (D) 2.43 KVAR
23. A hydro electric generating station is supplied from a reservoir of capacity 6×10^6 cubic meters at a head of 300 meters. The total energy available in kwh is ($\eta_{\text{overall}} = 85\%$, $\eta_{\text{turbine}} = 80\%$)
(A) 4.16×10^6 kwh (B) 6×10^9 kwh
(C) 138.975×10^3 kwh (D) 425×10^3 kwh

24. The y_{BUS} for the following circuit is



- (A) $J \begin{bmatrix} 15 & 10 & 5 & 0 \\ 10 & 20 & 10 & 0 \\ 5 & 10 & 20 & 5 \\ 0 & 0 & 5 & 5 \end{bmatrix}$
- (B) $-J \begin{bmatrix} 0.067 & 0.1 & 0.2 & 0 \\ 0.1 & 0.05 & 0.1 & 0 \\ 0.2 & 0.1 & 0.05 & 0.2 \\ 0 & 0 & 0.2 & 0.2 \end{bmatrix}$
- (C) $J \begin{bmatrix} 20 & 10 & 0 \\ 10 & 5 & 5 \\ 0 & 5 & 10 \end{bmatrix}$
- (D) $-J \begin{bmatrix} 15 & 5 & 10 & 0 \\ 5 & 20 & 0 & 5 \\ 10 & 0 & 20 & 10 \\ 0 & 5 & 10 & 15 \end{bmatrix}$

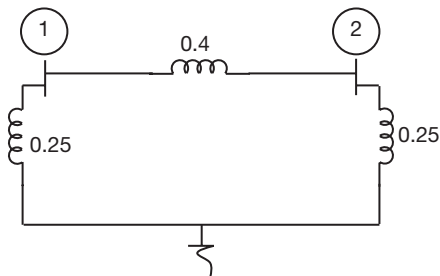
25. The power system network is characterized as $P_e = \frac{EV}{X} \sin \delta$. The voltages of the generator and infinite buses are 1.2 and 1 pu. The reactance of the system is 0.6 pu. The degree of magnetic coupling when the generator initially operating at a power angle of 60° is _____ pu.
26. A 33kV, 3-phase overhead transmission line has a resistance and reactance per phase of 6Ω and 24Ω respectively. The load at the receiving end is 20 MW at a power factor of 0.85 lagging calculate the current supplied by the synchronous conductor to maintain the receiving end voltage 33 kV.
 (A) 408.85A (B) 349.90A
 (C) 230.52A (D) 419.01A
27. A diesel power station has the following data
 Fuel consumption/day = 2000kg
 Units generated/day = 6000kwh
 Calorific value of fuel = 12000 k cal/kg
 Calculate the overall efficiency of the plant if the alternator efficiency is 92%
 (A) 21.5% (B) 33%
 (C) 36.5% (D) 34.20%
28. A 3-phase 33kV/11kV star/delta transformer is protected by circulating current scheme with 600/5 CT on the 33 kV side. The CT ratio on 11 kV side is
 (A) 207.84 : 1 (B) 360 : 1
 (C) 69.28 : 1 (D) 600 : $5\sqrt{3}$
29. A fuse wire of circular cross-section has a radius of 0.6 mm. The wire blows off at a current of 6A. The radius of the wire that will blow off at a current of 2A is
 (A) 0.2 mm (B) 8 mm
 (C) 0.008 mm (D) 0.28 mm
30. A motor is receiving 30% of the power from an infinite bus if the load on the motor is two times, Calculate the maximum value of δ .
 (A) 30° (B) 60°
 (C) 45° (D) 50°
31. A system of 132kV, the line to Ground capacitance is $0.02 \mu\text{F}$ and inductance is 6H. Calculate the voltage appearing across the pole of C.B, if a magnetizing current of 10A and $10\text{k}\Omega$ Resistance across the breaker.
 (A) 132.40 kV (B) 152.50 kV
 (C) 100 kV (D) 173.20 kV
- Common Data for Questions 32 and 33:**
 A delta connected load is supplied from a 3-phase supply. The fuse in the B line is removed and current in the other two lines is 50A.
32. The positive sequence current in the R-phase is
 (A) 86.60A (B) 28.86A
 (C) 0A (D) 49.98A
33. The negative sequence current in the B-phase is
 (A) 0A (B) 86.60A
 (C) 28.86A (D) 49.98A
- Statement for Linked Answer Questions 34 and 35:**
 A single core cable used on a 3-phase, 50Hz, 66 kV system. The cable is 2km long having a core diameter of 20 cm and an impregnated paper in thickness 8 cm. The relative permittivity of the insulation as 4.
34. The capacitance value is
 (A) $0.328 \mu\text{F}$ (B) $0.082 \mu\text{F}$
 (C) $0.164 \mu\text{F}$ (D) $1.312 \mu\text{F}$
35. The charging current is
 (A) 0.981A (B) 15.70A
 (C) 7.34A (D) 3.92A

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 2. C | 3. C | 4. A | 6. A | 7. B | 8. C | 9. A | 10. C | 11. A | 12. A |
| 13. A | 14. A | 15. D | 16. A | 18. C | 19. A | 20. A | 21. D | 22. B | 23. A |
| 24. A | 26. A | 27. A | 28. A | 29. D | 30. B | 31. D | 32. B | 33. C | 34. A |
| 35. D | | | | | | | | | |

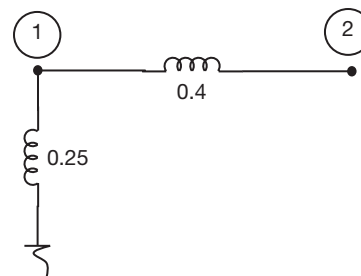
HINTS AND EXPLANATIONS

1. Step (1)



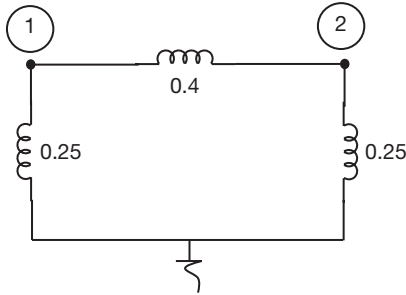
$$Z_{\text{bus}} = [0.25]$$

Step (2)



$$Z_{bus} = \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.65 \end{bmatrix}$$

Step (3)



$$Z_{bus_{new}} = \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.65 \end{bmatrix} - \frac{1}{0.65 + 0.25} \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.65 \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.65 \end{bmatrix} - \frac{1}{0.9} \begin{bmatrix} 0.0625 & 0.1625 \\ 0.1625 & 0.4225 \end{bmatrix}$$

$$Z_{bus} = \begin{bmatrix} 0.1806 & 0.0695 \\ 0.0695 & 0.1806 \end{bmatrix}$$

$$Z_{22} = 0.1806$$

2. Minimum voltage $= V - \frac{IR}{8}$ volts Choice (C)

3. Total Resistance/phase $R = 0.2 \times 300 = 60\Omega$
 Total Reactance/phase $X_L = 0.5 \times 300 = 150\Omega$
 Total Shunt admittance/phase $Y = j3 \times 10^{-6} \times 300$
 $= 0.0009 \angle 90^\circ$
 Series Impedance/phase $Z = 60 + j150 = 161.55 \angle 68.19$

$$\text{Characteristic impedance} = \sqrt{\frac{Z}{Y}}$$

$$= \sqrt{\frac{(161.55 \angle 68.19)}{(0.009 \angle 90)}}$$

$$= 423.67 \angle -10.90$$

Choice (C)

4. Per unit impedance $\propto \frac{MVA}{(kv)^2}$

$$\frac{X_1}{X_2} = \frac{MVA_1}{(kv_1)^2} \times \frac{(kv_2)^2}{MVA_2}$$

$$= \frac{MVA_1}{(kv_1)^2} \times \frac{(kv_1)^2}{4 \times 2 MVA_1}$$

$$\Rightarrow X_2 = 8X_1 \Rightarrow X_2 = 8X$$

Choice (A)

5. Steady state stability margin $= \frac{P_{max} - P_o}{P_{max}} \times 100$

$$P_o = P_{max} \sin \delta$$

$$40 = \frac{P_{max} - P_{max} \sin \delta}{P_{max}} \times 100$$

$$1 - \sin \delta = 0.4$$

$$\sin \delta = 0.6$$

$$\delta = 36.86^\circ$$

$$P_o = \frac{+EV}{X} \sin \delta$$

$$= \frac{1.2 \times 1}{0.4} \times 0.6$$

$$P_o = 1.8 \text{ pu.}$$

6. Choice (A)

$$7. V^1(s) = \text{refracted voltage} = \frac{2 \cdot \frac{1}{CS}}{z + \frac{1}{CS}} \cdot \frac{V}{S}$$

$$= \frac{2V}{S} \cdot \frac{1}{1 + ZCS}$$

$$= \frac{2V}{S} \cdot \frac{\frac{1}{ZC}}{S + \frac{1}{ZC}} = 2V \left(\frac{1}{S} - \frac{1}{S + \frac{1}{ZC}} \right)$$

$$V^1(t) = 2V(1 - e^{-t/zc})$$

Choice (B)

8. The P, V Quantities specified for voltage controlled bus and Q, δ Quantities are to be calculated. Choice (C)

9. Choice (A)

10. Choice (C)

$$11. \text{PSM} = \frac{\text{secondary current}}{\text{Relay current setting}}$$

$$= \frac{5000}{6.25 \times 80} = 10$$

Choice (A)

12. Isolating switch is Generally used under no load condition only. Choice (A)

13. Choice (A)

$$14. \text{Average RRRV} = \frac{200 \text{ kV}}{100} = 2 \text{ kV}/\mu\text{sec} \quad \text{Choice (A)}$$

$$15. \text{Rated making current} = 2.55 \times \frac{1500 \times 10^6}{\sqrt{3} \times 33 \times 10^3}$$

$$= 66.92 \text{ KA}$$

Choice (D)

16. Length $l = 10,000\text{m}$

$$\text{Insulation Resistance } R = 0.5 \times 10^6 \Omega$$

$$\text{Conductor radius } r_1 = \frac{30}{2} = 15\text{mm}$$

$$\text{Internal sheath radius } r_2 = \frac{80}{2} = 40\text{mm}$$

$$R = \frac{\rho}{2\pi l} \ln \frac{r_2}{r_1}$$

$$0.5 \times 10^6 = \frac{\rho}{2\pi \times 10,000} \ln \left(\frac{40}{15} \right)$$

$$\rho = 0.5 \times 10^6 \times 2\pi \times 10,000 \ln\left(\frac{40}{15}\right)$$

$$\rho = 3.08 \times 10^{10} \Omega\text{m}$$

Choice (A)

$$\begin{aligned} 17. \text{ Line current } I &= \frac{P}{\sqrt{3} V \cos \phi} \\ &= \frac{15 \times 10^6}{\sqrt{3} \times 132 \times 10^3 \times 0.8} \\ &= 82 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{Line losses} &= 3I^2 R = \frac{7.5}{100} \times \frac{15 \text{ m}}{2} \\ &= 562.5 \text{ kW} \end{aligned}$$

$$\begin{aligned} R &= \frac{562.5 \times 10^3}{3 \times 82^2} \\ &= 27.88 \Omega \end{aligned}$$

$$\text{Line length} = \frac{27.88}{0.205} = 136 \text{ km.}$$

18. Receiving end power = 300kW = 3,00,000 watts
Transmission efficiency = 0.8

$$\text{Sending end power} = \frac{3,00,000}{0.8} = 3,75,000 \text{ watts}$$

$$\text{Line losses} = 3,75,000 - 3,00,000 = 75,000 \text{ W}$$

$$\text{Line current } I = \frac{300 \times 10^3}{3000} = 100 \text{ A}$$

$$\text{Line losses} = 2 I^2 R$$

$$75,000 = 2 \times (100)^2 \times R \Rightarrow R = \frac{75,000}{2 \times (100)^2} = 3.75 \Omega$$

$$R = \frac{\rho L}{a}$$

$$l = \frac{3.75 \times 0.8}{1.8 \times 10^{-6}} = 1.67 \times 10^6 \text{ cm} \Rightarrow 16.67 \text{ km} \quad \text{Choice (C)}$$

19. Conductor radius $r = \frac{5}{2} = 2.5 \text{ cm}$

$$\text{Conductor spacing } d = 2 \text{ m} = 200 \text{ cm}$$

$$\text{Dielectric strength } g_o = 21.2 \text{ kV(rms) per cm}$$

$$\text{Disruptive critical voltage}$$

$$V_c = m_o g_o \delta r \ln\left(\frac{d}{r}\right) \text{ kV/phase}$$

$$= 0.9 \times 21.2 \times 0.98 \times 1 \times \ln\left(\frac{200}{2.5}\right)$$

$$= 81.936 \text{ kV/phase}$$

$$\text{Line voltage} = \sqrt{3} \times 81.93 = 141.91 \text{ kV} \quad \text{Choice (A)}$$

20. Phase voltage of cable = $\frac{66}{\sqrt{3}} = 38.10 \text{ kV}$

$$\text{Peak value of phase voltage } V = 38.01 \times \sqrt{2} = 53.73 \text{ kV}$$

$$\text{Maximum permissible stress } g_{\max} = 50 \text{ kV/cm}$$

Most economical conductor diameter is

$$d = \frac{2V}{g_{\max}} = \frac{2 \times 53.75}{50} = 2.15 \text{ cm}$$

Choice (A)

21. Length of cable $\ell = 1 \text{ km} = 1000 \text{ m}$

$$\text{Cable insulation Resistance } R = 500 \times 10^6 \Omega$$

$$\text{Conductor radius} = \frac{3}{2} = 1.5 \text{ cm}$$

$$\rho = 5 \times 10^{12} \Omega\text{m}$$

$$r_2 = \text{internal sheath radius}$$

$$R = \frac{\rho}{2\pi l} \ln\left(\frac{r_2}{r_1}\right)$$

$$\ln\left(\frac{r_2}{r_1}\right) = \frac{2\pi l R}{\rho} = \frac{2\pi \times 1000 \times 500 \times 10^6}{5 \times 10^{12}} = 0.628$$

$$\frac{r_2}{r_1} = 1.87 \Rightarrow r_2 = 1.5 \text{ cm} \times 1.87 = 2.81 \text{ cm}$$

$$\text{Insulation thickness} = r_2 - r_1 = 1.31 \text{ cm} \quad \text{Choice (D)}$$

22. $\cos \phi_1 = 0.8 \text{ lag} \Rightarrow \phi_1 = 36.86^\circ$

$$P = 10 \text{ kW}$$

$$\cos \phi_2 = 0.95 \text{ lag} \Rightarrow \phi_2 = 18.19^\circ$$

$$\text{Leading KVAR taken by the condenser bank}$$

$$= P(\tan \phi_1 - \tan \phi_2)$$

$$= 10(\tan 36.86 - \tan 18.19)$$

$$= 4.21 \text{ KVAR}$$

$$\therefore \text{Rating of capacitors connected in each phase}$$

$$= \frac{4.211}{3} = 1.40 \text{ KVAR} \quad \text{Choice (B)}$$

23. Weight of water available is

$$W = \text{volume of water} \times \text{density}$$

$$= 6 \times 10^6 \times 1000 (\because \text{mass of } 1 \text{ m}^3 \text{ of water is } 1000 \text{ kg})$$

$$= 6 \times 10^9 \text{ kg} = 6 \times 10^9 \times 9.81 \text{ N}$$

$$\text{Electrical energy available} = \omega \times H \times \eta_{\text{overall}}$$

$$= 9.81 \times 6 \times 10^9 \times 300 \times 0.85 \text{ watt sec}$$

$$= \frac{6 \times 10^9 \times 300 \times 9.81 \times 0.85}{3600 \times 1000}$$

$$= 4.16 \times 10^6 \text{ kWh}$$

Choice (A)

24. By Inspection method the y_{bus} matrix is

$$y_{\text{BUS}} = J \begin{bmatrix} 15 & 10 & 5 & 0 \\ 10 & 20 & 10 & 0 \\ 5 & 10 & 20 & 5 \\ 0 & 0 & 5 & 5 \end{bmatrix}$$

Choice (A)

25. Degree of magnetic coupling = Synchronous power

$$= \frac{EV}{X} \sin \delta$$

$$= \frac{1.2 \times 1}{0.6} \cos 60^\circ$$

$$= 1 \text{ pu.}$$

26. Load current $I_L = \frac{20 \times 10^6}{\sqrt{3} \times 33000 \times 0.85} = 411.65 \text{ A}$

Sending end voltage/phase V_s = receiving end voltage/phase V_r

$$= \frac{33 \times 10^3}{\sqrt{3}} = 19053 \text{ V}$$

$$V_s^2 = [V_r + I_L \cos \phi \cdot R - (I_c - I_L \sin \phi) X]^2 +$$

$$[I_L \cos \phi \cdot x + (I_c - I_L \sin \phi) R]^2$$

$$(19053)^2 = [19053 + 411.65 \times 0.85 \times 6 - (I_c - 411.65 \times 0.52) 24]^2 + [411.65 \times 0.85 \times 24 + (I_c - 411.65 \times 0.52) 6]^2$$

$$(19053)^2 = [19053 + 2099.41 - 24I_c + 5137.39]^2 + [8397.66 + 6I_c - 1284.34]^2 \quad \text{————— (1)}$$

On solving equation (1) $I_c = 408.85 \text{ A}$, 1513.60 the synchronous condenser current $I_c = 408.85 \text{ A}$ Choice (A)

27. Fuel consumption = $\frac{2000}{6000} = 0.34 \text{ kg/kwh}$

Heat produced by fuel per day = $2000 \times 1200 = 24 \times 10^6 \text{ k cal}$

$$\text{Overall efficiency} = \frac{\text{electrical output in heat units per day}}{\text{Heat produced by fuel per day}}$$

$$= \frac{6000 \times 860}{24 \times 10^6} = 21.5\% \quad \text{Choice (A)}$$

28. Phase current of delta connected CTs on 33kV side = 5A

Line current of delta connected CTs on 33kV side = $5\sqrt{3} \text{ A}$

Phase current of star connected CTs on 11000V side = $5\sqrt{3} \text{ A}$

if I is the line current on 11000V side then primary apparent power = secondary apparent power

$$\sqrt{3} \times 33000 \times 600 = \sqrt{3} \times 11000 \times I$$

$$I = 1800 \text{ A}$$

Turn-ratio of CTs on 11000 V side

$$= 1800 : 5\sqrt{3} = 207.84 : 1 \quad \text{Choice (A)}$$

29. $I a(r)^{\frac{3}{2}}$

$$\frac{I_2}{I_1} = \left(\frac{r_2}{r_1} \right)^{\frac{3}{2}}$$

$$r_2 = r_1 \times \left(\frac{I_2}{I_1} \right)^{\frac{2}{3}} = 0.6 \times \left(\frac{2}{6} \right)^{\frac{2}{3}} = 0.28 \text{ mm} \quad \text{Choice (D)}$$

30. $\sin \delta_o = 0.3 \Rightarrow \delta_o = 17.46$
 $\sin \delta_c = 0.6 \Rightarrow \delta_c = 36.86$

$$\delta_m = ?$$

$$(\delta_m - \delta_o) 0.6 = \int_{\delta_o}^{\delta_m} \sin \delta d\delta$$

$$0.6(\delta_m - 0.304) = \cos \delta_o - \cos \delta_m$$

$$0.6\delta_m = \cos \delta_o - \cos \delta_m + 0.1824$$

$$0.6\delta_m + \cos \delta_m = \cos(17.46) + 0.1824$$

$$0.6\delta_m + \cos \delta_m = 0.9539 + 0.1824 = 1.1363$$

Guess $\delta_m > 30$, solving above equation $\delta_m = 60$

Choice (B)

31. Voltage $V = i \sqrt{\frac{L}{C}}$

$$= 10 \sqrt{\frac{6}{0.02 \mu}} = 10 \times 10^3 \sqrt{\frac{600}{2}}$$

$$= 10 \times 10^4 \sqrt{3} = 173.20 \text{ kV}$$

Choice (D)

32. Given Data $I_R = 50 \angle 0$

$$I_Y = 50 \angle 180^\circ$$

$$I_B = 0 \text{ A}$$

$$I_{R1} = \frac{1}{3} (I_R + a I_Y + a^2 I_B)$$

$$= \frac{1}{3} [50 \angle 0 + 1 \angle 120 \times 50 \angle 180] = 28.86 \angle -30$$

$$I_{R2} = \frac{1}{3} [I_R + a^2 I_Y + a I_B]$$

$$= \frac{1}{3} [50 \angle 0 + 1 \angle 240 \times 50 \angle 180 + 0]$$

$$= 28.86 \angle 30^\circ$$

Choice (B)

33. $I_{b2} = a^2 I_{R2} = 1 \angle 240 \times 28.86 \angle 30$

$$= 28.86 \angle -90^\circ$$

Choice (C)

34. Capacitance of cable $C = \frac{\epsilon_r l}{41.4 \ln \left(\frac{D}{d} \right)} \times 10^{-9} \text{ F}$

Here $\epsilon_r = 4$, $l = 2000 \text{ m}$

$$d = 20 \text{ cm } D = 20 + 2 \times 8 = 36 \text{ cm}$$

$$C = \frac{4 \times 2000}{41.4 \ln \left(\frac{36}{20} \right)} \times 10^{-9} = 0.328 \mu\text{F}$$

Choice (A)

35. Voltage between core and sheath is

$$V_{ph} = \frac{66}{\sqrt{3}} = 38.10 \times 10^3 \text{ V}$$

$$= 38.10 \text{ kV}$$

$$\text{Charging current} = \frac{V_{ph}}{X_c} = 2\pi f c V_{ph}$$

$$= 2\pi \times 50 \times 0.328 \times 10^{-6} \times 38.1 \times 10^3$$

$$= 3.92 \text{ A}$$

Choice (D)

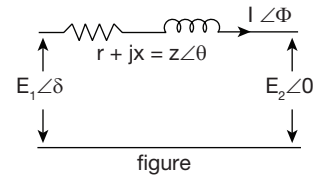
POWER SYSTEMS TEST 2

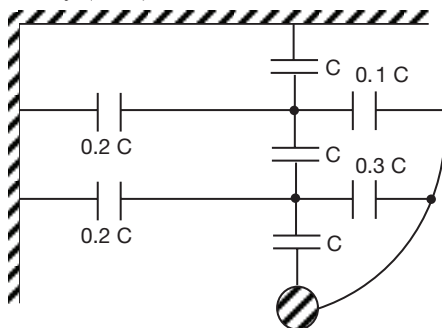
Number of Questions: 35

Section Marks: 90

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. The voltage appears across the contacts of the circuit breaker during the arcing period is
(A) Arc voltage (B) Restriking voltage
(C) Recovery Voltage (D) None of the above
2. In low oil circuit breakers the conducting parts are insulated by
(1) Oil
(2) Air
(3) Porcelain
(4) Organic insulating material
(A) 1, 2 and 3 (B) 1, 2 and 4
(C) 2, 3 and 4 (D) all of the above
3. If the length of arc increases the arc resistance
(A) Increases (B) decreases
(C) remains same (D) None of the above
4. In a 3- ϕ system the sum of line currents is equal to
(A) neutral current
(B) fault current
(C) zero sequence current
(D) $\sqrt{3}$ times of any phase current
5. All phase sequence currents are equal in
(A) shunt faults (B) L-G fault
(C) LLL fault (D) series faults
6. A conductor consists of seven identical strands each having a radius of 0.4cm. The self GMD of the conductor is
(A) 2.8 cm (B) 0.87 cm
(C) 1.2 cm (D) 0.8 cm
7. The conductor material properties are
(1) high electrical conductivity
(2) high tensile strength
(3) low specific gravity
(A) 1 and 2 (B) 2 and 3
(C) 1 and 3 (D) 1, 2 and 3
8. The number of wires are present in four layer stranded conductor is
(A) 60 (B) 61
(C) 36 (D) 37
9. In D.C distribution system the value of $\frac{\text{volume of copper for 3-wire system}}{\text{volume of copper for 2-wire system}}$ is
(A) 0.31 (B) 3.2
(C) 0.8 (D) 1.2
10. In the following single line diagram the maximum power received at load end is

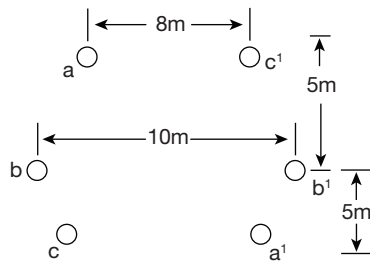


- (A) $\alpha + \delta = 90^\circ$ (B) $\theta + \phi = 90^\circ$
(C) $\alpha - \delta = 90^\circ$ (D) $\theta - \phi = 90^\circ$
11. An alternator is connected to an infinite bus bar and delivers 1.0 p.u current at 1.0 p.u voltage and 0.9 p.f lagging. The real power is
($\because$ reactance of the alternator $X_d = 1.0$ p.u)
(A) 1.0 p.u (B) 0.9 p.u
(C) 0.435 p.u (D) 0.897 p.u
12. For the figure shown below, the value of string efficiency (in %) is 
13. A generator is connected through a 3-cycle C.B to a transformer is rated 20 MVA, 11kV with reactance's of $X_d^{11} = 15\%$, $X_d^1 = 20\%$, and $X_d = 80\%$. It is operating at no load and rated voltage when a 3-phase short circuit occurs between the breaker and transformer. The initial symmetrical rms current in the breaker is
(A) 7 kA (B) 1.04 kA
(C) 1.81 kA (D) 12.12 kA
14. A system of 220kV, the line to ground capacitance is $0.05 \mu\text{F}$ and the inductance is 5H. Calculate the voltage appearing across the pole in a C.B if a magnetizing current of 10A.
(A) 100 kV (B) 10 kV
(C) 220 kV (D) 127 kV
15. The string efficiency can be improved by using
(A) longer cross arm
(B) guard rings
(C) Grading of insulator discs
(D) All of the above
16. A 3-phase, 50Hz, 20km long overhead line supplies 2000kw at 11kV, 0.9 p.f lagging. The line resistance is 0.04Ω per phase per km and line inductance is 0.8mH

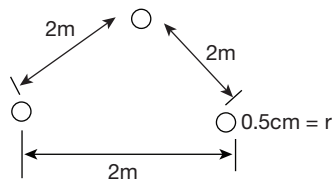
per phase per km. The voltage regulation of the transmission system is

- (A) 5.32% (B) 2.46%
(C) 4.36% (D) 8.26%

17. An insulator string consists of three units, each having a safe working voltage of 15kV. The ratio of self capacitance to shunt capacitance of each unit is 6:1. The string efficiency is
(A) 76.96% (B) 80.44%
(C) 72.55% (D) 86.54%
18. The A and B parameters of a 3-phase overhead transmission line are $0.95\angle 0$ and $150\angle 90$ respectively. To maintain the receiving end voltage equal to the sending end by connecting a shunt inductive reactor the value of shunt reactor should be
(A) $3000\angle 90$ (B) $157.89\angle 90$
(C) $6.34 \times 10^{-3}\angle -90$ (D) $3.34 \times 10^{-4}\angle -90$
19. The inductance per km of a transposed double circuit 3- ϕ line as shown in the figure is (diameter of each conductor is 3 cm)



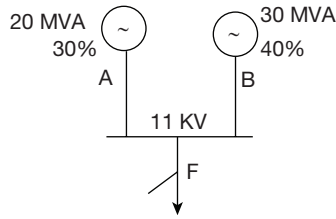
- (A) 0.608 mH/km/phase (B) 0.264 mH/km/phase
(C) 0.608 H/km/phase (D) 0.264 H/km/phase
20. A 3- ϕ , 132kV, 50Hz overhead transmission line as shown in the figure. The charging current per km is



- (A) 0.2 A/km (B) 0.2 mA/km
(C) 0.2 kA/km (D) 0.2 kA/km
21. A conductor with 3cm diameter is passed centrally through a porcelain bushing $\epsilon_r = 5$ having internal and external diameters of 4 cm and 10 cm respectively. Calculate the maximum gradient on the surface of the conductor, if 20kV r.m.s voltage applied between the conductor and earth.
(A) 34.32 kV/cm (B) 37.48 kV/cm
(C) 32.42 kV/cm (D) 28.32 kV/cm
22. A 50Hz overhead line has line to earth capacitance of $2\mu\text{F}$. It is decided to use an earth fault neutralizer.

Calculate the reactance to neutralize the capacitance of 85% length of the line.

- (A) 624.45 ohms (B) 1873.36 ohms
(C) 936.68 ohms (D) 1248.90 ohms
23. Two overhead lines are connected in parallel to supply a load of 20 MW at 0.8 pf (lag) and 30 kV. The reactance's of line A and B are 15Ω and 20Ω respectively. The power carried by transmission line B is _____ kW.
24. A 3-phase, 400kV, 50Hz transmission line consists of 2cm radius conductor spaced 4 meters apart in equilateral triangular formation, if the temperature is 40°C and atmospheric pressure is 76cm. The corona loss per km of the line is _____ ($m_o = 0.85$)
(A) 2 kw/km (B) 4 kw/km
(C) 1 kw/km (D) 0.5 kw/km
25. A 3-phase, 20 MVA, 11kV alternator has internal reactance of 6% and negligible resistance. Calculate the external percentage reactance per phase to be connected in series with the alternator so that steady current on short circuit does not exceed 10 times the full load current
(A) 10% (B) 4%
(C) 6% (D) 8%
26. A balanced star connected load takes 80A from a balanced 3-phase, 4-wire supply. If the fuses in the Y and B are removed. The Y-phase current before the fuses are removed is
(A) 0A (B) $80\angle 0$
(C) $80\angle 120$ (D) $80\angle 240^\circ$
27. A star-connected 3-phase, 20 MVA, 6.6kV alternator is protected by merz-price circulating current principle using 500/5 amperes current transformers. The star point of the alternator is earthed through a resistance of 8Ω , if the minimum operating current of the relay is 1.25A. The percentage of each phase of the stator winding which is unprotected against earth faults is
(A) 26.24% (B) 31.23%
(C) 16.24% (D) 8.32%
28. A two wire d.c distributor cable 1500 meters long is loaded with 0.8 A/meter, Resistance of each conductor is $0.05\Omega/\text{km}$. The maximum voltage drop of distributor, if the distributor is fed from both ends with equal voltages of 200V is.
(A) 22 V (B) 45 V
(C) 90 V (D) 60 V
29. The percentage reactance of each alternator on its own capacity as shown in the figure. Calculate the short circuit current when a 3-phase short circuit occurs at point F. (Base MVA = 50)



- (A) 8747.7 A (B) 2624.3 A
(C) 5248.6 A (D) 9499.8 A

30. Calculate the distance over which a load of 20MW at a p.f 0.8 lagging can be delivered by a 3-phase transmission line having conductors each of resistance 2Ω per kilometer. The voltage at the receiving end is to be 220 kV and the transmission loss is to be 8%.
(A) 62 km (B) 124 km
(C) 372 km (D) 186 km
31. A 50Hz generator is delivering 40% of the power that it is capable of delivering through a transmission line to an infinite bus. A fault occurs that increases the reactance between the generator and the infinite bus to 400% of the value before the fault when the fault is isolated, the maximum power that can be delivered is 60% of the original maximum value. The critical clearing angle of the machine is
(A) 69.23° (B) 23.57°
(C) 138.19° (D) 41.81°

Common Data Questions 32 and 33:

An alternator and a synchronous motor each rated for 100 MVA, 11 kV having sub transient reactance of 25% are connected through a transmission line of reactance 15% on the base of machine ratings. The motor acts as a load of 30MW at 0.8 p.f lead and terminal voltage 10kV when a 3-phase fault takes place at the motor terminals.

32. The p.u fault current supplied by generator
(A) 3.63 p.u (B) 2.27 p.u
(C) 5.90 p.u (D) 5.24 p.u
33. The p.u fault current supplied by motor
(A) 3.63 p.u (B) 2.27 p.u
(C) 5.24 p.u (D) 5.90 p.u

Linked Answer Question 34 and 35:

A 50Hz, 11kV, 3-phase alternator with earthed neutral has a reactance of 8 ohms and is connected to a bus-bar through a circuit breaker. The capacitance between phase and neutral is $0.05\mu\text{F}$.

34. The frequency of oscillation is
(A) 9.00 kHz (B) 4.50 kHz
(C) 28.28 kHz (D) 48.23 Hz
35. The average RRRV upto first peak is
(A) 8.98 kV/sec
(B) 161.82×10^3 kV/sec
(C) 80.91 kV/sec
(D) 114.42×10^3 kV/sec

ANSWER KEYS

- | | | | | | | | | | |
|-------|------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. A | 4. A | 5. B | 6. B | 7. D | 8. B | 9. A | 10. A |
| 11. D | 12. 94.8 to 95.8 | 13. A | 14. A | 15. D | 16. A | 17. B | 18. A | 19. A | |
| 20. A | 21. D | 22. A | 24. A | 25. B | 26. D | 27. A | 28. A | 29. A | 30. A |
| 31. A | 32. B | 33. A | 34. B | 35. B | | | | | |

HINTS AND EXPLANATIONS

1. The voltage appears across the contacts of the circuit breaker during the arcing period is Arc voltage.
Choice (A)
2. In low oil circuit breaker oil is used only for arc extinction. So, the conducting parts are insulated by air, porcelain and organic insulating material. Choice (C)
3. Choice (A)
4. $I_a + I_b + I_c = I_n$
 $\Rightarrow I_a, I_b, I_c$ are line currents of phases a, b, c respectively
 $\Rightarrow$ if there is no path neutral current. The sum of line currents is zero. Choice (A)

5. In L-G-fault:-

$$I_{RO} = I_{R1} = I_{R2} = \frac{I_R}{3}$$

Choice (B)

6. Self GMD = $2.176 \times 0.4 \text{ cm}$
 $= 0.8704 \text{ cm}$

Choice (B)

7. Choice (D)

8. The total number of individual wires
 $= 3n(n+1) + 1$
 $= 3 \times 4(4 \times 1) + 1 = 61$

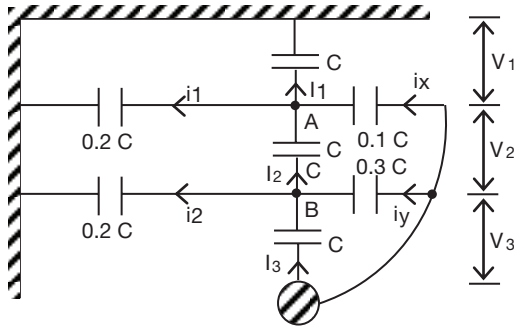
Choice (B)

11. $E = 1.0 + j1.0(0.9 - j0.435)$
 $= 1.435 + j0.9$
 $= 1.693 \angle 32.09$

$$P = \frac{1.69 \times 1.0}{1.0} \sin 32.09 = 0.897 \text{ p.u}$$

Choice (D)

12. Redrawing the circuit with the currents.



KCL @ A & B

$$I_2 + i_x = I_1 + i_1$$

$$WCV_2 + 0.1C(V_2 + V_3)$$

$$= WCV_1 + W(0.2C)V_1$$

$$1.2V_1 = 1.1V_2 + 0.1V_3$$

$$I_3 + i_y = I_2 + i_2$$

$$WCV_3 + W0.3C V_3 = WCV_2 + W0.2C(V_1 + V_2)$$

$$1.3V_3 = 0.2V_1 + 1.2V_2$$

Solve equation (1) and (2) we get

$$V_2 = \frac{154}{155}V_1 \text{ and } V_3 = \frac{166}{155}V_1$$

$$\text{String efficiency} = \frac{V_1 + V_2 + V_3}{3V_3} \times 100$$

$$= \frac{V_1 + \frac{154}{155}V_1 + \frac{166}{155}V_1}{3 \times \frac{166}{155}V_1} \times 100$$

$$= \frac{155 + 154 + 166}{3 \times 166} \times 100$$

$$= 95.38\%$$

13. short circuit MVA = $\frac{20}{15} \times 100 = 133.34 \text{ MVA}$

$$\text{short circuit current} = \frac{133.34}{\sqrt{3} \times 11} = 6.99 \text{ kA} = 7 \text{ kA}$$

Choice (A)

14. $v = i\sqrt{\frac{L}{C}} = 10\sqrt{\frac{5}{0.05}} = 100 \text{ kV}$

Choice (A)

15. Choice (D)

16. $R = 0.04 \times 20 = 0.8 \Omega$

$$X_L = 2\pi fL \times 20 = 2\pi \times 50 \times 0.8 \times 10^{-3} \times 20$$

$$= 5.02 \Omega$$

Receiving end voltage/phase

$$= V_R = \frac{11 \times 10^3}{\sqrt{3}} = 6351 \text{ V}$$

Load Power Factor $\cos\phi_R = 0.9$ lagging

Line current

$$= \frac{2000 \times 10^3}{\sqrt{3} \times V_L \times \cos\phi} = \frac{2000 \times 10^3}{\sqrt{3} \times 11 \times 10^3 \times 0.9} = 116.63 \text{ A}$$

$$V_s = V_R + IR \cos\phi_R + IX_L \sin\phi_R$$

$$= 6351 + 116.63 \times 0.8 \times 0.9 + 116.63 \times 5.02 \times \sin(\cos^{-1}(0.9)) = 6689.17 \text{ V}$$

$$\text{Percentage voltage regulation} = \frac{V_s - V_R}{V_R} \times 100$$

$$= \frac{6689 - 6351}{6351} \times 100 = 5.32\%$$

Choice (A)

17. The maximum voltage appears across the lowest unit in the string

$$V_1 = 15 \text{ kV}$$

$$K = \frac{1}{6} = 0.167$$

$$V_2 = V_3(1 + k) \Rightarrow V_3 = \frac{V_2}{1 + k}$$

$$V_3 = \frac{V_2}{1 + 0.167} = 0.85V_2$$

$$V_1 = V_2 + (V_3 + V_2)k$$

$$= V_2 + (0.85V_2 + V_2) \times 0.167$$

$$V_1 = 1.308V_2$$

$$\Rightarrow V_2 = \frac{15 \text{ kV}}{1.308} = 11.46 \text{ kV}$$

$$\Rightarrow V_3 = 9.74 \text{ kV}$$

$$\text{Voltage across the string} = V_1 + V_2 + V_3$$

$$= 9.74 + 11.46 + 15$$

$$= 36.20 \text{ kV}$$

$$\text{String efficiency} = \frac{36.20}{3 \times 15} \times 100 = 80.44\%$$

Choice (B)

18. $V_s = AV_R + BI_R$

$$\text{When } V_R = V_s \Rightarrow V_R(1 - A) = BI_R$$

$$\frac{V_R}{I_R} = \frac{B}{1 - A}$$

$$= \frac{150 \angle 90}{1 - 0.95 \angle 0} = 3000 \angle 90$$

Choice (A)

19. self GMD of each conductor

$$= 1.5 \times 0.7788 \text{ cm} = 0.01168 \text{ cm}$$

$$D_{bc} = D_{ab} = \sqrt{5^2 + 1^2} = 5.099 \text{ m}$$

$$D_{ab^1} = \sqrt{5^2 + 9^2} = 10.295 \text{ m}$$

$$D_{aa^1} = \sqrt{10^2 + 8^2} = 12.806 \text{ m}$$

$$D_{m1} = \sqrt[4]{5.099 \times 10 \times 8 \times 10.295} = 8.050 \text{ m}$$

$$D_{m2} = \sqrt[4]{5.099 \times 5.099 \times 10.295 \times 10.295} = 7.245 \text{ m}$$

$$D_{m3} = D_{m1} = 8.05 \text{ m}$$

$$D_m = \sqrt[3]{D_{m1} \times D_{m2} \times D_{m3}} = 7.772\text{m}$$

$$\text{Self GMD of each phase } D_{s1} = \sqrt{0.01168 \times 12.806}$$

$$= 0.3867\text{m} = D_{s3}$$

$$D_{s2} = \sqrt{0.01168 \times 10} = 0.3417\text{m}$$

$$D_s = \sqrt[3]{D_{s1} \times D_{s2} \times D_{s3}} = 0.3710\text{m}$$

$$\text{Inductance} = 2 \times 10^{-4} \ell n \left(\frac{7.772}{0.3710} \right)$$

$$= 0.608 \text{ mH/km/phase} \quad \text{Choice (A)}$$

20. The radius of conductors $r = 0.5\text{cm}$

$$\text{The mutual GMD of conductors } D_m = 2\text{m}$$

$$\text{Capacitance per phase per meter} = \frac{2\pi \epsilon_o}{\ln \left(\frac{200}{0.5} \right)} \frac{F}{m}$$

$$= 9.285 \times 10^{-9} \text{ F/km}$$

$$\text{The charging current} = \frac{132 \times 1000}{\sqrt{3}} \times 9.285 \times 10^{-9} \times 314$$

$$= 0.222 \text{ A/km} \quad \text{Choice (A)}$$

21. $g_{1\max} = \frac{q}{2\pi \epsilon_o r}$

$$g_{2\max} = \frac{q}{2\pi \epsilon_o \epsilon_r r_1}$$

$$\Rightarrow g_{1\max} r = g_{2\max} \times 5 \times 2$$

$$\Rightarrow g_{1\max} 1.5 = g_{2\max} \times 5 \times 2$$

$$g_{1\max} = 6.67 g_{2\max}$$

$$\text{now } g_{2\max} = \frac{g_{1\max}}{6.67} = 0.15 g_{1\max}$$

$$20 = g_{1\max} r \ln \left(\frac{2}{1.5} \right) + g_{2\max} \times 2 \ln \left(\frac{5}{2} \right)$$

$$= 1.5 g_{1\max} \cdot \ln \left(\frac{2}{1.5} \right) + 0.15 g_{1\max} \times 2 \ln \left(\frac{5}{2} \right)$$

$$= 0.431 g_{1\max} + 0.274 g_{1\max} = 0.706 g_{1\max}$$

$$\Rightarrow g_{1\max} = \frac{20}{0.706} = 28.32 \text{ kV/cm.} \quad \text{Choice (D)}$$

22. The inductive reactance for neutralizing 85% of the capacitance

$$L = \frac{1}{3c} = \frac{10^6}{3 \times 314 \times 2 \times 0.85}$$

$$= 624.45 \text{ ohms} \quad \text{Choice (A)}$$

23. $S_B = \frac{P}{P.f} \times \frac{Z_A}{Z_A + Z_B}$

$$= \frac{20 \times 10^3}{0.8} \times \frac{j15}{j35} \angle -36.86$$

$$= 10714.28 \angle -36.86 \text{ KVA}$$

$$P_B = 10714.28 \times \cos(-36.86)$$

$$= 8572.53 \text{ KW.}$$

24. $P = 241 \times 10^{-5} \left(\frac{f+25}{\delta} \right) \sqrt{\frac{r}{d}} (V - V_c)^2 \text{ kw/km/phase}$

$$\delta = \frac{3.926b}{273+t} = \frac{3.92 \times 76}{273+40} = 0.952$$

$$g_o = 21.1 \text{ kV/cm (r.m.s)}$$

$$V_c = m g_o \delta r \ln \left(\frac{d}{r} \right) \text{ KV}$$

$$= 0.85 \times 21.1 \times 0.952 \times 2 \times \ln \left(\frac{400}{2} \right)$$

$$= 180.92 \text{ kV}$$

$$\text{Supply voltage per phase} = V = \frac{400}{\sqrt{3}} = 230.94 \text{ kV}$$

$$P = 241 \times 10^{-5} \left(\frac{50+25}{0.952} \right) \times \sqrt{\frac{2}{400}} \times (230.94 - 180.92)$$

$$= 0.6715 \text{ kw/km/phase}$$

$$\text{Total corona loss per km} = 3 \times 0.6715$$

$$= 2.014 \text{ kw/km}$$

$$\text{Choice (A)}$$

25. Full load current $I = \frac{20 \times 10^6}{\sqrt{3} \times 11 \times 10^3}$

$$\text{Voltage per phase } V = \frac{11 \times 10^3}{\sqrt{3}}$$

$$\text{Total percentage reactance required}$$

$$= \frac{\text{Full load current}}{\text{short circuit current}} \times 100$$

$$= \frac{1}{10} \times 100 = 10\%$$

$$\text{External percentage reactance required}$$

$$= 10\% - 6\% = 4\%$$

$$\text{Choice (B)}$$

26. Before the fuses are removed from Y and B lines, the system is balanced and current in each line is 80A

$$I_Y = 80 \angle 240^\circ \text{ A}$$

$$\text{Choice (D)}$$

27. X% of the winding be unprotected $r = 8\Omega$

$$V_{ph} = \frac{6.6 \times 10^3}{\sqrt{3}} = 3810 \text{ V}$$

$$\text{Minimum fault current which will operate the relay}$$

$$= \frac{500}{5} \times 1.25 = 125$$

$$\text{e.m.f induced in } x\% \text{ winding}$$

$$= V_{ph} \times \left(\frac{x}{100} \right) = 3810 \times \frac{x}{100}$$

$$\text{The current } 125 = \frac{38.1x}{8}$$

$$\text{Unprotected winding } x = \frac{125 \times 8}{38.1} = 26.24\%$$

Choice (A)

28. $i = 0.8 \text{ A/m,}$

$$r = 2 \times 0.05/1000 = 0.1 \times 10^{-3} \Omega/\text{m}$$

$$l = 1500 \text{ m}$$

$$I = il = 0.8 \times 1500 = 1200 \text{ A}$$

$$R = rl = 0.1 \times 10^{-3} \times 1500 = 0.15 \Omega$$

$$\text{Maximum voltage drop} = \frac{IR}{8} = \frac{1200 \times 0.15}{8} = 22.5 \text{ V}$$

Choice (A)

29. $\%X_A = \frac{50}{20} \times 30 = 75\%$

$$\%X_B = \frac{50}{30} \times 30 = 50\%$$

$$\text{Line current } I = \frac{50,000 \times 10^3}{\sqrt{3} \times 11 \times 10^3} = 2624.31 \text{ A}$$

$$\%X = \frac{X_A X_B}{X_A + X_B} = \frac{75 \times 50}{75 + 50} = 30\%$$

$$\text{Short circuit current } I_{SC} = I \times \frac{100}{\%X}$$

$$= 2624.31 \times \frac{100}{30}$$

$$= 8747.7 \text{ A}$$

Choice (A)

30. $I = \frac{20 \times 10^6}{\sqrt{3} \times 220 \times 10^3 \times 0.8} = 65.60 \text{ A}$

$$\text{Line loss} = 8\% \text{ of power delivered}$$

$$= 0.08 \times 20,000 = 1600 \text{ kW}$$

$$\Rightarrow 3I^2 R = 1600 \times 10^3$$

$$3 \times 65.60^2 \times R = 1600 \times 10^3$$

$$\text{Total resistance } R = 123.93 \Omega$$

$$r = 2 \Omega/\text{km}$$

$$\text{length of line} = \frac{R}{r} = \frac{123.93}{2} = 61.96 \text{ km}$$

Choice (A)

31. $P_m \sin \delta_o = 0.4 P_m \Rightarrow \delta_o = \sin^{-1}(0.4) = 23.57^\circ$

During fault the reactance is 400% of the value before the fault

$$r_1 = 0.25 \text{ and } r_2 = 0.6$$

$$\delta_c = \cos^{-1} \left[\frac{\left(\frac{P_s}{P_m} \right) (\delta_m - \delta_o) + r_2 \cos \delta_m - r_1 \cos \delta_o}{r_2 - r_1} \right]$$

$$\Rightarrow 0.4 P_m = 0.6 P_m \sin \delta_m \Rightarrow \delta_m = \sin^{-1} \left(\frac{0.4}{0.6} \right) = 41.81$$

$$\delta_m = 180 - 41.81 = 138.19$$

$$\delta_m = 2.412 \text{ rad}$$

$$\delta_c = \cos^{-1}$$

$$\left(\frac{0.4(2.412 - 0.411) - 0.6 \times 0.74547 - 0.25 \times 0.916}{0.6 - 0.25} \right)$$

$$= \cos^{-1}(0.354)$$

$$= 69.23^\circ$$

Choice (A)

32. Base MVA = 100 MVA

$$\text{Base kv} = 11 \text{ kV}$$

$$\text{The base current} = \frac{100 \times 1000}{\sqrt{3} \times 11} = 5248.63 \text{ A}$$

$$\text{The prefault voltage} = \frac{10}{11} = 0.909 \text{ p.u.}$$

$$\text{The fault impedance} = j0.4 \times \frac{j0.25}{j0.65} = j0.154 \text{ p.u.}$$

$$\therefore \text{The fault current} = \frac{0.909}{j0.154} = -j5.902 \text{ p.u.}$$

The full load current before the fault takes place

$$= \frac{30 \times 1000}{\sqrt{3} \times 10 \times 0.8} = 2165.06 \text{ A}$$

$$\text{P.U load current} = \frac{2165.06}{5248.63} = 0.412 \angle 36.8$$

The p.u fault current supplied by the generator

$$-j5.902 \times \frac{25}{65} = -j2.27 \text{ p.u.}$$

Choice (B)

33. The p.u fault current supplied by the motor

$$= -j5.902 \times \frac{4}{6.5} = -j3.632 \text{ p.u.}$$

Choice (A)

34. Inductance per phase $= L = \frac{X_L}{2\pi f} = \frac{8}{2\pi \times 50}$

$$= 0.025 \text{ H}$$

$$\text{Capacitance per phase } C = 0.05 \mu\text{F}$$

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.025 \times 0.05 \times 10^{-6}}}$$

$$= 4501.58 \text{ Hz}$$

Choice (B)

35. $t = \frac{1}{2f_n} = 1.11 \times 10^{-4} \text{ sec}$

$$\text{Average RRRV} = \frac{\text{Peak restriking voltage}}{\text{Time upto first peak}}$$

$$\frac{\sqrt{2} \times \frac{11}{\sqrt{3}}}{1.11 \times 10^{-4}} = 161.82 \times 10^3 \text{ kV/sec}$$

Choice (B)

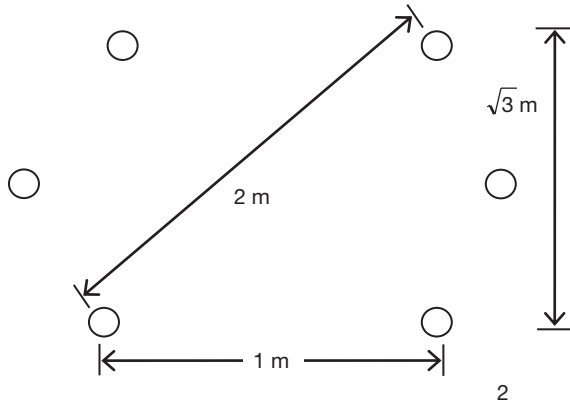
POWER SYSTEMS TEST 3

Number of Questions: 25

Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

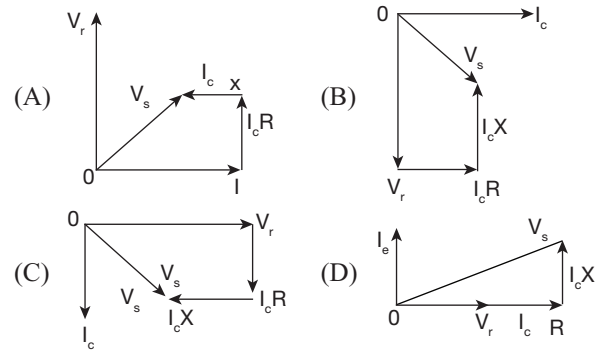
1. A sending end bus transfers power through the following transmission line to receiving end. The value of capacitance per ph per km in μF is _____ (if radius = 2cm).



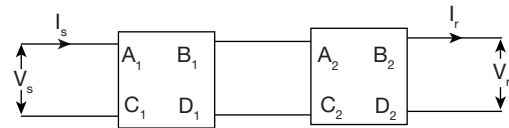
2. The velocity of travelling wave through cable at relative permittivity 4 is
 (A) $3 \times 10^8 \text{m/sec}$ (B) $1.5 \times 10^8 \text{m/sec}$
 (C) $0.75 \times 10^8 \text{m/sec}$ (D) $12 \times 10^8 \text{m/sec}$
3. Which of the following methods are used for improving the string efficiency of suspension type insulators
 (P) Grading of insulator discs
 (Q) Longer cross arm
 (R) Guard rings
 (A) Only P (B) Only Q
 (C) Both P and Q (D) P, Q and R
4. For transmitting power a solid conductor is replaced by bundle conductor consisting of five smaller conductors per phase. If the radius of each conductor in bundle conductor is 10mm, then the diameter of solid conductor is _____
 (A) 22.36 mm (B) 67.08 mm
 (C) 44.72 mm (D) 100 mm
5. If δ is the loss angle of the cable then the power factor of the cable is
 (A) $\cos \delta$ (B) $\sin \delta$
 (C) $\tan \delta$ (D) None of the above
6. Proximity effect is due to
 (A) The current flowing through its own conductor
 (B) The field produced by the current flowing through the adjacent (nearest) conductor
 (C) The load supplied by the conductor
 (D) The atmosphere surrounding the conductor
7. In Hydro power plant spill way is used as
 (A) to reduce pressure of water in penstock
 (B) to discharge excess of water in reservoir

- (C) to reduce water pressure at turbine valve
 (D) to discharge excess of water in surge tank

8. The phasor diagram of Ferranti effect is

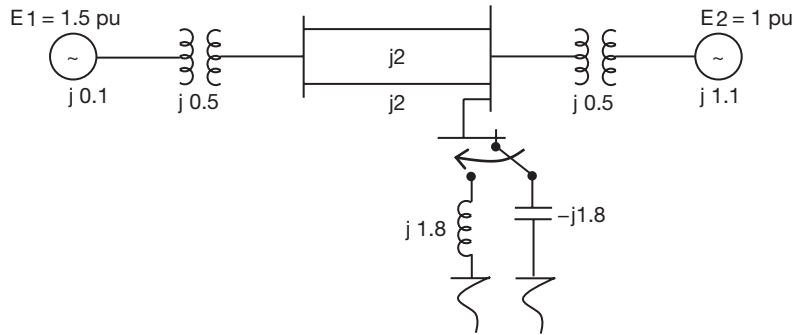


9. The resultant A, B, C, D parameters of the following network is

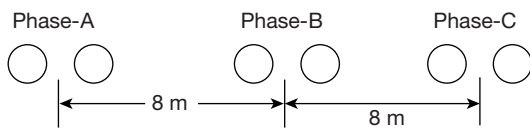


- (A) $A = A_1 A_2 + B_1 B_2$; $B = A_1 B_2 + A_2 B_1$;
 $C = B_1 C_2 + B_2 C_1$; $D = B_2 C_1 + D_1 D_2$
- (B) $A = A_1 A_2 + B_1 C_1$; $B = A_1 B_2 + B_1 D_1$;
 $C = A_2 C_1 + C_1 D_2$; $D = B_2 C_1 + D_1 D_2$
- (C) $A = A_1 A_2 + B_1 C_2$; $B = A_1 B_2 + B_1 D_2$;
 $C = A_2 C_1 + C_2 D_1$; $D = B_2 C_1 + D_1 D_2$
- (D) $A = A_1 A_2 + B_2 C_1$; $B = A_2 B_1 + B_1 D_2$;
 $C = A_2 C_1 + C_2 D_1$; $D = B_2 C_1 + D_1 D_2$
10. When multiplication factor (K) is greater than one in nuclear reactor means
 (A) Number of neutrons in one generation is greater than the number of neutrons of immediately preceding generation.
 (B) Number of neutrons in one generation is greater than the number of neutrons of next generations.
 (C) The control rods of nuclear reactor is taken out
 (D) None of the above
11. A 3-phase, 50Hz, transmission line has its conductors at the corners of an equilateral triangle with side 4m. The diameter of each conductor is 1.8 cm. The inductive reactance of the circuit per phase is
 (A) 0.398 m Ω /m (B) 1.27 mH/ph/m
 (C) 0.398 Ω /km (D) 3.98 m Ω /m
12. A two disc insulator string has each disc capacitance C and shunt capacitance between metal link and ground is 0.4C, the top most unit voltage is 11kV. The voltage across the unit near to the conductor is _____
 (A) 11 kV (B) 15.4 kV
 (C) 26.41 kV (D) 19.05 kV

13. For the system shown in figure below the value of change in the steady state stability limit in pu is _____ if change in switch is occurred

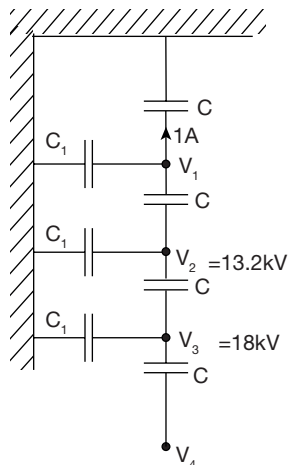


14. Calculate the capacitance between any two conductors in a 3 - phase, 3 - core metal sheathed cable. The capacitance between two conductors bunched with the sheath and the third conductor is $0.5\mu\text{F/km}$, the capacitance between bunched conductors and sheath is $0.65\mu\text{F/km}$
- (A) $0.214\mu\text{F/km}$ (B) $0.642\mu\text{F/km}$
(C) $0.963\mu\text{F/km}$ (D) $0.321\mu\text{F/km}$
15. A three-phase, 60Hz, conductors as shown in the figure with each sub conductor of the bundle has a diameter of 30mm and spacing between sub conductors is 0.4m



Calculate the line inductance per kilometer of the circuit.

- (A) $9.98 \times 10^{-4}\text{mH}$ (B) 0.998mH
(C) $99.86 \times 10^{-2}\text{mH}$ (D) $0.99 \times 10^{-3}\text{mH}$
16. Each conductor of a 3 phase HV line is suspended by four disc type insulators as shown in figure and the currents and voltages are also given. When $X_c = 1\Omega$. Calculate the total line voltage



- (A) 119 kV (B) 238 kV
(C) 68 kV (D) 136 kV

17. Match the following

	List - I		List II
P	Coolant Rods	1	A fissile fuel of a reactor
Q	Thorium - 232	2	Boron
R	Heavy water	3	Graphite
		4	A fertile fuel of nuclear reactor
		5	Used as a moderator

	P	Q	R		P	Q	R
(A)	2	1	5	(B)	2	1	3
(C)	4	1	3	(D)	4	1	5

18. A three phase, 765 kV, 60 Hz, 300 km, fully transposed line has the following positive sequence impedance and admittance:

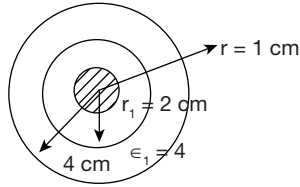
$$Z = 0.0165 + j0.3306 \Omega/\text{km}$$

$$Y = j4.674 \times 10^{-6} \text{ S/km}$$

Neglecting line losses, find the theoretical steady state stability limit for this line, given that surge impedance is 266.1Ω and wavelength is 5000 km

- (A) 2200 MW (B) 5974 MW
(C) 2987 MW (D) 8985 MW
19. A short transmission line having its line impedance angle as 30° . For getting zero regulation the receiving end lagging power factor angle should be
- (A) 30° (B) 60°
(C) 120° (D) 45°
20. A 3-phase, 132kV, 50Hz, system has 200 km long transmission line of diameter 1.05 cm, 2.5 delta spacing, air temperature 25° and approximate barometric pressure of 70 cm. calculate the corona power loss (Assume surface irregularity factor of 0.85)
- (A) 8kW / Phase / km
(B) 14.03kW / Phase / km
(C) 2.69kW / Phase / km
(D) 4.67kW / Phase / km
21. A 10 km transmission line is operating at a frequency of 1000 Hz. The transmission line is a
- (A) electrically short
(B) electrically long transmission
(C) any of the above
(D) both (a) and (b)

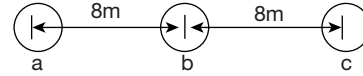
22. A string insulator has 5 units, the voltage across the bottom most unit is 35%. Its string efficiency is
 (A) 57.14% (B) 30.76%
 (C) 69.24% (D) 42.86%
23. Calculate the maximum safe working voltage of the cable as shown in the figure (Assume the peak voltage gradient in the air 35kV/cm)



- (A) 42.87 kV peak (B) 30.32 kV peak
 (C) 21.43 kV peak (D) 37.13 kV peak

24. Determine the reactance of a 130 kV overhead line where the conductors are located in a plane and the distance between two adjacent conductors is 4m. Conductor diameter is 20 mm
 (A) 0.82 Ω /km/phase (B) 0.205 Ω /km/phase
 (C) 0.41 Ω /km/phase (D) 0.615 Ω /km/phase

25. Three phase transposed line is as shown:



GMR of each conductor is 1.515 cm. Determine the inductance per phase per kilometer of the line.

- (A) 0.85 mH/km
 (B) 2.6 mH/km
 (C) 2.6 H/km
 (D) 1.3 mH/km

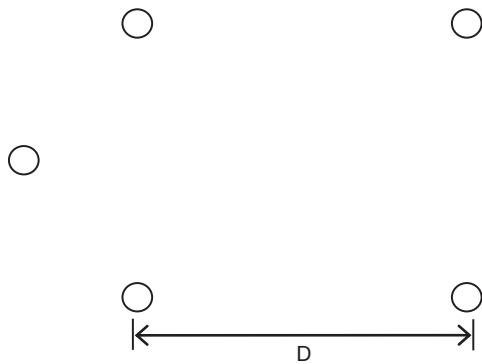
ANSWER KEYS

2. B 3. D 4. C 5. B 6. B 7. B 8. B 9. C 10. A 11. A
 12. B 14. D 15. B 16. A 17. D 18. B 19. B 20. D 21. B 22. A
 23. B 24. C 25. D

HINTS AND EXPLANATIONS

1. If we find the GMD and GMR for the given.

Configuration we will get as



$$GMD = 3^{1/4} \quad D = 3^{1/4}$$

$$GMR = \sqrt{r \times 2D} = \sqrt{2 \times 10^{-2} \times 1}$$

$$C/\text{ph/m} = \frac{2\pi\epsilon}{\ln \frac{GMD}{GMR}}$$

$$= \frac{2\pi \times 8.854 \times 10^{-12}}{\ln \frac{3^{1/4}}{\sqrt{0.02}}}$$

$$C_n = 0.0249 \mu\text{F/ph/km.}$$

2. Surge impedance loading α (Voltage)²

$$\frac{SIL_1}{SIL_2} = \frac{V_1^2}{4V_1^2}$$

$$\Rightarrow SIL_2 = 4SIL_1$$

Choice (B)

3. Choice (D)

4. Solid conductor Radius = R_1

One conductor Radius in Bundle conductor = r

$$\pi R^2 = n \pi r^2$$

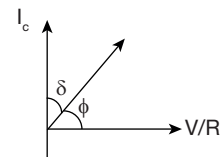
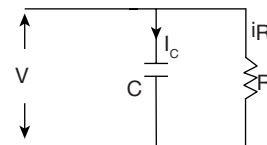
$$R = \sqrt{n} r = \sqrt{5} \times 10$$

$$= 22.36\text{mm}$$

Diameter of solid conductor = $2R$

$$= 44.72\text{mm} \quad \text{Choice (C)}$$

5. The cable equivalent diagram



The phasor diagram

$$\text{Power factor } \cos\phi = \cos(90 - \delta) = \sin\delta$$

Choice (B)

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6. Choice (B)

7. Spill way acts as safety valve for down and it can be used to discharge excess of water in reservoir.

Choice (B)

8. During Ferranti effect the sending voltage less than the receiving end voltage and capacitor current leading the voltage.
Choice (B)

9. Choice (C)

$$10. K = \frac{\text{Number of neutrons of any one generation}}{\text{Number of neutrons of immediately preceding generation}}$$

$\therefore K > 1$ means number of neutrons of any one generation is greater than the number of neutrons of immediately preceding generation. Choice (A)

$$11. L_{ph} = 0.2 \ln \left(\frac{4 \times 100}{0.7788 \times 0.9} \right)$$

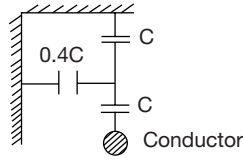
$$= 1.27 \times 10^{-6} \text{H/ph/m}$$

$$X_L = 1.27 \times 10^{-6} \times 2\pi \times 50$$

$$= 0.398 \text{m}\Omega/\text{m}$$

Choice (A)

$$12. K = \frac{\text{Shunt capacitance}}{\text{self capacitance}} = 0.4$$



The voltage across the unit near to the conductor

$= (1 + K)$ voltage across top most unit

$= (1 + 0.4) \times 11 = 15.4 \text{kV}$

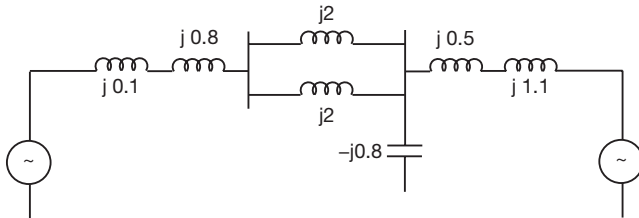
Choice (B)

$$13. \text{ With capacitor SSSL} = \frac{E_1 E_2}{X_{eq} b/n E_1 \& E_2}$$

$$X_{eq} = j1.9 + j1.6 + \frac{j1.9 \times j1.6}{-j1.8}$$

$$= j3.5 - j1.688$$

$$= j1.811 \text{ pu}$$



$$P_{\max_1} = \frac{1.5 \times 1}{1.81} = 0.828 \text{ pu}$$

With inductor connection

$$P_{\max_2} = \frac{E_1 E_2}{X_{eq}}$$

$$X_{eq} = j1.9 + j1.6 + \frac{j1.9 \times j1.6}{j1.8}$$

$$= j5.188$$

$$P_{\max_2} = \frac{1.5}{5.18} = 0.289 \text{ pu}$$

$$\text{Change in SSSL} = P_{\max_1} - P_{\max_2} = 0.539$$

14. Given $C_x = 3C_1 = 0.65$

$$C_y = C_1 + 2C_2 = 0.5$$

$$\text{Capacitance between any two conductors} = \frac{C_0}{2}$$

$$C_0 = \frac{3}{2}C_y - \frac{C_x}{6} = \frac{3}{2} \times 0.5 - \frac{0.65}{6}$$

$$= 0.642 \mu\text{F/km}$$

$$\text{Capacitance between any two conductors} = 0.321 \mu\text{F/km}$$

Choice (D)

15. $D = 8 \text{m}$

$$S = 0.4 \text{m}$$

$$R = 15 \text{mm}$$

$$\text{Self GMR of each group } D_{SR} = (r^1 s)^{1/2}$$

$$= \sqrt{0.7788 \times 15 \times 10^{-3} \times 0.4} = 0.0683 \text{m}$$

GMD between bundle conductors

$$= D_M^R = \sqrt[3]{(D_{AB} D_{BC} \cdot D_{CA})} = \sqrt[3]{8 \times 8 \times 16} = 10.07 \text{m}$$

Inductance of the bundle conductor line

$$L_R = 2 \times 10^{-7} \ln \left(\frac{D_M^R}{D_{SR}^R} \right)$$

$$= 2 \times 10^{-7} \ln \left(\frac{10.07}{0.0683} \right)$$

$$= 0.998 \text{mH/Km}$$

Choice (B)

16. Let $X_{C_1} = \frac{1}{k} \Omega$

$$V_2/V_1 = \frac{k+1}{1} \Rightarrow V_2 = V_1(k+1)$$

$$V_3/V_1 = k^2 + 3k + 1$$

$$\Rightarrow \frac{V_3}{V_2} = \frac{1+3k+k^2}{k+1} \Rightarrow \frac{18}{13.2} = \frac{1+3k+k^2}{k+1}$$

$$\Rightarrow k = 0.2$$

$$V_1 = V_2/k + 1 = 13.2/1.2 = 11 \text{ kV}$$

$$V_4 = V_1(1 + k^3 + 5k^2 + 6k) = 26.49 \text{ kV}$$

$$V_{\text{Total}} = (26.49 + 18 + 13.2 + 11) \text{ kV} = 68.69 \text{ kV per phase}$$

$$\therefore V_{\text{line}} = \sqrt{3} \times 68.69 \text{ kV} = 119 \text{ kV} \quad \text{Choice (A)}$$

17. Thorium - 232 is a fertile fuel and Heavy water is used as a moderator in nuclear reactor. Choice (D)

$$18. \text{ SIL} = \frac{(765)^2}{266.1} = 2199 \text{ MW}$$

$$l = 300, \lambda = 5000$$

$$P_{\max} = \frac{1 \times 1 \times 2199}{\sin\left(\frac{2\pi \times 300}{5000}\right)} = 5974 \text{ MW} \quad \text{Choice (B)}$$

19. For zero regulation power factor angle

$$\phi = \pi/2 - \text{impedance angle}$$

$$\phi = 90^\circ - 30^\circ$$

$$= 60^\circ \quad \text{Choice (B)}$$

$$20. \text{ Radius of conductor} = \frac{1.05}{2} = 0.525 \text{ cm}$$

$$\text{The ratio } \frac{d}{r} = \frac{2.5}{0.525} \times 100 = 476.19$$

$$\sqrt{\frac{r}{d}} = \sqrt{\frac{1}{476.19}} = 0.0458$$

$$\delta = \frac{3.92b}{273+t} = \frac{3.92 \times 70}{273+25} = 0.92$$

$$\text{Critical disruptive voltage } V_d = 21.1 \times 0.85 \delta r \ell_n \left(\frac{d}{r} \right)$$

$$= 21.1 \times 0.85 \times 0.92 \times 0.525 \ell_n (476.19)$$

$$= 53.41 \text{ kV line to neutral}$$

$$\text{Power loss} = 241 \times 10^{-5} \left(\frac{f+25}{\delta} \right) \sqrt{\frac{r}{d}} (V - V_d)^2$$

$$= 241 \times 10^{-5} \left(\frac{75}{0.92} \right) (0.0458) (76.21 - 53.41)^2$$

$$= 4.67 \text{ kW/phase/km} \quad \text{Choice (D)}$$

21. The transmission line may be electrically long or short. For a electrically short line indicates that BI is shorter angle in degree generally below 10 electrical degree.

$$\text{BI} = \frac{2\pi f}{|} \text{ radians}$$

$$= \frac{2\pi \times 1000}{3 \times 10^5} \times 10 \times \frac{180}{\pi} \text{ (elect degree)}$$

$$= 12^\circ.$$

$$\text{BI} > 10^\circ \Rightarrow \text{So electrically long line.} \quad \text{Choice (B)}$$

22. Voltage across bottom unit = $0.35 \times$ Voltage across the string

$$\text{String efficiency} = \frac{\text{Voltage across string}}{n \times 0.35 \times \text{Voltage across the string}}$$

$$= \frac{1}{5 \times 0.35} \times 100 = 57.14\% \quad \text{Choice (A)}$$

23. The maximum working voltage

$$V = r g_{1\max} \ell_n \left(\frac{r_1}{r_2} \right) + r_1 g_2 \max \ell_n \left(\frac{R}{r_1} \right) 1$$

$$g_{1\max} = \frac{q}{2\pi\epsilon_0 r}$$

$$g_{2\max} = \frac{q}{2\pi\epsilon_0 \epsilon_r r_1}$$

$$r g_{1\max} = g_{2\max} \epsilon_r r_1$$

$$\Rightarrow g_{2\max} = \frac{r g_{1\max}}{\epsilon_r r_1}$$

$$G_{2\max} = \frac{1 \times 34}{4 \times 2} = 4.375 \text{ kV/cm}$$

$$V = 1 \times 35 \ell_n \left(\frac{2}{1} \right) + 2 \times 4.375 \ell_n \left(\frac{4}{2} \right)$$

$$= 30.32 \text{ kV peak} \quad \text{Choice (B)}$$

24. $a_{12} = a_{23} = 4$; $a_{13} = 8$; $d/2 = 0.01 \text{ m}$

$$a = \sqrt[3]{4 \times 4 \times 8} = 5.04$$

$$x = 2\pi 50 \times 2 \times 10^{-4} \ell_n \left(\frac{5.04}{0.01} + \frac{1}{4} \right)$$

$$= 0.0628 \ell_n (504 + 0.25) = 0.41 \text{ } \Omega/\text{km/phase} \quad \text{Choice (C)}$$

25. $\text{GMD} = \sqrt[3]{8 \times 8 \times 16} = 10.0794 \text{ m}$

$$L = \frac{0.2 \times 10.0794}{1.515 \times 10^{-2}} = 1.3 \text{ mH/km} \quad \text{Choice (D)}$$

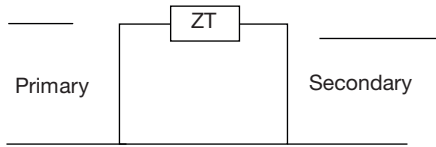
POWER SYSTEMS TEST 4

Number of Questions: 25

Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. A zero sequence network of a 3 - phase transformer as shown in the below figure
The corresponding transformer connection is



- (A) $Y : Y$ (B) $Y : Y$ with ground symbols
(C) $\Delta : \Delta$ (D) $\Delta : Y$ with ground symbols
2. By increasing the voltage level two times, the surge impedance loading of a transmission line is
(A) Increased by two times
(B) Increased by four times
(C) Remains same
(D) None of the above
3. The Bundle conductor
(A) reduces corona loss
(B) increases corona loss
(C) increases surge impedance
(D) reduces power loss
4. The daily energy produced in a thermal power station is 800Mwh and maximum demand on the station is 100MW. Then the load factor of the station is ———
(A) 0.80 (B) 0.34
(C) 0.12 (D) 0.22
5. Characteristic Impedance of a loss less transmission line is
(A) $\sqrt{\frac{r - j\omega C}{g + j\omega L}}$
(B) $\sqrt{\frac{r + j\omega L}{g + j\omega C}}$
(C) $\sqrt{(r + j\omega L)(g + j\omega C)}$
(D) $\sqrt{(r - j\omega C)(g + j\omega L)}$
6. Given equation
 $x_1 + x_1 x_2 = 10$
 $x_1 + x_2 = 6$
and initial estimates $x_1^{(0)} = 1; x_2^{(0)} = 1$, what are the x_1, x_2 values after third iteration of Gauss – seidel method?
(A) 1, 5 (B) 1, 1
(C) 5, 5 (D) 5, 1

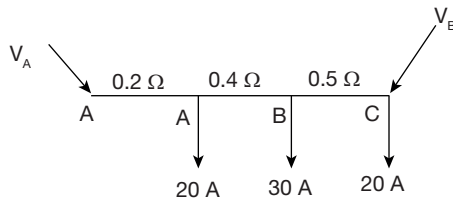
7. Calculate the surge impedance loading of a loss less 400kV, 3 - phase, 60Hz over head line if the average value of surge impedance is 500ohms.
(A) 500MW (B) 400MW
(C) 320MW (D) 300MW
8. Which of the following distribution system is more reliable
(A) Radial system
(B) Ring main system
(C) Inter connected system
(D) 2 - Wire system
9. Given system of equations
 $x_1^2 - 2x_1 - x_2 = 3$
 $x_1^2 + x_2^2 = 41$
What is the last element of the jacobian matrix for this system?
(A) $3x_2$ (B) -1
(C) $2x_1$ (D) $2x_2$
10. In power system load flow studies bus admittance matrix is
(A) Null matrix (B) Full matrix
(C) Sparse matrix (D) Diagonal matrix
11. Let the voltages in three phases be $V_a = 8\angle 0^\circ, V_b = 6\angle -90^\circ$ and $V_c = 16\angle 143.1^\circ$, the positive, negative and zero sequence voltages are
(A) $\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 9.8\angle 18.4^\circ \\ 2\angle 143.1^\circ \\ 4.3\angle -86.2^\circ \end{bmatrix}$
(B) $\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2\angle 143.1^\circ \\ 9.8\angle 18.4^\circ \\ 4.3\angle -86.2^\circ \end{bmatrix}$
(C) $\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2\angle 143.1^\circ \\ 4.3\angle -86.2^\circ \\ 9.8\angle 18.4^\circ \end{bmatrix}$
(D) $\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 4.3\angle -86.2^\circ \\ 9.8\angle 18.4^\circ \\ 2\angle 143.1^\circ \end{bmatrix}$
12. The surge impedance for the overhead line and cable are 450 ohms and 50 ohms respectively. If the 30kV surge is transmitted in to the cable calculate the surge travels along an overhead line
(A) 3kV (B) 150kV
(C) 24kV (D) 37.5kV

13. Z_{bus} for a given system is

$$Z_{bus} = j \begin{bmatrix} 0.24 & 0.14 & 0.2 & 0.2 \\ 0.14 & 0.2275 & 0.175 & 0.175 \\ 0.2 & 0.175 & 0.310 & 0.310 \\ 0.2 & 0.175 & 0.310 & 0.5 \end{bmatrix}$$

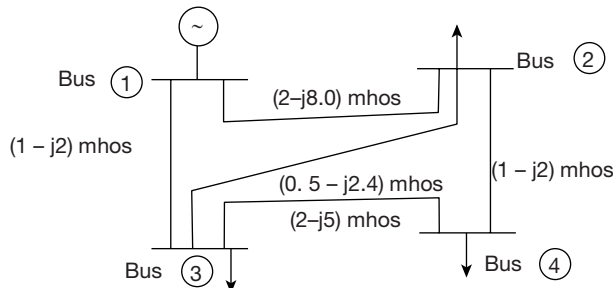
A belted three phase fault occurs at bus 4. What is the bus voltage during fault?

- (A) 0.84 p.u. (B) 1.24 p.u.
(C) 0 p.u. (D) 0.5 p.u.
14. The incremental cost characteristics of the two units in a plant are given by
 $I_{C1} = 0.2P_1 + 6.0Rs/Mwh$
 $I_{C2} = 0.4P_2 + 5.0Rs/Wh$
 Calculate the total load 500MW supplied by each plant
 (A) $P_1 = 168.34MW, P_2 = 331.66MW$
 (B) $P_1 = 250MW, P_2 = 250MW$
 (C) $P_1 = 331.66MW, P_2 = 168.34MW$
 (D) $P_1 = 300MW, P_2 = 200MW$
15. A d. c distribution system as shown in figure



The two ends of the feeder are fed by voltage sources such that $V_A - V_B = 5V$. Calculate the value of the voltage V_A for a minimum voltage of 300V at any point along the feeder

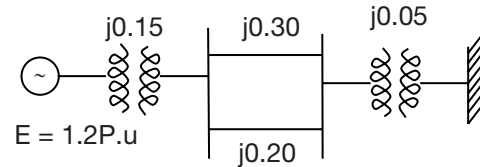
- (A) 298.90V (B) 287.27V
(C) 312.72V (D) 301.09V
16. For the following single line diagram, power flows as given in the below table then calculate the second bus voltage at the end of first iteration using Gauss – seidal method. (Assume acceleration factor is 1.5)



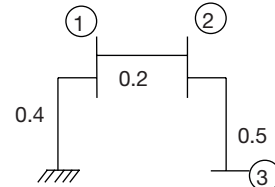
Bus code	P	Q	V
1	-	-	1.05
2	0.4	0.2	1.0
3	0.3	0.1	1.0
4	0.2	0.1	1.0

- (A) $1.0\angle-1.65^\circ$ (B) $1.0\angle-4.12^\circ$
(C) $1.0\angle-1.85^\circ$ (D) $1.0\angle1.65^\circ$

17. The 20 MVA generator supplying rated power at 0.8 pf (lag) when a fault reduces the electric power output by 50%. The power input to the generator may be assumed constant. Losses may be neglected. The accelerating torque at the time of the fault is _____ N-m.
18. When the following system is operating under normal condition, the maximum power transmitted from generating station to infinite bus is _____
 Infinite bus $V = 1.0\angle0^\circ$

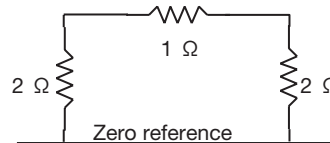


- (A) 1.71 p.u. (B) 1.2 p.u.
(C) 2.67 p.u. (D) 3.0 p.u.
19. An overhead line with surge impedance of 500Ω is terminated through a resistance of Z . A surge travelling over the line has a reflection coefficient of 0.8 then the value of Z is
 (A) 500Ω (B) 900Ω
 (C) 4500Ω (D) 1250Ω
20. The bus admittance matrix of the following network is _____ [For the given per unit impedance values]



- (A) $\begin{bmatrix} -5.25 & 5 & 0 \\ 5 & -7 & 2 \\ 0 & 2 & -2 \end{bmatrix}$ (B) $\begin{bmatrix} 5.25 & -5 & 0 \\ -5 & 7 & -2 \\ 0 & -2 & 2 \end{bmatrix}$
 (C) $\begin{bmatrix} -7.25 & 5 & 2 \\ 5 & -7 & 2 \\ 2 & 2 & -4 \end{bmatrix}$ (D) $\begin{bmatrix} -0.6 & 0.2 & 0 \\ 0.2 & -0.7 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$

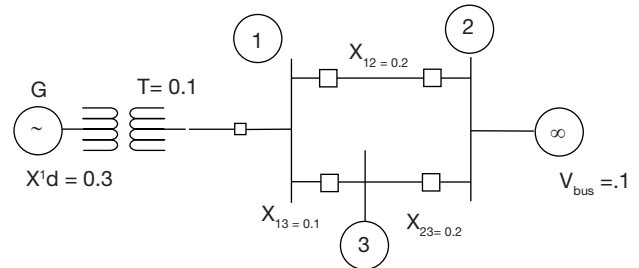
21. With the usual notations the impedance matrix for the system shown in the figure is



- (A) $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 3/8 & -1/3 \\ -1/8 & 3/8 \end{bmatrix}$
 (C) $\begin{bmatrix} 6/5 & 4/5 \\ 4/5 & 6/5 \end{bmatrix}$ (D) $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$

22. A $1-\phi$ transformer is having primary winding impedance in pu $Z_{1(\text{pu})} = (0.1 + j0.8)$, secondary winding impedance in pu $Z_{2(\text{pu})} = (0.2 + j1.8)$ pu. What is the per unit equivalent impedance referred to secondary side of transformer.
 (A) $0.1 + j0.8$ (B) $0.2 + j1.8$
 (C) $0.3 + j2.6$ (D) Insufficient data
23. The maximum voltage drop of uniformly loaded d, c distributor fed at one end is 40V. If the same distributor fed at both end with equal voltages then the corresponding maximum voltage drop is
 (A) 40 V (B) 20 V
 (C) 160 V (D) 10 V
24. The per unit impedance of an alternator corresponding to base values 132kV and 100MVA is 0.4pu. calculate the per unit impedance value for the base value of 66kV and 250MVA.

- (A) 0.8 (B) 0.2
 (C) 2 (D) 4
- 25.



Given the above 60 Hz system (all values in p.u), if the infinite bus draws real power of 1 p.u at 0.95 pf lagging, the internal voltage of the generator is _____.
 (A) 1.05V (B) 1.18V
 (C) 1.28V (D) 1.32V

ANSWER KEYS

1. C 2. B 3. A 4. B 5. B 6. D 7. C 8. C 9. D 10. C
 11. B 12. B 13. C 14. C 15. C 16. B 17. 50929.52 N-m 18. C 19. C
 20. B 23. D 24. D 25. C

HINTS AND EXPLANATIONS

1. Choice (C)
2. Surge impedance loading α (Voltage)²

$$\frac{SIL_1}{SIL_2} = \frac{V_1^2}{4V_2^2}$$

$$\Rightarrow SIL_2 = 4SIL_1$$
 Choice (B)
3. Choice (A)
4. Load factor = $\frac{\text{average demand}}{\text{maximum demand}}$

$$= \frac{800 \times 10^6}{24 \times 100 \times 10^6} = 0.34$$
 Choice (B)
5. Choice (B)
6. $x_1 = \frac{10}{1+x_2}$; $x_2 = 6 - x_1$
- | | |
|----------------------------------|----------------------------------|
| $x_1^{(1)} = \frac{10}{1+1} = 5$ | $x_1^{(2)} = \frac{10}{1+1} = 5$ |
| $x_2^{(1)} = 6 - 5 = 1$ | $x_2^{(2)} = 6 - 5 = 1$ |
- and the same for third iteration. Choice (D)
7. Surge impedance loading $P = \frac{V^2}{Z_c} = \frac{400 \times 400}{500}$

$$= 320 \text{ MW}$$
 Choice (C)
8. Choice (C)

9. $J = \begin{bmatrix} 2x_1 - 2 & -1 \\ 2x_1 & 2x_2 \end{bmatrix}$ Jacobian = $\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$
 Hence, last element = $2x_2$ Choice (D)
10. In load flow studies BUS admittance matrix is sparse matrix because most of the elements are zero.
 Choice (C)
11. $V_s = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} [V_p]$

$$\Rightarrow V_s = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 8 \angle 0^\circ \\ 6 \angle -90^\circ \\ 16 \angle -143.1^\circ \end{bmatrix}$$

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \angle 143.1^\circ \\ 9.8 \angle 18.4^\circ \\ 4.3 \angle -86.2^\circ \end{bmatrix}$$
 Choice (B)
12. The voltage in overhead line = $30 \times \frac{500}{2 \times 50}$

$$= 150 \text{ kV}$$
 Choice (B)

$$13. I_4 = \frac{V_4(0)}{Z_{44}} = \frac{1.0}{j0.5} = -j2 \text{ p.u.}$$

$$V_4(f) = V_4(0) - Z_{44} I_4(f) = 1 - j0.5 \times (-j2) = 0 \text{ p.u.}$$

(Since the bus is solidly connected to ground during fault voltage should anyway be zero volts)

Choice (C)

$$14. P_1 + P_2 = 500 \text{ MW} \quad \text{—————(1)}$$

$$0.2P_1 + 6 = 0.4P_2 + 5.0$$

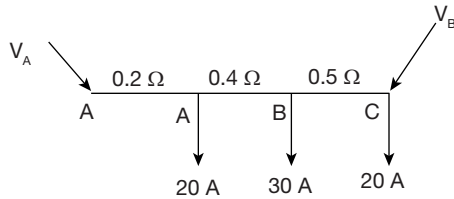
$$0.2P_1 - 0.4P_2 = -1$$

$$\Rightarrow 4P_2 - 2P_1 = 10 \quad \text{—————(2)}$$

Solving (1) and (2)

$$P_1 = 331.66 \text{ MW}, P_2 = 168.34 \text{ MW} \quad \text{Choice (C)}$$

15.



$$V_A - V_B = I \times 0.2 + (I - 20) \times 0.4 + (I - 50) \times 0.5$$

$$5 = I(1.1) - 8 - 25$$

$$I = \frac{38}{1.1} = 34.54 \text{ A}$$

Direction of current will be opposite in BC section

$$V_A = 300 + (34.54 \times 0.2) + (34.54 - 20) \times 0.4$$

$$= 312.724 \text{ V} \quad \text{Choice (C)}$$

$$16. V_2' = \frac{1}{y_{22}} \left[\frac{P_2 - jQ_2}{V_2^*} - y_{21} V_1^0 - y_{23} V_3^0 - y_{24} V_4^0 \right]$$

$$= \frac{1}{3.5 - j10.4} \left[\frac{-0.4 + j0.2}{1} - (-2 + j6)1.05 \right. \\ \left. - (-0.5 + j2.4)1.0 - (-1 + j2)1.0 \right]$$

$$= 1.0 \angle -1.6508^\circ$$

$$V_{2acc}' = (1.0 \angle -1.6508^\circ) + 1.5 (1.0 \angle -1.6508^\circ - 1.0 \angle 0^\circ)$$

$$= 1.0 \angle -4.123^\circ \quad \text{Choice (B)}$$

$$17. \text{Output power before fault } (P_s) = 20000 \times 0.8 \\ = 16000 \text{ kW.}$$

$$\text{Accelerating power } P_e = 0.5 \times 16000 \\ = 8000 \text{ kW}$$

$$P_a = P_s - P_e \\ = 16000 - 8000$$

$$T_a W_s = P_a \\ T_a = \frac{8000}{2\pi \times \frac{120 \times 50}{4 \times 60}}$$

$$= 50929.52 \text{ N-m.}$$

$$18. P_{\max} = \frac{EV}{X_{eq}}$$

$$X_{eq} = j0.15 + j0.05 + j0.25$$

$$= j0.45 \Omega$$

$$P_{\max} = \frac{1.2 \times 1.0}{0.45} = 2.67 \text{ p.u.}$$

Choice (C)

$$19. \text{Reflection coefficient} = \frac{Z - 500}{Z + 500}$$

$$0.8 = \frac{Z - 500}{Z + 500}$$

$$0.8Z + 0.8 \times 500 = Z - 500$$

$$400 + 500 = 0.2Z$$

$$Z = \frac{9000}{2} = 4500 \Omega$$

Choice (C)

$$20. Y_{11} = \frac{1}{0.4} + \frac{1}{0.2} = 5.25$$

$$Y_{12} = Y_{21} = -\frac{1}{0.2} = -5$$

$$Y_{13} = Y_{31} = 0$$

$$Y_{22} = \frac{1}{0.2} + \frac{1}{0.5} = 7$$

$$Y_{23} = Y_{32} = -\frac{1}{0.5} = -2$$

$$Y_{33} = \frac{1}{0.5} = 2$$

Choice (B)

$$21. [Z]_{1 \times 1} = [2]$$

$$[Z]_{2 \times 2} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$$

$$[Z]_{2 \times 2} = \left[\begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} - \left(\frac{1}{3+2} \right) \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \end{bmatrix} \right]$$

$$[Z]_{2 \times 2} = \left[\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} - \left(\frac{1}{5} \right) \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix} \right]$$

$$[Z]_{2 \times 2} = \left[\begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4/5 & 6/5 \\ 6/5 & 9/5 \end{bmatrix} \right]$$

$$[Z]_{2 \times 2} = \begin{bmatrix} 6/5 & 4/5 \\ 4/5 & 6/5 \end{bmatrix}$$

(or) By using Z parameters of two port networks.

Choice (C)

$$\begin{aligned}
 22. \quad Z_{02(\text{PU})} &= \left(\frac{Z_1 K^2 + Z_2}{Z_{\text{base}2}} \right) \\
 &= \left(\frac{Z_1 K^2}{Z_{\text{base}2}} + \frac{Z_2}{Z_{\text{base}2}} \right) [\text{In transformer } Z_{2\text{base}} = K^2 Z_{1\text{base}}] \\
 &= \left(\frac{Z_1}{\frac{Z_{\text{base}2}}{K^2}} + \frac{Z_2}{Z_{\text{base}2}} \right) \\
 &= \left(\frac{Z_1}{Z_{\text{base}1}} + \frac{Z_2}{Z_{\text{base}2}} \right) \\
 &= Z_{1(\text{PU})} + Z_{2(\text{PU})} \\
 \therefore Z_{01(\text{PU})} &= Z_{02(\text{PU})} = Z_{1(\text{PU})} + Z_{2(\text{PU})} \\
 &= (0.3 + j2.6) \quad \text{Choice (C)}
 \end{aligned}$$

23. The maximum voltage drop fed at both ends

$$= \frac{1}{4} \times \text{The maximum voltage drop fed at one end}$$

$$= \frac{1}{4} \times 40 = 10\text{V}$$

Choice (D)

24. The new per unit value

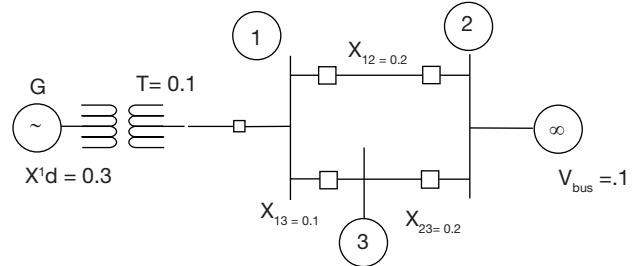
$$Z_{\text{new}} = 0.4 \left(\frac{MVA_{\text{new}}}{MVA_{\text{old}}} \right) \left(\frac{kV_{\text{old}}}{kV_{\text{new}}} \right)^2$$

$$\begin{aligned}
 &= 0.4 \left(\frac{250}{100} \right) \left(\frac{132}{66} \right)^2 \\
 &= 4\text{pu.}
 \end{aligned}$$

Choice (D)

25. Equivalent circuit;

$$X_{\text{eq}} = X_d^1 + X_{TR} + X_{12} \parallel (X_{13} + X_{23}) = 0.52 \text{ pu}$$



Current to the infinite bus is

$$I = \frac{P}{V_{\text{bus}} (p.f)} \angle \cos^{-1}(p.f) = \frac{1}{1 \times 0.95} \angle -\cos^{-1}(0.95)$$

$$= 1.0526 \angle -18.195^\circ \text{ pu}$$

$$E \angle \delta = V_{\text{bus}} + jX_{\text{eq}} I$$

$$= 1 \angle 0^\circ + j0.52(1.0526 \angle -18.195^\circ)$$

$$= 1.2812 \angle 23.946^\circ \text{ pu}$$

Choice (C)

POWER SYSTEMS TEST 5

Number of Questions: 25

Section Marks: 90

Directions for questions 1 to 25: Select the correct alternative from the given choices.

1. The self inductance of a long cylindrical conductor due to its internal flux linkages is KH/m . If the diameter of the conductor is double, then self inductance of the conductor due to its internal flux linkages would be
(IAS 2001)

(A) 0.5 K H/m (B) 1 K H/m
(C) 1.414 K H/m (D) 4 K H/m

2. Which of the following statements are correct for static relays

(P) Power consumption is low
(Q) Large overload capacity
(R) Greater sensitivity
(S) Fast in operation

(A) P, Q and R (B) P, Q and S
(C) P, R and S (D) P, Q, R and S

3. A system of 66kV, the line to ground capacitance is $0.02\mu F$ and the inductance is 4H. When the magnetizing current of 8A, find the voltage appearing across the pole in a circuit breaker

(A) 113.13kV (B) 1.13kV
(C) 113.13V (D) 56.56kV

4. A 25KVA, 2000V/500V single phase transformer with 20% impedance draws a steady short circuit line current of

(A) 50A (B) 12.5A
(C) 62.5A (D) 250A

5. Match the following:

	List – I		List – II
P	Translay relay	1	Generators
Q	Negative sequence relay	2	Bus bars
R	Differential relay	3	Long transmission lines

$\begin{matrix} P & Q & R \\ \text{(A)} & 3 & 5 & 1 \\ \text{(C)} & 4 & 5 & 2 \end{matrix}$
 $\begin{matrix} P & Q & R \\ \text{(B)} & 3 & 4 & 2 \\ \text{(D)} & 4 & 5 & 1 \end{matrix}$

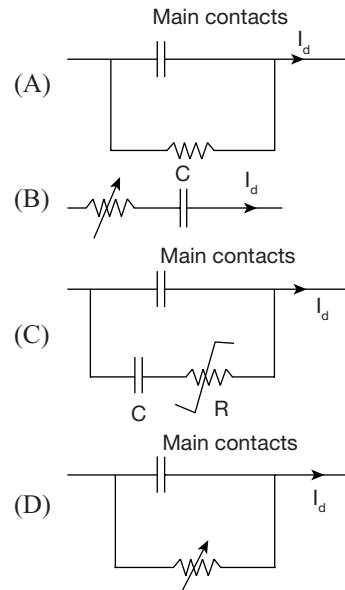
6. A 132 kV, 3-phase system circuit breaker has natural frequency of 20 KHz during fault. Calculate the time taken to reach peak value of restricting voltage.

(A) 0.5m sec (B) 0.05m sec
(C) 0.25m sec (D) 0.025m sec

7. A 3-phase, 132kV, oil circuit breaker is rated at 1500A, 5000MVA, 3S. The symmetrical breaking current is

(A) 1.5kA (B) 55.743kA
(C) 25.98kA (D) 21.86kA

8. Which of the following represent the schematic diagram of d. c. switch



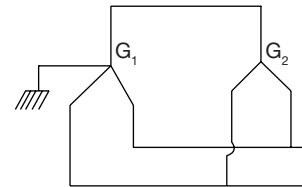
9. If the percentage reactive of the system up to fault point is 40% and the base MVA is 25. Then the short circuit KVA is

(A) 62.5 (B) 62500
(C) 625 (D) 10000

10. A 60Hz overhead line has line to earth capacitance of $2\mu F$. It is decided to use an earth fault neutralizer. Calculate the Inductance value to neutralize the capacitance.

(A) 442.09H (B) 4.42H
(C) 1.40H (D) 1.17H

11. Two 11kV, 40MVA, three phase star connected generators operate in parallel. The positive, negative and zero sequence reactances of each being $j0.20p.u.$, $j0.15p.u.$, $j0.08p.u.$ respectively.



Calculate the voltage of the healthy phase for a line to line fault on the terminals of the generators. [Assume fault impedance is equal to zero].

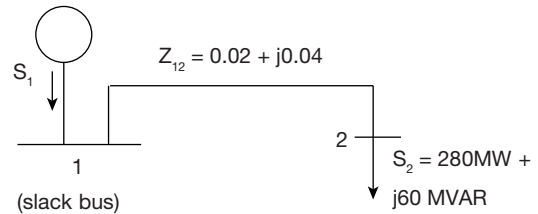
(A) 0.571p.u. (B) 0.285p.u.
(C) 0.642p.u. (D) 0.428p.u.

12. A four pole, 50Hz, 100MVA, turbo generator has a moment of inertia of $9 \times 10^3 Kg - m^2$, then calculate the inertia constant M value

(A) 0.11MJ/elect degree
(B) 1.11MJ/elect degree

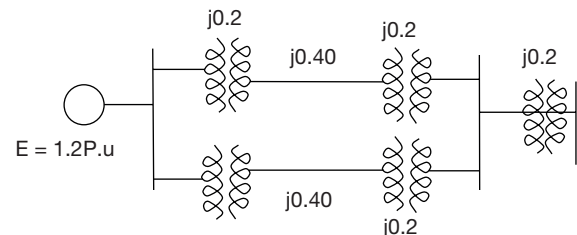
- (C) 0.012MJ/elect degree
(D) 111.03MJ/elect degree

13. A 60 Hz generator supplied 60% of maximum power to an infinite bus through a reactive network. A fault occurs, due to which total reactance increases by 400%. When the fault is cleared, Maximum power that can be delivered is 80% original maximum value. The critical clearing angle for this condition is
(A) 73.74° (B) 61.64°
(C) 38.02° (D) 45.64°
14. The inertia constants of two groups of machine which do not swing together are 4P.u and 6P.u. Then the equivalent inertia constant of the system is
(A) 6P.u (B) 4P.u
(C) 10P.u (D) 2.4P.u
15. A 50Hz, generator is delivering 40% of the power that it is capable of delivering through a transmission line to an infinite bus. A fault occurs that increases the reactance between generator and the infinite bus to 400% of the value before the fault. When the fault is isolated the maximum power that can be delivered is 80% of the original maximum value. The critical clearing angle will be
(A) 23.57° (B) 30°
(C) 53.54° (D) 85.92°
16. An interconnected 50 Hz power system consists of one area with three turbine generator units rated 1000, 750 and 600 MVA respectively. The regulation constant of each unit is $R = 0.05$ pu based on its own rating. Each unit initially operates at $\frac{1}{2}$ of its own rating, when the system load suddenly increases by 200 MW. The steady state drop in frequency is
(A) 0.212 Hz (B) 0.424 Hz
(C) 0.122 Hz (D) 0.425 Hz
17. A 60 Hz four pole turbo generator valid at 500 MVA, 22 kV has the inertia constant, $H = 7.5$ MJ/MVA. Find the angular acceleration, if electrical power developed = 400 MW and input – Rotational Losses = 740,000 Hp
(A) 21.89 rpm/s²
(B) 43.78 rpm/s²
(C) 437.8 rpm/s²
(D) 36.5 rpm/s²
18. In a short circuit test on a 66kV, 3 - phase system, the breaker gave the following results i.e power factor of the fault 0.5 and recovery voltage 0.90 of full line voltage. Then calculate the active recovery voltage.
(A) 48.5kV (B) 42.0kV
(C) 53.8kV (D) 38.0kV
19. For the given system, If $V_2 = 0.9 - j0.1$, What is the $|S_1|$.



- (A) 300 MVA (B) 100 MVA
(C) 316 MVA (D) 116 MVA

20. A substation supplies an industrial load of 8MW. If an capacitor bank of 5MVAR is installed to maintain the load power factor at 0.95. Then calculate the reactive power supplied by the substation to an industry
(A) 8.6 MVAR (B) 2.6 MVAR
(C) 12.6 MVAR (D) 7.6 MVAR
21. A 132 kV, 200Km transmission line is designed with a line reactance of 0.8Ω/km. Then calculate the transmission capacity of the line
(A) 216MW (B) 325MW
(C) 108MW (D) Insufficient data
22. Two alternators rated for 150MW and 250MW have governor drop characteristics of 4 percent from no load to full load, they are connected in parallel to share a load of 300MW. Calculate the load shared by 250MW alternator.
(A) 214.29MW (B) 85.71MW
(C) 125.34MW (D) 171.42MW
23. A 132kV, 50Hz system has line to ground capacitance of 0.02μF. If the resistance of 12KΩ is connected across the circuit breaker to eliminate the transient voltage in the system. Then calculate the inductance present in the system
(A) 5.76H (B) 4.8mH
(C) 11.52H (D) 9.6mH
24. For the following single line diagram, If the L-G fault occurs at middle of the one transmission line. Then what is the value of equivalent reactance during the fault



- (A) 1.4 (B) 0.4
(C) 0.1 (D) 0.23
25. Two generating units rated for 250 MW and 400 MW respectively have regulations of 6% and 6.4% respectively (from no load to full load). They operate in parallel and share 500 MW load. The power given by second (400 MW) unit is
(A) 200 MW (B) 400 MW
(C) 250 MW (D) 300 MW

ANSWER KEYS

2. C	3. A	4. C	5. B	6. D	7. D	8. C	9. B	10. D	11. D
12. C	13. B	14. D	15. D	16. A	17. D	18. B	19. C	20. D	21. C
22. A	23. C	24. D	25. D						

HINTS AND EXPLANATIONS

$$1. L_{\text{internal}} = \frac{\mu_o \mu_r}{8\pi} \text{ H/m}$$

Internal inductance is independent on diameter

Choice (B)

2. The static relays have small over load capacity.

Choice (C)

$$3. \text{ Voltage } V = i \sqrt{\frac{L}{C}}$$

$$= 8 \sqrt{\frac{4}{0.02 \times 10^{-6}}} = 113.13 \text{ kV}$$

Choice (A)=

$$4. \text{ Line current} = \frac{25000}{2000}$$

$$I_{\text{base}} = 12.5 \text{ A}$$

$$\text{Short circuit current} = \frac{\text{Base current}}{\text{percentage impedance}}$$

$$I_{\text{sc}} = \frac{1.25}{0.2} = 62.5 \text{ A}$$

Choice (C)

5. Translay relay is generally used for the protection of long transmission lines.

Choice (B)

$$6. \text{ The time taken to reach maximum value} = \frac{1}{2f_n}$$

$$= \frac{1}{2 \times 20 \times 10^3} = 0.025 \text{ m sec}$$

Choice (D)

$$7. \text{ The symmetrical breaking current} = \frac{MVA}{\sqrt{3} \times kV}$$

$$= \frac{5000 \times 10^6}{\sqrt{3} \times 132 \times 10^3} = 21.86 \text{ kA}$$

Choice (D)

8. Choice (C)

9. Base MVA = 25MVA

$$\text{Short circuit KVA} = \frac{\text{Base KVA}}{\text{Percentage reactance}}$$

$$= \frac{25,000}{0.4} = 62,500 \text{ KVA}$$

Choice (B)

$$10. \omega L = \frac{1}{3\omega C}$$

$$\omega L = \frac{1}{3 \times 2\pi \times 60 \times 2 \times 10^{-6}}$$

$$\omega L = 442.09$$

$$\Rightarrow L = 1.17 \text{ H}$$

Choice (D)

$$11. I_{a1} = \frac{E_a}{X_{1eq} + X_{2eq}} = \frac{1}{j0.10 + j0.75}$$

$$= -j5.714$$

$$V_{a1} = V_{a2} = E_a - I_{a1} X_{1eq}$$

$$= 1.0 - j(-5.714)(j0.10)$$

$$= 0.4286 \text{ P.u}$$

Choice (D)

$$12. \text{ Kinetic energy stored} = \frac{1}{2} J \omega^2$$

$$= \frac{1}{2} J \left(\frac{2\pi N}{60} \right)^2$$

$$= \frac{1}{2} \times 9 \times 10^3 \left(\frac{2\pi \times 1500}{60} \right)^2 = 111.03 \text{ MJ}$$

$$M = \frac{\text{Kinetic energy stored in MJ}}{180f}$$

$$= \frac{111.03}{180 \times 50} = 0.012 \text{ MJ/elect degree}$$

Choice (C)

$$13. P_{\text{max}} \sin \delta_o = 0.6 P_{\text{max}}$$

$$\Rightarrow \delta_o = 36.87^\circ \text{ (or } 0.6435 \text{ rad)}$$

$$r_1 = 0.25$$

$$r_2 = 0.8$$

$$r_2 P_{\text{max}} \sin \delta_{\text{max}} = P_m$$

$$P_m / P_{\text{max}} = 0.6$$

$$\Rightarrow \sin \delta_{\text{max}} = 0.6 / 0.8 = 0.75$$

$$\delta_{\text{max}} = 180 - 48.59^\circ = 131.41^\circ = 2.294 \text{ rad}$$

$$\cos \delta_{cr} = \{0.6(2.294 - 0.6435) +$$

$$0.8 \cos(131.4^\circ) - 0.25$$

$$\cos 36.87^\circ\} / (0.8 - 0.25) = 0.475$$

$$\delta_{cr} = \cos^{-1}(0.475) = 61.64^\circ$$

Choice (B)

14. If the inertia constant of two machines M_1 and M_2 which do not swing together.

$$\text{The equalent inertia constant } M = \frac{M_1 M_2}{M_1 + M_2}$$

$$= \frac{6 \times 4}{6 + 4} = 2.4 \text{ P.U}$$

Choice (D)

$$15. P_m \sin \delta_o = 0.4 P_m$$

$$\Rightarrow \delta_o = 23.57^\circ \text{ (or } 0.411 \text{ radians)}$$

$$r_1 = \frac{100}{400} = 0.25$$

$$r_2 = 0.8$$

$$\delta_c = \cos^{-1} \left[\frac{\left(\frac{P_s}{P_m} \right) (\delta_m - \delta_0) + r_2 \cos \delta_m - r_1 \cos \delta_0}{r_2 - r_1} \right]$$

$$\Rightarrow \delta_m = 30^\circ \text{ (or) } \delta_m = 180^\circ - 30^\circ = 150^\circ$$

$$\delta_m = 2.617 \text{ rad}$$

$$\delta_c = \cos^{-1} \left[\frac{0.4(2.617 - 0.411) - 0.8 \times 0.866 - 0.25 \times 0.9167}{0.55} \right]$$

$$= \cos^{-1}(0.071) = 85.92^\circ$$

Choice (D)

16. $R_{p.u.new} = R_{p.u.old} \frac{S_{new}}{S_{old}}$

$$R_{1\text{ new}} = R_{1\text{ old}} = 0.05$$

$$R_{2\text{ new}} = 0.05 \times \frac{1000}{750} = 0.06667$$

$$R_{3\text{ new}} = 0.05 \times \frac{1000}{600} = 0.08333$$

$$\beta = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{0.05} + \frac{1}{0.0667} + \frac{1}{0.0833}$$

$$\approx 20 + 15 + 12 = 47 \text{ pu}$$

$$\Delta f = \left(\frac{-1}{\beta} \right) \Delta P_m = \frac{-1}{47} \times 0.2 \text{ p.u.}$$

$$= -4.25 \times 10^{-3} \times 50 \text{ Hz}$$

$$= -0.212 \text{ Hz}$$

Choice (A)

17. Input power = $740000 \times 746 \times 10^{-6} = 552 \text{ MW}$

$$\frac{7.5}{180 \times 60} \frac{d^2 \delta}{dt^2} = \frac{552 - 400}{500} \text{ (swing equation)}$$

$$\Rightarrow \frac{d^2 \delta}{dt^2} = 437.8 \text{ elec degrees/S}^2$$

Since machine has 4 poles

$$\frac{d^2 \delta}{dt^2} = \frac{437.8}{2} = 218.9 \text{ mech degrees/S}^2$$

$$\text{or } 60 \times \frac{218.9}{360} = 36.5 \text{ rpm/s}^2$$

Choice (D)

18. The peak value of line to neutral voltage

$$= \frac{66}{\sqrt{3}} \times \sqrt{2} = 53.88 \text{ kV}$$

Recovery voltage = $0.9 \times 53.88 = 48.49 \text{ kV}$

Active recovery voltage = $V_m \sin \theta$

Power factor $\cos \phi = 0.5$

$$\Rightarrow \phi = 60^\circ$$

$$\sin 60 = 0.866$$

Active recovery voltage

$$= 48.49 \text{ kV} \times 0.866 = 42 \text{ kV}$$

Choice (B)

19. $Y_{12} = \frac{1}{0.02 + j0.04} = 10 - j20$

In p.u.; $S_2 = \frac{-(280 + j60)}{100} = -2.8 - j0.6 \text{ p.u.}$

$$I_{12} = y_{12}(V_1 - V_2) = (10 - j20)[1 + j0 - (0.9 - j0.1)]$$

$$= 3 - j1$$

$$I_{21} = -I_{12} = -3 + j1$$

$$S_{12} = V_1 I_{12}^* = (1 + j0)(3 + j1) = 3 + j1 \text{ p.u.}$$

$$= 300 \text{ MW} + j100 \text{ MVAR}$$

$$|S| = \sqrt{300^2 + 100^2} = \sqrt{100000} = 316.22 \text{ MVA}$$

Choice (C)

20. Load active power = $P = 8 \text{ MW}$

Load reactive power = Q

Capacitor reactive power = $Q_c = 5 \text{ MVAR}$

Reactive power supplied by Sub-station after capacitor installation = $Q - Q_c$

Power factor = $\cos \phi = 0.95$

$$\frac{Q - Q_c}{P} = 0.95$$

$$Q - 5 = 0.95 \times 8$$

$$Q = 12.6 \text{ MVAR}$$

MVAR supplied by substation after capacitor installation = 7.6 MVAR

Choice (D)

21. Transmission capacity = $\frac{(132 \times 10^3)^2}{0.8 \times 200}$

$$= 108.9 \text{ MW}$$

Choice (C)

22. Power supplied by 150MW unit is x

The percent drop in speed is $\frac{4x}{100}$

Percent drop in speed of 250MW is $\frac{4}{250}(300 - x)$

$$\frac{4x}{100} = \frac{4}{250}(300 - x)$$

$$2.5x = 300 - x$$

$$3.5x = 300$$

$$X = 85.71 \text{ m}$$

The load shared by 250MW alternator is $= 300 - 85.71$

$$= 214.29 \text{ MW}$$

Choice (A)

23. For eliminating the transient voltages in the system

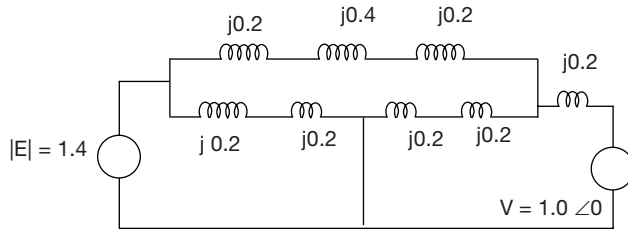
when $R = \frac{1}{2} \sqrt{\frac{L}{C}}$

$$12 \times 10^3 = \frac{1}{2} \sqrt{\frac{L}{0.02 \times 10^{-6}}}$$

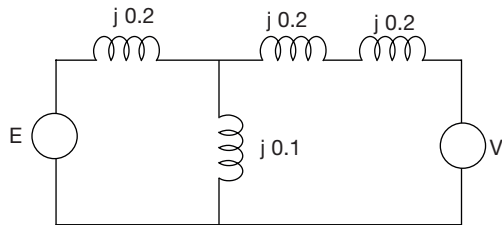
$$L = 11.52 \text{ H}$$

Choice (C)

24. The positive sequence reactance diagram during the fault



The reactance diagram simplified



The equivalent reactance during fault $X = 0.23$

Choice (D)

25. For common MVA base of 1000;

$$R_1 = \frac{1000}{250} (0.06) = 0.24$$

$$R_2 = \frac{1000}{400} (0.064) = 0.16$$

$R_1 P_1 = R_2 P_2$ (since frequency is equal)

$$\Rightarrow P_2 = \frac{R_1 P_1}{R_2}$$

$$\text{and } P_1 + P_2 = \frac{500}{1000} (\text{in p.u.})$$

$$P_2 = \frac{0.24 P_1}{0.16} \Rightarrow P_2 = 1.5 P_1$$

$$P_1 + 1.5 P_1 = 0.5$$

$$\Rightarrow P_1 = \frac{0.5}{2.5} = 0.2 \text{ p.u.}$$

$$P_2 = 1.5 \times 0.2 = 0.3 \text{ p.u.} = 0.3 \times 1000 = 300 \text{ MW}$$

Choice (D)